DEWETRON

TRION™/TRION3™ Series Modules

TECHNICAL REFERENCE MANUAL

WELCOME TO THE WORLD OF DEWETRON!

Congratulations on your new device! It will supply you with accurate, complete and reproducible measurement results for your decision making.
Look forward to the easy handling and the flexible and modular use of your DEWETRON product and draw upon more than 25 years of DEWETRON expertise in measurement engineering.

ISO9001

EN ISO 14001

THE MEASURABLE DIFFERENCE.

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Thank you!

Thank you very much for your investment in DEWETRON's unique data acquisition systems. These are top-quality instruments which are designed to provide you years of reliable service. This guide has been prepared to help you get the most from your investment, starting from the day you take it out of the box, and extending for years into the future.

This guide includes important startup notes, as well as safety notes and information about keeping your DEWETRON system in good working condition over time.

We strongly suggest that you read this entire manual, especially the safety and care sections, as well as to avoid damaging your DEWETRON system.

Intended use

What are TRION™/TRION3™ series modules?

TRION™/TRION3™ series data acquisition modules are used for measuring of different physical and / or electrical measurands (depending on model) and fit only into DEWE2/DEWE3 all-in-one instruments, mainframes and front-ends. Modules are available for binary / counter / timing inputs, for bus interfaces and for analog inputs.

The connection is depending on model or configuration and takes place via safety banana plugs, BNC connectors, D-SUB connectors, SMB connetcors, µdot connectors, LEMO® connectors or RJ-45 connectors.

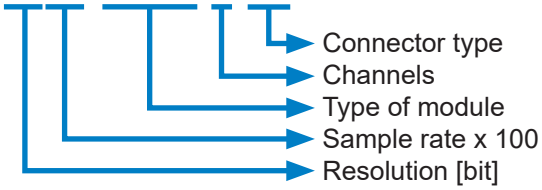
This product is designed for use under industrial environments, both indoor and outdoor. Any unspecified usage other than outlined in these specifications is not considered intended use.

The following general safety precautions must be observed during all phases of operation, service, and repair of this product. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of design, manufacture, and intended use of the product. DEWETRON GmbH assumes no liability for the customer's failure to comply with these requirements.

Labeling of TRION™ series modules

TRION™ series modules do have different names and contain important information about the module itself. The labeling gives information about the kind of module, channels, resolution, sample rate and type of connector used. The example below will show you on how to decode your TRION™ series module:

TRION-2402-dSTG-8-RJ



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CE-Certificate of conformity **C1**

Training

DEWETRON offers training at various offices around the world several times each year. DEWETRON headquaters in Austria have a very large and professional conference and seminar center, where training classes are conducted on a regular basis starting with sensors and signal conditioning, A/D technology and software operation. For more information about training services, please visit:
http://www.dewetron.com/services/dewetron-academy/

Dewetron Inc. in the USA also has a dedicated training facility connected to its headquarters, located in Rhode Island. For more information about training services in the US, please visit:
http://www.dewetron.us/service-support/system-training-usa/

Calibration

Every instrument needs to be calibrated at regular intervals. The standard norm across nearly every industry is annual calibration. Before your DEWETRON data acquisition system is delivered, it is calibrated at our DEWETRON headquater. Each of this system is delivered with a certificate of compliance with our published specifications. Detailed calibration reports from our calibration system are available for purchase with each order. We retain them for at least one year, so calibration reports can be purchased for up to one year after your system was delivered.

Support

DEWETRON has a team of people ready to assist you if you have any questions or any technical difficulties regarding the system. For any support please contact your local distributor first or DEWETRON directly.

For Asia and Europe, please contact:

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Web: http://www.dewetron.us

The telephone hotline is available
Monday to Friday between
08:00 and 04:30 EST

Service/Repair Policy

We are very sorry that your DEWETRON system is not operating properly. Our team is here to ensure that your DEWETRON product is returned to peak performance as quickly as possible.
Please help us to help you by following the RMA policy.

Some problems can be solved remotely by our support team. To facilitate a quicker resolution to the problem and save unnecessary shipping costs, we ask you to first have your problem investigated by our technical support before sending your product. Contact details for our support can be found on our website. Please describe the error accurately and with as much detail as possible. This helps expedite the repair process.

If a repair is necessary, please complete our online RMA form. You will then receive an RMA (Return Material Authorization) number and detailed instructions that identify where to ship the damaged product.

Please note: Products arriving at our repair department without RMA require follow-up calls and investigation, which lead to longer turnaround. Only the team of DEWETRON is allowed to perform any kinds of repairs to your system to assure a safe and proper operation in future.

Any spare parts (screws, backplanes, cables,...) must be obtained from DEWETRON only.

NOTICE

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Warranty Information

A copy of the specific warranty terms applicable to your DEWETRON product and replacement parts can be obtained from your local sales and service office.

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Parkring 4
8074 Grambach / Austria

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Safety convention

Observe precautions for handling electrostatic sensitive devices!

This icon denotes a caution, which advises you of precautions to take to avoid injury, data loss, or a system crash. When this symbol is marked on the product, refer to the technical reference manual.

Indicates hazardous voltages.

WARNING	*Calls attention to a procedure, practice, or condition that could cause bodily injury or death.*
CAUTION	*Calls attention to a procedure, practice, or condition that could possibly cause damage to equipment or permanent loss of data.*

Electromagnetic compatibility

Class A

Federal communications commission

This equipment has been tested and found to comply with the limits stated in EN55011 for Class A products. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

Operation of this equipment in a residential area is likely to cause harmful interference in which case the user is required to correct the interference at their own expense.

SAFETY INSTRUCTIONS

Safety instructions for all TRION™ modules

- The TRION™ modules may only be installed by experts.
- Read your manual before operating the module.
- Observe local laws when using the module.
- DO NOT operate the product in an explosive atmosphere or in the presence of flammable gases or fumes.
- DO NOT operate damaged equipment: Whenever it is possible that the safety protection features built into this product have been impaired, either through physical damage, excessive moisture, or any other reason, REMOVE POWER and do not use the product until safe operation can be verified by service-trained personnel. If necessary, return the product to a DEWETRON sales and service office for service and repair to ensure that safety features are maintained.
- Keep away from live circuits: Operating personnel must not remove equipment covers or shields. Procedures involving the removal of covers or shields are for use by service-trained personnel only. Under certain conditions, dangerous voltages may exist even with the equipment switched off. To avoid dangerous electrical shock, DO NOT perform procedures involving cover or shield removal unless you are qualified to do so.
- DO NOT try to service or adjust the module.
- DO NOT substitute parts or modify equipment.
- Before opening the instrument or computer (experts only) disconnect power!
- Don’t touch internal wiring (electrostatic damage is possible).
- Don’t use higher supply voltage than specified!
- Use only original plugs and cables for harnessing.
- Safety of the operator and the unit depend on following these rules.
- Using the board for medical applications only at owner’s risk

This product has left the factory in safety-related flawless and proper condition.
In order to maintain this condition and guarantee safety use, the user has to consider the security advices and warnings in this manual.

EN 61326-3-1:2008

IEC 61326-1 applies to this part of IEC 61326 but is limited to systems and equipment for industrial applications intended to perform safety functions as defined in IEC 61508 with SIL 1-3.

The electromagnetic environments encompassed by this product family standard are industrial, both indoor and outdoor, as described for industrial locations in IEC 61000-6-2 or defined in 3.7 of IEC 61326-1. Equipment and systems intended for use in other electromagnetic environments, for example, in the process industry or in environments with potentially explosive atmospheres, are excluded from the scope of this product family standard, IEC 61326-3-1.

Devices and systems according to IEC 61508 or IEC 61511 which are considered as “operationally well-tried”, are excluded from the scope of IEC 61326-3-1.

Fire-alarm and safety-alarm systems, intended for protection of buildings, are excluded from the scope of IEC 61326-3-1.

Environmental Considerations

Information about the environmental impact of the product.

Product End-of-Life Handling

Observe the following guidelines when recycling a DEWETRON system:

System and Components Recycling

Production of these components required the extraction and use of natural resources. The substances contained in the system could be harmful to your health and to the environment if the system is improperly handled at it's end of life! Please recycle this product in an appropriate way to avoid an unnecessary pollution of the environment and to keep natural resources.

This symbol indicates that this system complies with the European Union's requirements according to Directive 2002/96/EC on waste electrical and electronic equipment (WEEE). Please find further informations about recycling on the DEWETRON web site
www.dewetron.com

Restriction of Hazardous Substances

This product has been classified as Monitoring and Control equipment, and is outside the scope of the 2002/95/EC RoHS Directive. This product is known to contain lead.

GENERAL MODULE INFORMATION

General Module Specifications

Environmental (unless otherwise noted)

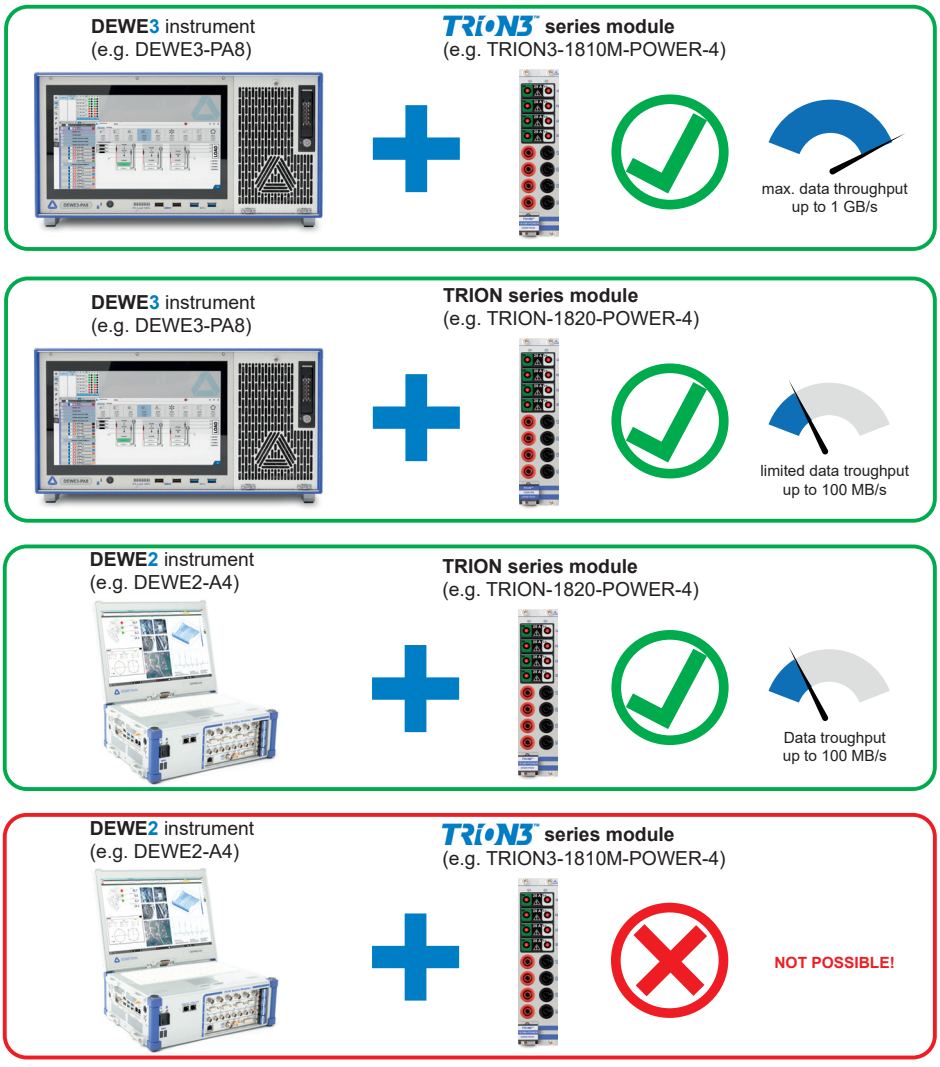
Operating temperature:	0 °C to +50 °C, down to -20 °C with prewarmed unit
Storage temperature:	-20 °C to +70 °C
Humidity (operating):	10 % to 80 %, non condensing 5 % to 95 % rel. humidity

TRION™/TRION3™ modules/OXYGEN compatibility

Since not all TRION modules are supported by previous OXYGEN versions, please refer to OXYGEN release notes and version history for TRION module compatibility.

DEWE2 - DEWE3 - TRION™ - TRION3™ hardware compatibility

In 2019 DEWETRON introduced a new family of data acquisition systems, the DEWE3 and TRION3™ express series. The DEWE3 chassis feature a PXIe hybrid backplane and support any TRION3™ series modules. Our DEWE3 chassis are also backward compatible and for simplification, the illustrations below will give you an overview of the hardware compatibility:



DEWE2/DEWE3 SYSTEMS OVERVIEW

Signal Conditioning Amplifier

A/D Conversion

Computer inside

Display

POWER ANALYZER

> up to 16 power channels
> 0.04 % power accuracy from 1 - 1000 Hz
> Mixed Signal Analyzer
> Multi-touch screen (up to 11.6”)

DEWE3-PA8

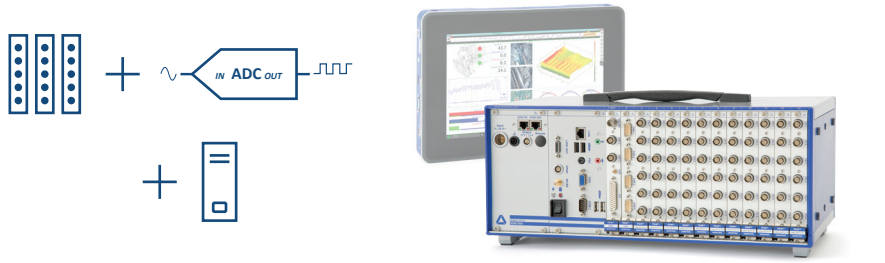
ALL-IN-ONE

> Built-in display
> Compact and flexible configuration
> Powerful PC inside for fast online displays and analysis
> Convenient for mobile applications

DEWE2-A4

MAINFRAME

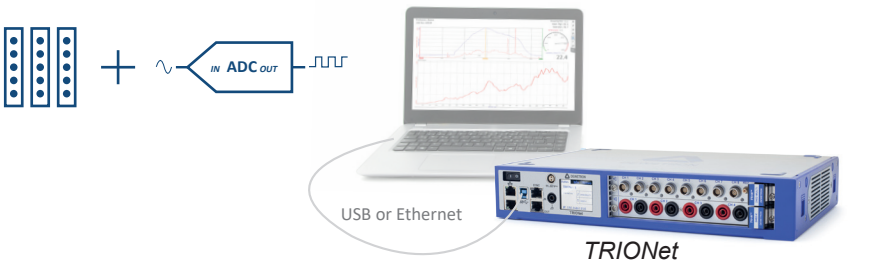
> Powerful PC inside for fast online displays and analysis
> Can be used with external display
> Very popular for applications where the instrument is installed in a poorly accessible place for the user



DEWE2-M13s

FRONT-END

> Used with an external computer
> Expansion for All-in-one or Mainframe instruments
> Multiple units can be daisy-chained
> Connected via USB3.0 or GBit-Ethernet



TRIONet

DEWE2/DEWE3 SYSTEMS OVERVIEW

Overview of DEWE2/DEWE3 systems

DEWE2 All-In-One Instruments

	DEWE2-A4	DEWE2-A7	DEWE2-A13
Slots for TRION™ acquisition modules	4	**7**	**13**

DEWE2 Mainframes

	DEWE2-M4s	DEWE2-M4	DEWE2-M7s	DEWE2-M7
Slots for TRION™ acquisition modules	4	4	7	7

	DEWE2-M13s	DEWE2-M13	DEWE2-M18
Slots for TRION™ acquisition modules	13	13	18

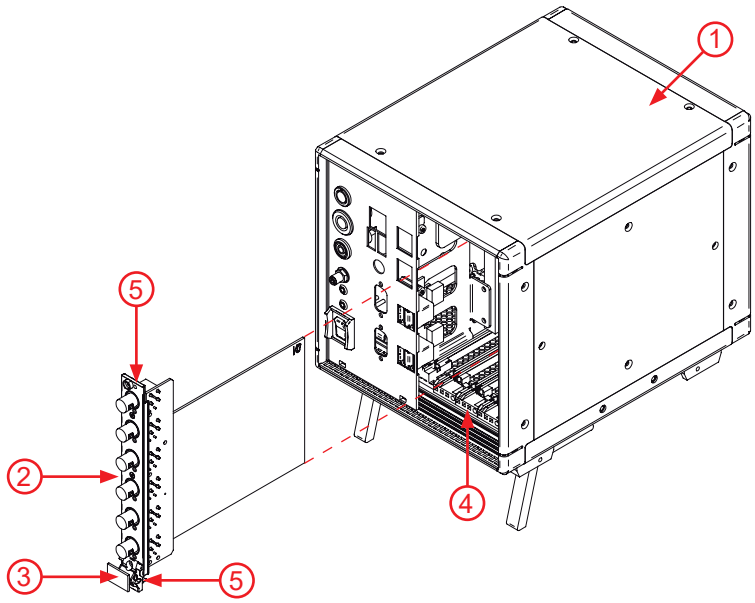
DEWE2 Front End

	TRIONet
Slots for TRION™ acquisition modules	2

DEWE2/DEWE3 Power Analyzer

	DEWE2-PA7	DEWE3-PA8
Slots for TRION™ acquisition modules	7	8

Installing a TRION™ module into the chassis



1 DEWE2/DEWE3 chassis
2 TRION™/TRION3™ series module
3 Injector/ejector handle
4 Module guides
5 Mounting screws

Step 1: Proper ESD precautions must be taken to avoid any damage to the unit.

Step 2: Power off and unplug all connected cables including sensors from the DEWE2/DEWE3 chassis and TRION™/TRION3™ series modules.

Step 3: Identify a supported TRION peripheral slot.
Some modules require a TRION STAR-slot. For more information please see chapter: "STAR-slot for TRION modules.

Step 4: Remove the filler panel of an unused TRION peripheral or STAR-slot.

Step 5: Place the module edges of the TRION™/TRION3™ modules into the module guides at the top and bottom of the chassis.

Step 6: Insert the TRION™/TRION3™ module to the rear of the chassis until a resistance appears.

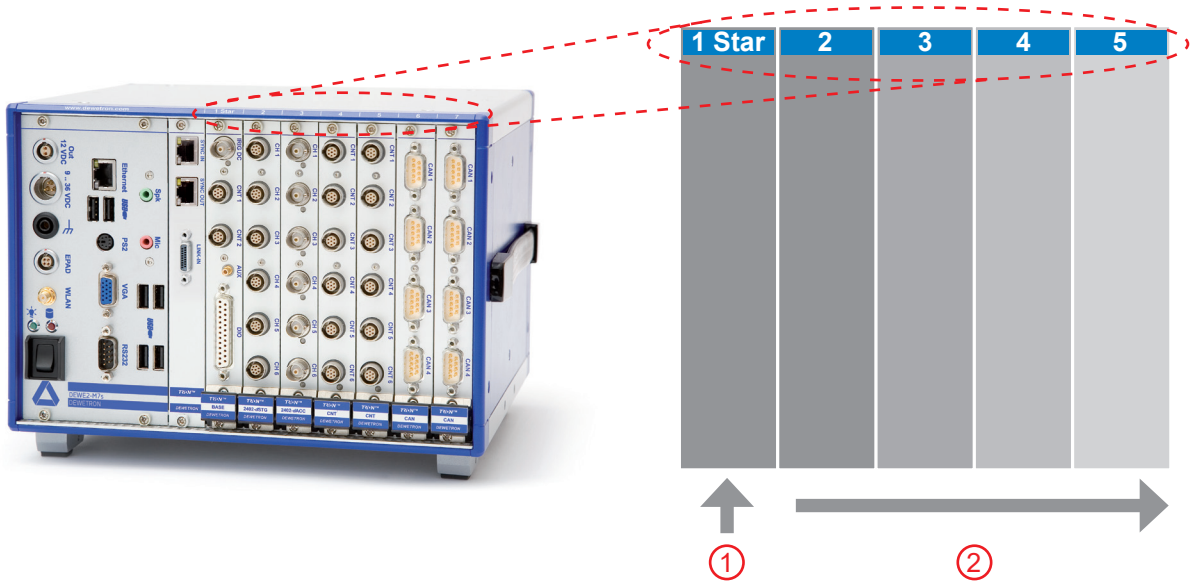
Step 7: Pull up on the injector/ejector handle to latch the device

Step 8: Secure the installed TRION™/TRION3™ front panel to the chassis using the mounting screws.

WARNING: *Unused TRION slots must not remain uncovered! Make sure to reinstall the filler panels of unused TRION slots to guarantee proper cooling of the installed modules. WARRANTY VOID if the modules overheat due to missing filler panels!*

INSTALL/REMOVE A TRION™ MODULE

STAR-slot for TRION timing/sync modules



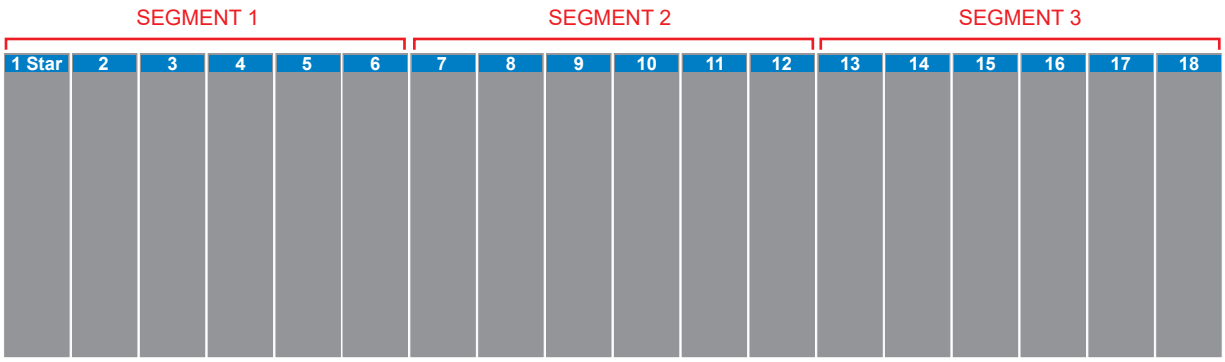
1 TRION system timing slot
2 TRION peripheral slots

The TRION system timing slot is either slot "1" or labelled as "STAR". This slot accepts any TRION™/ TRION3™ modules. Timing/Sync/GPS modules have to be installed in this slot!

NOTE: *If the system is equipped with a TRION-BASE, TRION-TIMING or TRION-VGPS-20/-100 module, it has to be installed in the "star slot"! This is the only slot a module is able to override the system 10 MHz clock with its PPS-synced 10 MHz, and thus providing the system with a timebase of higher accuracy.*

Owners of a DEWE2-M13/M13s/F13s/F18/M18

The slots for TRION™ series modules at the DEWE2-M13/M13s/F13s/F18/M18 are divided into three segments. The reason for this is because of the internal bus system to connect and collect all the data from the installed TRION™ series modules. The figure below will show you the three different segments of the DEWE2-M13/M13s/F13s/F18/M18:



NOTE: *A TRION™ module has to be installed in the first segment (Slot 1 to 6) for the following ones to become activated.*

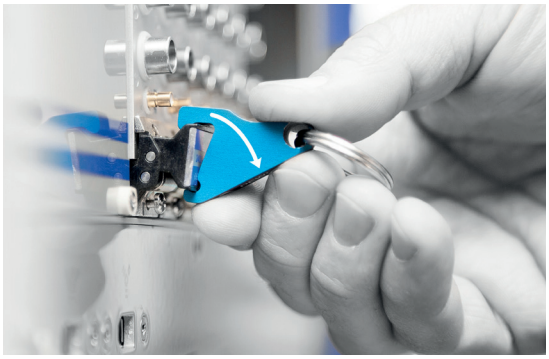
Removing a TRION™ module

Step 1: Proper ESD precautions must be taken to avoid any damage to the unit.

Step 2: Power off and unplug all connected cables including sensors from the DEWE2/DEWE3 chassis and TRION™/TRION3™ series modules.

Step 3: Loosen the screws at the top and bottom of the TRION™/TRION3™ module front panel.

Step 4: Pull down the injector/ejector handle or use the TRION™ extraction tool to release the module.



Step 5: Remove the TRION™/TRION3™ module and reinstall the filler panel for the empty slot.

WARNING:

Unused TRION slots must not remain uncovered! Make sure to reinstall the filler panels of unused TRION slots to guarantee proper cooling of the installed modules. WARRANTY VOID if the modules overheat due to missing filler panels!

INSTALL/REMOVE A TRION™ MODULE

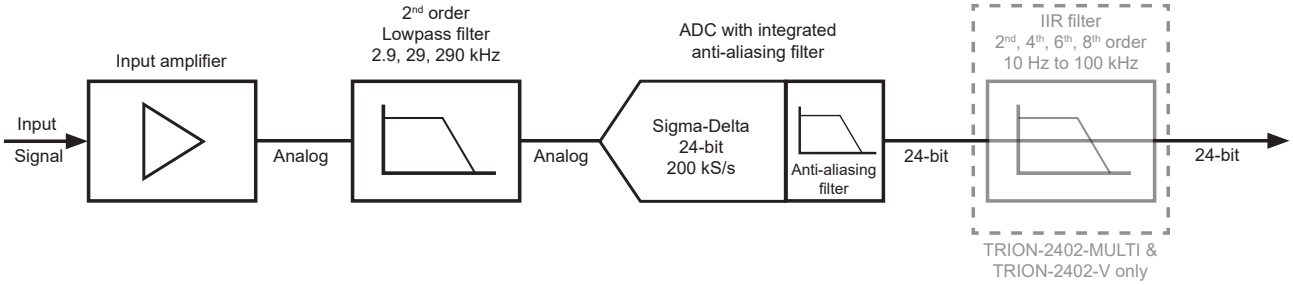
Note

Analog to digital conversion of TRION-2402 series modules

Any **TRION-2402 series** module uses up to 8 delta-sigma A/D converters. If you sample with a data rate of 102.4 kS/s, the ADC actually samples the input signal with 13.1072 MS/s (multiply the data rate with 128) and produces 1-bit samples which are applied to the digital filter. The filter expands the data to 24-bits and rejects signal parts greater than 51.2 kHz (Nyquist frequency). It also re-samples the data to the desired rate of 102.4 kS/s.

A 1-bit quantizer introduces many quantization errors to the signal. The 1-bit, 13.1072 MS/s from the ADC carry all information to produce 24-bit samples at 102.4 kS/s. The delta-sigma ADC converts from high speed to high resolution by adding random noise to the signal. In this way the resulting quantization noise is restricted to frequencies above 100 kHz. This noise is not correlated with the useful signal and is rejected by the digital filter.

TRION-2402 sample system architecture



Anti-aliasing filter

ADCs can only represent signals of a limited bandwidth. The maximum frequency you can represent is the half of the sampling rate. This maximum frequency is also called Nyquist frequency. The bandwidth between 0 Hz and the Nyquist frequency is called Nyquist bandwidth. Signals exceeding this frequency range can not be converted correctly.

For example, if the sample rate is 1000 S/s, the Nyquist frequency is 500 Hz. If the input signal is a 375 Hz sine wave, the resulting samples represent a 375 Hz sine wave. If a 625 Hz sine wave is sampled, the resulting samples represent a 375 Hz sine wave too. This happens because signals exceeds the Nyquist frequency (500 Hz). The represented frequency of the sine wave is the absolute value of the difference between the input frequency and the closest integer multiple of the sampling rate (in this case 1000 Hz).

Some examples:

Input sine wave 2280 Hz, sampling frequency 1000 Hz: |2280 - 2 * 1000| = 280 Hz

Input sine wave 3890 Hz, sampling frequency 1000 Hz: |3890 - 4 * 1000| = 110 Hz

The effect that frequencies above the Nyquist frequency appear as low frequency inside the Nyquist bandwidth is called aliasing. Signals which are not a pure sine wave can have many components (harmonics) above the Nyquist frequency. These harmonics are erroneously aliased back to the baseband, added to parts of the accurately sampled signal and produces a distorted data set. To block frequencies outside the Nyquist bandwidth, a lowpass filter is applied to the signal before it reaches the ADC.

If aliasing is caused by a clipped or overranged waveform, (exceeding the voltage range of the ADC) it can not be rejected with any filter. The ADC assumes the closest value to the actual value of the signal in its digital range when the signal is clipping. The result of clipping is also a sudden change in the signal slope and results in corrupt digital data with high-frequency energy. This energy is spread over the complete frequency spectrum and is aliased back into the baseband. Do not allow the signal to exceed the input range to avoid this.

ANALOG TO DIGITAL CONVERSION

Each input channel has three analog 2nd order low pass filters in front of the ADC. Depending on the sample rate the TRION™ board automatically selects the best suitable filter. The analog sampling rate of a sigma delta converter is much higher than the data output rate. This is called oversampling. That's why in contrast to a traditional anti aliasing filter the cut-off frequency of this analog filter could be very high. So there is almost no attenuation or phase shift within the bandwidth of interest because of this filter.

Sample rate	Max. analog filter bandwidth	Digital filter bandwidth	Oversampling
100 S/s to 1 kS/S	2.9 kHz	0.494 *fs	256 *fs
>1 k to 10 kS/S	29 kHz	0.494 *fs	256 *fs
>10 to 51.210 kS/S	290 kHz	0.494 *fs	256 *fs
>51.2 to 102.410 kS/S	290 kHz	0.5 *fs	128 *fs
>102.4 to 204.810 kS/S	290 kHz	0.38 *fs	64 *fs

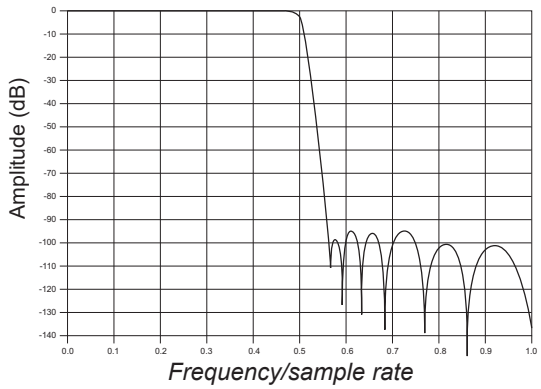
The programmable low pass filter allows aliasing free measurement also at low sample rates. For example at 100 S/s the oversampling frequency would be 25.6 kHz. That means that any noise signal around this frequency would be mapped into your measured data, but the 2.9 kHz filter attenuates most of this noise signal.

After conversion the 1-bit oversampled data is passed to a digital anti-aliasing filter. This filter has no phase error and an extremely flat frequency response. It also has an extremely sharp roll-off near the cutoff frequency (0.38 to 0.494 times the sample rate) and the rejection above 0.5465 times the sample rate is greater than 92 dB. The output stage of the digital filter finally resamples higher frequencies to 24-bit samples.

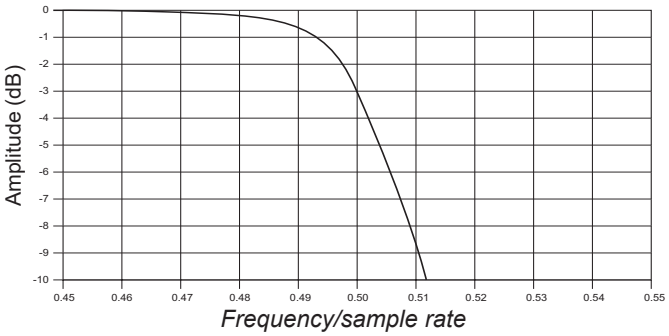
The digital filter passes only signal components within the Nyquist bandwidth or within the Nyquist bandwidth at 64, 128 or 256 times (depending on the sampling rate) the sample rate and multiples of it. The upstream analog lowpass filter rejects most noise near these multiples. The following diagrams show the frequency response of the input circuitry. In the following diagrams the y-axis shows the amplitude attenuation in dB whilst the x-axis shows the coefficient between signal frequency and sample rate.

Sample rate 100 S/s to 51.2 kS/s

Input frequency response

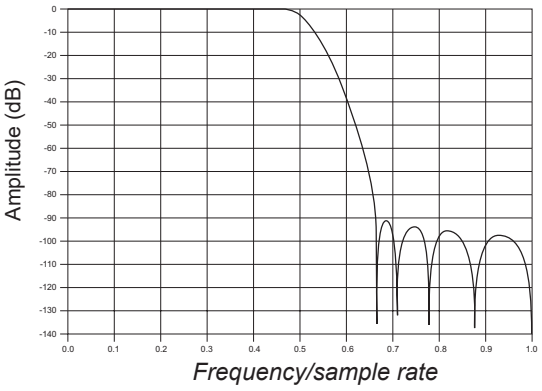


Input frequency response near the cutoff

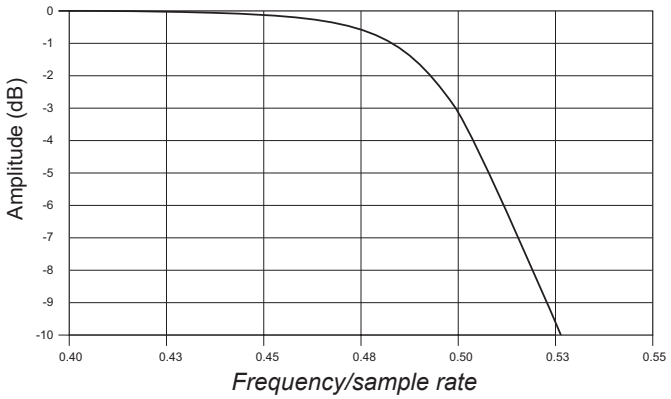


Sample rate 51.2 kS/s to 102.4 kS/s

Input frequency response

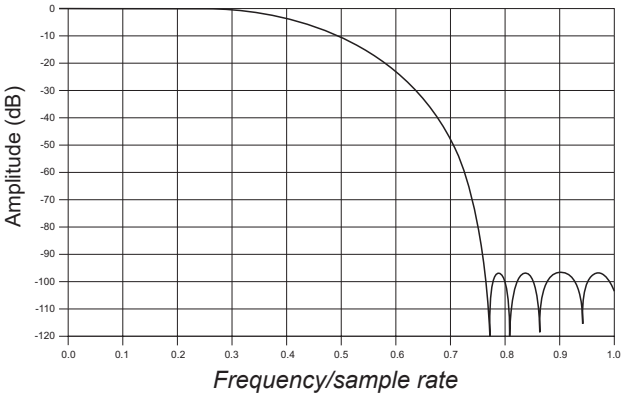


Input frequency response near the cutoff

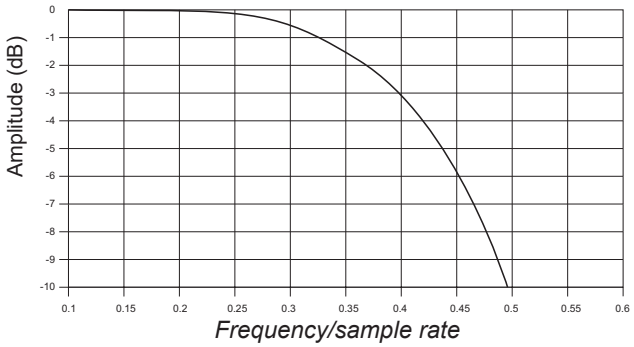


Sample rate 102.4 kS/s to 204.8 kS/s

Input frequency response



Input frequency response near the cutoff



IIR filtering

The TRION-2402-MULTI & TRION-2402-V series uses a combination of an analog and a digital filter. Whenever the user selects a filter, the TRION board configures both of them. The advantage of an digital IIR filter in comparison to an analog filter is, that they do not have any component related tolerances. So even for high filter orders, there is no signal delay or phase shift between the channels. The disadvantage is, they cannot distinguish between aliased signals and real signals. In order to block all aliased signal components, the cut-off frequency of the analog filter is set at least 3 times higher than the digital one.

Selected filter frequency	Auto selected analog filter
100 kHz to >10 kHz	290 kHz
10 kHz to >300 Hz	29 kHz
<300 Hz	2.9 kHz

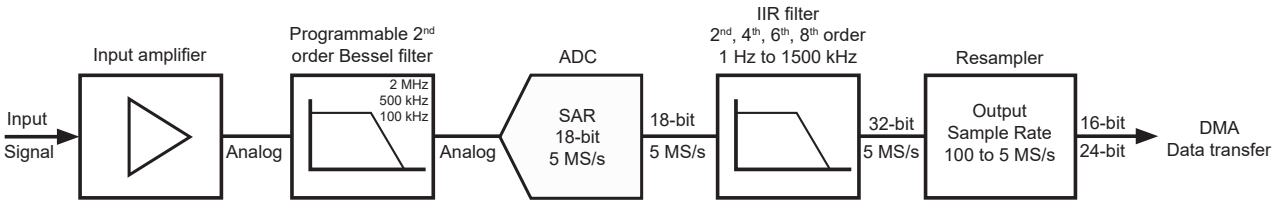
ANALOG TO DIGITAL CONVERSION

Note:

Analog to digital conversion of TRION(3)-18xx-MULTI series modules

The TRION(3)-18xx-MULTI series utilizes one 18-bit 5 MS/s ultra-low noise successive approximation ADC per channel. This allows measuring a signal bandwidth of up to 2 MHz with very low noise and excellent accuracy. By using frequency compensated analog amplifiers, the TRION(3)-18xx-MULTI series can keep the phase shift between the channels very low. The channel-to-channel phase error within one board is typically below 5 nsec. In other words, a 1 kHz sine wave signal can be measured with a maximum phase error of 0.002°.

TRION(3)-1850 sample system architecture



TRION(3)-18xx series overview

Versions					
Board	**Max. sample rate [S/s]**	**Resolution [bit]**	**Oversampling**	**Channels**	**Data Bus Bandwidth**
TRION3-1850-MULTI-4-D	5 M	16/24	yes	4	200 MB/s per card
TRION3-1850-MULTI-8-L0B	5 M	16/24	yes	8	200 MB/s per card
TRION3-1820-MULTI-4-D	2 M	16/24	yes	4	200 MB/s per card
TRION3-1820-MULTI-8-L0B	2 M	16/24	yes	8	200 MB/s per card
TRION-1850-MULTI-4-D	5 M	16/24	yes	4	80 MB/s per system
TRION-1820-MULTI-4-D	2 M	16/24	yes	4	80 MB/s per system

High speed mode (TRION(3)-1850-MULTI series only)

Between 2 and 5 MS/s the card is in the high-speed mode. The data transfer in that mode switches from 24 to 16-bit per channel per default. However if its needed user can still select a higher resolution. There is no need to do any rescaling in software. This is important to keep the CPU load on the host system as low as possible.

Oversampling mode

Most physical measurement applications do not require a signal bandwidth of 2 MHz. Therefore, the needed sample rate is much lower. In that case, traditional signal conditioning systems simply decrease the ADC clock to reduce the amount of data. The TRION(3)-18xx-MULTI series automatically activates the oversampling mode at 2.5 MS/s and below. That means by using averaging and filtering techniques the resolution increases while the sample rate drops. In theory, the benefit of oversampling is 0.5 bit more per every half of the sample rate. The table on the next page shows the performance of the TRION-1850-MULTI. To optimize the performance it is recommended, to use sample rates that can be divided from 5 MS/s without remainder. In that case, the ADC runs on the highest possible setting, 5 MS/s.

	Typical specifications TRION(3)-18xx-MULTI					
Signal-to-Noise-Ratio	100 mV		10 V		100 V	
Effective Number Of Bits	SNR	ENOB	SNR	ENOB	SNR	ENOB
Sample rate	[dB]	[Bit]	[dB]	[Bit]	[dB]	[Bit]
1 kS/s	114	18.7	125.6	20.6	123	20.1
10 kS/s	103	16.8	119.3	19.5	117	19.1
100 kS/s	94.7	15.4	110.5	18.1	106.4	17.4
200 kS/s	91.9	15	106.6	17.4	103.9	17
1000 kS/s	84.9	13.8	101	16.3	98.1	16
2000 kS/s	81.9	13	97	15.8	95.4	15.6

Filtering

The TRION(3)-18xx-MULTI series uses a combination of an analog and a digital filter. Whenever the user selects a filter, the card sets up both of them. The advantage of an IIR filter in comparison to an analog filter is, that they do not have any component related tolerances. So even for high filter orders, there is no signal delay or phase shift between the channels. The disadvantage is, they cannot distinguish between aliased signals and real signals. Therefore they normally are not used as anti-aliasing filter. Consequently an analog filter in front of the ADC is needed to minimize the aliasing components in the measured signal. However, in case of oversampling the aliased bandwidth of the ADC starts much higher than the output bandwidth. In that mode, the IIR Filter works as an excellent aliasing filter. The cut-off frequency of the analog filter is around

3 times higher than the digital one. This keeps the influence of components tolerances very small and as a result attenuates the aliased signal bandwidth of the ADC. In the high-speed mode this does not work. For measuring a 2 MHz input bandwidth, in high-speed mode the IIR filter could be deactivated. Consequently there is a 2 MHz 2nd order Bessel filter left, which is purely analog. The user has to be aware of aliasing effects in that mode.

Selected filter frequency	Analog filter
1500 kHz to >167 kHz	2 MHz
600 kHz to >167 kHz	2 MHz
167 kHz to >30 kHz	500 kHz
<30 kHz	100 kHz

Step Response

In comparison to delta-sigma converters, SAR converters have a perfect step response.
Whenever a square signal has to be measured in the time domain, SAR or flash ADCs should be used.
In order to eliminate the overshoot of the measurement system a Bessel filter characteristic should be used.

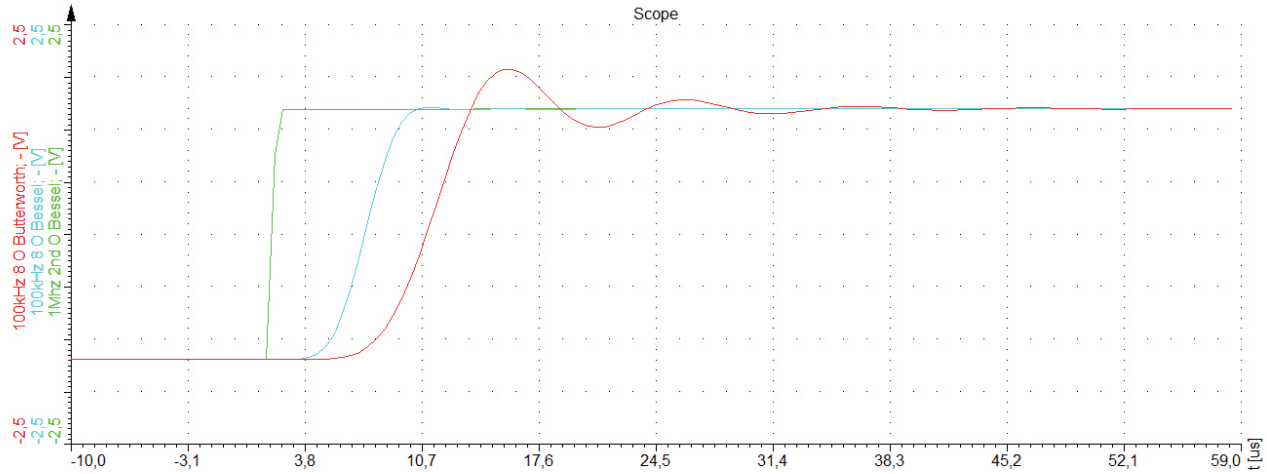


Figure 2: Step response

Aliasing protection

As mentioned in the filtering section, in the oversampling mode the IIR filter is used as an anti-aliasing filter. When using the TRION(3)-18xx-MUTLI series, DEWETRON recommends setting up the sample rate at least 3 times higher than the bandwidth of interest. Therefore usually the auto selected filter is 30 % of the sample rate. In case of 8th order Butterworth filter, this gives a stop-band attenuation of 60 dB (0.1 % of the input signal). If a higher stop-band attenuation is required they user may select a higher ratio between sample rate and filter frequency. For the highest aliasing protection, the user should also consider the TRION-2402 series.

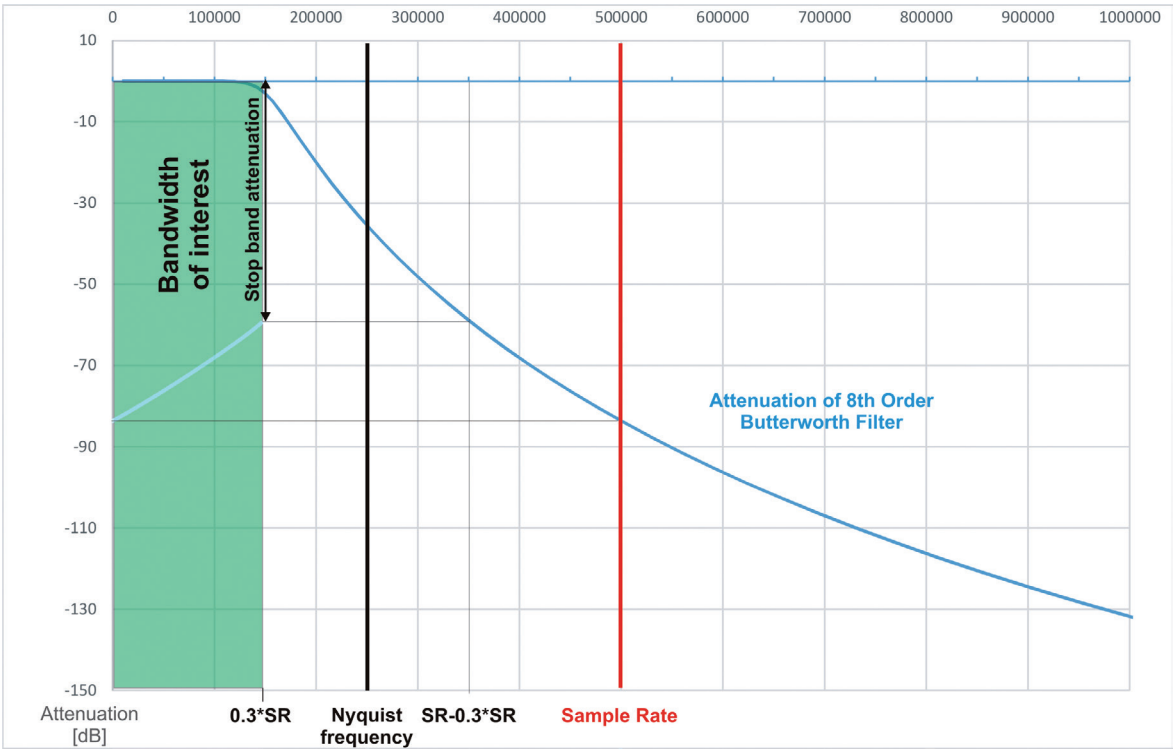


Figure 3: Stop-band attenuation

Stop-band attenuation		
Selected bandwidth	**8th order Bessel**	**8th order Butterworth**
[% of SR]	[dB]	[dB]
30	20	60
20	52	90
10	107	150

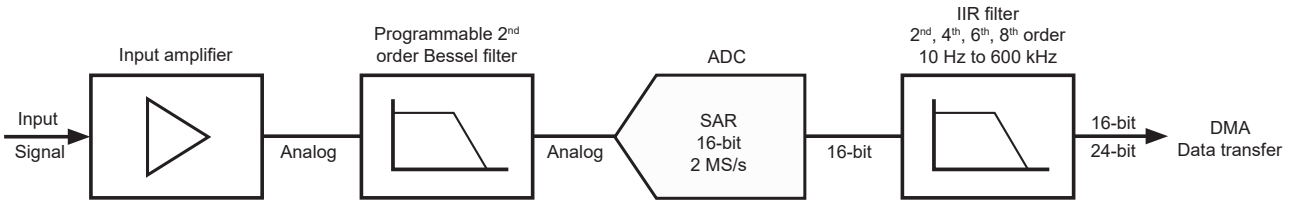
CPU load

The TRION(3)-18xx-MULTI series computes all mathematic calculations on the board itself. Consequently no calculation for filtering or calibration is required from the host CPU. This frees the CPU for other purposes, and allows a very high data throughput.

Analog to digital conversion of TRION-16xx/18xx series modules

The TRION-1620 series utilizes one 16 bit 2 MS/s ultra-low noise successive approximation ADC per channel. This allows measuring a signal bandwidth of up to 1 MHz with very low noise and excellent accuracy. By using frequency compensated analog amplifiers, the TRION-1620 series can keep the phase shift between the channels very low.
The channel-to-channel phase error within one board is typically below 20 nsec.
In other words, a 1 kHz sine wave signal can be measured with a maximum phase error of 0.007°.

TRION-1620 sample system architecture



TRION-16xx series overview

Versions						
Board	**Max. sample rate [S/s]**	**Resolution [bit]**	**Oversampling**	**IEPE**	**Counter**	**TEDS**
TRION-1603-LV-6-BNC	250 k	16	yes	-	-	-
TRION-1603-LV-6-L1B	250 k	16	yes	-	-	yes
TRION-1620-LV-6-BNC	2 M	16/24	yes	-	-	-
TRION-1620-LV-6-L1B	2 M	16/24	yes	-	-	yes
TRION-1620-ACC-6-BNC	2 M	16/24	yes	4/8 mA	CH0	-
TRION-1620-ACC-6-L1B	2 M	16/24	yes	4/8 mA	CH0	yes

High speed mode (TRION-1620 series only)

Between 1 and 2 MS/s the card is in the high-speed mode. The data transfer in that mode switches from 24 to 16-bit per channel. Since the gain and offset correction is done analog, not a single bit gets lost because of board calibration. There is also no need to do any rescaling in software. This is important to keep the CPU load on the host system as low as possible.

Oversampling mode (TRION-1620 series only)

Most physical measurement applications do not require a signal bandwidth of 1 MHz. Therefore, the needed sample rate is much lower. In that case, traditional signal conditioning systems simply decrease the ADC clock to reduce the amount of data. The TRION-1620 series automatically activates the oversampling mode at 1 MS/s and below. That means by using averaging and filtering techniques the resolution increases while the sample rate drops. In theory, the benefit of oversampling is 0.5 bit more per every half of the sample rate. The table on the next page shows the performance of the TRION-1620-ACC in the 2 V range. To optimize the performance it is recommended, to use sample rates that can be divided from 2 MS/s without remainder. In that case, the ADC runs on the highest possible setting, 2 MS/s.

Sample rate [kS/s]		2000	1000	500	250	200	100	10	1	0,1
Oversampling active		no	yes	yes	yes	yes	yes	yes	yes	yes
Filter		1 MHz	300 kHz	150 kHz	75 kHz	65 kHz	30 kHz	3 kHz	300 Hz	30 Hz
Data Word Length	[bit]	16	24	24	24	24	24	24	24	24
SNR	[dB]	88	93.2	99.5	102.7	103.7	106	111	123	130
Spurious Free Dynamic	[dB]	88	130	140	140	142	142	150	150	155
ENOB	[bit]	14.3	15.2	16.2	16.8	16.9	17.3	18.1	20.1	21.3

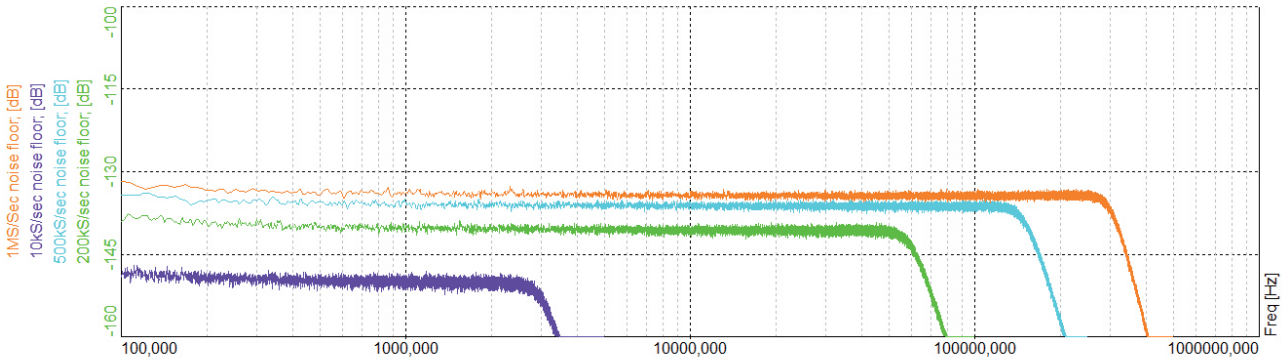


Figure 1: Noise floor of TRION-1620-ACC at different sample rates

Filtering

The TRION-16xx series uses a combination of an analog and a digital filter. Whenever the user selects a filter, the card sets up both of them. The advantage of an IIR filter in comparison to an analog filter is, that they do not have any component related tolerances. So even for high filter orders, there is no signal delay or phase shift between the channels. The disadvantage is, they cannot distinguish between aliased signals and real signals. Therefore they normally are not used as anti-aliasing filter. Consequently an analog filter in front of the ADC is needed to minimize the aliasing components in the measured signal. However, in case of oversampling the aliased bandwidth of the ADC starts much higher than the output bandwidth. In that mode, the IIR Filter works as an excellent aliasing filter. The cut-off frequency of the analog filter is around 3 times higher than the digital one. This keeps the influence of components tolerances very small and as a result attenuates the aliased signal bandwidth of the ADC. In the high-speed mode this does not work. For measuring a 1 MHz input bandwidth, in high-speed mode the IIR filter could be deactivated. Consequently there is a 1 MHz 2nd order Bessel filter left, which is purely analog. The user has to be aware of aliasing in that mode.

Selected filter frequency	Analog filter
600 kHz to >100 kHz	1 MHz
100 kHz to >30 kHz	333 kHz
30 kHz to >10 kHz	100 kHz
10 kHz to >3 kHz	33 kHz
<3 kHz	10 kHz

Step Response

In comparison to delta-sigma converters, SAR converters have a perfect step response.
Whenever a square signal has to be measured in the time domain, SAR or flash ADCs should be used.
In order to eliminate the overshoot of the measurement system a Bessel filter characteristic should be used.

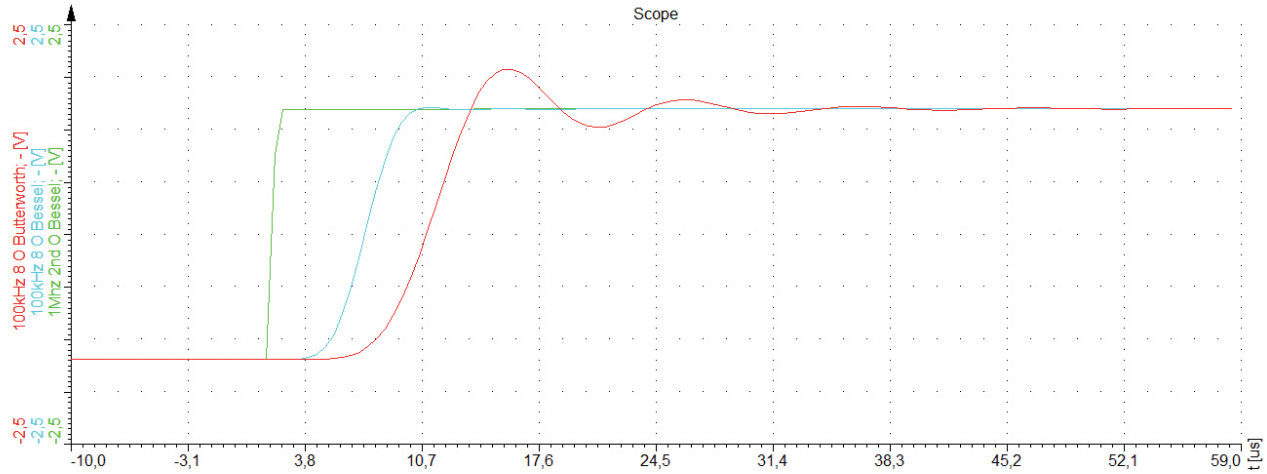


Figure 2: Step response

Aliasing protection

As mentioned in the filtering section, in the oversampling mode the IIR filter is used as an anti-aliasing filter. When using the TRION-16xx series, DEWETRON recommends setting up the sample rate at least 3 times higher than the bandwidth of interest. Therefore usually the auto selected filter is 30 % of the sample rate. In case of 8th order Butterworth filter, this gives a stop-band attenuation of 60 dB (0.1 % of the input signal). If a higher stop-band attenuation is required they user may select a higher ratio between sample rate and filter frequency. For the highest aliasing protection, the user should also consider the TRION-2402 series.

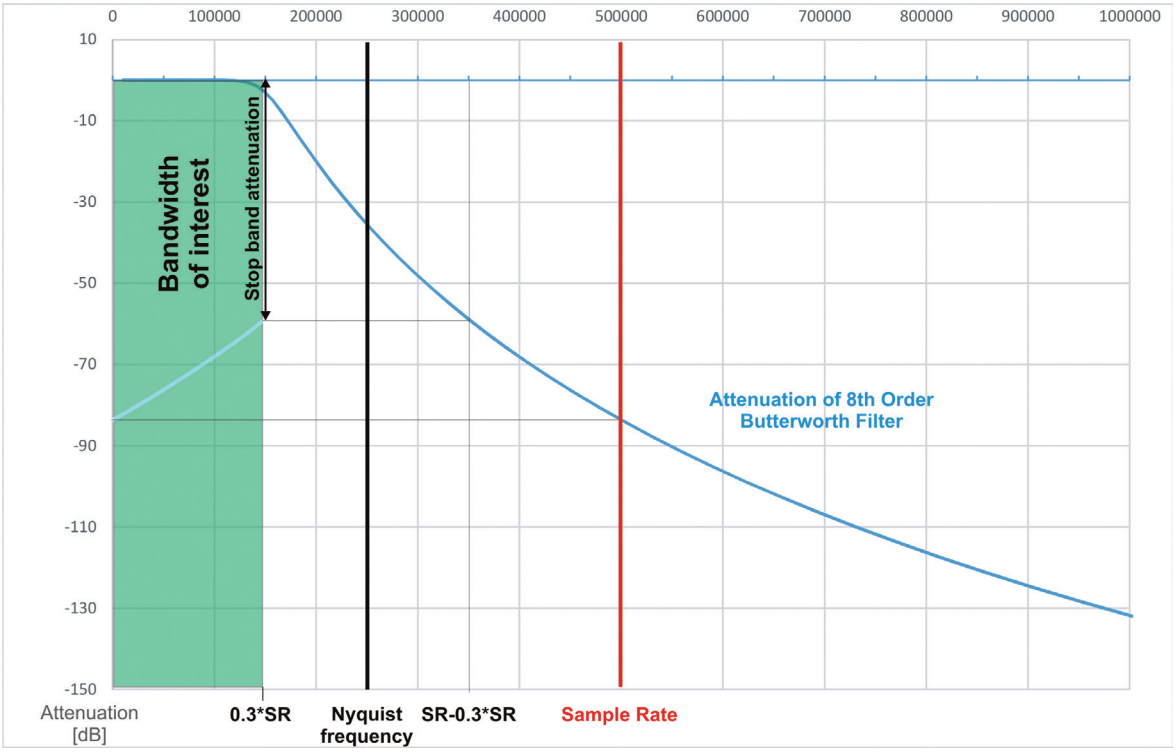


Figure 3: Stop-band attenuation

Stop-band attenuation		
Selected bandwidth	**8th order Bessel**	**8th order Butterworth**
[% of SR]	[dB]	[dB]
30	20	60
20	52	90
10	107	150

CPU load

The TRION-1620 series computes all mathematic calculations on the board itself. Consequently no calculation for filtering or calibration is required from the host CPU. This frees the CPU for other purposes, and allows a very high data throughput.

Isolated voltage input module

- Sampling: 2 MS/s per channel at 16-bit; 24-bit in oversampling mode
- ADC: Low noise, SAR
- Input types: Voltage: ±5 mV to ±100 V[1)]
  IEPE®: ±5 mV to ±50 V
  Current: 10 mA to 100 mA[4)]
- Isolation: 1.5 kV

Module specifications

TRION-1620-series specifications										
Input channels		TRION-1620-LV-6-BNC: 6 channels BNC; voltage input TRION-1620-ACC-6-BNC: 6 channels BNC; voltage input; IEPE®; 1 counter TRION-1620-LV-6-L1B: 6 channels 1B LEMO; voltage input; 1 to 28 V sensor supply; TEDS TRION-1620-ACC-6-L1B: 6 channels 1B LEMO; voltage input; IEPE®; 1 counter; sensor supply; TEDS								
Sampling Rate / Resolution										
High speed mode		>1 to 2 MS/s, 16-bit								
Over Sampling mode		100 S/s to 1MS/s, 24-bit								
Data Transfer		16-bit / 24-bit								
ADC type		SAR (Successive Approximation Register)								
Data rate DMA transfer		6 analog channels: max 24 MB/s; 1 x counter: max. 16 MB/s								
Input ranges	Voltage	±5, ±10, ±20, ±50, ±100, ±200, ±500 mV, ±1 V, ±2 V, ±5 V, ±10 V, ±20 V, ±50 V, ±100 V[1)],								
	IEPE®	±5, ±10, ±20, ±50, ±100, ±200, ±500 mV, ±1 V, ±2 V, ±5 V, ±10 V, ±20 V, ±50 V								
	Current[4)]	10, 20, 50, 100 mA								
Input noise (5 mV range)		0 to 10 Hz : 1.5 μV_{pp} Noise density: 6.4 nV/SQRT(Hz)								
Input impedance		1 MΩ shunted by 18 pF								
Current input		Internal 10 Ω shunt; max. 100 mA protected with resettable fuse								
Input bias current		<1 nA								
Input coupling		DC; AC: 0.16Hz[2)]								
Accuracy[3)]	Voltage	DC to 1 kHz: ±0.02 % of reading ± 0.02 % of range ±20 µV >1 kHz to 5 kHz: ±0.2 % of reading ± 0.02 % of range ±20 µV >5 kHz to 10 kHz: ±0.5 % of reading ± 0.02 % of range ±20 µV >10 kHz to 50 kHz: ±1.00 % of reading ± 0.02 % of range ±20 µV >50 kHz to 100 kHz: ±3.00 % of reading ± 0.02 % of range ±20 µV								
	Current[4)]	DC to 1 kHz: ±0.1 % of reading ± 0.02 % of range ±10 µA >1 kHz to 5 kHz: ±0.2 % of reading ± 0.02 % of range ±10 µA >5 kHz to 10 kHz: ±0.5 % of reading ± 0.02 % of range ±10 µA >10 kHz to 50 kHz: ±1.00 % of reading ± 0.02 % of range ±10 µA >50 kHz to 100 kHz: ±2.00 % of reading ± 0.02 % of range ±20 µA								
Gain drift		typical 10 ppm/°C max. 20 ppm/°C								
Offset drift		typical 0.3 µV/°C + 10 ppm of range/°C, max 15 µV/°C + 20 ppm of range/°C								
Linearity		typical 0.01 %								
Input configuration Isolation impedance Isolation voltage		Isolated Isolation resistance >1 GΩ; Isolation capacitance typically 15 pF 1500 V								
IEPE® excitation[2)]		4 mA, 8 mA ±10 % @ 1 % ±1 mV accuracy @ 24V compliance voltage								
Voltage excitation[4)]		1 to 28 V @ 1 % ±1 mV accuracy freely programmable (max. 100 mA, max. 1 W) per channel								
Typical Signal-to-noise ratio, Spurious free SNR, Effective number of Bits[5)]		20 mV range			2 V range			100 V range		
		SNR	SFDR[6)]	ENOB[7)]	SNR	SFDR[6)]	ENOB[7)]	SNR	SFDR[6)]	ENOB[7)]
	Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]
	0.1 kS/s	104	125	17.0	130	155	21.3	130	155	21.3
	1 kS/s	97	125	15.8	123	150	20.1	122	145	20.0
	10 kS/s	91	122	14.8	111	150	18.1	112	135	18.3
	100 kS/s	82	116	13.3	106	142	17.3	105	130	17.1
	200 kS/s	78.7	116	12.8	103.7	142	16.9	102	125	16.7
	500 kS/s	74	114	12.0	99.5	140	16.2	98	121	16.0
	1000 kS/s	71	87	11.5	93.2	130	15.2	93	116	15.2
	2000 kS/s	56	56	9.0	88	88	14.3	88	88	14.3
Typical THD		-97 dB								
Typical CMRR	≤2 V range	>140 dB @ 50 Hz: >120 dB @ 1 kHz								
	>2 V range	>90 dB @ 50 Hz: >60 dB @ 1 kHz								

continued on next page ...

TRION-1620-ACC /-LV

Low pass Filter (-3 dB, digital)	10 Hz, 30 Hz, 100 Hz, 300 Hz, 1 kHz, 3 kHz, 10 kHz, 30 kHz, 100 kHz, 300 kHz, 600 kHz
Characterisic	Bessel or Butterworth
Filter order	2^{nd}, 4^{th}, 6^{th}, 8^{th}
Analog antialiasing filter	2^{nd} order Bessel, automatically selected
Bandwidth (-3 dB, deacivated digital filter)	1 MHz 2^{nd} order Bessel filter
Crosstalk fin 1 kHz [10 kHz]	≤2 V Range: 120 dB [105 dB]
Channel to channel phase mismatch	typically <10 nsec when using the same range; <60 nsec for using different ranges
Board to board phase mismatch	<30 nsec
Over voltage protection	±300 V_{DC}
Counter	1 x counter channel linked to analog channel #1; trigger level 70 % of actual analog input range
Counter modes	Event counting, period, frequency, pulse width, duty cycle
Counter input bandwidth	1 MHz to 10 kHz depending on analog filter of CH1
ESD protection	IEC61000-4-2: ±8 kV air discharge, ±4 kV contact discharge
Supported TEDS chips (LEMO only)	DS2406, DS2430A, DS2431, DS2432, DS2433
Power consumption	Voltage mode: 6 W; IEPE® mode: 7.5 W

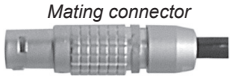
[1] For safety reasons maximum allowed voltage: 70 V_{DC} (46.7 V_{RMS} AC)
[2] TRION-1620-ACC only
[3] 1 year accuracy 23 °C ±5 °C
[4] TRION-1620-LV-6-L1B only
[5] LP Filter in auto mode
[6] SFDR excluding harmonics
[7] ENOB calculated from SNR

TRION-1620-ACC/LV-6-BNC module

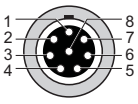
Hot: IN +
Shield: IN -

TRION-1620-ACC/LV-6-L1B module

Mating connector



FGG.1B.308.CLAD52Z



8-pin LEMO connector female
EGG.1B.308.CLL

1	EXC
2	GND
3	IN+
4	IN-
5	Excitation Sense
6	RESERVED (GND)
7	Current+
8	TEDS
Housing	GND

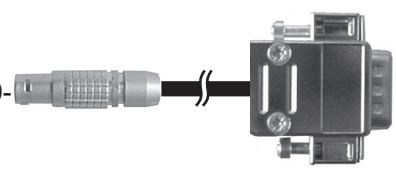
Optional accessory

TRION-ADAP-CLAMP-L1B8-D9F

High quality adaptor cable from Lemo 1B.308 plug to SUBD-9 socket, 0.5 meters.

For connecting PNA-CLAMP-1800-DC or SE-CUR-HALL-6 sensors to TRION-1620-LV-6-L1B modules.

Other sensors on request.

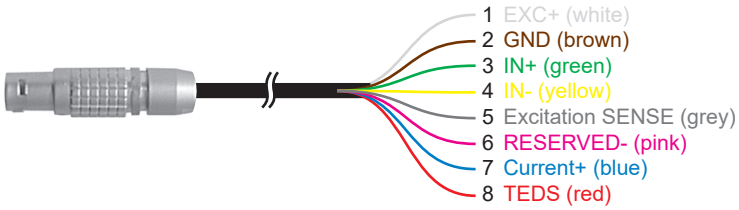


1	TEDS
2	IN+
3	n.c.
4	GND
5	EXT+
6	n.c.
7	IN-
8	n.c.
9	n.c.

For connecting PNA clamps or hall sensors

TRION-CBL-L1B8-OE-05-00

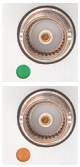
High quality cable from Lemo 1B.308 plug to open end, 5 meters



LED function

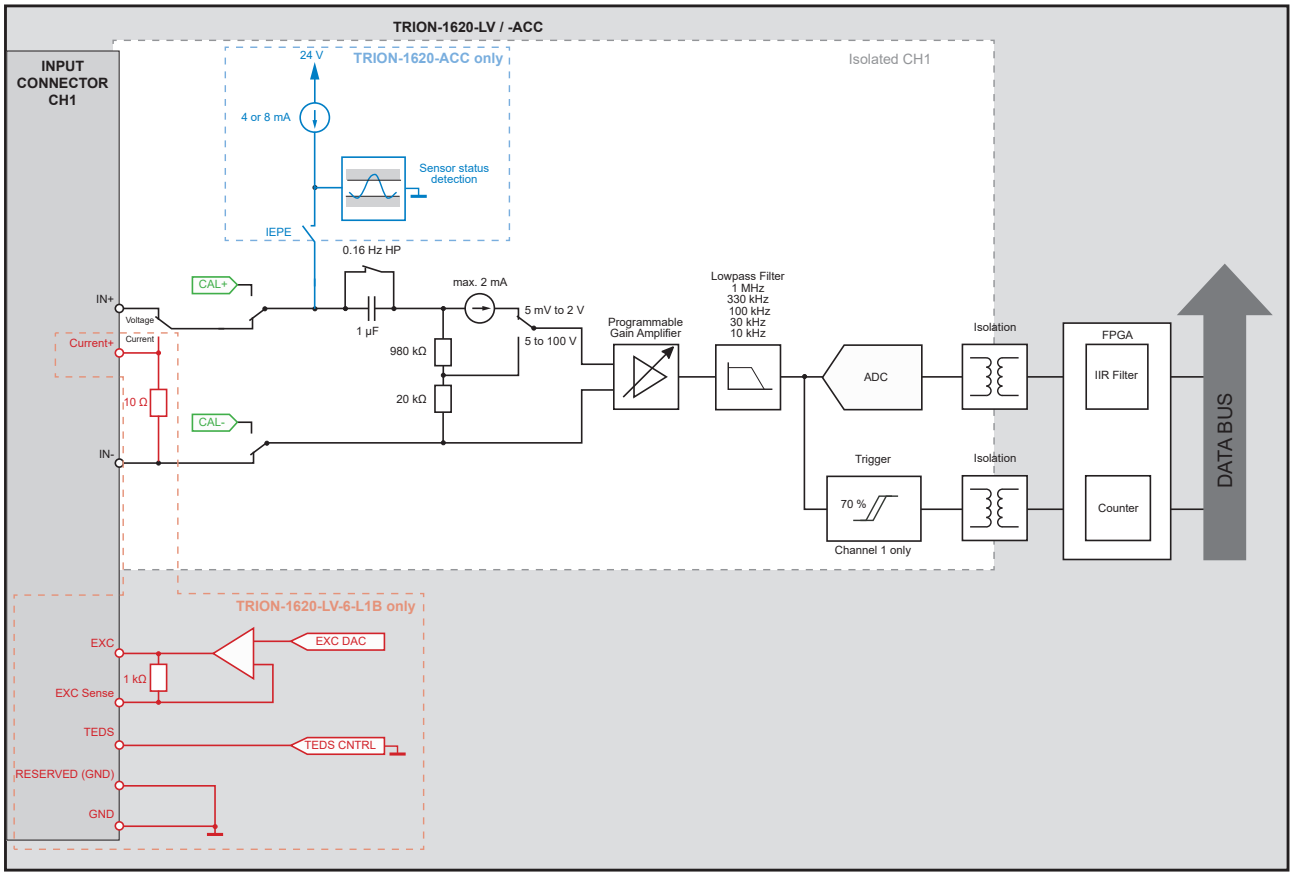
Green: Normal operation

Orange: Channel ID function. Typically active during channel setup



Block diagram

Base block diagram of the TRION-1620-ACC & -LV module:



TRION-1620-ACC /-LV

The TRION-16xx series is a highly accurate, isolated, 16-bit voltage digitizer. Each channel is separately isolated and has its own AD converter. For ranges above 2 V, a temperature compensated resistance divider attenuates the input signal. In lower ranges, the signal is directly routed to the programmable gain amplifier via a current limiting circuit. This architecture allows measuring voltages from a few µV to 100 V with an excellent signal-to-noise ratio and accuracy. The current limiting circuit can easily withstand 300 V_{DC}. So also the 5 mV range will not be damaged when 300 V are applied by accident. After the gain amplifier, the conditioned signal passes a programmable low pass filter before getting to the ADC.
Please refer to chapter "TRION-16xx analog to digital conversion" for more details about bandwidth and filtering.

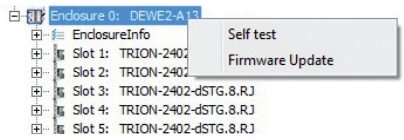
TRION-16xx series function

Short

The short function switches IN+ to IN- via the calibration circuit. It can be used to check the offset stability of the input amplifier.

Self test

The TRION-16xx series has an integrated special self-test circuit. It consist of a programmable high precision voltage source on the first isolated channel and a relay matrix. It is used to check the analog input path of the voltage amplifier by applying 0 V and 90 % of the input range to the input. This test can be performed in the channel setup for the actual range. During the board self test, which is available in the DEWE2 Explorer, this test is performed for all ranges and channels automatically



by right clicking the board in the DEWE2 explorer you can perform a self test.

IEPE® (TRION-1620-ACC only)

The TRION-1620-ACC also supports IEPE® sensors. This board is equipped with ultra-low noise constant current source. The excitation can be selected between 4 and 8 mA. The compliance voltage is 24 V.

Caution: TEDS functionality is not available in that mode!

Counter function (TRION-1620-ACC only)

The first channel of the TRION-1620-ACC module, can be used beside the normal functionality also as a counter input. It has a fixed trigger level at 75% of the actual analog input range. This makes the input perfectly suitable for all kind of tacho probes. By activating the IEPE® supply it is even possible using probes without any additional sensor supply, just with a BNC cable.

Supported counter functions are:

- Simple event counting
- Period measurement
- Pulse width measurement
- Frequency
- Duty cycle

For detailed information about this function please refer to the chapter "Functional description of advanced counters" of the TRION-CNT module.

Caution: It is not possible to change the analog input settings out of the counter dialog. This has to be done in the channel setup of the analog input!

Isolation

The isolation of the module has many advantages:

- It allows very high common mode voltages even in the 5 mV range. This is especially required for current measurement with shunt technology.
- High input protection.
- DC offset errors because of ground loops are eliminated.
- Eliminating current loops; noise reduction.

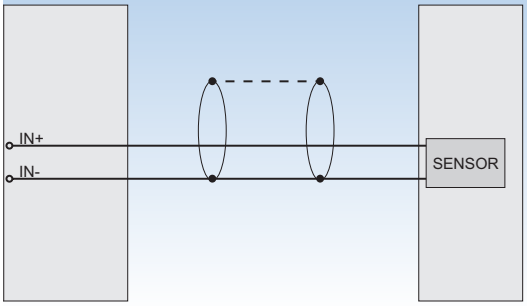
Ground connection

The TRION-16xx series is fully isolated and has high impedance inputs, with very high sensitivity. For achieving the highest signal-to-noise ration it is strongly recommended to connect the DEWE2 system to a structural ground potential. This could be for example the chassis of the car or train, in case of vehicle measurements. With that simple method, you can avoid catching noise signals like the 50 / 60 Hz interference. Sometimes the power supply cable already provides this connection. If the system runs on battery or with an isolated DC power supply, the operator should take care of the ground connection.

Signal connection

Voltage measurement

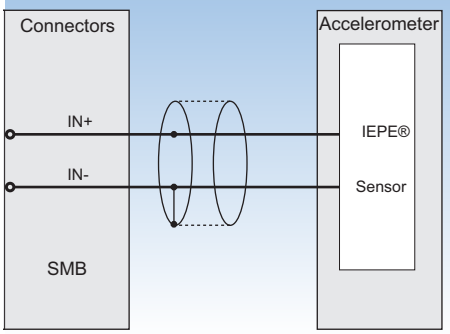
Isolated Sensors
Battery powered sensors
Sensors with differential output



Software settings for voltage measurement

Mode:	Voltage
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Excitation:	0 V

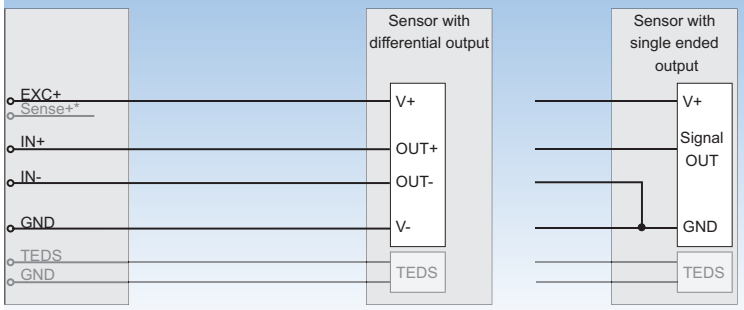
Current measurement (TRION-1620-ACC only)



Software settings for acceleration sensors

Mode:	IEPE®
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Excitation:	4 mA

Sensors with sensor supply and voltage output

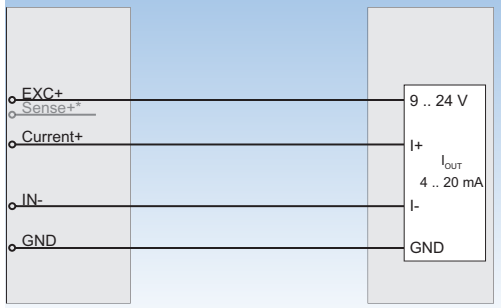


Software settings for sensors with sensor supply and voltage output

Mode:	Voltage
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Excitation:	15 V

*) Sense as well as TEDS connection is optional.

Sensors with sensor supply and current output

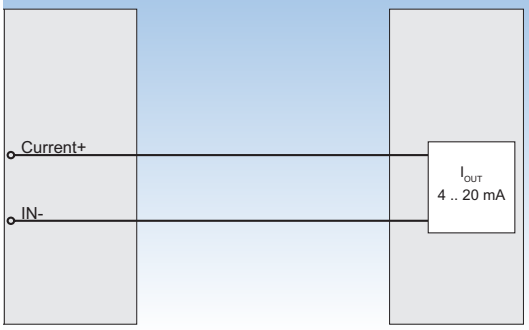


Software settings for sensors with sensor supply and current output

Mode:	Current
Range:	20 mA
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
SHUNT resistance:	10 Ω
Type:	Internal
Excitation:	15 V

*) Sense connection is optional.

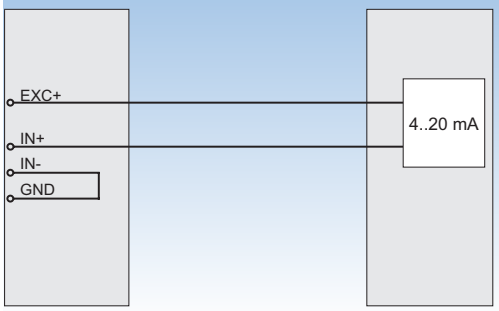
Current measurement



Software settings for current measurement

Mode:	Current
Range:	20 mA
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
SHUNT resistance:	10 Ω
Type:	Internal
Excitation:	0 V

Loop powered sensors with 4 to 20 mA output



Software settings for loop powered sensors with 4 to 20 mA output

Mode:	Current
Range:	20 mA
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
SHUNT resistance:	10 Ω
Type:	Internal
Excitation:	28 V

Notes

Isolated voltage input module

- Sampling: 250 kS/s per channel at 16-bit;
- ADC: Low noise, SAR
- Input types: Voltage: ±5 mV to ±100 V[1)]
  Current: 10 mA to 100 mA
- Isolation: 1.5 kV

Module specifications

TRION-1603-LV series specifications										
Input channels		TRION-1603-LV-6-BNC TRION-1603-LV-6-L1B	6 channels BNC; voltage input 6 channels LEMO; voltage input; 5/12 V sensor supply; TEDS							
Sampling Rate / Resolution		100 S/s to 250 kS/s	16-bit							
Data Transfer		16-bit								
ADC type		SAR (Successive Approximation Register)								
Data rate DMA transfer		6 analog channels: max 3 MB/s								
Input ranges	Voltage	±5, ±10, ±20, ±50, ±100, ±200, ±500 mV, ±1 V, ±2 V, ±5 V, ±10 V, ±20 V, ±50 V, ±100 V[1)]								
	Current[2)]	10, 20, 50, 100 mA								
Input noise (5 mV range)		0 to 10 Hz : Noise density:	1.5 μV_{PP} 6.4 nV/SQRT(Hz)							
Input impedance		1 MΩ shunted by 18 pF								
Input bias current		<1 nA								
Input coupling		DC								
Accuracy[3)]	Voltage	DC to 1kHz >1 kHz to 5 kHz >5 kHz to 10 kHz	±0.02 % of reading ± 0.02 % of range ±20 µV ±0.2 % of reading ± 0.02 % of range ±20 µV ±0.5 % of reading ± 0.02 % of range ±20 µV							
	Current[2)]	DC to 1kHz >1 kHz to 5 kHz >5 kHz to 10 kHz	±0.1 % of reading ± 0.02 % of range ±10 µA ±0.2 % of reading ± 0.02 % of range ±10 µA ±0.5 % of reading ± 0.02 % of range ±10 µA							
Gain drift		typical 10 ppm/°C max. 20 ppm/°C								
Offset drift		typical 0.3 µV/°C + 10 ppm of range/°C, max 15 µV/°C + 20 ppm of range/°C								
Linearity		typical 0.01 %								
Sensor excitation[2)]		1 to 28 V freely programmable @ 1 % ±1 mV accuracy (max. 100 mA, max 1 W) per channel								
Input configuration		Isolated								
Isolation impedance		Isolation resistance >1 GΩ; Isolation capacitance typically 15 pF								
Current input		Internal 10 Ω shunt; max. 100 mA protected with resettable fuse								
Isolation voltage		1500 V with TRION-1603-LV-6-BNC 800 V with TRION-1603-LV-6-L1B								
Typical Signal-to-noise ratio, spurious-free SNR, Effective number of Bits[4)]		20 mV range			2 V range			100 V range		
		SNR	SFDR[5)]	ENOB[6)]	SNR	SFDR[5)]	ENOB[6)]	SNR	SFDR[5)]	ENOB[6)]
	Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]
	1 kS/s	93	120	15.2	93	120	15.2	93	120	15.2
	10 kS/s	90	120	14.7	93	120	15.2	93	120	15.2
	100 kS/s	80	116	13.0	93	120	15.2	93	120	15.2
	250 kS/s	74	100	12.0	93	120	15.2	93	120	15.2
Typical THD		-97 dB								
Typical CMRR	≤2 V range	>140 dB @ 50 Hz	>120 dB @ 1 kHz							
	>2 V range	>90 dB @ 50 Hz	>60 dB @ 1 kHz							
Low pass Filter (-3 dB, digital)		10 Hz, 30 Hz, 100 Hz, 300 Hz, 1 kHz, 3 kHz, 10 kHz, 30 kHz, 100 kHz								
Characterisic		Bessel or Butterworth								
Filter order		2nd , 4th, 6th, 8th								
Analog antialiasing filter		2nd order Bessel, automatically selected								
Bandwidth (-3 dB, deactivated digital filter)		100 kHz 2nd order Bessel filter								
Crosstalk fin 1 kHz [10 kHz]		≤2 V Range: 120 dB [105 dB]								
Channel to channel phase mismatch		typically <60 nsec when using the same range; <200 nsec for using different ranges								
Board to board phase mismatch		<30 nsec								
Over voltage protection		±300 V_{DC}								
ESD protection		IEC61000-4-2: ±8 kV air discharge, ±4 kV contact discharge								
Supported TEDS chips[2)]		DS2406, DS2430A, DS2431, DS2432, DS2433								
Power consumption		6 W w/o sensor supply[2)]; absolute maximum with sensor supply[2)]: 13 W								

[1)] For safety reasons maximum allowed voltage: 70 V_{DC} (46.7 V_{RMS} AC)
[2)] TRION-1603-LV-6-L1B only
[3)] 1 year accuracy 23 °C ±5 °C
[4)] LP Filter in auto mode
[5)] SFDR excluding harmonics
[6)] ENOB calculated from SNR

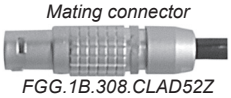
TRION-1603-LV

TRION-1603-LV-6-BNC module

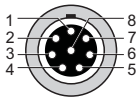
Hot: IN +
Shield: IN -

TRION-1603-LV-6-L1B module

Mating connector



FGG.1B.308.CLAD52Z



8-pin LEMO connector female
EGG.1B.308.CLL

1	EXC
2	GND
3	IN+
4	IN-
5	Excitation Sense
6	RESERVED (GND)
7	Current+
8	TEDS
Housing	GND

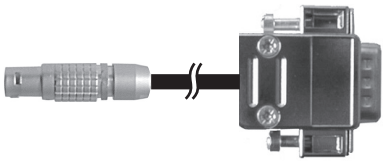
Optional accessory

TRION-ADAP-CLAMP-L1B8-D9F

High quality adaptor cable from Lemo 1B.308 plug to SUBD-9 socket, 0.5 meters.

For connecting a PNA-CLAMP-1800-DC or SE-CUR-HALL-6 sensor to TRION-1603-LV-6-L1B modules.

Other sensors on request.

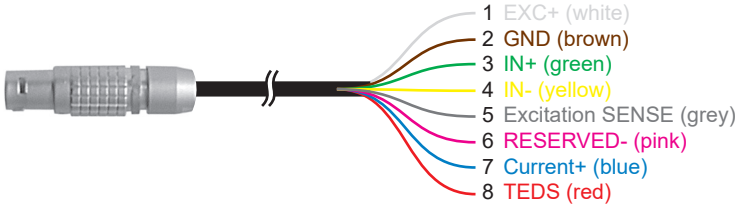


1 TEDS
2 IN+
3 n.c.
4 GND
5 EXT+
6 n.c.
7 IN-
8 n.c.
9 n.c.

For connecting
PNA clamps or hall
sensors

TRION-CBL-L1B8-OE-05-00

High quality cable from Lemo 1B.308 plug to open end, 5 meters



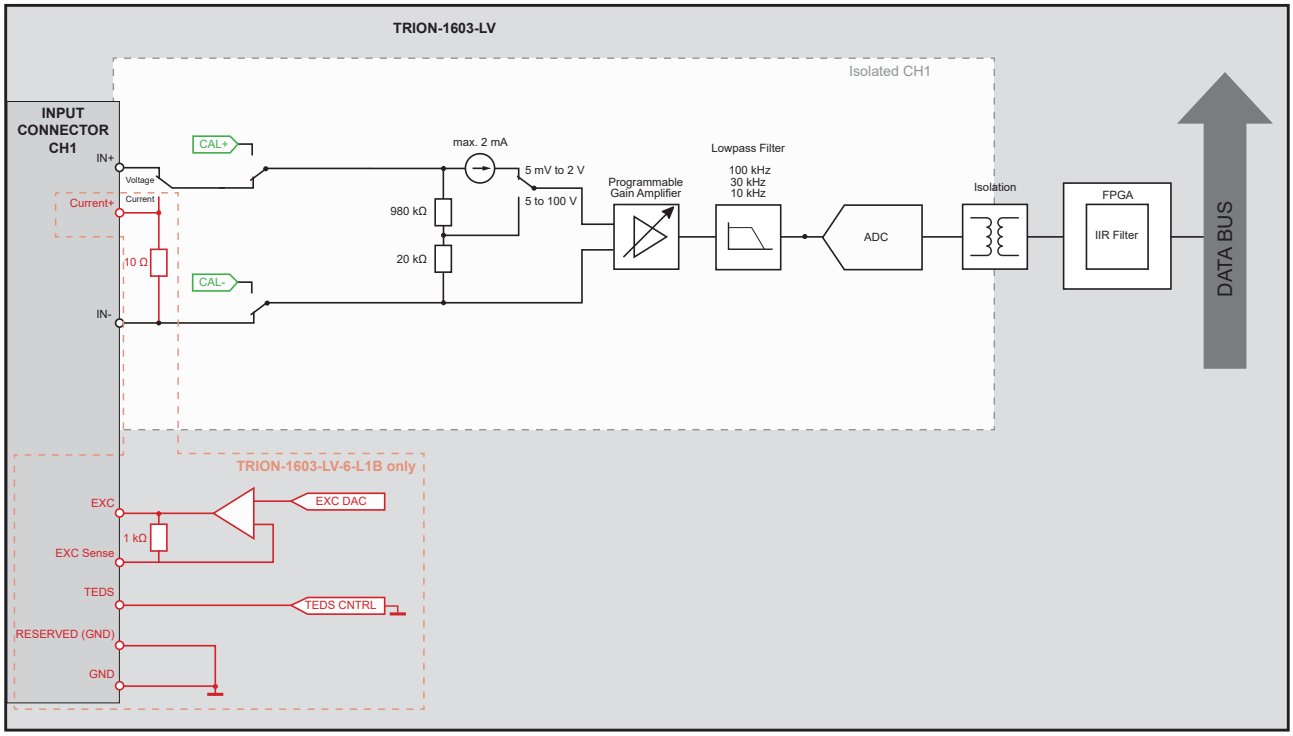
LED function

Green: Normal operation

Orange: Channel ID function. Typically active during channel setup

Block diagram

Base block diagram of the TRION-1603-LV module:



The TRION-16xx series is a highly accurate, isolated, 16-bit voltage digitizer. Each channel is separately isolated and has its own AD converter. For ranges above 2 V, a temperature compensated resistance divider attenuates the input signal. In lower ranges, the signal is directly routed to the programmable gain amplifier via a current limiting circuit. This architecture allows measuring voltages from a few µV to 100 V with an excellent signal-to-noise ratio and accuracy. The current limiting circuit can easily withstand 300 V_{DC}. So also the 5 mV range will not be damaged when 300 V are applied by accident. After the gain amplifier, the conditioned signal passes a programmable low pass filter before getting to the ADC.
Please refer to chapter “TRION-16xx analog to digital conversion” for more details about bandwidth and filtering.

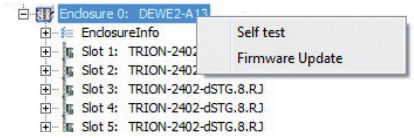
TRION-16xx series function

Short

The short function switches IN+ to IN- via the calibration circuit. It can be used to check the offset stability of the input amplifier.

Self test

The TRION-16xx series has an integrated special self-test circuit. It consist of a programmable high precision voltage source on the first isolated channel and a relay matrix. It is used to check the analog input path of the voltage amplifier by applying 0 V and 90 % of the input range to the input. This test can be performed in the channel setup for the actual range. During the board self test, which is available in the DEWE2 Explorer, this test is performed for all ranges and channels automatically



by right clicking the board in the DEWE2 explorer you can perform a self test.

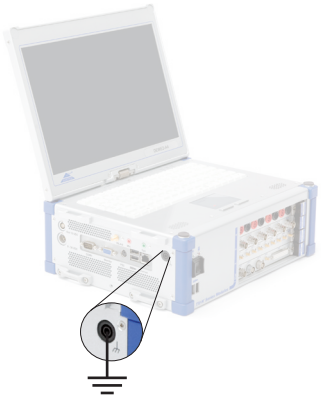
Isolation

The isolation of the module has many advantages:

- It allows very high common mode voltages even in the 5 mV range. This is especially required for current measurement with shunt technology.
- High input protection.
- DC offset errors because of ground loops are eliminated.
- Eliminating current loops; noise reduction.

Ground connection

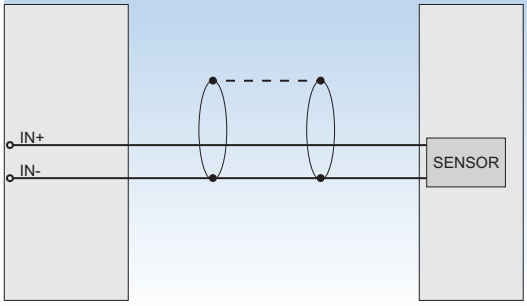
The TRION-16xx series is fully isolated and has high impedance inputs, with very high sensitivity. For achieving the highest signal-to-noise ration it is strongly recommended to connect the DEWE2 system to a structural ground potential. This could be for example the chassis of the car or train, in case of vehicle measurements. With that simple method, you can avoid catching noise signals like the 50 / 60 Hz interference. Sometimes the power supply cable already provides this connection. If the system runs on battery or with an isolated DC power supply, the operator should take care of the ground connection.



Signal connection

Voltage measurement

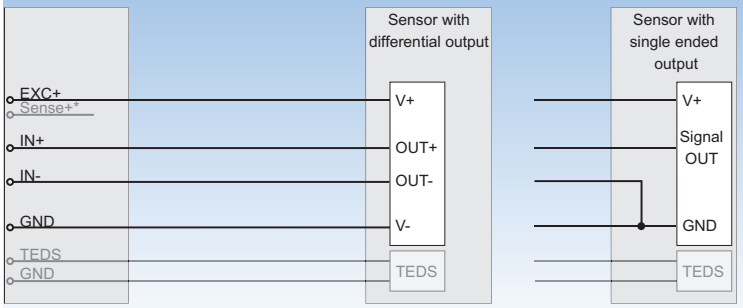
Isolated Sensors
Battery powered sensors
Sensors with differential output



Software settings for voltage measurement

Mode:	Voltage
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Excitation:	0 V

Sensors with sensor supply and voltage output

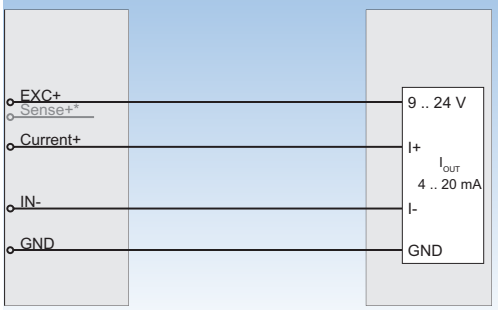


Software settings for sensors with sensor supply and voltage output

Mode:	Voltage
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Excitation:	15 V

*) Sense as well as TEDS connection is optional.

Sensors with sensor supply and current output

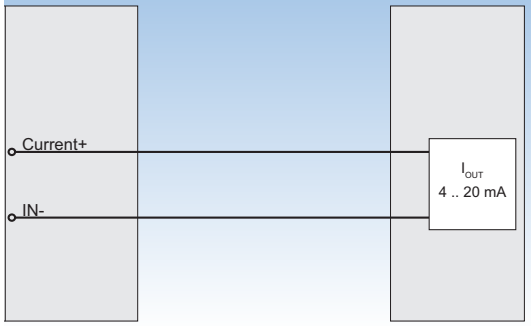


Software settings for sensors with sensor supply and current output

Mode:	Current
Range:	20 mA
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
SHUNT resistance:	10 Ω
Type:	Internal
Excitation:	15 V

*) Sense connection is optional.

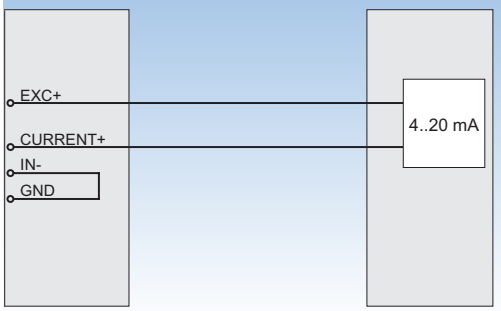
Current measurement



Software settings for current measurement

Mode:	Current
Range:	20 mA
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
SHUNT resistance:	10 Ω
Type:	Internal
Excitation:	0 V

Loop powered sensors with 4 to 20 mA output

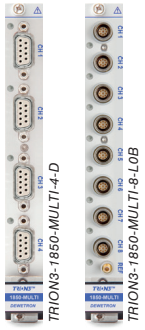


Software settings for loop powered sensors with 4 to 20 mA output

Mode:	Current
Range:	20 mA
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
SHUNT resistance:	10 Ω
Type:	Internal
Excitation:	28 V

Universal input module

- Sampling: 5 MS/s per channel
- Input types: Voltage, bridge, resistance, RTD, IEPE® current and counter
- Isolation: 350 V_{DC}
- CAN: High-speed CAN2.0 port
- Bandwidth: 2 MHz



Module specifications

General specifications									
Input channels	TRION-1820-MULTI-4-D	4 channels DSUB connector	CH1 Analog/CAN	CH2 Analog	CH3 Analog/CNT	CH4 Analog/CNT			
	TRION3-1820-MULTI-8-L0B	8 channels 0B LEMO connector	CH1 Analog CAN	CH2 Analog	CH3 Analog	CH4 Analog	CH5 Analog	CH6 Analog	CH7 Analog CNT	CH8 Analog CNT
	TRION3-1820-MULTI-4-D	4 channels DSUB connector	CH1 Analog/CAN	CH2 Analog	CH3 Analog/CNT	CH4 Analog/CNT			
	TRION3-1850-MULTI-8-L0B	8 channels 0B LEMO connector	CH1 Analog CAN	CH2 Analog	CH3 Analog	CH4 Analog	CH5 Analog	CH6 Analog	CH7 Analog CNT	CH8 Analog CNT
	TRION3-1850-MULTI-4-D	4 channels DSUB connector	CH1 Analog/CAN	CH2 Analog	CH3 Analog/CNT	CH4 Analog/CNT			
Sampling rate / Resolution	TRION-1820-MULTI TRION3-1820-MULTI	100 S/s to 2 MS/s 24-bit							
	TRION3-1850-MULTI	100 S/s to 2 MS/s 24-bit >2 MS/s to 5 MS/s 18-bit							
Onboard data buffer		512 MB							
Isolation		Channel to channel 350 V_{DC} [1] Channel to chassis 350 V_{DC} [1]							
Input types		Ranges	Supported sensors						
Voltage		±0.1 mV to ±100 V[1] freely programmable	-						
IEPE®		±100 mV to ±10 V freely programmable	IEPE® sensors						
Bridge		1 to 1000 mV/V	4-, 5-, 6-wire full bridge; 3-, 4-, 5-wire ½ bridge; 2-, 3-, 4-wire ¼ bridge; 120/350/1000 Ω internal ¼ bridge completion						
Resistance		10 Ω to 30 kΩ	Potentiometer, Resistance temperature detection: Pt100, Pt200, Pt300, Pt500, Pt2000 (2-, 3-, 4-wire)						
Current		±30 mA	4 to 20 mA sensors; loop powered sensors						
MSI		MSI2-CH-x: 500 to 50000 pC MSI2-TH-x: various TC ranges MSI2-LVDT	LVDT, RVDT, Charge output and thermocouple sensors						
REF connector		SMB connector to apply external calibration signal (LEMO version only)							
Input connector		9-pin LEMO EPG.0B.309 (TRION3-18xx-MULTI-8-L0B) 9-pin SUB-D connector (TRION(3)-18xx-MULTI-4-D)							
Environmental specifications	Operating temperature	32 to 113 °F (0 to +45 °C)							
	Storage temperature	-4 to 158 °F (-20 to +70 °C)							
	Humidity	10 to 80 % non cond., 5 to 95 % rel. humidity							

[1] for safety reasons it is not allowed to apply more than 47.2 V_{PEAK} or 70 V_{DC}

TRION(3)-18xx-MULTI

Input amplifier																	
Voltage input accuracy[1) 2)] <= 10 V	DC	±0.02 % of reading ± 0.02 % of range ±20 µV															
	0.1 Hz to 10 kHz	±0.02 % of reading ± 0.02 % of range ±20 µV															
	10 kHz to 50 kHz	±0.1 % of reading ± 0.02 % of range ±20 µV															
	50 kHz to 100 kHz	±0.5 % of reading ± 0.02 % of range ±20 µV															
>10 V Input divider on	DC	±0.02 % of reading ± 0.02 %															
	0.1 Hz to 5 kHz	±0.02 % of reading ± 0.02 %															
	5 kHz to 50 kHz	±0.5 % of reading ± 0.02 %															
	50 kHz to 100 kHz	±1.5 % of reading ± 0.02 %															
Amplifier drift	Gain drift	typical 10 ppm/°C max. 20 ppm/°C															
	Offset drift	typical 0.3 µV/°C+ 10 ppm of range/°C, max 2 µV/°C + 20 ppm of range/°C															
Linearity		typical <25 ppm															
Current input accuracy[1) 2)]		0.1 % of reading ± 10 µA															
Current input impedance		Direct input (IN- to GNDi): 75 Ω±25 Ω															
		Loop powered sensor: 120 Ω ±1 Ω															
Input impedance		Differential												Input (Single ended)			
	0 to 10 V range	100 MΩ or 1 MΩ (programmable) // 15 pF												50 MΩ // 28 pF			
	10 to 100 V range	2 MΩ // 35 pF												1 MΩ //65 pF			
Input bias current		<100 nA 100 mV range															
Input configuration		Single ended or differential (programmable)															
Input coupling		DC / AC (highpass filter 0.16 Hz to 100 Hz freely programmable)															
Common mode voltage		$350V_{DC}$ [3)]															
Overvoltage protection (IN+ to IN-)	0 to 10 V range	50 V continuous, 100 V for 60 seconds (maximum input current limited to 7 mA)															
	10 to 100 V range	250 V															
Low pass filter (-3 dB, digital)		1 Hz to 1.5 MHz freely programmable or OFF															
	Characteristic	Bessel or Butterworth															
	Filter order	2^{nd}, 4^{th}, 6^{th}, 8^{th}															
	Filter setting AUTO	30 % of samplerate with 8^{th} order Bessel															
Analog anti-aliasing filter		2nd order Bessel, automatically selected															
		100 kHz, 500 kHz, 2 MHz; <=1 V range bandwidth is limited to 1.8 MHz															
Channel to channel phase mismatch		<10 nsec between channels using the same range															
CMRR		135 dB @ 50 Hz; 110 dB @ 1 kHz; 90 dB @ 10 kHz; 90 dB @ 100 kHz;															
Typical crosstalk		-134 dB (10 V range; 0 to 100 kHz)															
Input noise (100 mV range)	0 to 10 Hz	0.3 μV_{PP}															
	Noise density	6.9 nV/SQRT(Hz)															
typical THD		10V range: -108 dB; 1V range: -102 dB for 1 kHz fundamental frequency															
Signal to noise ratio; Spurious free SNR;	100 mV range					1 V range				10 V range				100 V range			
Effective number of Bits[4)]; Noise V_{PP}	SNR	SFDR[5)]	ENOB[6)]	Noise	SNR	SFDR[5)]	ENOB[6)]	Noise	SNR	SFDR[5)]	ENOB[6)]	Noise	SNR	SFDR[5)]	ENOB[6)]	Noise	
Sample rate	[dB]	[dB]	[Bit]	[mV_{PP}]	[dB]	[dB]	[Bit]	[mV_{PP}]	[dB]	[dB]	[Bit]	[mV_{PP}]	[dB]	[dB]	[Bit]	[mV_{PP}]	
1 kS/s	113.5	130	18.6	0.001	112.4	135	18.4	0.010	127.2	140	20.8	0.018	120.1	140	19.7	0.400	
10 kS/s	103.0	130	16.8	0.003	109.0	135	17.8	0.017	119.5	140	19.6	0.055	114.7	140	18.8	0.950	
100 kS/s	94.7	130	15.4	0.011	103.9	130	17.0	0.038	109.8	140	17.9	0.190	106.6	140	17.4	2.700	
200 kS/s	91.4	130	14.9	0.016	101.4	130	16.6	0.051	107.4	140	17.6	0.260	104.1	140	17.0	3.800	
1000 kS/s	84.7	125	13.8	0.038	95.0	130	15.5	0.116	99.8	139	16.3	0.650	97.7	135	15.9	8.300	
2000 kS/s	81.4	120	13.2	0.058	91.0	128	14.8	0.170	95.4	132	15.6	1.100	94.1	132	15.3	14.000	
5000 kS/s	78.7	110	12.8	0.080	88.7	125	14.4	0.270	93.1	130	15.2	1.600	91.4	130	14.9	19.000	
Filter = OFF	76.2	105	12.4	0.110	86.5	120	14.1	0.330	90.5	130	14.7	2.000	89.0	130	14.5	23.000	

[1)] 1 year accuracy 23 °C ±5 °C
[2)] add 0.02 % of reading with filter settings OFF
[3)] for safety reasons it is not allowed to apply more than 47.2 V_{PEAK} or 70 V_{DC}
[4)] LP Filter in auto mode
[5)] SFDR excluding harmonics
[6)] ENOB calculated from SNR

Excitation		
Excitation voltage	0 to 24 V_{DC}; freely programmable separately for each channel; 1mV resolution; balanced around GNDi; remote sense support;	
1 year accuracy drift	±0.03 % ±1.5 mV	
Drift	±10 ppm/°C ±50 µV/°C	
Current limit	0.1 to 5 V:	100 mA
	>5 V to 24 V:	limited to 0.5 W
Protection	Continuous short	
Load and line regulation error	±0.002 % with sense line connected	
Excitation current	0.1 to 60 mA_{DC} (programmable, 16 bit DAC) 1 µA; balanced around GNDi	
1 year accuracy[1)]	0.1 to 5 mA: 0.05 % ±2 µA	>5 to 60 mA: 0.5% ±5 µA
Drift	15 ppm/°C	
Compliance voltage	0.1 to 20 mA:	24 V
	>20 mA:	10 V
Output impedance	>10 MΩ	
Load regulation bandwidth	100 kHz	
IEPE® excitation	2 to 20 mA; 10%; >21 V compliance voltage	

[1)] 1 year accuracy 23 °C ±5 °C

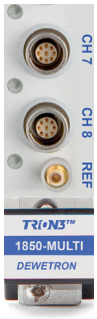
Bridge functions		
Supported bridge types	Full bridge	4-, 5- or 6-wire full bridge
		4-wire full bridge with constant current excitation (piezoresistive bridge sensors), Potentiometer
	Half bridge	3-, 4- or 5-wire ½ bridge with internal completion (software programmable)
	Quarter bridge	2-, 3- or 4-wire ¼ bridge with internal completion resistor for 120 Ω, 350 Ω and 1000 Ω (software programmable)
		2-wire ¼ with constant current excitation for dynamic measurement (AC coupled)
Internal quarter bridge completion	120 Ω; 350 Ω; 1000 Ω;	±0.05%
Bridge resistance	80 Ω to 10 kΩ @ ≤ 5 V_{DC} excitation;	the lower limit is caused by the maximum power supply.
Shunt calibration	4000 steps programmable shunt; shunt target can be programmed in mV/V;	
Completion resistor accuracy	0.05 % ±15 ppm/K	
Automatic bridge balance	±400% of range	
Bridge features	Bridge Balance; line-resistance compensation	

CAN functions	
CAN specification	CAN 2.0
CAN physical layer	High speed
CAN termination	Programmable: high impedance or 120 Ω
CAN bus protection	±36 V

Counter functions	
Counter	2 x counter channels linked to the last two analog channels; trigger level is 75% of range
Counter modes	Event counting, frequency
Timebase / resolution	5 MHz (200 nsec)

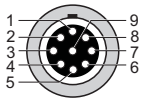
TRION(3)-18xx-MULTI

TRION3-18xx-MULTI-8-L0B module



Mating connector

FGG.0B.309.CLADxx



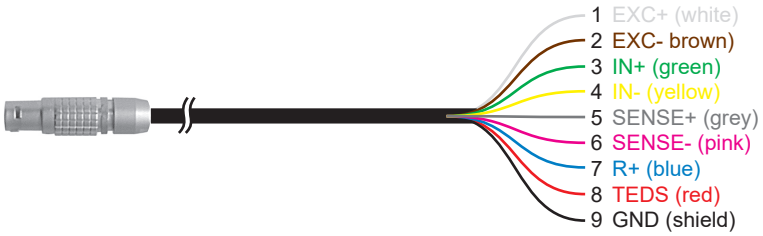
9-pin LEMO connector female
EPG.0B.309.HLN

1	EXC+ (CAN power supply 12 V, CH1 only)
2	EXC-
3	IN+
4	IN-
5	SENSE+ (CAN high, CH1 only)
6	SENSE- (CAN low, CH1 only)
7	R+
8	TEDS
9	$GND_{isolated}$ (CAN GND, CH1 only)
Housing	connected to Chassis GND

Optional accessory

TRION-CBL-L0B9-OE-xx-00

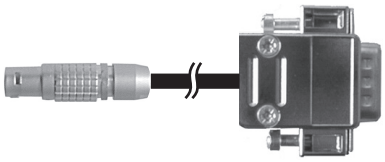
01 = 1 meter
05 = 5 meters



High quality cable from Lemo 0B.309 plug to open end, 1 or 5 meters.

TRION-CBL-L0B9-D9-0.5-02

High quality adaptor cable from Lemo 0B.309 plug to SUBD-9 socket, 0.5 meters.



1 EXC+
2 IN+
3 SENSE+
4 GNDi
5 R+
6 SENSE-
7 IN-
8 EXC-
9 TEDS
H: Shield

For connecting any MSI-BR series adaptors

TRION-CBL-L0B9-BNC-0.5-02

High quality adaptor cable from LEMO 0B.309 plug to BNC cable jack, 0.5 meters

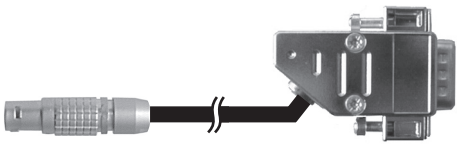
For connecting voltage signals and IEPE® sensors to TRION3-18xx-MULTI-8-L0B modules.

Hot: IN +
Shield: IN -

TRION-CBL-L0B9-CAN-0.5

Adaptor cable from LEMO 0B.309 plug to D-SUB-9 plug for CAN, 0.5 meters

For TRION3-18xx-MULTI-8-L0B modules channel 1 only!



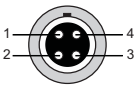
Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

TRION-CBL-L0B9-CPAD-01-00

Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION3-18xx-MULTI-8-L0B (CH 1).

For connecting 1x CPAD series module



4-pin LEMO connector male
FGG.1B.304.CLL

Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-L0B9-CPAD-01-01

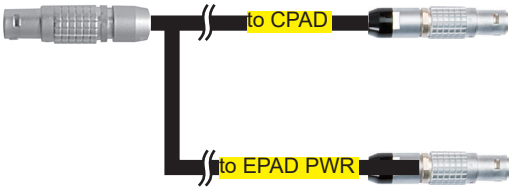
Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

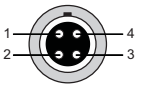
Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

For connecting the first CPAD-series module of a module-chain to a TRION3-18xx-MULTI-8-L0B.

For connecting more than one CPAD series module

Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND



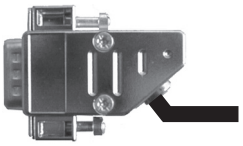


4-pin LEMO connector male
FGG.1B.304.CLL

Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

TRION(3)-18xx-MULTI

TRION(3)-18xx-MULTI-4-D module



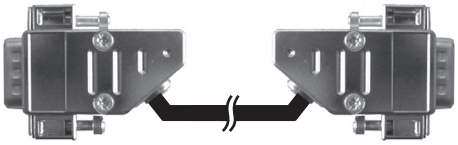
1 EXC+ (CAN power supply 12 V, CH1 only)
2 IN+
3 Sense+ (CAN low, CH1 only)
4 $GND_{isolated}$ (CAN GND, CH1 only)
5 R+
6 Sense- (CAN high, CH1 only)
7 IN-
8 EXC-
9 TEDS
Housing connected to Chassis GND

Optional accessory

TRION-CBL-D9-CAN-0.5

Adaptor cable from D-SUB-9 plug to D-SUB-9 plug for CAN, 0.5 meters.

For TRION(3)-18xx-MULTI-4-D modules.



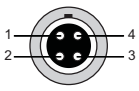
Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

TRION-CBL-D9-CPAD-01-00

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION(3)-18xx-MULTI-4-D (CH 1).

For connecting 1x CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

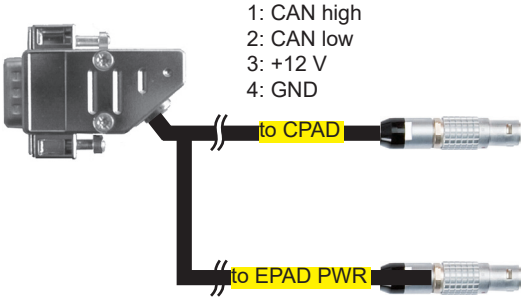
TRION-CBL-D9-CPAD-01-01

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

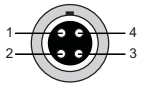
For connecting the first CPAD-series module of a module-chain to a TRION(3)-18xx-MULTI-4-D.

Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND



Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

For connecting more than one CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

LED function

Green: Normal operation

Orange: Channel ID function. Typically active during channel setup or CAN mode active.

Red: Error.

TRION(3)-18xx-MULTI function overview

Isolation

The TRION(3)-18xx-MULTI is fully isolated. That means every channel has a separate isolated excitation voltage and input amplifier. The main advantages of that configuration are:

> Very high common mode voltages of ±350 V.
> Overcurrent protection e.g. if the isolation of a strain gage on a 110 V power line fails.
> Ground loops are eliminated.
> Noise reduction.

Free variable gain and excitation

Amplifier parameters like gain, excitation voltage, excitation current and sensor offset can be varied for every channel individually. This allows a perfect match of each input channel to any sensor.

Sensor balance

Normally every strain gage sensor has a certain offset caused by manufacturer tolerances and sensor mounting. The "*Sensor balance*" function removes that offset automatically up to 400 % of the selected input range.

Quarter bridge features

> Completion resistors

The TRION-18xx-MULTI supports 3 different quarter bridge completions: 120, 350 and 1000 Ω.

> Internal shunt calibration

The purpose of this function is to check Quarter Bridge wiring and determining the sensitivity loss due to cable resistance. By applying a known resistor to the internal completion resistor, a known bridge unbalance can be simulated. In case of ideal wiring the measured unbalance will correlate exactly with the simulated unbalance. But in reality, cable resistance will decrease the measured value. By using the ratio between expected and measured unbalance this effect could be compensated. The TRION(3)-18xx-MULTI supports a programmable shunt. The user can directly enter the "mV/V" within certain limits. The module calculates the appropriate resistor and applies it on demand. Sensor failures during test could easily be checked with this function. Simply compare the Shunt Cal result before and after the test run.

> Compensate cable resistance

The TRION(3)-18xx-MULTI series uses an even more accurate way to determine the cable resistance than using the shunt. The internal routing matrix allows directly measuring the line resistance between R+ and the strain gage. A gain correction factor is calculated afterwards and automatically applied. This function is only available when bridge scaling is active.

> 4 wire quarter bridge

To fully compensate any cable related effects the module also supports 4 wire technology.

E.g. temperature drift:
Copper has a temperature drift of 0.4 %/°C. This is especially a problem at quarter bridges, because also the offset changes with the wire resistance. The following table shows the difference between the 3 wiring methods for a 120 Ω strain gage with a 50m cable at 0.25 mm² diameter.

	Initial error		Drift because of 10 °C warm-up	
	Offset	Sensitivity	Offset	Sensitivity
2-wire	25183 µm/m	-4.97 %	956 µm/m	-0.18 %
3-wire	0 µm/m	-2.6 %	0 µm/m	-0.01 %
4-wire	0 µm/m	0.0 %	0 µm/m	0.00 %

External calibration (REF input)

Instead of the internal calibration voltage, an external signal can be applied to every input channel by using the calibration relay matrix. This allows external calibration of the voltage input without disconnecting the input connector (TRION(3)-18xx-MULTI-8-L0B only).

Counter

The module supports two counter input channels. Both can be routed to the last two analog inputs. The trigger level is automatically 75% of the actual analog input range. Frequency measurement and event counting is supported.

CAN

The first channel of the TRION(3)-18xx-MULTI also has a CAN bus interface. Any CAN2.0B compatible device or bus can be connected. In CAN mode, the analog input function of the channel is deactivated. Sensor excitation is switched to ±12 V. For further information, refer to chapter "TRION-CAN module" within this manual.

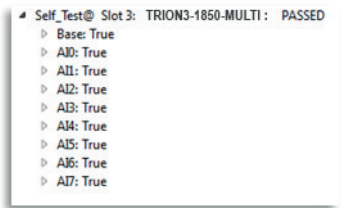
One single CPAD series module can be directly connected to that channel. If more modules are required an additional power supply is needed.

TEDS

Transducer Electronic Data sheet. The TEDS interface is used to identify MSI series adapters.

DEWETRON Explorer SELF TEST functions

The self test function is designed to verify all features of the board. It also includes a complex analog accuracy check.

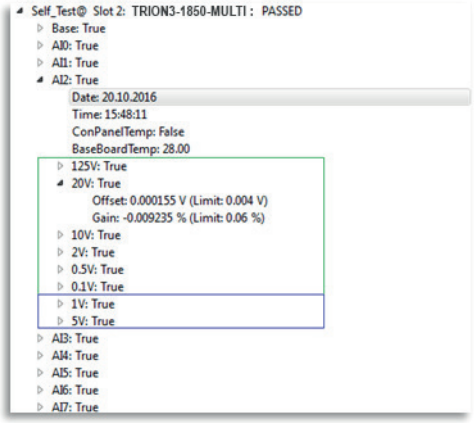


Test results

Base section

Here, the test results of all I²C devices, the PLL, EE-Prom and the SDRAM are displayed. This test checks the infrastructure of the board. If it passes, also the PCI bus is working fine. If anything in this section fails, the board is defective and has to be repaired.

AI0 to AI7 Analog test section



Input Range

These are the test results of the analog channels. For every channel and every range, an appropriate test voltage is applied by using the internal calibration source. If the measured voltage is within certain limits, this test passes. The detailed test results can be found by opening the tree structure in the result screen. By comparing the current measurement with the allowed limit, it can be categorized. If the measured result is slightly out of the limit a gain adjustment and auto zero will fix the problem. If the error is more than two or three times out of the limits, servicing the board should be considered.

Excitation Voltage

At this test 1 V and 5 V is applied to the excitation terminals. By reading back the excitation with the input amplifier, it is checked if the excitation circuit is working correctly.

Caution: *This test applies voltage to the sensor terminals. If the cabling or sensor short circuits the excitation for some reason, this test would fail even if the TRION(3)-18xx-MULTI is working correctly. Check if the connected sensors allow this test.*

Auto Zero

The "Auto Zero" or "Amplifier Balance" function eliminates automatically all internal amplifier offsets. It switches the differential amplifier inputs IN+ and IN- to the internal GND reference point. Consequently the offset of the module is adjusted to zero for all ranges. This function can take up to 4 seconds. It allows compensating the long term zero drift, as well as temperature drifts of the amplifier. The determined offset correction values are stored in the user memory section of the EE-Prom.

Gain Cal

The "Gain cal" function is an adjustment function. It applies the internal calibration source to all channels. The measured error is used as a calibration factor for correcting the measurements. These correction factors are stored in the user memory section of the EE-Prom.

Three Wire Offset

This function is not supported by the TRION(3)-18xx-MULTI series

Reset Default

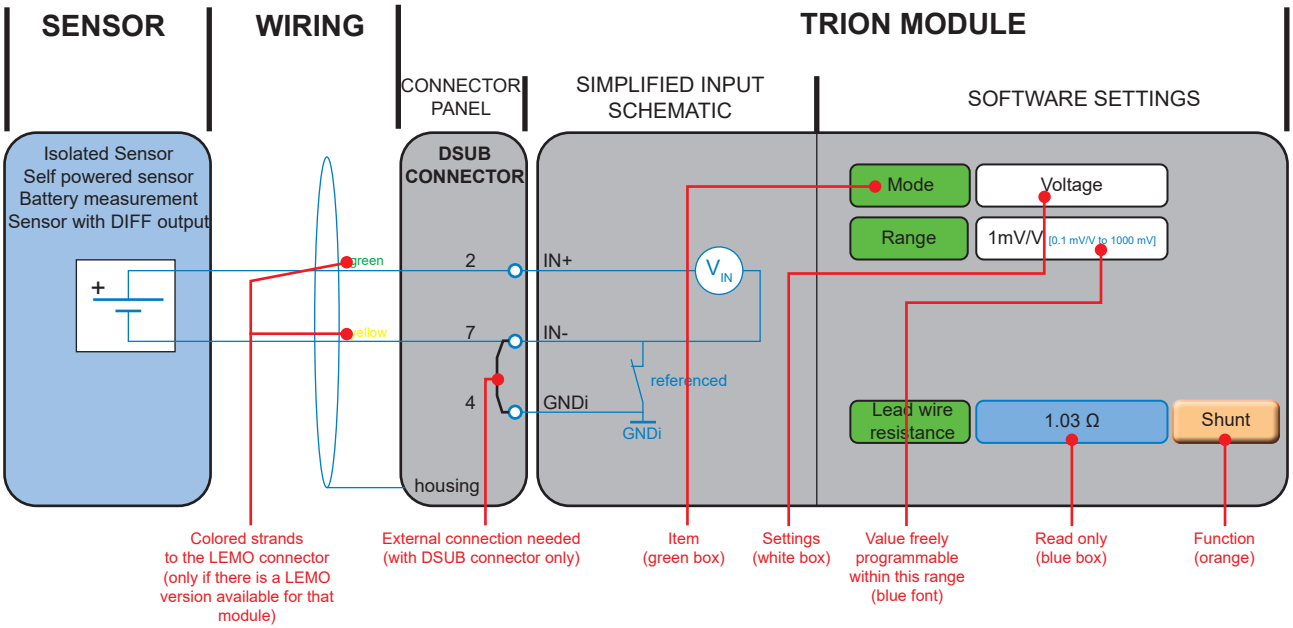
All correction values stored in the user memory will be cleared.

Save/Save all

Stores the test result in an .XML File on the HDD.

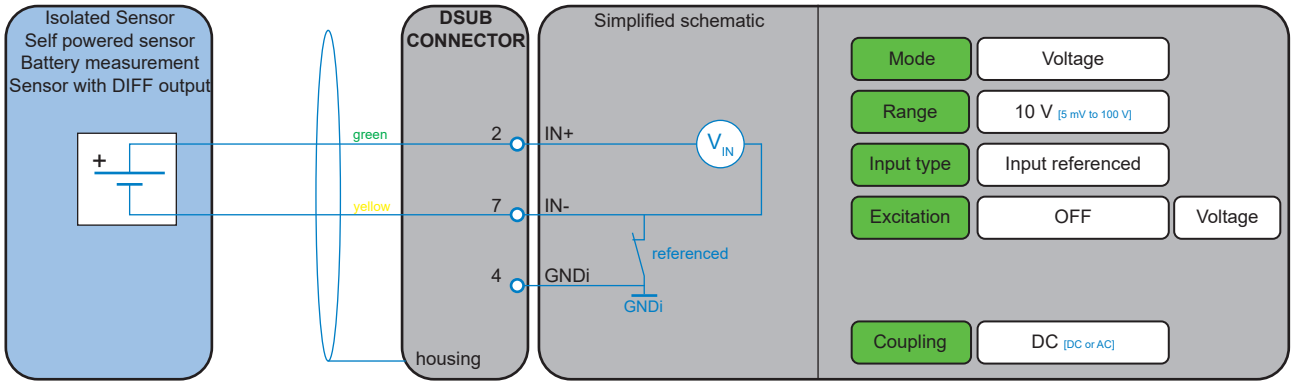
Signal connection

The following schematics will give you an overview on how to connect all the different sensors to the TRION-18xx-MULTI module. To make things easier, the example below will introduce you on how to read the schematics.

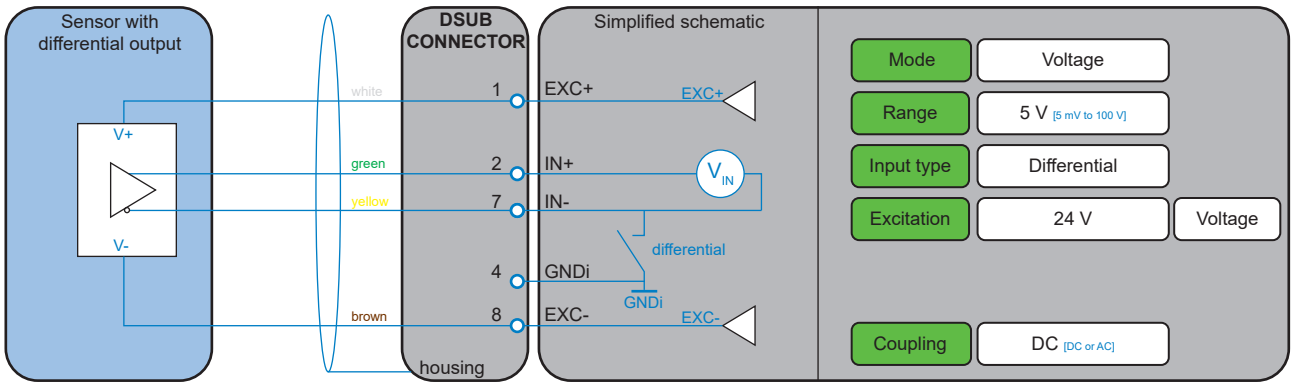


Voltage

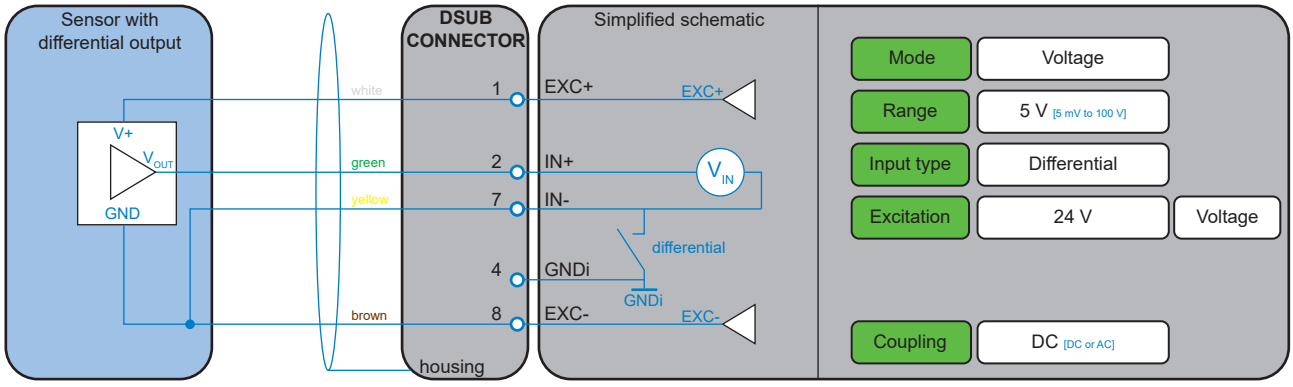
Voltage measurement



Differential output sensor powered by the TRION module

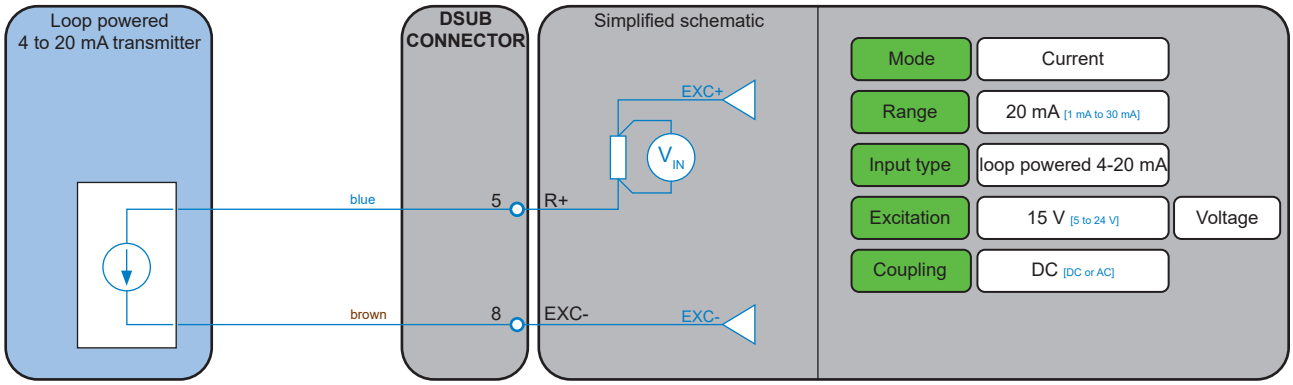


Single ended sensor powered by the TRION module

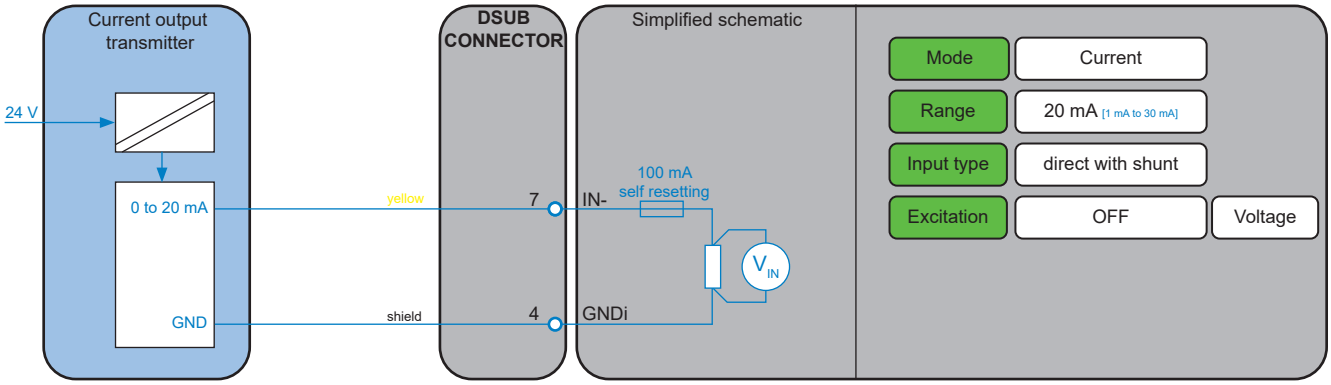


Current

Loop powered 4 to 20 mA transmitter

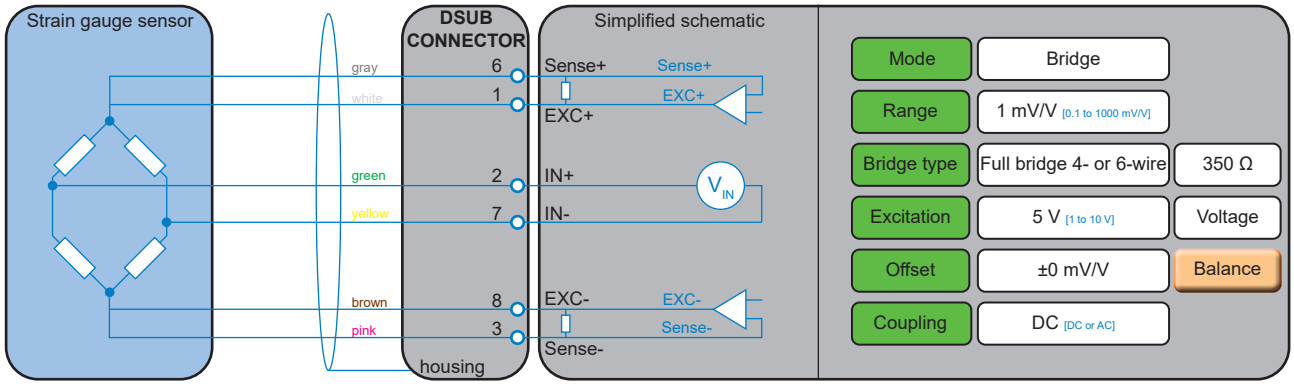


Current measurement (with external powered transmitter)



Strain gauge measurement

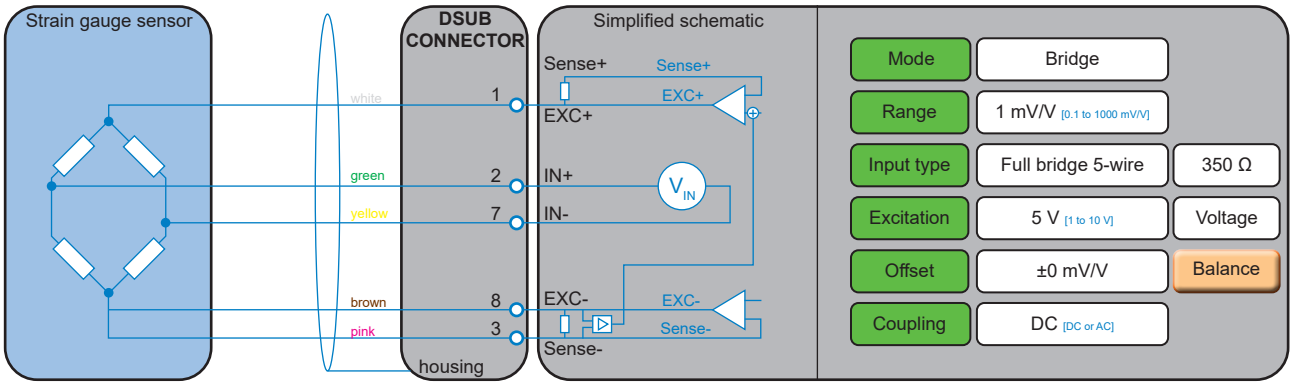
Full bridge 6-wire



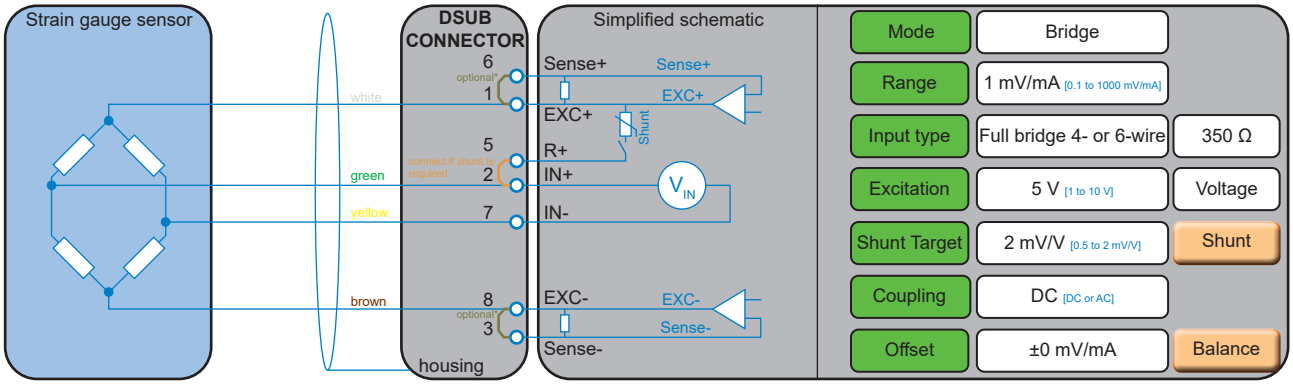
High speed strain gage measurement (>50 kHz)

For high speed applications (bandwidth >50 kHz) it is not recommended using the internal completion circuit, especially if you have long sensor cables. Usually it is better using external completion resistors nearby the strain gage or use full bridge sensors. The advantage is you will get a differential signal out of the sensor. Disturbances and sensor cable included noise will be minimized. Also lower resistance values of the strain gages reduces the noise because of lower thermal noise and signal source resistance. These resistors should have the same value as the srtain gage. They should also have a low temperature coefficient.
A value below 25 ppm/°C is recommended.

Full bridge 5-wire

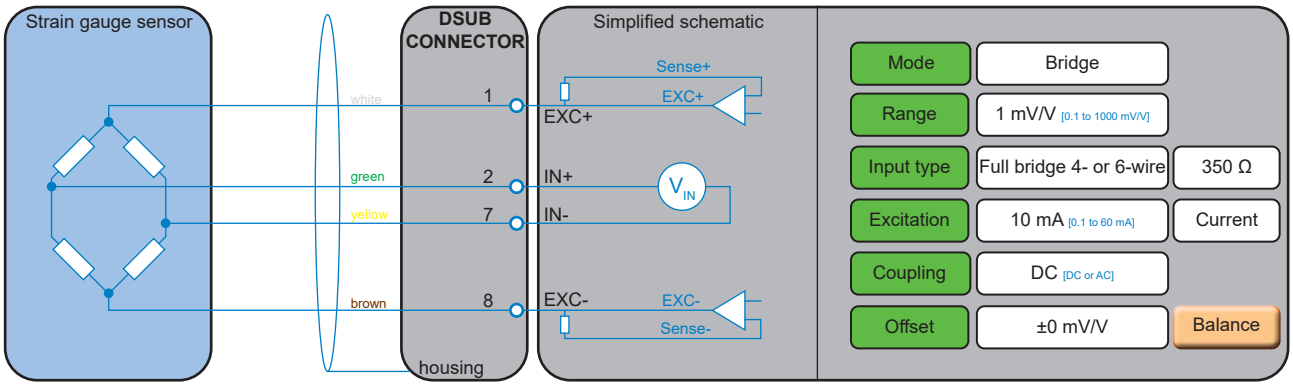


Full bridge 4-wire

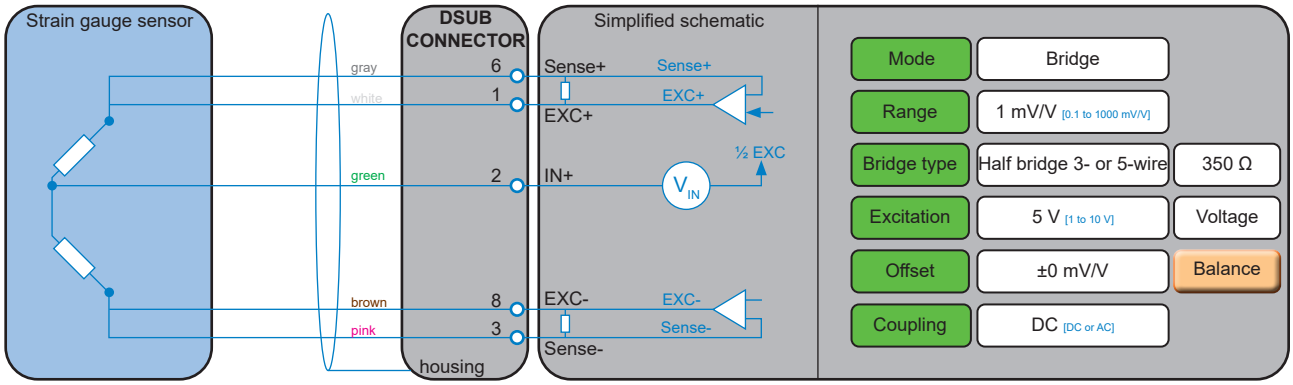


*) Optional: might be installed on existing sensor cables from previous amplifier series.

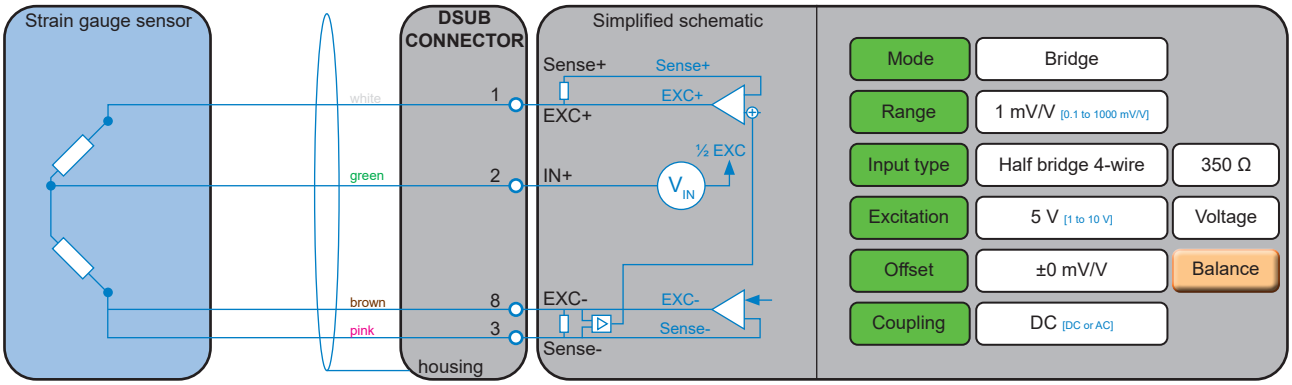
Full bridge 4-wire with constant current excitation



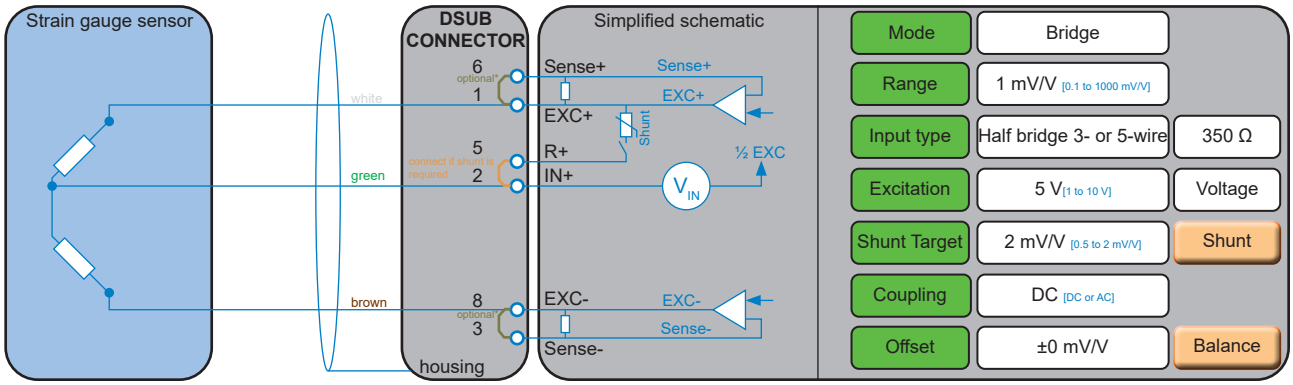
Half bridge 5-wire



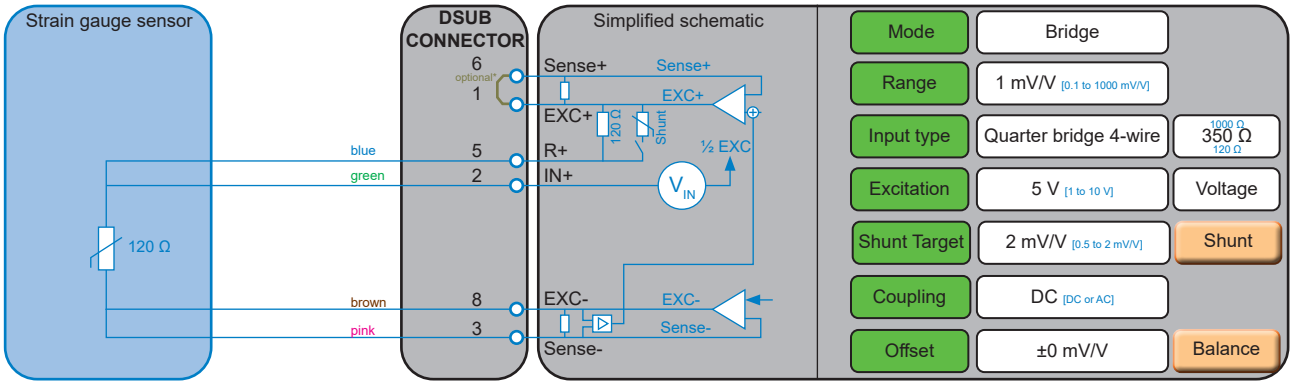
Half bridge 4-wire



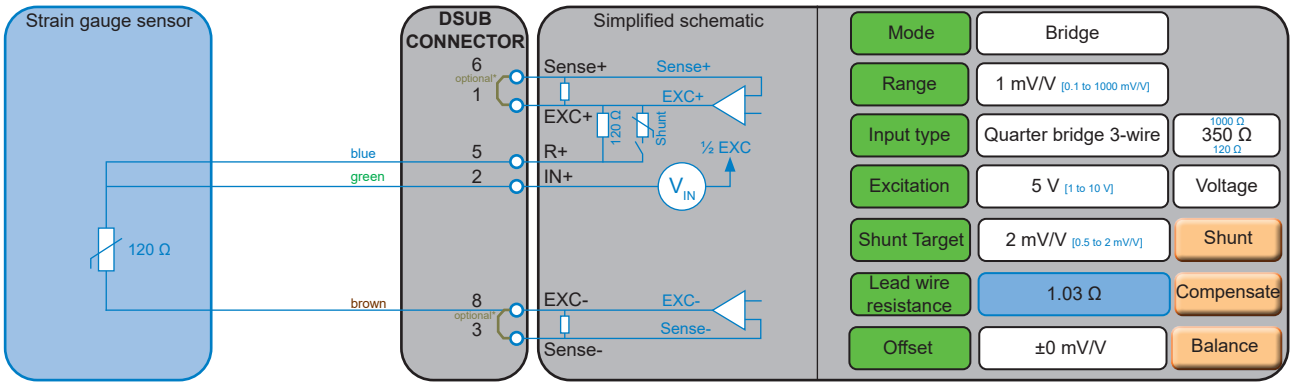
Half bridge 3-wire



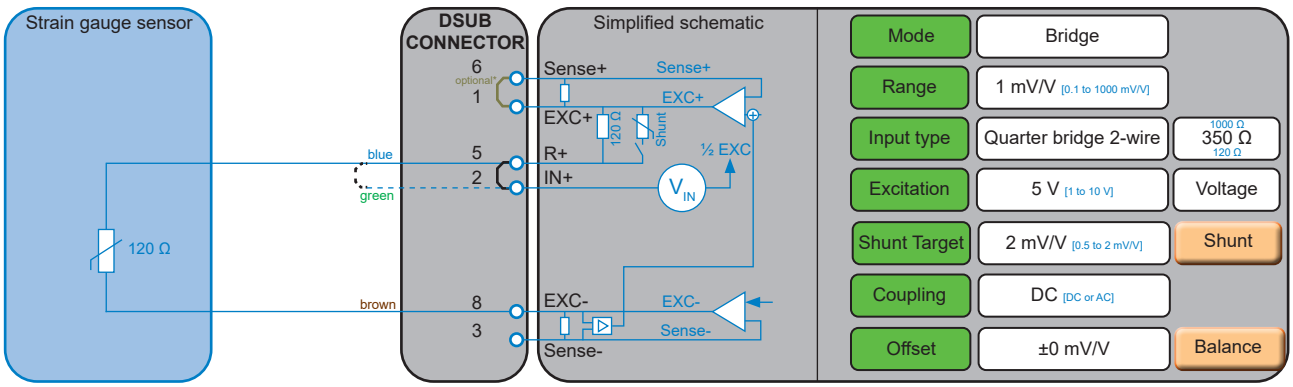
Quarter bridge 4-wire



Quarter bridge 3-wire

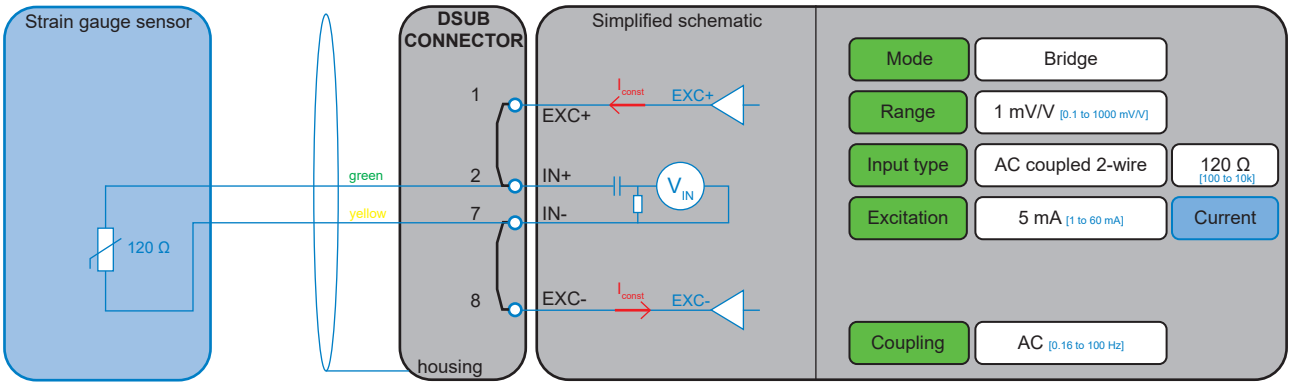


Quarter bridge 2-wire



TRION(3)-18xx-MULTI

Quarter bridge 2-wire with constant current supply and AC coupling

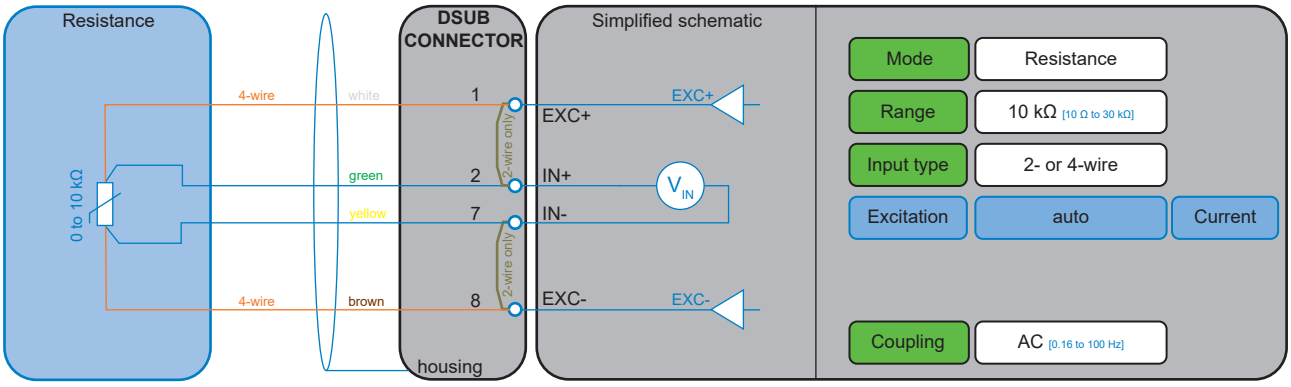


Resistance

TRION(3)-18xx-MULTI resistance accuracy

TRION(3)-18xx-MULTI resistance accuracy		
Range (Ω)	**Excitation current (mA)**	**Accuracy**
30 k	0.2	6 Ω ±1 % of reading
10 k	0.5	2 Ω ±0.45 % of reading
3000	1	0.6 Ω ±0.25 % of reading
1000	1	0.2 Ω ±0.25 % of reading
300	1	80 mΩ ±0.25 % of reading
100	1	40 mΩ ±0.25 % of reading
30	5	8 mΩ ±2 % of reading
10	5	4 mΩ ±2 % of reading

Resistance measurement

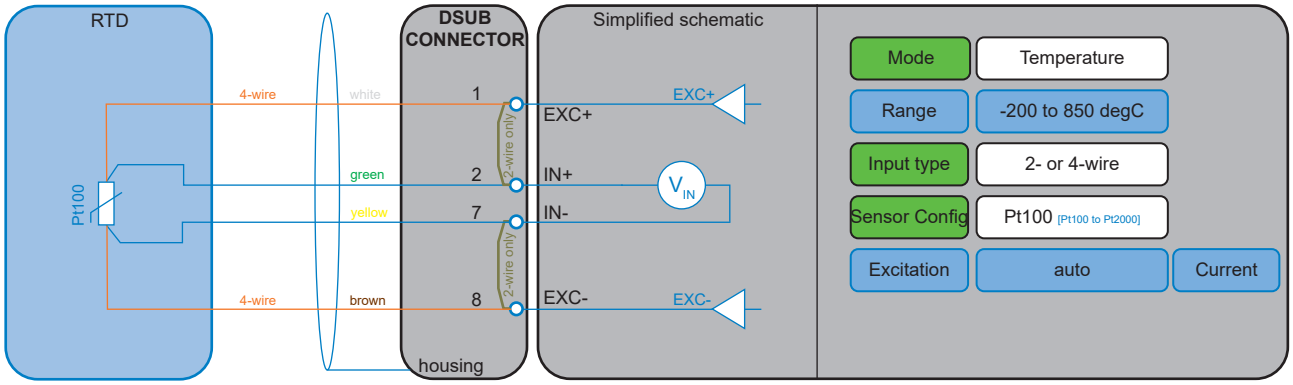


RTD (Pt100, Temperature)

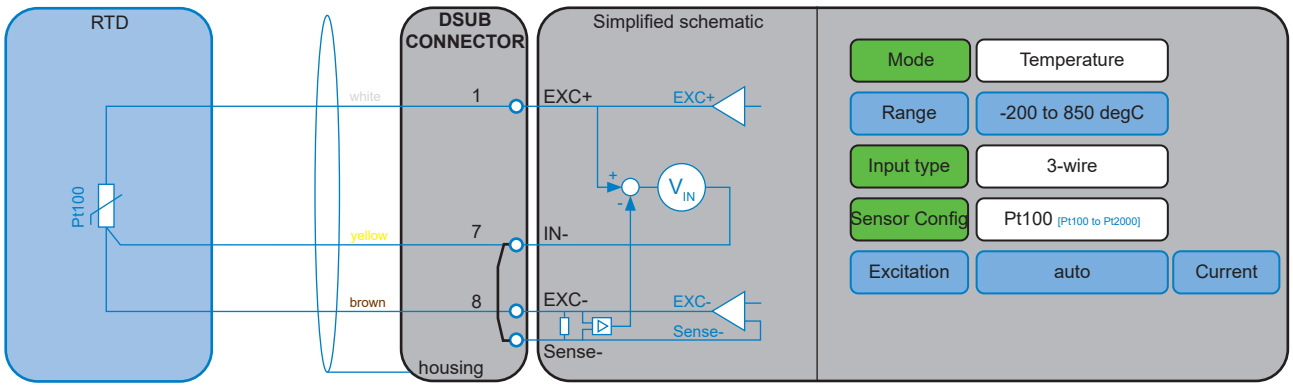
TRION(3)-18xx-MULTI RTD accuracy				
RTD (Type)	**Temperature range (°C)**	**Excitation current (mA)**	**Range (Ω)**	**Accuracy**
Pt100	-200 to 850	1	1000	0.9 °C ±0.33 % of reading
Pt200	-200 to 850	1	1000	0.7 °C ±0.33 % of reading
Pt500	-200 to 850	1	2000	0.7 °C ±0.33 % of reading
Pt1000	-200 to 850	0.5	10000	1.1 °C ±0.4 % of reading
Pt2000	-200 to 850	0.5	10000	1.1 °C ±0.4 % of reading

RTD temperature drift specification for TRION(3)-18xx-MULTI					
Range (Ω)	**Excitation current (mA)**	**Voltage range (V)**	**Accuracy**	**Temp drift (ppm /°C)**	**RTD sensor**
10 k	0.5	10	2 Ω ±0.45 %	100	Pt2000, Pt1000
3 k	1	10	0.6 Ω ±0.25 %	100	Pt500
1 k	1	2	0.2 Ω ±0.25 %	100	Pt200, Pt100

RTD 2- and 4-wire temperature measurement

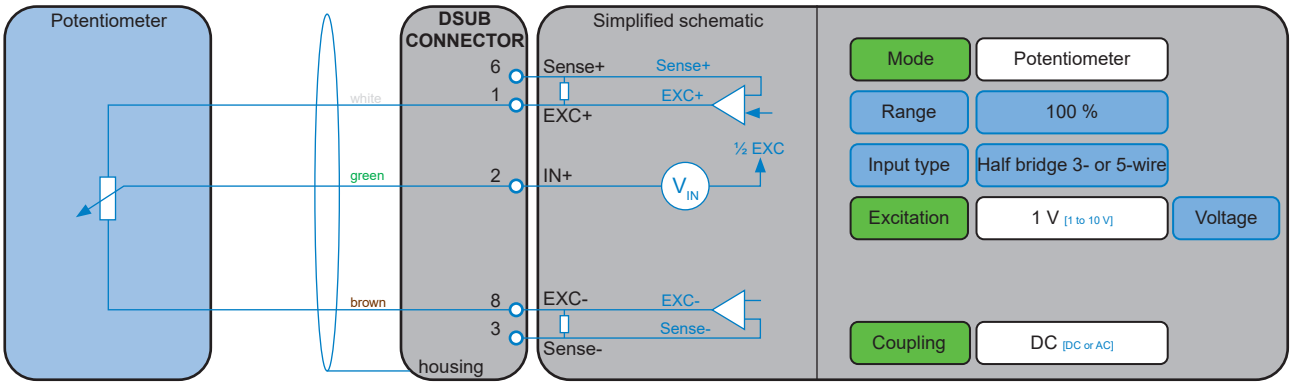


RTD 3-wire temperature measurement



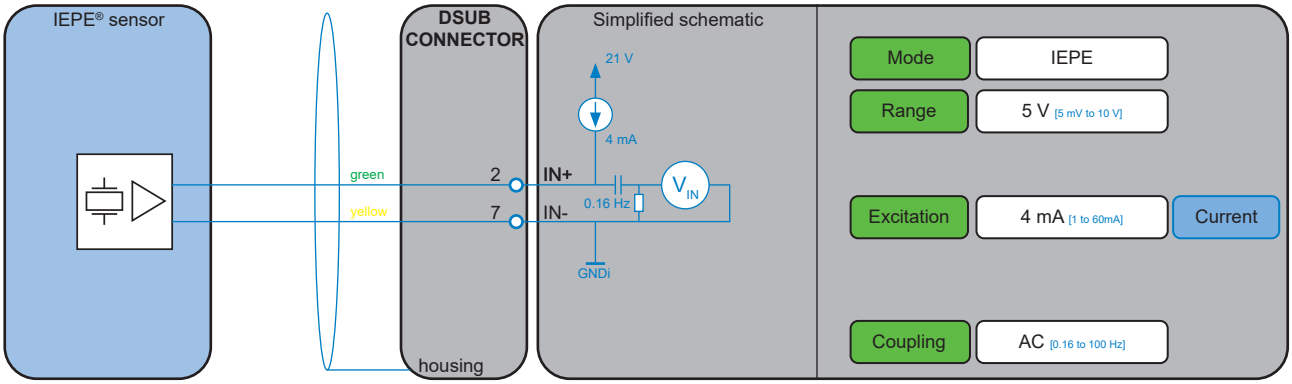
Potentiometer

Potentiometer



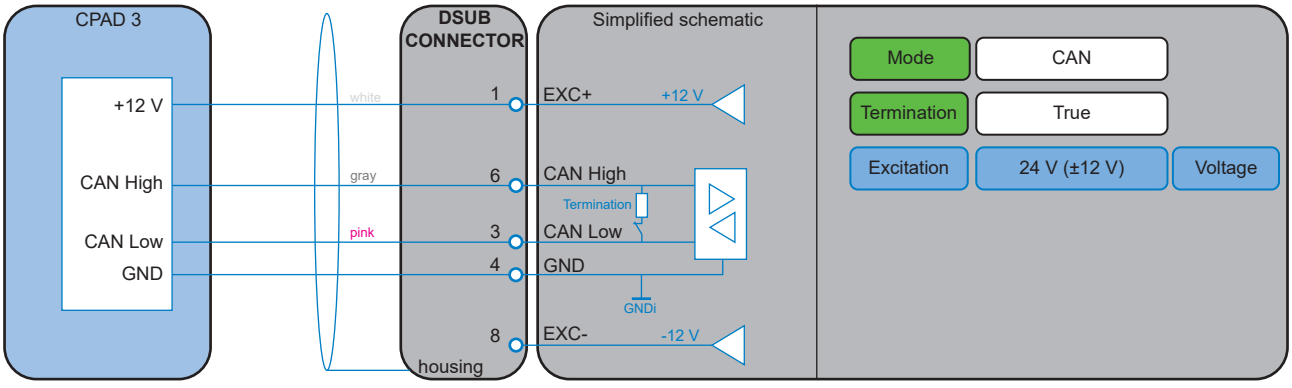
IEPE®

IEPE® sensor



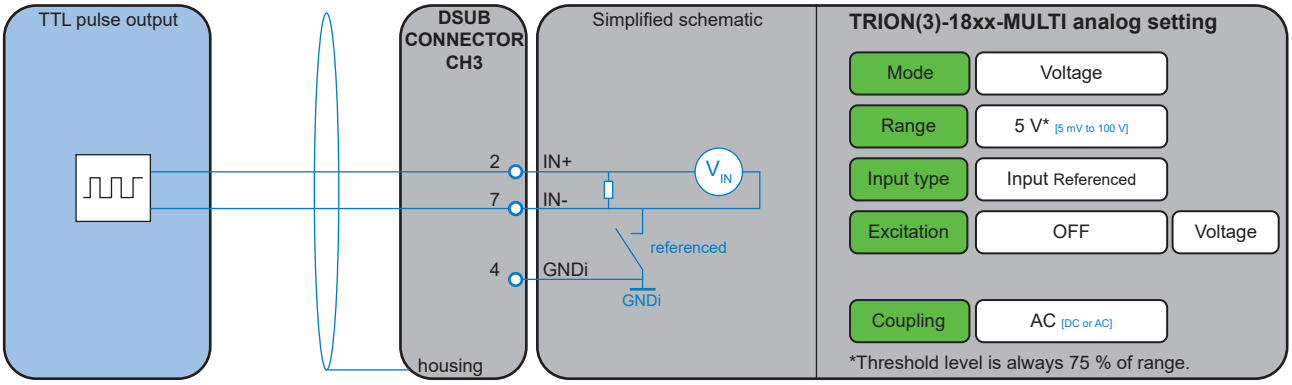
CAN

CAN



Counter

Counter measurement



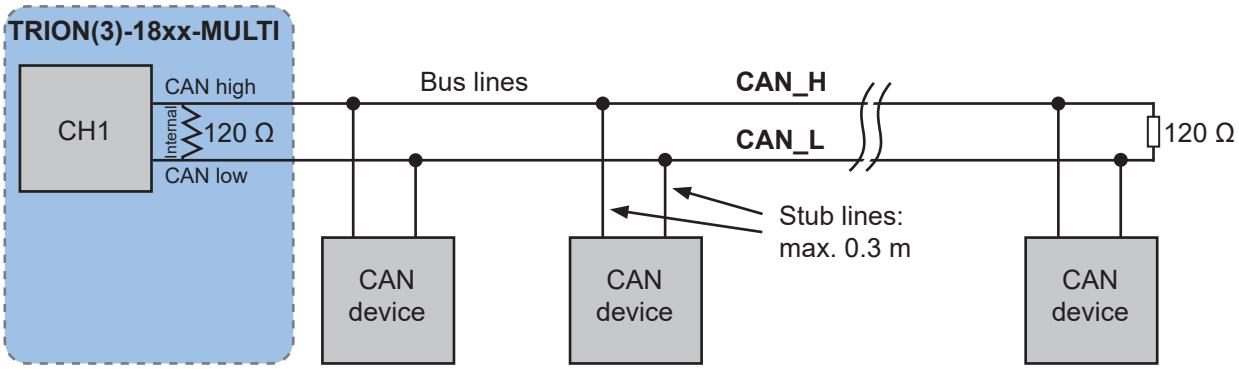
Hint: Every analog input mode can be used with the counter function, not just voltage ones. This is just an example.

Caution: *Set the analog input range small enough that the 75 % of range threshold level will be reached by your signal source. The input area can be freely programmed and so implicitly also the threshold level.*

TRION(3)-18xx-MULTI counter setting

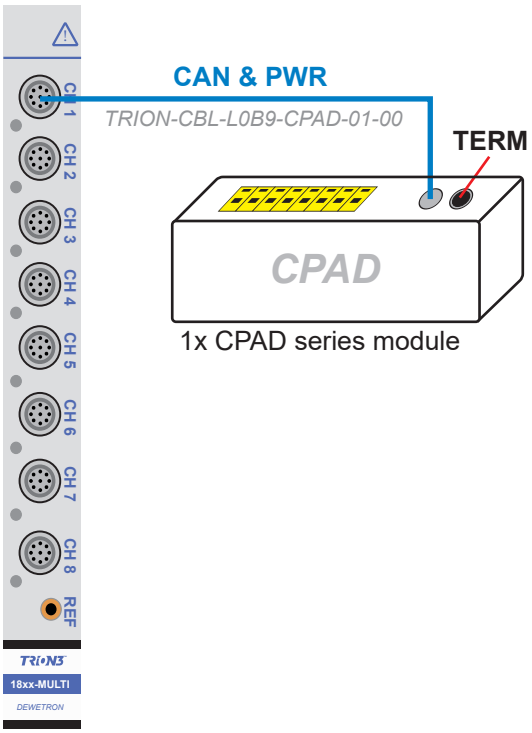
Mode	Frequency [Events]
Threshold	3.75 V
Retrigger	1.25 V
Filter	1 µs
Source	Input AI3

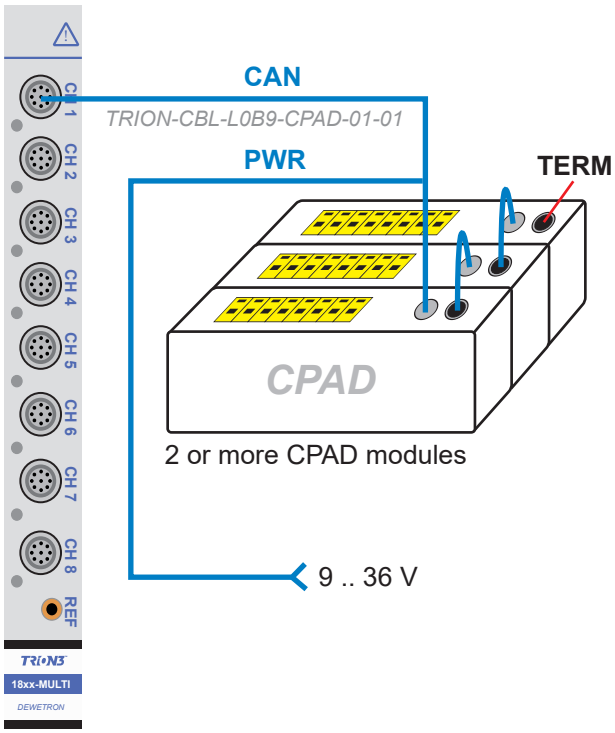
CAN bus connection



Connecting CPADs to the TRION(3)-18xx-MULTI

One single CPAD series module can be directly connected to channel 1 of the TRION(3)-18xx-MULTI module. If more modules are required an additional power supply is needed.

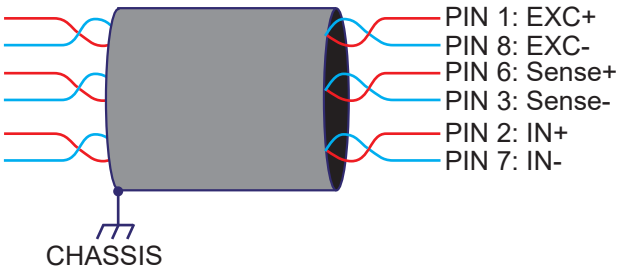




Cables and Shielding

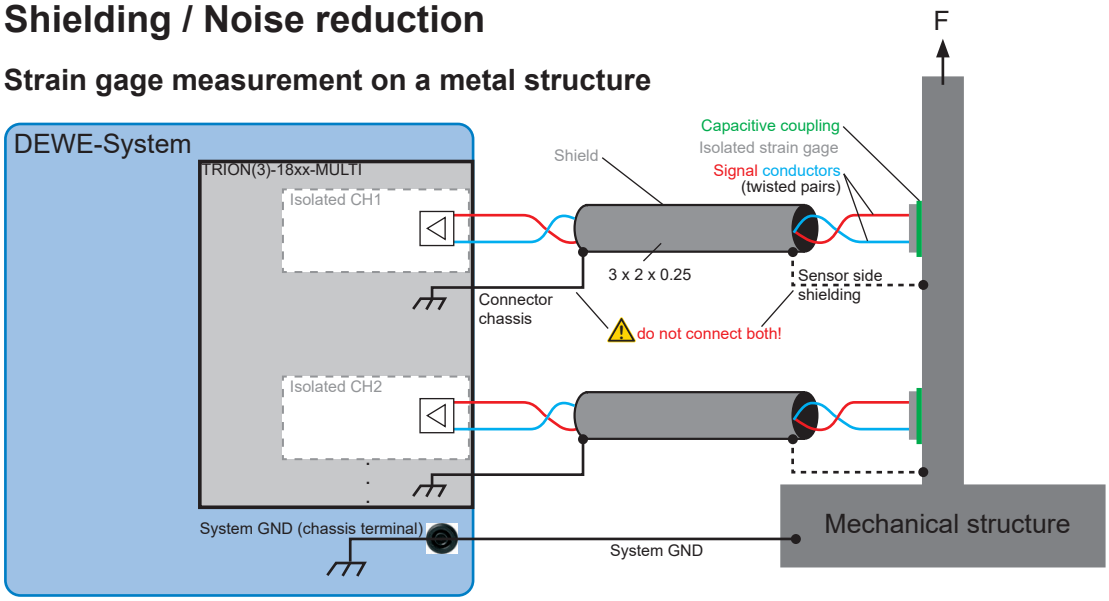
To keep the influence of electromagnetic disturbances as small as possible, shielded twisted pair cables are recommended. Connect the shield to the connector housing or to the mechanical structure.

The twisted pairs for **full bridge**, **half bridge**, **voltage** and **resistance** mode are:



Shielding / Noise reduction

Strain gage measurement on a metal structure

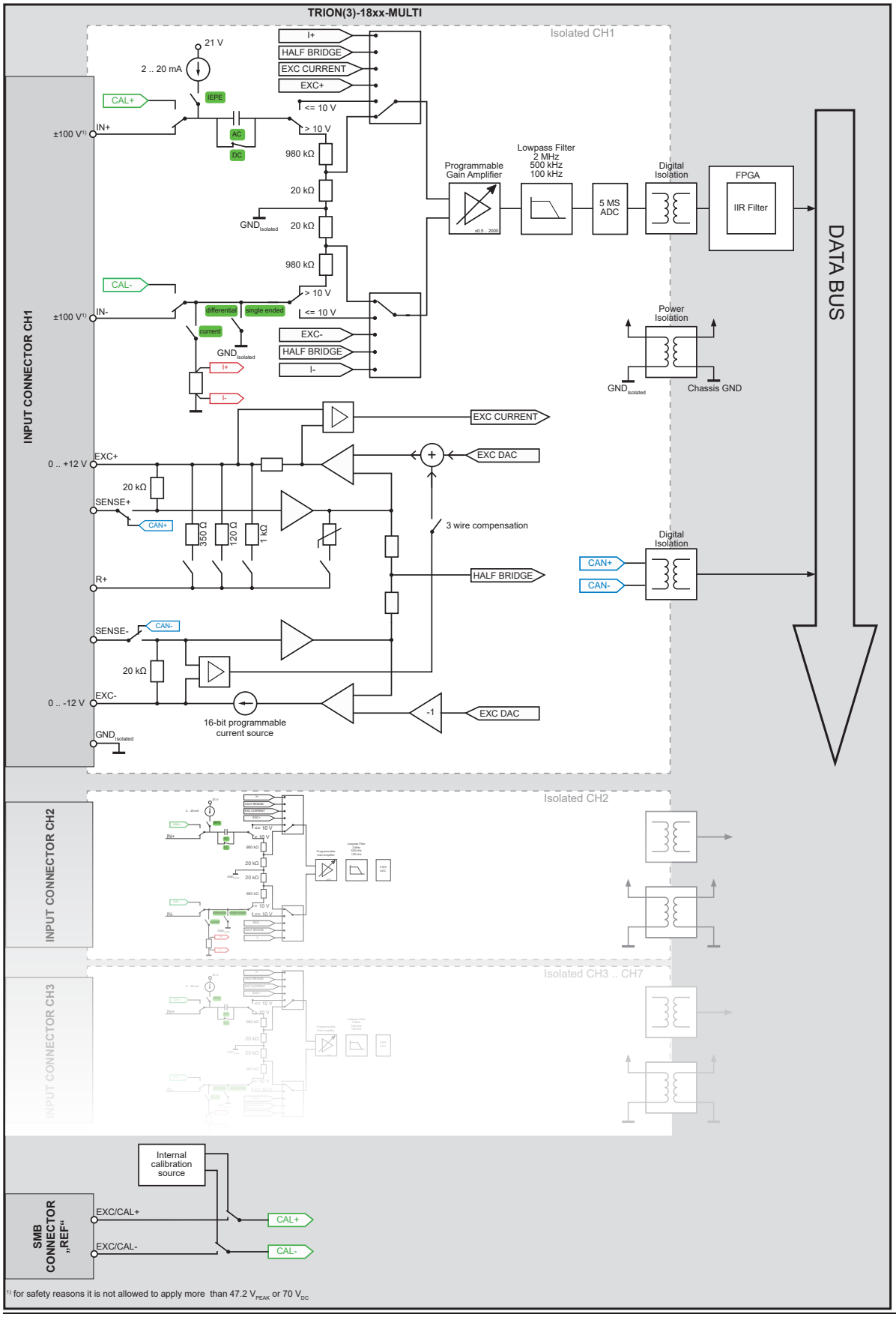


CAUTION: *Connect cable shield either to the connector chassis on the TRION side, **or** to the structure on the sensor side. Do **NOT** connect on both sides!*

It is always important that you connect your DEWETRON system ground (chassis terminal) to the ground potential of your measured object. This guarantees that the measurement system is not floating against the measured structure. It could simply be a connection to the metal structure of your proving ground. In case of an automotive application for example, it would be a connection to the cars chassis. Only if the DEWETRON system and the measured structure have an earth connection the system grounding line might not be needed.

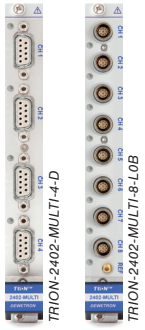
CAUTION: *Grounding concept has changed from TRION-2402-MULTI series.*

Block diagram



Universal analog input module

- Sampling: 24 bit; 204.8 kS/s per channel
- Input types: Voltage, bridge, resistance, RTD and IEPE®
- Isolation: 350 V_{DC} channel to channel and channel to chassis
- CAN: High-speed CAN2.0 port



Module specifications

TRION-2402-MULTI specifications		
Input channels	TRION-2402-MULTI-4-D TRION-2402-MULTI-8-L0B	4 channels SUB-D connector (CH1 can be used as CAN port) 8 channels 0B LEMO connector (CH1 can be used as CAN port)
ADC		
Resolution	24 bit	
Sampling rate	1 kS/s to 204.8 kS/s per channel	
Isolation[1)]	Channel to channel Channel to chassis	350 V_{DC}[1)] 350 V_{DC}[1)]
Input ranges		
Voltage	±5 mV to ±100 V[1)] freely programmable	
IEPE®	±100 mV to ±10 V freely programmable	
Bridge	1 to 1000 mV/V	
Resistance	10, 30, 100, 300 Ω, 1, 3, 10, 30 kΩ	
Accuracy[2)]	±0.02 % of reading ± 0.02 % of range ±20 µV	
Gain drift	typical 10 ppm/°C max. 20 ppm/°C	
Offset drift	typical 0.3 µV/°C+ 10 ppm of range/°C, max 2 µV/°C + 20 ppm of range/°C	
Linearity	typical ±0.01 %	
Input impedance	0 to 10 V range 10 to 100 V range	100 MΩ 2 MΩ
Input bias current	<5 nA	
Input configuration	Single ended or differential (programmable)	
Input coupling	DC / AC (high pass filter 0.16 Hz)	
Common mode voltage	±350 V_{DC}[1)]	
Over voltage protection	0 to 10 V range 10 to 100 V range	50 V continuous, 100 V for 60 seconds 250 V
Excitation voltage range	0 to 24 V_{DC} freely programmable; separately for each channel	
Resolution	1 mV	
1 year Accuracy	±0.03 % ±1.5 mV	
Drift	±10 ppm/°C ±50 µV/°C	
Current limit	0.1 to 5 V: 100 mA >5 V to 24 V: limited to 0.5 W	
Protection	Continuous short	
Load and line regulation error	±0.002 % with sense line connected	
Excitation current	0.1 to 60 mA_{DC} (programmable, 16 bit DAC)	
Resolution	1 uA	
1 year Accuracy	0.1 to 5 mA: 0.05 % ±2 µA	>5 to 60 mA: 2% ±5 µA
Drift	15 ppm/°C	
Compliance voltage	0.1 to 20 mA: 24 V >20 mA: 10 V	
Output impedance	>10 MΩ	

continued on next page ...

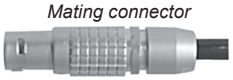
TRION-2402-MULTI

Supported sensors	4- 6-wire full bridge 3- 5-wire ½ bridge with internal completion (software programmable) 3- or 4-wire ¼ bridge with internal resistor for 120 Ω and 350 Ω (software programmable) 4-wire full bridge with constant current excitation (piezoresistive bridge sensors) Potentiometer Resistance Resistance temperature detection: Pt100, Pt200, Pt300, Pt500, Pt1000, Pt2000 (2-, 3-, 4-wire) IEPE®															
Bridge resistance	80 Ω to 10 kΩ @ ≤ 5 V_{DC} excitation															
Shunt calibration	Two internal shunt resistors 50 kΩ and 100 kΩ															
Shunt and completion resistor accuracy	0.05 % ±15 ppm/K															
Automatic bridge balance	±400 % of Range															
Typical Signal-to-noise ratio, Spurious free	10 mV range				100 mV range				1 V range				10 V range			
SNR, Effective number of Bits[3], Noise V_{PP}	SNR	SFDR[4]	ENOB[5]	Noise	SNR	SFDR[4]	ENOB[5]	Noise	SNR	SFDR[4]	ENOB[5]	Noise	SNR	SFDR[4]	ENOB[5]	Noise
Sample rate	[dB]	[dB]	[Bit]	[mV_{pp}]	[dB]	[dB]	[Bit]	[mV_{pp}]	[dB]	[dB]	[Bit]	[mV_{pp}]	[dB]	[dB]	[Bit]	[mV_{pp}]
1 kS/s	82	108	13.3	0.002	101	128	16.5	0.002	111	141	18.1	0.025	112	141	18.3	0.100
10 kS/s	82	108	13.3	0.005	101	123	16.5	0.005	106	134	17.3	0.030	112	140	18.3	0.120
100 kS/s	72	103	11.7	0.015	92	123	15.0	0.016	104	134	17.0	0.058	104	136	17.0	0.210
200 kS/s	69	99	11.2	0.022	88	120	14.3	0.025	88	133	14.3	0.230	96	135	15.7	0.950
200 kS/s; Filter = OFF	69	99	11.2	0.059	80	106	13.0	0.061	81	106	13.2	1.300	81	106	13.2	5.400
Typical THD	-100 dB															
Typcial Crosstalk	-125 dB (10 V range; 0 to 1 kHz)															
Typical CMRR	110 dB @ 50 Hz, 90 dB @ 1 kHz, 80 dB @ 10 kHz															
Self test (self calibration)	Each channel is able to perform a complex self test by using internal high precision references															
Low pass filter (-3 dB, digital)	1 Hz to 40 % of sample rate freely programmable or OFF															
Characteristic	Bessel or Butterworth															
Filter order	2^{nd} , 4^{th}, 6^{th}, 8^{th}															
Filter setting AUTO	30 % of sample rate with 8^{th} order Bessel															
Analog anti-aliasing filter	2^{nd} order Bessel,															
Sample rate > 10 kS/s	250 kHz (-3 dB), 150 kHz (-1 dB)															
ADC anti-aliasing filter	-3 dB @ Filter = OFF															
1 kS/s ≤ fs ≤ 51.2 kS/s	0.494 fs															
51.2 kS/s < fs ≤ 102.4 kS/s	0.49 fs															
102.4 kS/s < fs ≤ 204.8 kS/s	0.38 fs fs = sample frequency															
Channel to channel phase mismatch	typically <60 nsec between channels using the same range															
CAN specification	CAN 2.0															
CAN Physical Layer	High Speed															
CAN Termination	Programmable: High Impedance or 120 Ω															
CAN bus protection	±36 V															
Input connector	9-pin LEMO EPG.0B.309, 9-pin SUB-D connector															
REF connector	SMB															
Supported MSI	MSI-BR-TH-x, MSI-BR-CH-x, MSI2-TH-x, MSI2-CH-x, MSI2-LVDT															

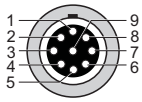
[1] for safety reasons it is not allowed to apply more than 47.2 V_{PEAK} or 70 V_{DC}
[2] 1 year accuracy 23 °C ±5 °C
[3] LP Filter in auto mode
[4] SFDR excluding harmonics
[5] ENOB calculated from SNR

TRION-2402-MULTI-8-L0B module

Mating connector



FGG.0B.309.CLADxx



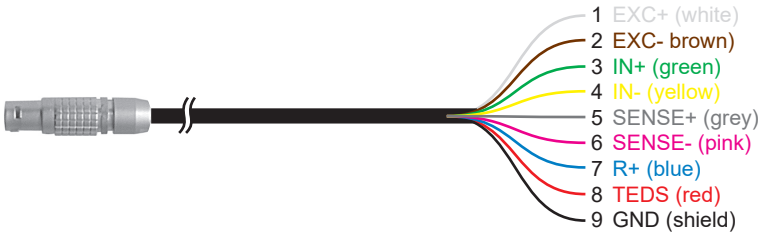
9-pin LEMO connector female
EPG.0B.309.HLN

1	EXC+ (CAN power supply 12 V, CH1 only)
2	EXC-
3	IN+
4	IN-
5	SENSE+ (CAN high, CH1 only)
6	SENSE- (CAN low, CH1 only)
7	R+
8	TEDS
9	$GND_{isolated}$ (CAN GND, CH1 only)
Housing	$GND_{isolated}$

Optional accessory

TRION-CBL-L0B9-OE-xx-00

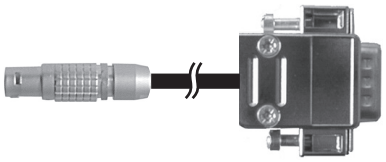
01 = 1 meter
05 = 5 meters



High quality cable from Lemo 0B.309 plug to open end, 1 or 5 meters.

TRION-CBL-L0B9-D9-0.5-01

High quality adaptor cable from Lemo 0B.309 plug to SUBD-9 socket, 0.5 meters.



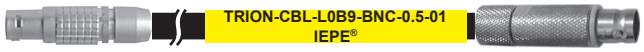
1 EXC+
2 IN+
3 SENSE+
4 GND
5 R+
6 SENSE-
7 IN-
8 EXC-
9 TEDS

For connecting any MSI-BR series adaptors

TRION-CBL-L0B9-BNC-0.5-01

High quality adaptor cable from LEMO 0B.309 plug to BNC cable jack, 0.5 meters

For connecting IEPE® sensors to TRION-2402-MULTI-8-L0B modules.



Hot: IN +
Shield: IN -

⚠ *NOTE: Do not use this cable for connecting voltage signals!*

TRION-CBL-L0B9-BNC-0.5-02

High quality adaptor cable from LEMO 0B.309 plug to BNC cable jack, 0.5 meters

For connecting voltage signals to TRION-2402-MULTI-8-L0B modules.

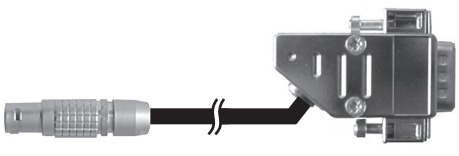
Hot: IN +
Shield: IN -

⚠ *NOTE: Do not use this cable for connecting IEPE® sensors!*

TRION-CBL-L0B9-CAN-0.5

Adaptor cable from LEMO 0B.309 plug to D-SUB-9 plug for CAN, 0.5 meters

For TRION-2402-MULTI-8-L0B modules channel 1 only!



Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

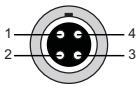
TRION-2402-MULTI

TRION-CBL-L0B9-CPAD-01-00

Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION-2402-MULTI-8-L0B (CH 1).

For connecting 1x CPAD series module



4-pin LEMO connector male
FGG.1B.304.CLL

Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-L0B9-CPAD-01-01

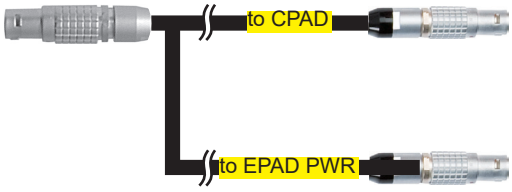
Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

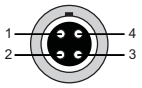
Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

For connecting the first CPAD-series module of a module-chain to a TRION-2402-MULTI-8-L0B.

Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND

For connecting more than one CPAD series module

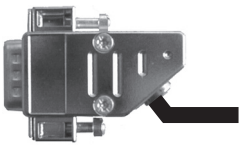




4-pin LEMO connector male
FGG.1B.304.CLL

Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

TRION-2402-MULTI-4-D module



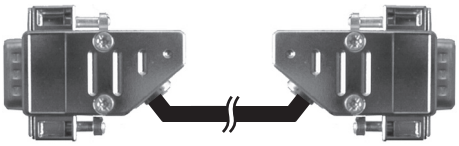
1 EXC+ (CAN power supply 12 V, CH1 only)
2 IN+
3 Sense+ (CAN low, CH1 only)
4 $GND_{isolated}$ (CAN GND, CH1 only)
5 R+
6 Sense- (CAN high, CH1 only)
7 IN-
8 EXC-
9 TEDS

Optional accessory

TRION-CBL-D9-CAN-0.5

Adaptor cable from D-SUB-9 plug to D-SUB-9 plug for CAN, 0.5 meters.

For TRION-2402-MULTI-4-D modules.

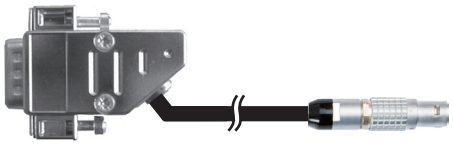


Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

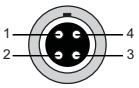
TRION-CBL-D9-CPAD-01-00

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION-2402-MULTI-4-D (CH 1).



For connecting 1x CPAD series module



4-pin LEMO connector male
FGG.1B.304.CLL

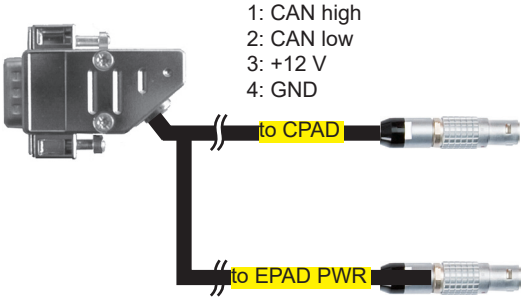
Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-D9-CPAD-01-01

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

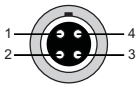
Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

For connecting the first CPAD-series module of a module-chain to a TRION-2402-MULTI-4-D.



Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND

For connecting more than one CPAD series module



4-pin LEMO connector male
FGG.1B.304.CLL

Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

LED function

Green:	Normal operation
Orange:	Channel ID function. Typically active during channel setup or CAN mode active.
Red:	Error.

TRION-2402-MULTI function overview

Isolation

The TRION-2402-MULTI is fully isolated. That means every channel has a separate isolated excitation voltage and input amplifier. The main advantages of that configuration are:

- Very high common mode voltages of ±350 V.
- Overcurrent protection e.g. if the isolation of a strain gage on a 110 V power line fails.
- Ground loops are eliminated.
- Noise reduction.

Free variable gain and excitation

Amplifier parameters like gain, excitation voltage, excitation current and sensor offset can be varied for every channel individually. This allows a perfect match of each input channel to any sensor.

Amplifier balance (amplifier zero)

The amplifier balance function eliminates automatically all internal amplifier offsets. It switches the differential amplifier inputs IN+ and IN- to the internal GND reference point. Consequently the offset of the module is adjusted to zero for all ranges. This function can take up to 4 seconds. It allows compensating the long term zero drift, as well as temperature drifts of the amplifier. It can be performed for one individual channel or for all channels at once.

Sensor balance

Normally every strain gage sensor has a certain offset caused by manufacturer tolerances and sensor mounting. The “*Sensor balance*” function removes that offset automatically up to 400 % of the selected input range.

Input Short

This function switches both differential amplifier inputs IN+ and IN- from the input terminals to the internal half bridge reference of the module. With this function, the absolute sensor offset can be determined.

Internal calibration voltage

The TRION-2402-MULTI has an internal, ultra-stable, programmable reference voltage generator. The voltage could be applied to every input channel via a relay matrix. Therefore the complete input signal path - from the analog input amplifier to the ADC can be checked. Eventually existing gain drifts can be discovered and compensated by utilizing this internal reference voltage.

External calibration (REF input)

Instead of the internal calibration voltage, an external signal can be applied to every input channel by using the calibration relay matrix. This allows external calibration of the voltage input without disconnecting the input connector (TRION-2402-MULTI-8-L0B only).

CAN

The first channel of the TRION-2402-MULTI also has a CAN bus interface. Any CAN2.0B compatible device or bus can be connected. In CAN mode, the analog input function of the channel is deactivated. Sensor excitation is switched to ±12 V. For further information, refer to chapter
"TRION-CAN module" within this manual.

One single CPAD series module can be directly connected to that channel. If more modules are required an additional power supply is needed.

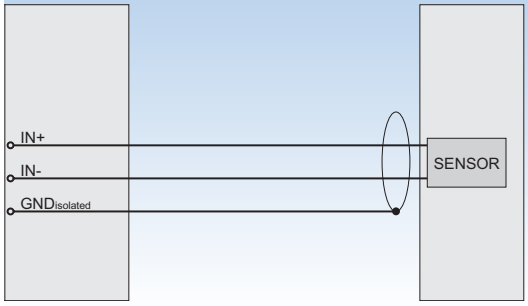
TEDS

The maximum distance between module and TEDS chip: 20 m.

Signal connection

Voltage measurement

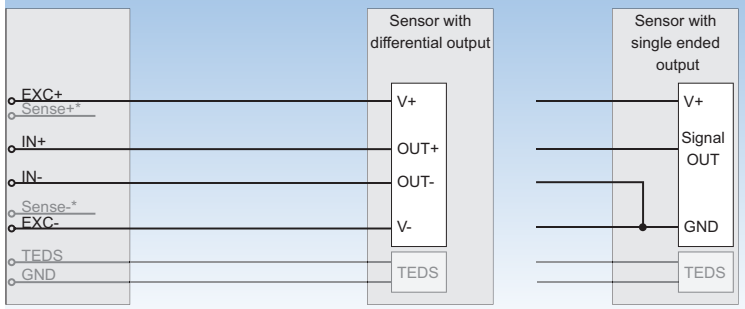
Isolated Sensors
Battery powered sensors
Sensors with differential output



Software settings for voltage measurement

Mode:	Voltage
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	Single ended
Excitation:	0 V
Type:	Voltage

Sensors with sensor supply and voltage output

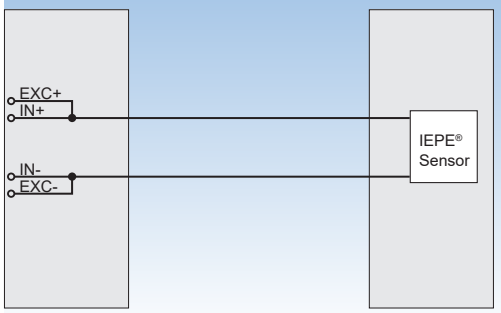


Software settings for sensors with sensor supply and voltage output

Mode:	Voltage
Range:	10 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	Differential
Excitation:	15 V
Type:	Voltage

*) Sense as well as TEDS connection is optional.

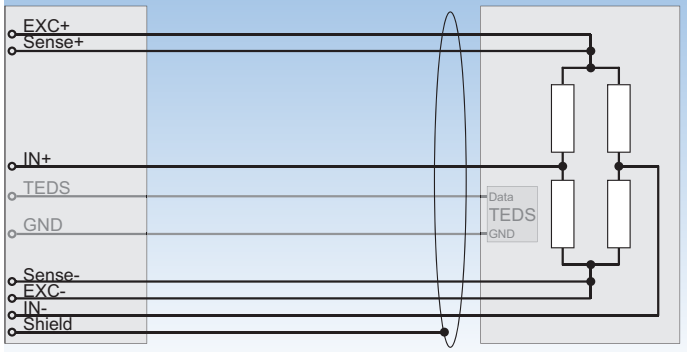
IEPE®



Software settings for IEPE® sensors

Mode:	IEPE
Range:	5 V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Coupling:	AC 0.16 Hz
Excitation:	8 mA
Type:	Current

Full bridge 6 wire sensor connection

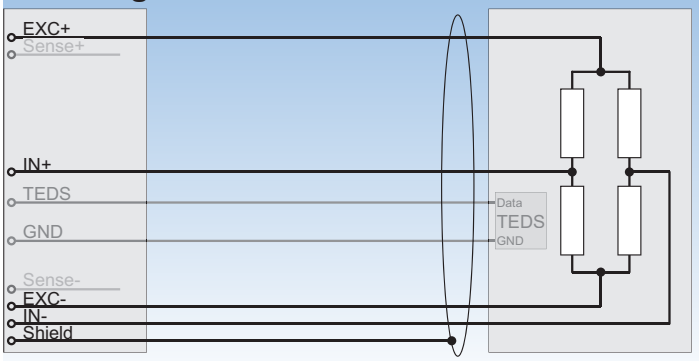


Software settings for full bridge 6 wire sensor connection

Mode:	Bridge
Range:	1 mV/V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	Full bridge
Excitation:	1 V
Type:	Voltage

Voltage or current excitation is allowed. TEDS connection is optional.

Full bridge 4 wire sensor connection

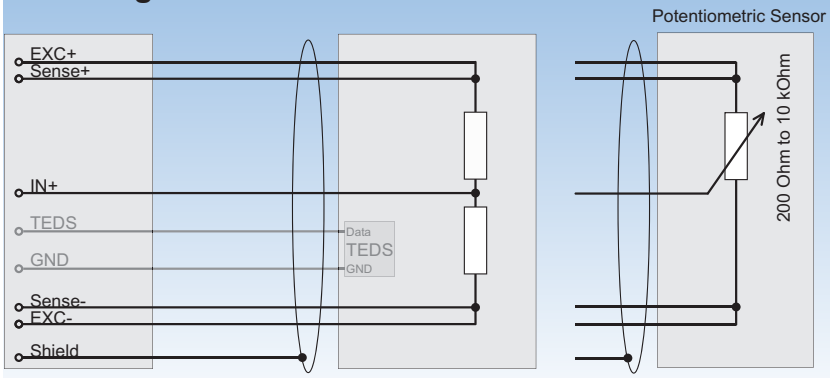


Software settings for full bridge 4 wire sensor connection

Mode:	Bridge
Range:	1 mV/V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	Full bridge
Excitation:	1 V
Type:	Voltage

Voltage or current excitation is allowed. TEDS connection is optional.

Half bridge or Potentiometer 5 wire sensor connection

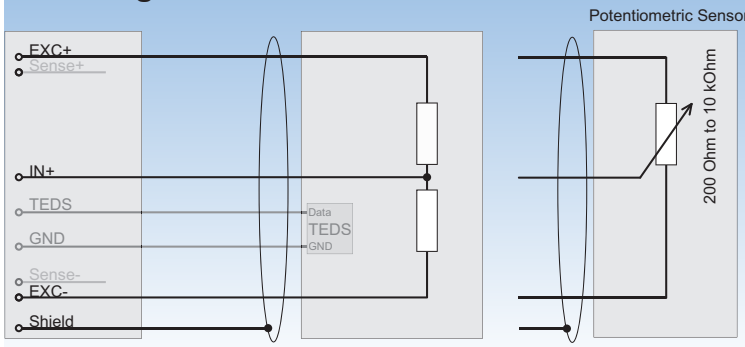


Software settings for half bridge or Potentiometer 5 wire sensor connection

Mode:	Bridge
Range:	3 mV/V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	Half bridge
Excitation:	1 V
Type:	Voltage

Voltage and current excitation is allowed. A potentiometer can be seen similar to a half bridge sensor with ±500 mV/V sensitivity. TEDS connection is optional.

Half bridge or Potentiometer 3 wire sensor connection

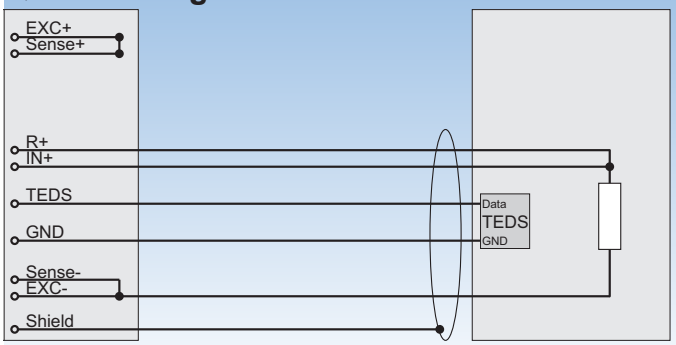


Software settings for half bridge or Potentiometer 3 wire sensor connection

Mode:	Potentiometer
Range:	100 %
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Excitation:	1 V
Type:	Voltage

TEDS connection is optional.

Quarter bridge 3 wire sensor connection

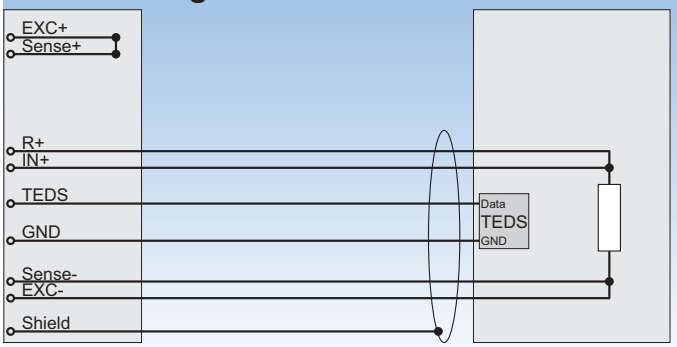


Software settings for quarter bridge 3 wire sensor connection

Mode:	Bridge
Range:	1 mV/V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	BRQUATER3W
Bridge Res:	120R
Excitation:	1 V
Type:	Voltage

The 3-wire quarter bridge is only able to compensate symmetric wire resistance.
Sense connection is optional.

Quarter bridge 4 wire sensor connection

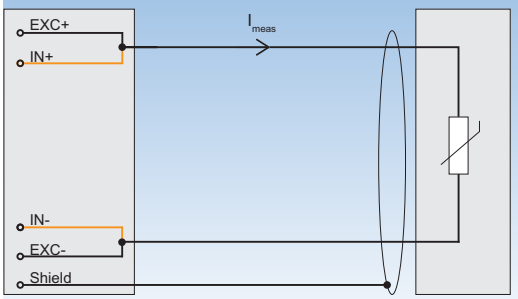


Software settings for quarter bridge 4 wire sensor connection

Mode:	Bridge
Range:	1 mV/V
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	BRQUATER4W
Bridge Res:	120R
Excitation:	1 V
Type:	Voltage

The 4 wire connection provides full lead wire resistance compensation. Sense connection is optional.

Resistance and RTD measurement 2 wire connection

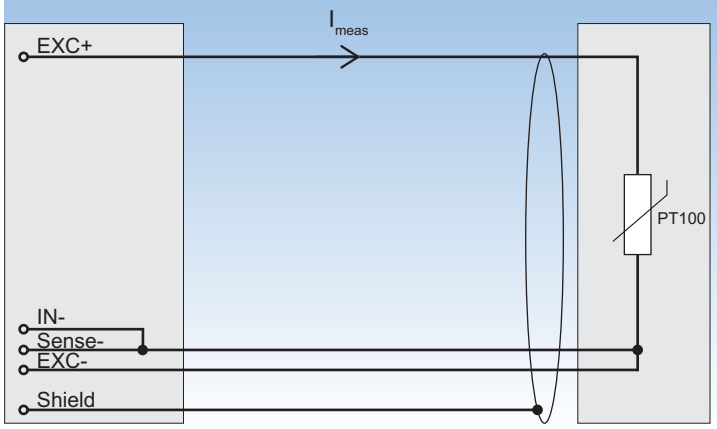


Software settings for resistance and RTD measurement 2 wire connection

Mode:	Resistance
Range:	10000 Ω
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Input type:	2 wire or 4 wire

The 2 wire technology does not compensate any lead wire resistance. For accurate temperature or resistance measurement the 4 wire technology is strongly recommended.

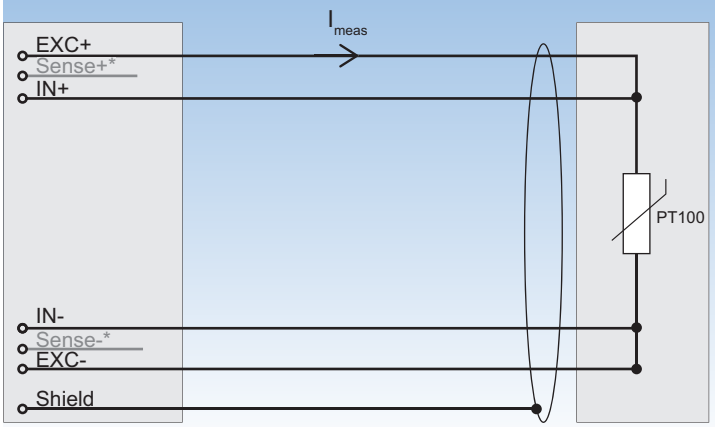
RTD measurement 3 wire connection



Software settings for resistance and RTD measurement 3 wire connection

Mode:	Temperature
Range:	-200 .. 850 °C
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Sensor type:	PT100
Input type:	3 wire

Resistance and RTD measurement 4 wire connection

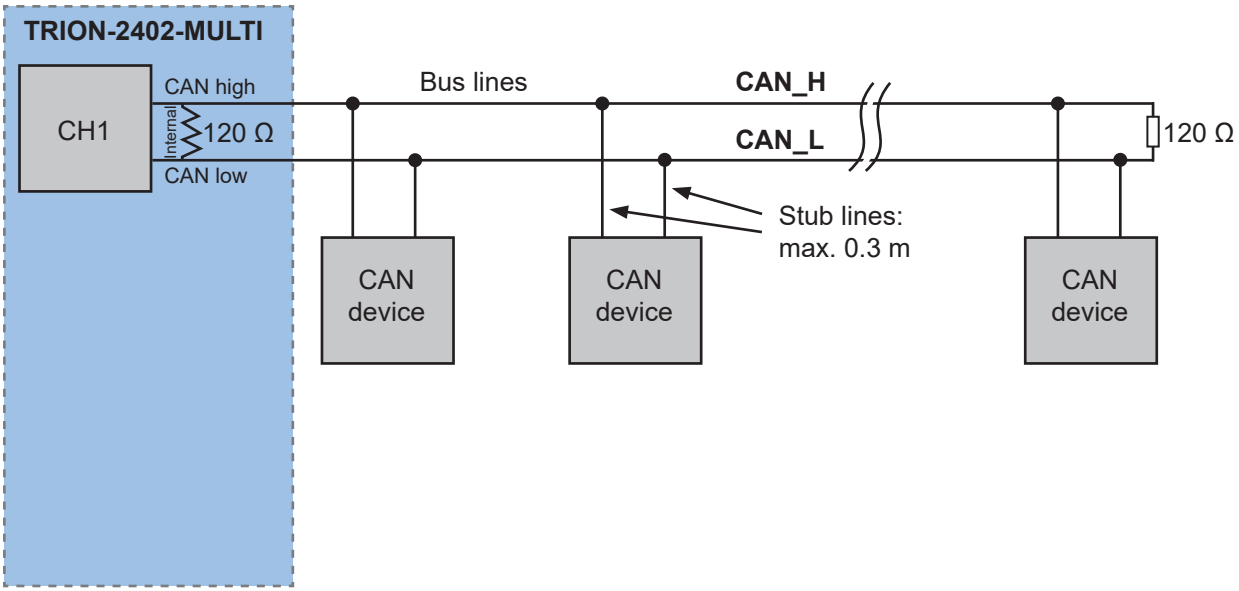


Software settings for resistance and RTD measurement 4 wire connection

Mode:	Temperature
Range:	-200 .. 850 °C
Lowpass:	10 kHz
Type:	Bessel
Order:	8th
Sensor type:	PT100
Input type:	2 wire or 4 wire

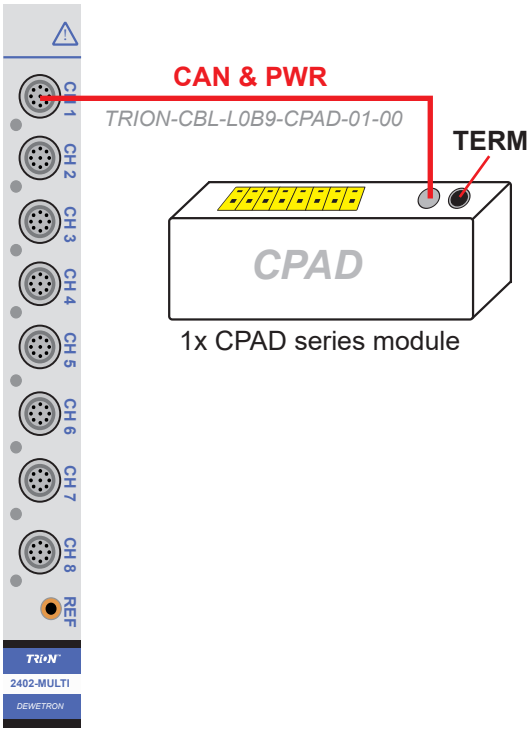
*) Sense connection is optional.

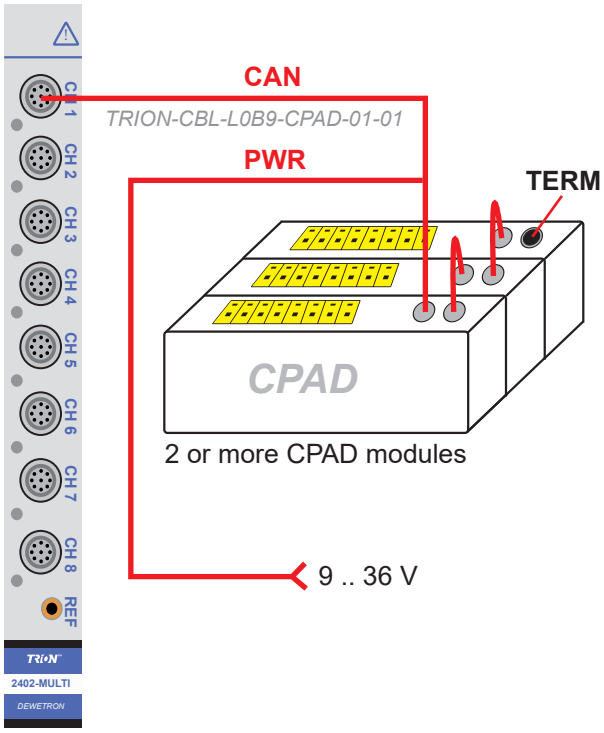
CAN bus connection



Connecting CPADs to the TRION-2402-MULTI

One single CPAD series module can be directly connected to channel 1 of the TRION-2402-MULTI module. If more modules are required an additional power supply is needed.





TRION-2402-MULTI resistance accuracy

TRION-2402-MULTI resistance accuracy		
Range (Ω)	**Excitation current (mA)**	**Accuracy**
30 k	0.2	6 Ω ±1 % of reading
10 k	0.5	2 Ω ±0.45 % of reading
3000	1	0.6 Ω ±0.25 % of reading
1000	1	0.2 Ω ±0.25 % of reading
300	1	80 mΩ ±0.25 % of reading
100	1	40 mΩ ±0.25 % of reading
30	5	8 mΩ ±2 % of reading
10	5	4 mΩ ±2 % of reading

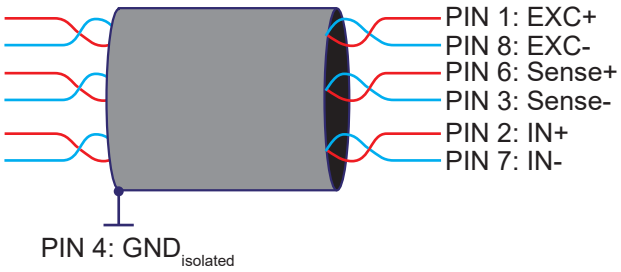
TRION-2402-MULTI resistance accuracy				
RTD (Type)	**Temperature range (°C)**	**Excitation current (mA)**	**Range (Ω)**	**Accuracy**
PT100	-200 to 850	1	1000	0.9 °C ±0.33 % of reading
PT200	-200 to 850	1	1000	0.7 °C ±0.33 % of reading
PT500	-200 to 850	1	2000	0.7 °C ±0.33 % of reading
PT1000	-200 to 850	0.5	10000	1.1 °C ±0.4 % of reading
PT2000	-200 to 850	0.5	10000	1.1 °C ±0.4 % of reading

RTD temperature drift specification for TRION-2402-MULTI					
Range (Ω)	**Excitation current (mA)**	**Voltage range (V)**	**Accuracy**	**Temp drift (ppm /°C)**	**RTD sensor**
10 k	0.5	10	2 Ω ±0.45 %	100	Pt2000, Pt1000
3 k	1	10	0.6 Ω ±0.25 %	100	Pt500
1 k	1	2	0.2 Ω ±0.25 %	100	Pt200, Pt100

Cables and Shielding

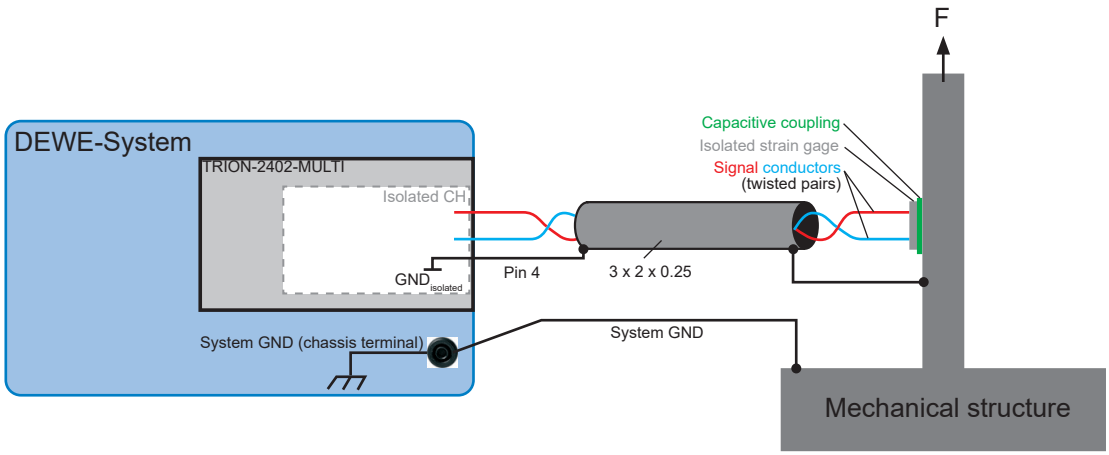
To keep the influence of electromagnetic disturbances as small as possible, shielded twisted pair cables are recommended. Connect the shield to the isolated GND (Pin4) to get the best result.

The twisted pairs for **full bridge**, **half bridge**, **voltage** and **resistance** mode are:



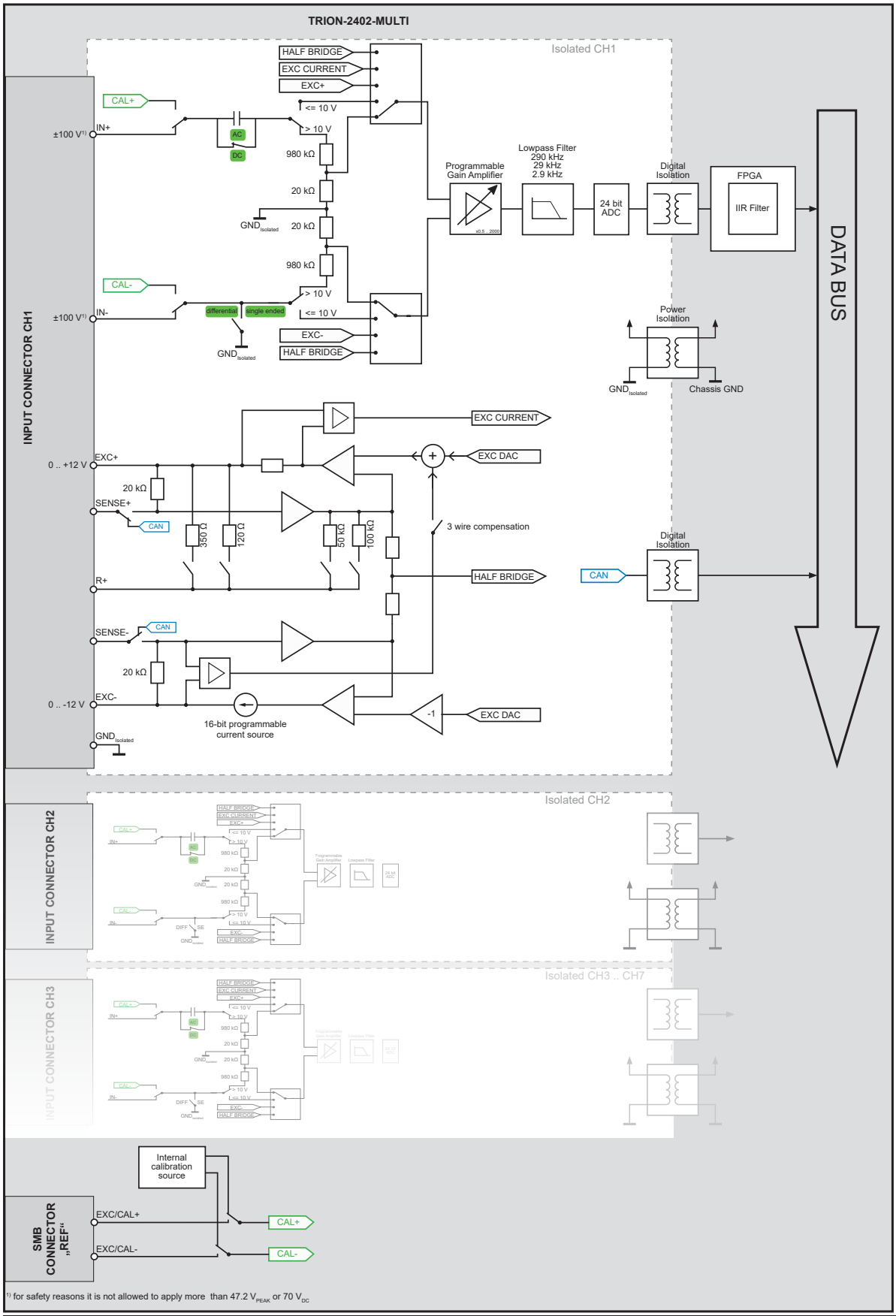
Shielding / Noise reduction

Strain gage measurement on a metal structure

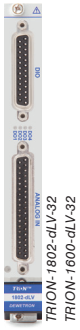


It is always important that you connect your DEWETRON system ground (chassis terminal) to the ground potential of your measured object. This guarantees that the measurement system is not floating against the measured structure. It could simply be a connection to the metal structure of your proving ground. In case of an automotive application for example, it would be a connection to the cars chassis. Only if the DEWETRON system and the measured structure have an earth connection the system grounding line might not be needed.

Block diagram



Multi-function module with voltage inputs, digital I/Os, counter and CAN



- Channels: 32 single ended or 16 differential, synchronous channels
- Sampling: TRION-1802-dLV-32: 18 bit; 200 kS/s per channel
  TRION-1600-dLV-32: 16 bit; 20 kS/s per channel
- Input types: 5 V/10 V
- Features: 2x Counter; CAN bus; RS-485; 8x DI; 4x DO; 2x Alarm Out

Module specifications

TRION-1802/1600-dLV series specifications					
Configuration	TRION-1802-dLV-32	32 channels single ended or 16 channels fully differential			
	TRION-1802-dLV-32-CAN	32 channels single ended or 16 channels fully differential + CAN			
	TRION-1600-dLV-32	32 channels single ended or 16 channels fully differential			
	TRION-1600-dLV-32-CAN	32 channels single ended or 16 channels fully differential + CAN			
Sampling Rate / Resolution					
TRION-1802-dLV-32					
High speed mode	>50 to 200 kS/s,	18-bit			
Over Sampling mode	100 S/s to 50 kS/s,	24-bit			
TRION-1600-dLV-32	100 S/s to 20 kS/s,	16-bit			
Data Transfer					
TRION-1802-dLV-32	16-bit / 24-bit / 32-bit				
TRION-1600-dLV-32	16-bit				
Onboard data buffer	512 MByte				
ADC type	18-bit SAR[2] (Successive Approximation Register)				
Data rate DMA transfer	32 analog channels: max 28 MB/s; 2x counter: max. 6 MB/s				
Input ranges Voltage	±5 V; ±10 V				
Input noise	0 to 10 Hz:	10 μV_{PP}			
	full bandwidth:	1.35 mV_{PP}			
Input impedance	1 MΩ single ended, 2 MΩ differential				
Input bias current	<25 pA				
Input coupling	DC				
Accuracy[1] Voltage	DC to 1 kHz	±0.02 % of reading ± 0.01 % of range ±20 µV			
	>1 kHz to 5 kHz	±0.5 % of reading ± 0.01 % of range ±20 µV			
	>5 kHz to 10 kHz[2]	±1 % of reading ± 0.01 % of range ±20 µV			
Gain drift	typical 10 ppm/°C max. 20 ppm/°C				
Offset drift	typical 0.3 µV/°C + 10 ppm of range/°C, max 15 µV/°C + 20 ppm of range/°C				
Typical Signal-to-noise ratio, Spurious-free SNR, Effective number of Bits, V_{PP} [2]	10 V range				
	SNR	SFDR[3]	ENOB[4]	Noise peak to peak	
Sample rate	[dB]	[dB]	[Bit]	[mV_{PP}]	
0.1 kS/s	127	130	20.8	0.015	
1 kS/s	118	130	19.3	0.055	
10 kS/s	109	130	17.8	0.22	
20 kS/s	106	130	17.3	0.33	
50 kS/s[2]	102[2]	130[2]	16.7	0.52[5]	
100 kS/s[2]	99[2]	130[2]	16.2	0.66[5]	
200 kS/s[2]	96[2]	125[2]	15.7	1.00[5]	
Linearity	<20 ppm				
Input configuration	differential or single ended with GND Sense				
Typical THD	-95 dB				
Typical CMRR in differential mode	100 dB @ 50 Hz; >70 dB @ 1 kHz				
Low pass Filter (-3 dB, IIR)	1 Hz to 40 % of sample rate freely programmable or OFF				
Characteristic	Bessel or Butterworth				
Filter order	2^{nd}, 4^{th}, 6^{th}, 8^{th}				
Analog antialiasing filter	3^{rd} order Butterworth				
Bandwidth (-3 dB, deactivated IIR filter)	70 kHz 3^{rd} order Butterworth filter				
Crosstalk fin 1 kHz [10 kHz]	>108 dB				
Channel to channel phase mismatch	typically <30 nsec when using the same input range				
Board to board phase mismatch	<30 nsec				
Common mode input voltage range	±12.5 V				
Overvoltage protection (IN+, IN-, Sense)	±50 V_{DC}				

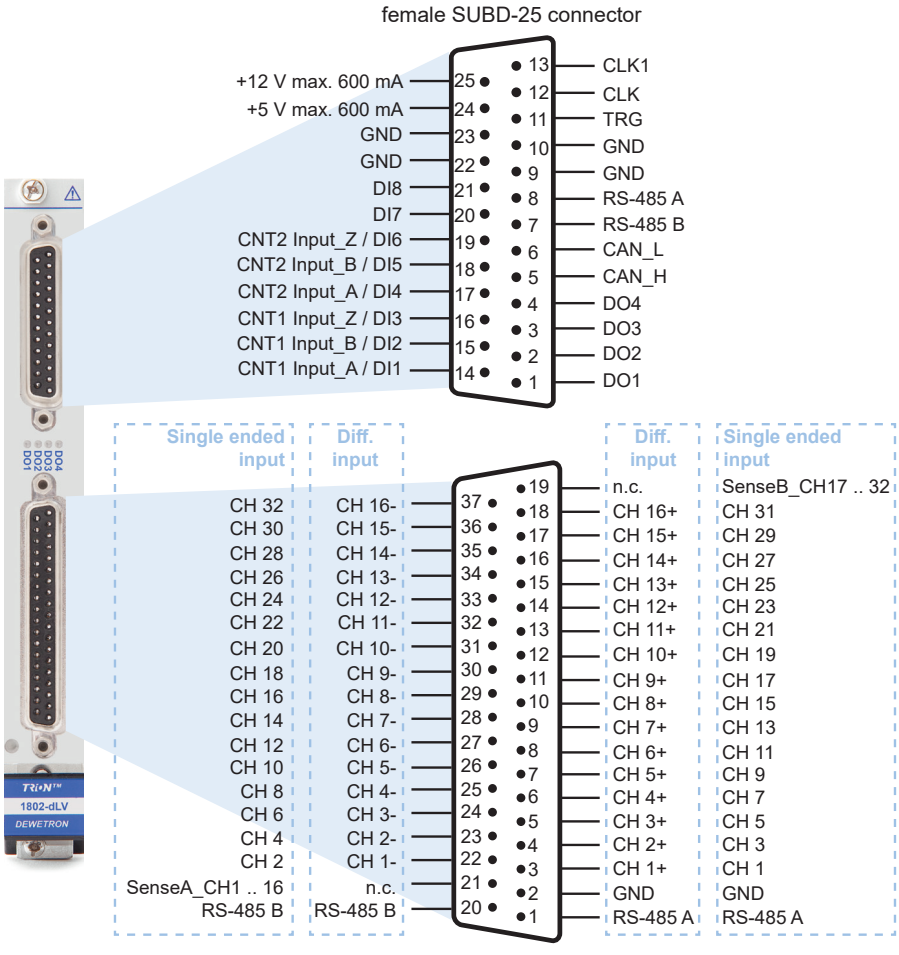
continued on next page ...

TRION-1802/1600-dLV-32

Digital IN specification	
Digital Input	8 CMOS/TTL compatible digital inputs; weak pullup via 100 kΩ
Overvoltage protection	±30 V permanent, 50 V_{PEAK} (for 100 ms)
Counter	2 counter channels; TTL input; shared with digital inputs
Counter modes	
Event counting	Basic event counting, gated counting, up/down counting and encoder mode (X1, X2 and X4)
Waveform timing	Period, frequency, pulse width duty cycle and edge separation
Sensor modes	Encoder (angle and linear)
Digital OUT specification	
Digital output	4 DO; TTL
Output indication	LED (green = high; off = low)
Maximium current	25 mA continousely
Power-on default	Low
Interfaces	
CAN bus	1 CAN Bus; not isolated; routed to SUBD-25
CAN specification	CAN 2.0B
CAN Physical Layer	High Speed
CAN Bus fault protection	±36 V
Termination	Programmable: High impedance or 120 Ω
RS485	1 RS485 interface dedicated to DAQP and HSI series modules
General specification	
Sensor power supply (per module)	5 V (600 mA) and 12 V (600 mA)
ESD protection	IEC61000-4-2: ±8 kV air discharge, ±4 kV contact discharge
Power consumption	Voltage mode: 6 W

[1] 1 year accuracy 23 °C ±5 °C
[2] LP Filter in auto mode
[3] SFDR excluding harmonics
[4] ENOB calculated from SNR
[5] TRION-1802-dLV-32 only

TRION-1802/1600-dLV-32 module



Optional accessory

TRION-x-dLV-CB16-D9

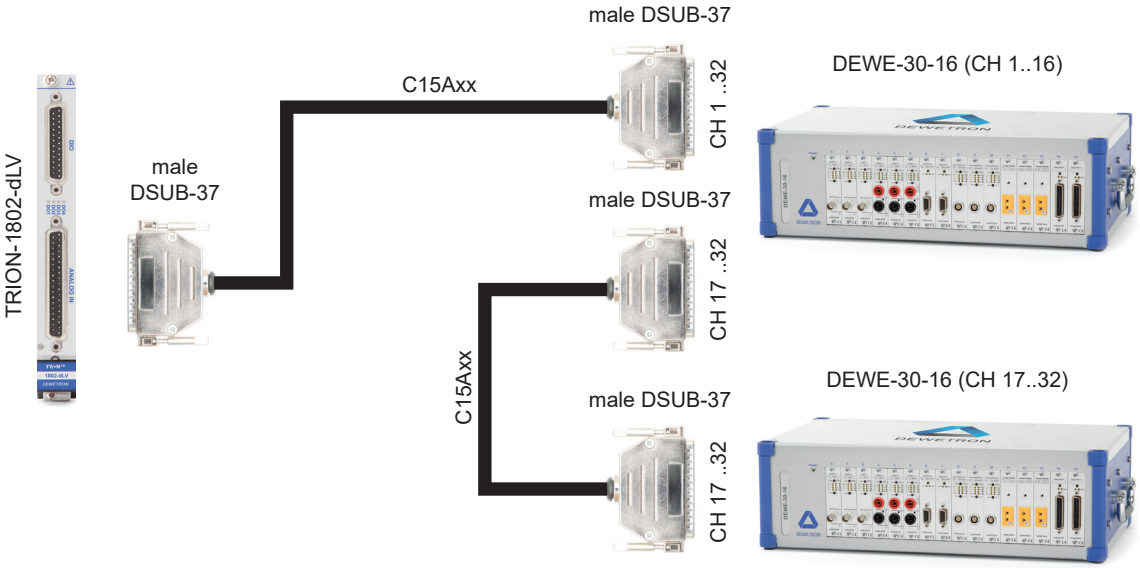
- 16 channel sensor connection box
- Precision ±5 V excitation voltage with remote sense per channel
- MSI support (Modular Smart Interface)
- Auxiliary sensor supply

MODULAR SMART INTERFACES	INPUT	SENSOR EXCITATION	BANDWIDTH (max.) consider limit of used TRION module	ACCURACY (typ.)	SENSOR CONNECTION
MSI2-STG	Bridge-type sensors Full-bridge, half-bridge, quarter bridge 120 Ω and 350 Ω	5 V and 10 V	100 kHz	±0.1 %	Miniature spring terminals
MSI2-LVDT	LVDT and RVDT sensors, 5 wire connection	AC supply 1 to 3.5 V at 2.5, 10 or 20 kHz	1 kHz	±0.1 %	Soldering pads
MSI-BR-ACC	IEPE® sensors, typ. accelerometer, microphone	4 mA	1.4 Hz to 100 kHz	±0.2 %	BNC
MSI2-CH-5 MSI2-CH-100	Charge type sensors up to 100 000 pC	n/a	0.08 Hz to 45 kHz	±0.5 %	BNC
MSI2-TH-K MSI2-TH-J MSI2-TH-T MSI2-TH-x	Thermocouple sensors Standard models for type K, J, T, others on request	n/a	DC to 100 kHz	±1 °C	Mini TC socket
MSI-BR-V-200	Voltage up to 200 V	n/a	DC to 100 kHz	±0.1 %	BNC
MSI-BR-RTD	RTD sensors PT100, Pt200, Pt500, PT1000, PT2000; 2, 3 and 4 wire connection	1.25 mA	DC to 10 kHz	±0.1 %	Binder 712 series 5-pin socket
MSI2-250R-20mA	4 to 20 mA sensors	5 to 48 V AUX PWR	DC to 100 kHz	±0.1 %	Miniature spring terminals

For further information please refer to the '*TRION-x-dLV-CB16-D9 / MSI2 Series*' technical reference manual.
https://ccc.dewetron.com/dl/TRION-x-dLV-CB16-D9

C15Axx

Cable for connecting 16 or 32 channels to a TRION-1802-dLV module in **single ended** configuration.
Two C15Axx cables are required for 32 channels.



LED function

Status LED

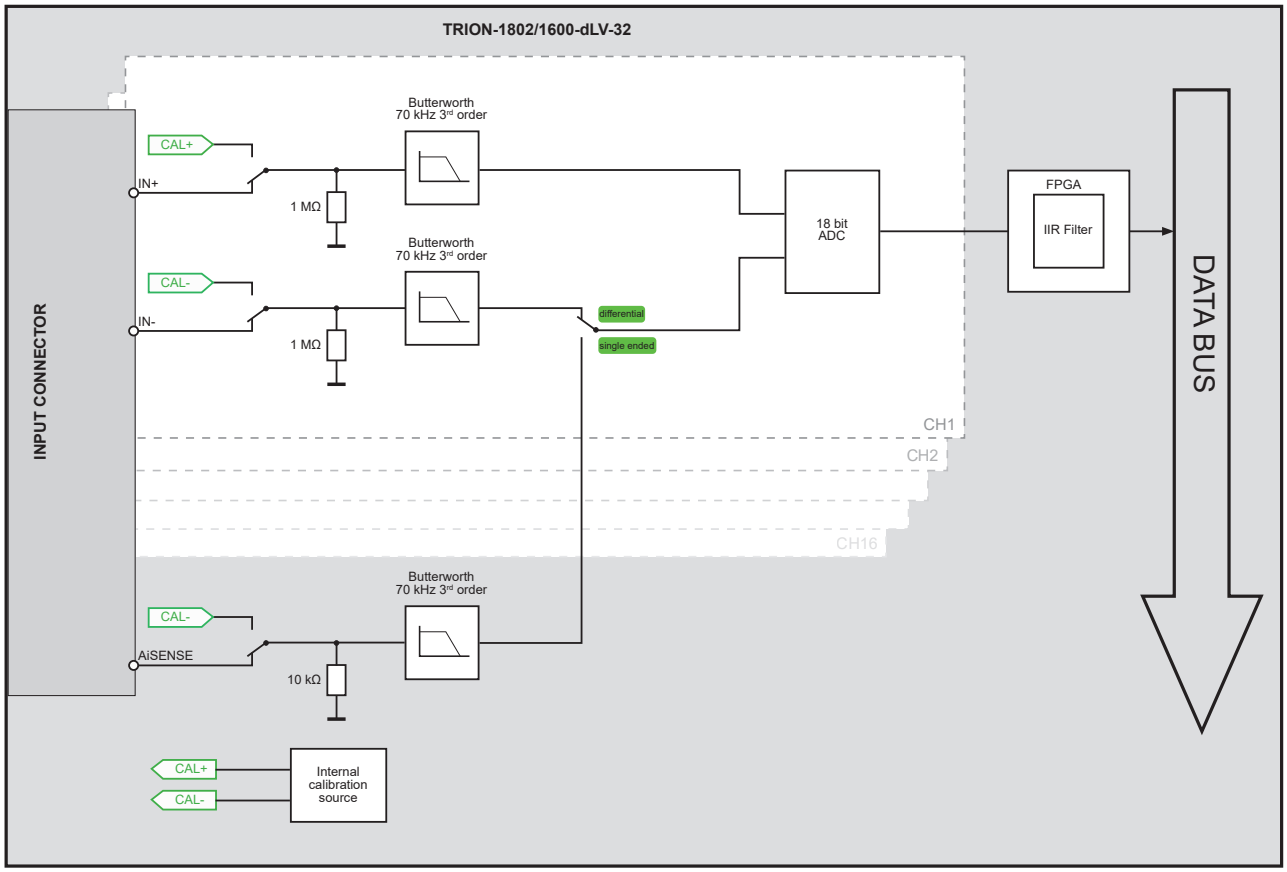
Green: Normal operation
Off: Board not initialized.

DO LED

Green: DO is active (output is 5V)
Off: DO is deactivated (output is 0V)

Block diagram

Base block diagram of the TRION-1802/1600-dLV-32 module:



The TRION-1802/1600-dLV series is a highly accurate, 18-bit voltage digitizer. Each channel has its own AD converter. Please refer to chapter “TRION-16xx sample system architecture” for more details about bandwidth and filtering.

TRION-1802/1600-dLV-32 function overview

Short

The short function switches IN+ to IN- via the calibration circuit. It can be used to check the offset of the input amplifier.

Auto Zero

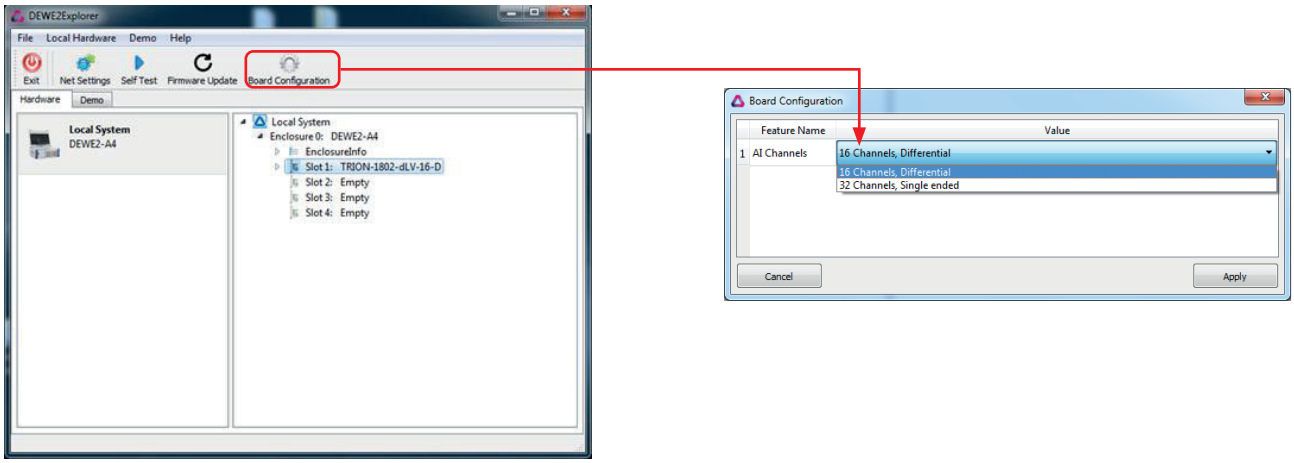
Uses the short function to compensate the input offset. This allows eliminating long-term offset drifts as well as compensating environmental temperature related offsets.

Self Test

The TRION-1802/1600-dLV series has an integrated special self-test circuit. It consists of a programmable high precision voltage source on the first channel and a relay matrix. It is used to check the analog input path of the voltage amplifier by applying 0 V and 90 % of the input range to the input. During the board self test, which is available in the DEWE2 Explorer, this test is performed for all ranges and channels automatically.

Single ended / differential mode

Use DEWE2 Explorer to setup the TRION-1802/1600-dLV series board as 16 channel differential or 32 channel single ended. The information is stored on the board.



Counter functions

Supported counter functions are:

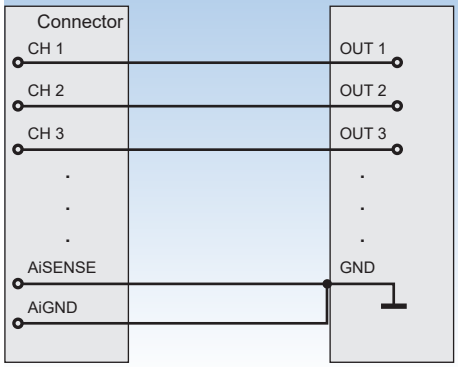
- Simple event counting
- Period measurement
- Pulse width
- Frequency
- Duty cycle

For detailed information about this functions please refer to the chapter "Functional description of advanced counter" of the TRION-CNT module.

Signal connection

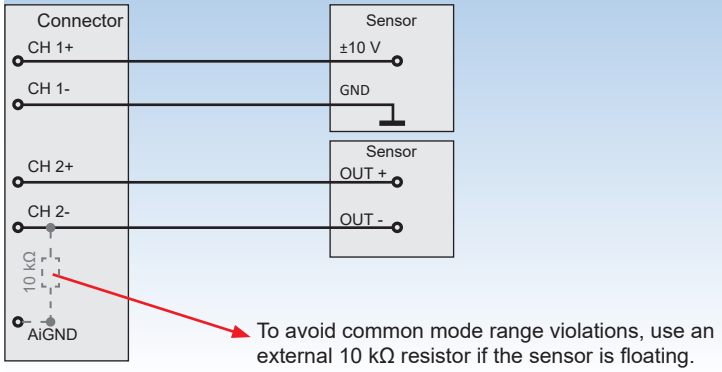
Single ended

This is recommended to use with DEWE-30 series instruments or any other multichannel output device with common ground.



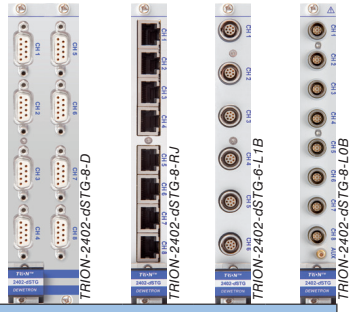
Differential input

This is recommended to use with multiple separated sensors without common ground or differential output.



Differential universal input module

- Sampling: 24 bit; 204.8 kS/s per channel
- Input types: Voltage up to ±10 V
  Strain gauge, bridge sensor, piezoresistive bridge
  IEPE®
  Resistance, potentiometer



TRION-dSTG specifications												
Input channels	8 using RJ-45 sockets (TRION-2402-dSTG-8-RJ)											
	8 using SUB-D-9 sockets (TRION-2402-dSTG-8-D, double-wide slotpanel)											
	6 using LEMO 1B sockets (TRION-2402-dSTG-6-L1B)											
	8 using LEMO 0B sockets (TRION-2402-dSTG-8-L0B)											
Sampling rate	204.8 kS/s per channel											
Resolution	24 bit											
Rated input voltage (max.)	±10 V											
Input ranges												
Voltage	±10, 30, 100, 300 mV, 1 V, 3 V, 10 V											
Bridge	1, 3, 10, 30, 100, 300, 1000 mV/V or mV/mA											
IEPE®	±100, 300 mV, 1V, 3V, 10V											
Resistance	10, 30, 100, 300 Ω, 1, 3, 10, 30 kΩ											
Current	Depending on external Shunt											
Voltage input accuracy[1]	±0.02 % of reading ± 0.02 % of range ±20 µV											
Gain drift	typical 10 ppm/°C max. 20 ppm/°C											
Offset drift	typical 0.3 µV/°C+ 10 ppm of range/°C, max 2 µV/°C + 20 ppm of range/°C											
linearity	typical 0.01 %											
Input impedance	100 MΩ											
Input bias current	< 1 nA											
Input configuration	Single ended or differential (programmable)											
Input coupling	DC, AC (0.16 Hz, 0.5 Hz, 3.4 Hz, 10 Hz); max. DC voltage when AC coupled: 50 V											
Excitation voltage	0 to 13.5 V_{DC} (programmable, 1 mV steps), 100 mA max. current, max 8 W per module											
Accuracy[1]	±0.03 % ±1 mV											
Drift	±10 ppm/K ±50 µV/K											
Current limit	100 mA											
Protection	Continuous short to ground											
Excitation current	0.2 to 20 mA_{DC} (pogrammable, 1 µA steps)											
Accuracy[1]	0.05% ±2 µA											
Drift	15 ppm/°C											
Compliance voltage	10 V											
Output impedance	>10 MΩ											
IEPE® Excitation	4 mA ±10 %											
Compliance voltage	22 V											
Supported sensors	4- or 6-wire full bridge 3- or 5-wire ½ bridge with internal completion 3- or 4-wire ¼ bridge with internal resistor for 120 and 350 Ω 4-wire full bridge with constant current excitation (piezoresistive bridge sensors) Potentiometric; Resistance IEPE® (fixed 4 mA excitation)											
Bridge resistance	80 Ω to 10 kΩ @ ≤ 5 V_{DC} excitation											
Shunt calibration	Two internal shunt resistors 50 kΩ and 100 kΩ											
Shunt and completion resistor accuracy[1]	0.05 % ±15 ppm/K											
Automatic bridge balance	±250 % of Range											
Typical Signal-to-noise ratio, spurious free SNR Effective number of Bits[2]	10 mV range			100 mV range			1 V range			10 V range		
	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]
Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]
1 kS/s	82	108	13.3	101	128	16.5	111	141	18.1	112	141	18.3
10 kS/s	78	106	12.7	98	126	16.0	108	136	17.6	109	138	17.8
100 kS/s	72	103	11.7	92	123	15.0	104	134	17.0	107	136	17.5
200 kS/s	69	99	11.2	80	120[5]/106	13.0	81	133[5]/106	13.2	81	135[5]/106	13.2
Typical THD	-97 dB											
Typical CMRR	100 dB @ 50 Hz; 90 dB @ 1 kHz; 80 dB @ 10 kHz											
Analog anti aliasing filter	2nd order Bessel, automatically set by sample rate											
Sample rate ≤ 1k S/s	2.5 kHz (-3 dB), 1.5 kHz (-1 dB)											
Sample rate ≤ 10 kS/s	25 kHz (-3 dB), 15 kHz (-1 dB)											
Sample rate > 10kS/s	250 kHz (-3 dB), 150 kHz (-1 dB)											
Bandwidth (-3 dB digital filter)												
1 kS/s ≤ fs ≤ 51.2 kS/s	0.494 fs											
51.2 kS/s < fs ≤ 102.4 kS/s	0.49 fs											
102.4 kS/s < fs ≤ 204.8 kS/s	0.38 fs											
Crosstalk fin 1 kHz [10 kHz]	120 dB [105 dB]											
Channel to channel phase mismatch	typically <60 nsec between channels using the same range											
Common mode voltage	±10 V_{DC}											
Over voltage protection	±50 V_{DC}											
Supported TEDS chips	DS2406, DS2430A, DS2431, DS2432, DS2433											
Supported MSI adaptors	MSI adaptors are NOT supported!											

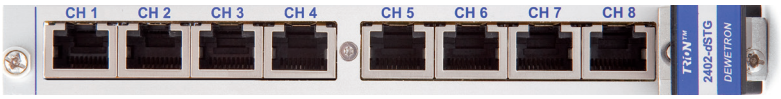
continued on next page ...

TRION-2402-dSTG

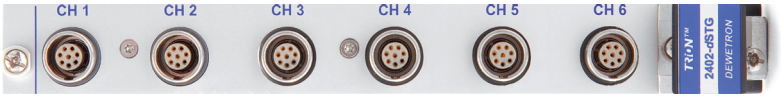
Typical power consumption[6]	
Voltage mode; no excitation	7 W
IEPE® mode	7 W
350 Ω full bridge (5 V / 10 V)	7 W / 9.5 W
120 Ω quarter bridge 5 V excitation	8 W
Bridge mode without connected sensor	11.5 W[7]
Weight	Appr. 200 g (RJ45 version), appr. 250 g (LEMO version)

[1] 1 year accuracy 23 °C ±5 °C
[2] LP Filter in auto mode
[3] SFDR excluding harmonics
[4] ENOB calculated from SNR
[5] below 0.22 fs
[6] Consider maximum power supply of your DEWE2 chassis
[7] Do not switch to bridge mode if the input is open!

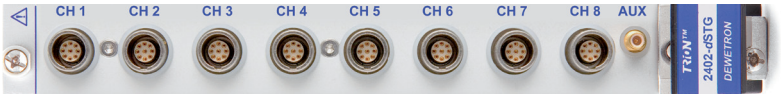
TRION-2402-dSTG model overview



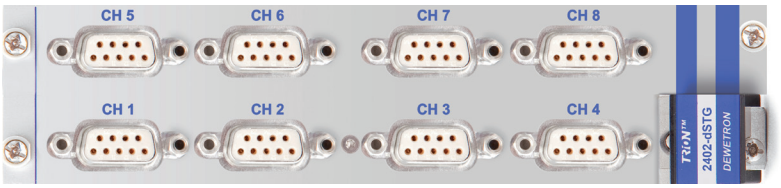
TRION-2402-dSTG-8-RJ



TRION-2402-dSTG-6-L1B



TRION-2402-dSTG-8-L0B



TRION-2402-dSTG-8-D

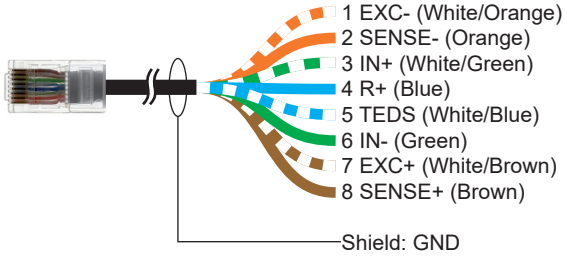
TRION-2402-dSTG-8-RJ module

1	EXC-
2	SENSE-
3	IN+
4	R+
5	TEDS
6	IN-
7	EXC+
8	SENSE+
Housing	GND

Optional accessory

TRION-CBL-RJ-OE-05-00

High quality cable from RJ45 plug to open end, 5 meters

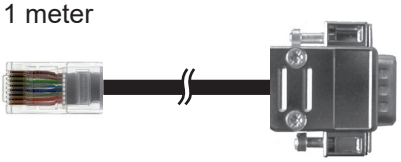


TRION-CBL-RJ-BNC-01-00

High quality cable from RJ45 plug to BNC socket, 1 meter

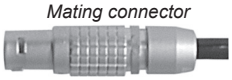
TRION-CBL-RJ-D9-01-00

High quality adaptor cable from RJ45 plug to DSUB-9 socket, 1 meter

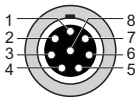


TRION-2402-dSTG-6-L1B module

Mating connector



FGG.1B.308.CLAD52Z



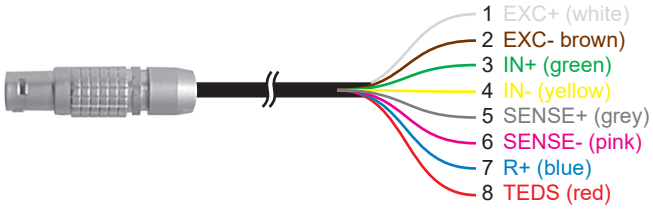
8-pin LEMO connector female
EGG.1B.308.CLL

1	EXC+
2	EXC-
3	IN+
4	IN-
5	SENSE+
6	SENSE-
7	R+
8	TEDS
Housing	GND

Optional accessory

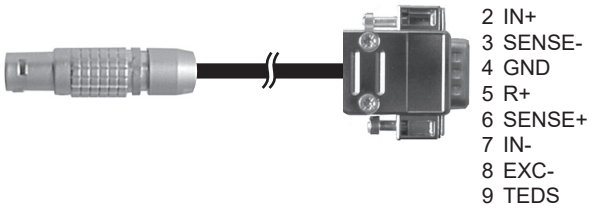
TRION-CBL-L1B8-OE-05-00

High quality cable from Lemo 1B.308 plug to open end, 5 meters



TRION-CBL-L1B8-D9-0.5-00

High quality adaptor cable from LEMO 1B.308 plug to DSUB-9 socket, 0.5 meters



LEMO-FGG.1B.308.CLAD52Z

LEMO FGG.1B.308 mating connector, for cable diameter 4.2 to 5.2 mm

LEMO-FGG.1B.308.CLAD62Z

LEMO FGG.1B.308 mating connector, for cable diameter 5.2 to 6.2 mm

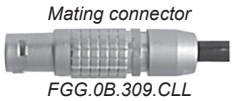
LEMO-FGG.1B.308.CLAD72Z

LEMO FGG.1B.308 mating connector, for cable diameter 6.2 to 7.2 mm

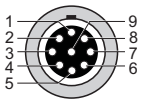
TRION-2402-dSTG

TRION-2402-dSTG-8-L0B module

Mating connector



FGG.0B.309.CLL



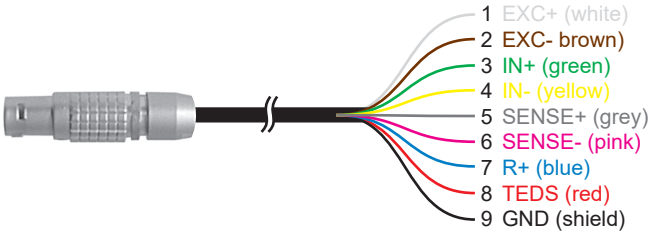
9-pin LEMO connector female
EPG.0B.309.HLN

1	EXC+
2	EXC-
3	IN+
4	IN-
5	SENSE+
6	SENSE-
7	R+
8	TEDS
9	GND
Housing	GND

Optional accessory

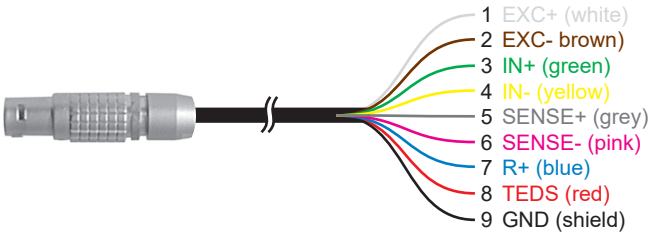
TRION-CBL-L0B9-OE-05-00

High quality cable from Lemo 0B.309 plug to open end, 5 meters.



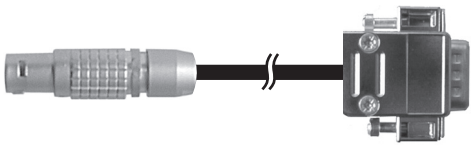
TRION-CBL-L0B9-OE-01-00

High quality cable from Lemo 0B.309 plug to open end, 1 meter.



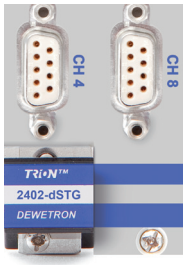
TRION-CBL-L0B9-D9-0.5-00

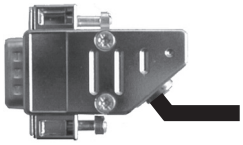
High quality adaptor cable from Lemo 0B.309 plug to SUBD-9 socket, 0.5 meters.



1 EXC+
2 IN+
3 SENSE-
4 GND
5 R+
6 SENSE+
7 IN-
8 EXC-
9 TEDS

TRION-2402-dSTG-8-D module





1 EXC+
2 IN+
3 Sense -
4 GND
5 R+
6 Sense +
7 IN-
8 EXC-
9 TEDS

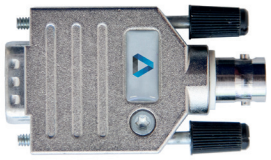
NOTE: This module requires two empty module slots in your DEWE2-system!

Optional accessory

ADAP-D9M-BNCF

Adapter to directly connect voltage
or IEPE® signals to TRION-2402-dSTG-8-D

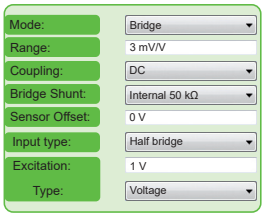
1 EXC+
2 IN+
3 Sense -
4 GND
5 R+
6 Sense +
7 IN-
8 EXC-
9 TEDS



Hot: IN+
Shield: IN-

TRION-2402-dSTG functions

Free variable gain and excitation



Amplifier parameters like gain, excitation voltage, excitation current and sensor offset are free programmable for every channel individually. That allows to perfectly match each input channel to any sensor. Customized programming of the amplifier could be simply done by entering the desired value in the appropriate field.

Excitation

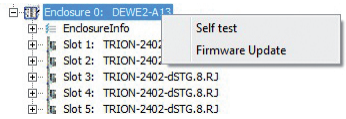
The excitation circuit of the dSTG provides constant current and constant voltage excitation. The voltage mode also supports remote sense. The Sense wires have to be connected all the time, even if no remote sense is required.
Beside the programmable excitation current there is also a fixed 4 mA excitation for IEPE® sensors available. This IEPE® current has a higher bias voltage of 22 V and is directly applied to the IN+ terminal.

Amplifier balance (Amplifier zero)

The amplifier balance allows eliminating automatically all internal amplifier offsets. It switches the differential amplifier inputs IN+ and IN- to the internal GND reference point. Then the offset of the module is automatically adjusted to zero for all ranges. This function takes up to 4 seconds.
It allows compensating the long term zero drift, as well as temperature drifts of the amplifier.
It could be performed for a single channel.

Self test

The dSTG module has a special self test circuit integrated. The first part of it is a high precision voltage source and a temperature compensated divider. It is used to check the analog input path of the voltage amplifier by applying 0 V and ±98 % of the input range to the input. This test could be performed in the Channel setup for the actual range. During the board self test which, is available in the DEWE2 Explorer, this test is performed for all ranges and channels automatically.



by right clicking the board in the DEWE2 explorer you can perform a self test.

The second part is a signal routing matrix that allows checking the sensor power supply of the acquisition channel. The driven current and the exact voltage are monitored. If the connected sensor exceeds the maximum power consummation or the nominal excitation value is wrong this test fails. That is an indication that either the sensor is damaged or the connection is wrong. Also a broken sensor cable could produce a negative result. This test could be performed in the channel setup.

Sensor balance

Typically every strain gage sensor has a certain offset. This offset is on the one hand caused by manufacturing tolerances and on the other influenced by the sensor mounting. By performing a “sensor balance” this sensor offset could be automatically removed up to 250% of range.

Internal completion resistors

The dSTG has an internal half bridge completion and two internal quarter bridge completions for 120 Ω and 350 Ω strain gages. The used high precision resistors with low temperature drift allow a long-time stable measurement of almost every strain gage type without using an external completion network.

Internal shunt

With the Shunt function a 100 kΩ or a 50 kΩ shunt could be applied to the bridge sensor. That allows lead wire compensation for 3-wire quarter bridge sensors. It could also be used for checking the sensor connection of half bridge and full bridge sensors.

Input short

It switches both differential amplifier inputs IN+ and IN- from the input terminals to the internal half bridge reference of the module. With this function the absolute sensor offset could be determined.

Filter

Refer to Trion-2402 series general description.

AC coupling

The dSTG has four different input high pass filters available for AC coupling:

Frequency	Time constant
0.16 Hz	1 sec
0.5 Hz	320 msec
3.4 Hz	47 msec
10 Hz	16 msec

That allows removing DC components of the signal and using a much smaller input range. The maximum DC input voltage should not exceed ±50 V_{DC}.

TEDS

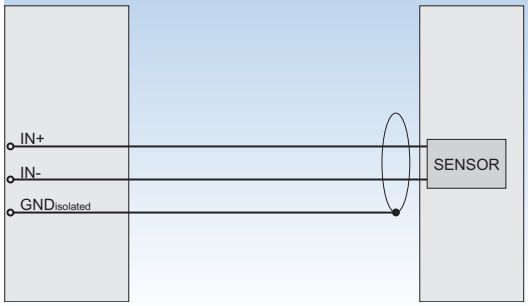
The dSTG uses a separate terminal for TEDS communication. All common TEDS chips are supported.

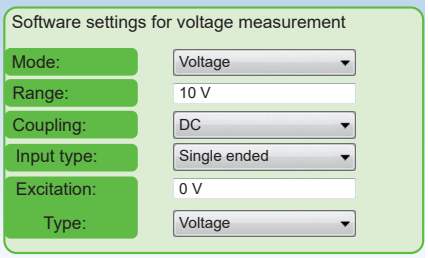
Caution: In IEPE® mode the TEDS function is not available because IEPE® Sensors uses IN+ for TEDS communication.

Signal connection

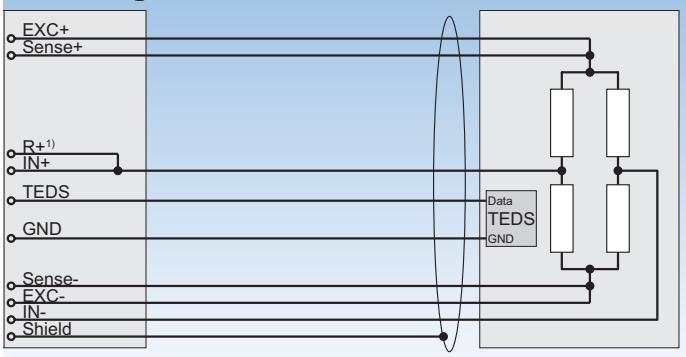
Voltage measurement

Isolated Sensors
Battery powered sensors
Sensors with differential output





Full bridge 6 wire sensor connection

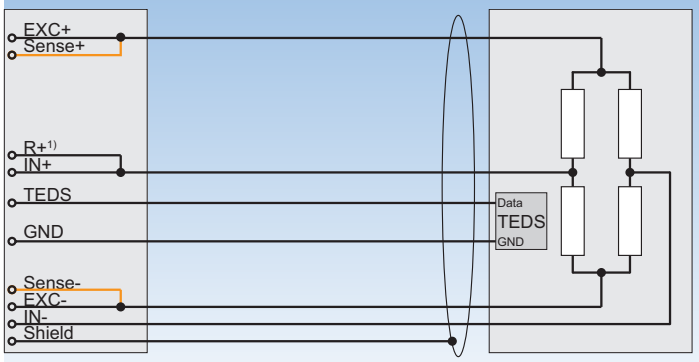


Software settings for full bridge 6 wire sensor connection

Mode:	Bridge
Range:	1 mV/V
Coupling:	DC
Bridge Shunt:	Internal 50 kΩ
Input type:	Full bridge
Excitation:	1 V
Type:	Voltage

Voltage or current excitation is allowed. TEDS connection is optional.

Full bridge 4 wire sensor connection

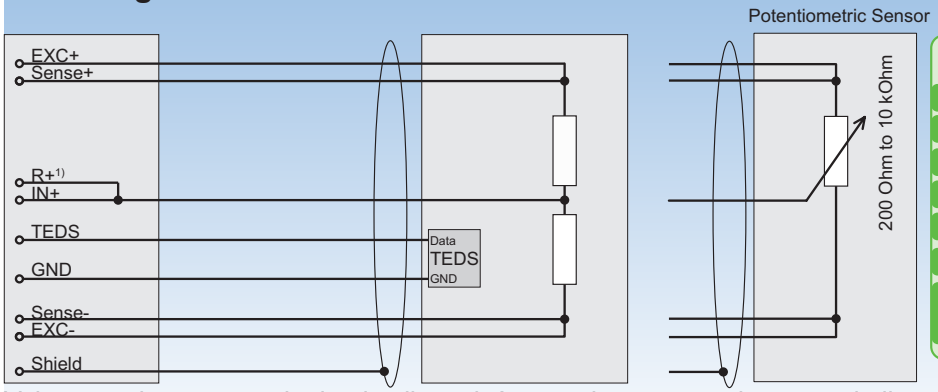


Software settings for full bridge 4 wire sensor connection

Mode:	Bridge
Range:	1 mV/V
Coupling:	DC
Bridge Shunt:	Internal 50 kΩ
Sensor Offset:	0 V
Input type:	Full bridge
Excitation:	1 V
Type:	Voltage

Voltage or current excitation is allowed. Senses terminals have to be connected to the excitation also when 4-wire connection is used.

Half bridge or Potentiometer 5 wire sensor connection



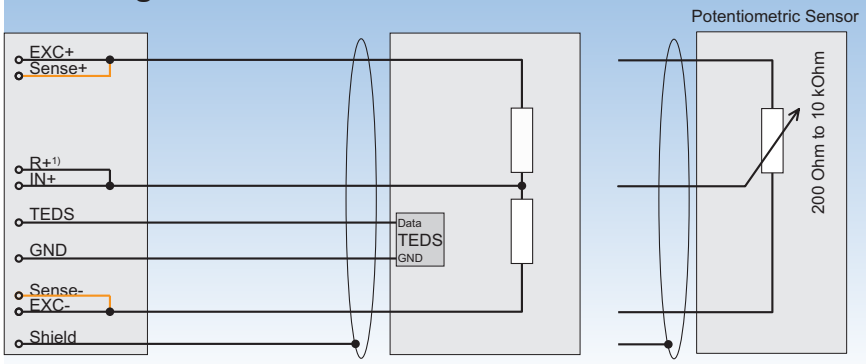
Software settings for half bridge or Potentiometer 5 wire sensor connection

Mode:	Bridge
Range:	3 mV/V
Coupling:	DC
Bridge Shunt:	Internal 50 kΩ
Sensor Offset:	0 V
Input type:	Half bridge
Excitation:	1 V
Type:	Voltage

Voltage and current excitation is allowed. A potentiometer can be seen similar to a half bridge sensor with ±500 mV/V sensitivity.

[1] 'R+' has to be connected only if shunt calibration is required, otherwise it can be left unconnected.

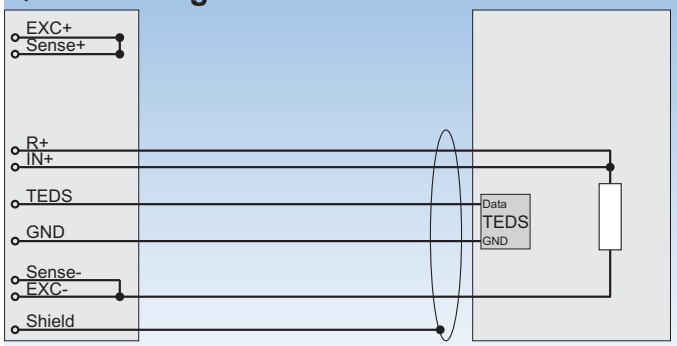
Half bridge or Potentiometer 3 wire sensor connection



Software settings for half bridge or Potentiometer 3 wire sensor connection	
Mode:	Bridge
Range:	3 mV/V
Coupling:	DC
Bridge Shunt:	Internal 50 kΩ
Sensor Offset:	0 V
Input type:	Half bridge
Excitation: Type:	1 V / Voltage

Senses terminals have to be connected to the excitation also when 4-wire connection is used.

Quarter bridge 3 wire sensor connection

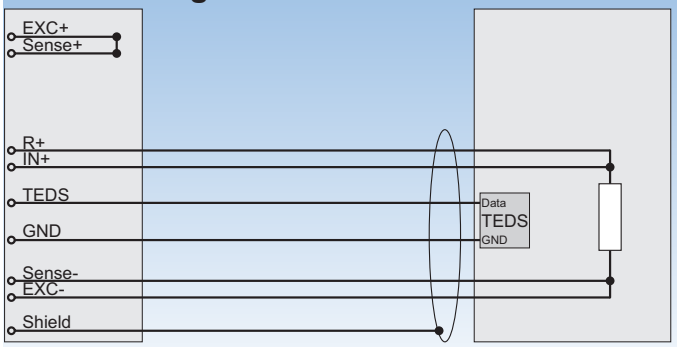


Software settings for quarter bridge 3 wire sensor connection	
Mode:	Bridge
Range:	3 mV/V
Coupling:	DC
Bridge Shunt:	Internal 50 kΩ
Sensor Offset:	0 V
Input type:	BRQUATER3W
Bridge Res:	120R
Excitation: Type:	1 V / Voltage

CAUTION: Sense+ has to be connected to EXC+!

The 3-wire quarter bridge is only able to compensate symmetric wire resistance.

Quarter bridge 4 wire sensor connection

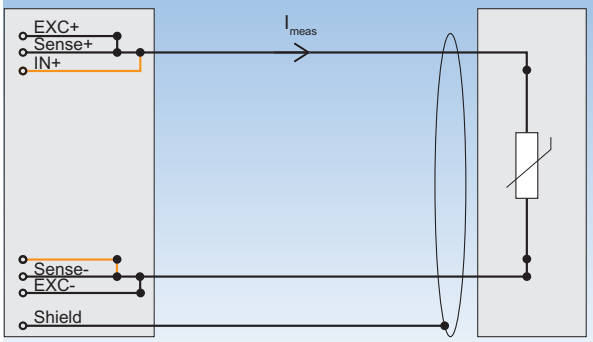


Software settings for quarter bridge 4 wire sensor connection	
Mode:	Bridge
Range:	3 mV/V
Coupling:	DC
Bridge Shunt:	Internal 100 kΩ
Sensor Offset:	0 V
Input type:	BRQUATER4W
Bridge Res:	120R
Excitation: Type:	1 V / Voltage

CAUTION: Sense+ has to be connected to EXC+!

The 4 wire connection provides full lead wire resistance compensation.

Resistance measurement 2 wire connection



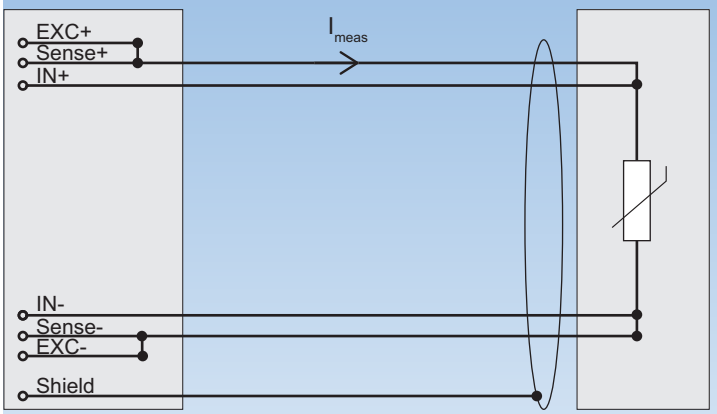
Software settings for resistance measurement 2 wire connection

Mode: Resistance

Range: 10000 Ω

The 2 wire technology does not compensate any lead wire resistance.
For accurate resistance measurement the 4 wire technology is strongly recommended.
[1] 'R+' has to be connected only if shunt calibration is required, otherwise it can be left unconnected.

Resistance measurement 4 wire connection



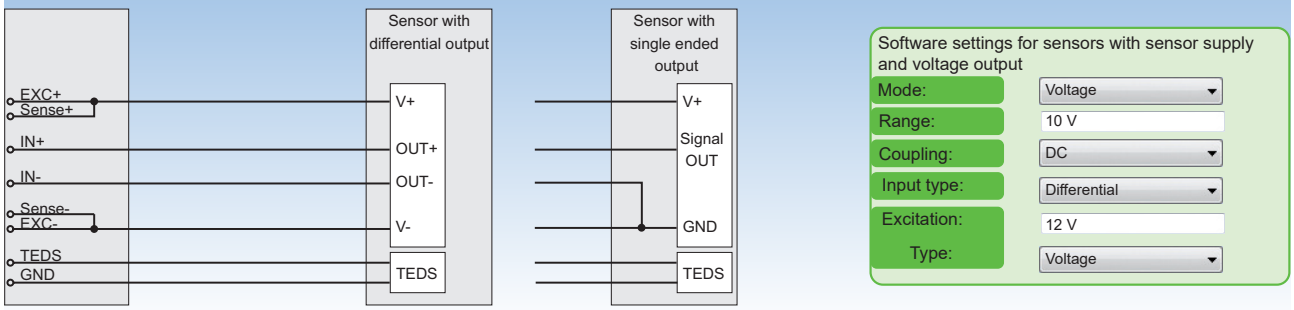
Software settings for resistance measurement 2 wire connection

Mode: Resistance

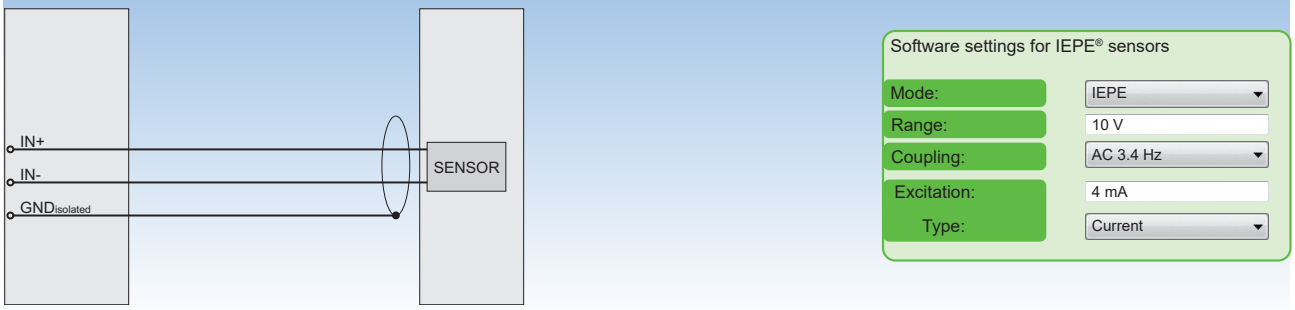
Range: 10000 Ω

TRION-2402-dSTG resistance accuracy		
Range (Ω)	**Excitation current (mA)**	**Accuracy**
30 k	0.2	6 Ω ±1 % of reading
10 k	0.5	2 Ω ±0.45 % of reading
3000	1	0.6 Ω ±0.25 % of reading
1000	1	0.2 Ω ±0.25 % of reading
300	1	80 mΩ ±0.25 % of reading
100	1	40 mΩ ±0.25 % of reading
30	10	8 mΩ ±0.07 % of reading
10	10	4 mΩ ±0.07 % of reading

Sensors with sensor supply and voltage output



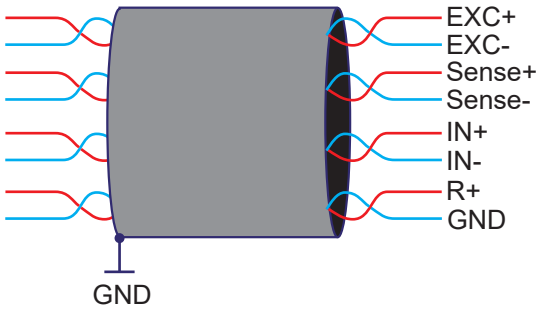
IEPE® Sensor



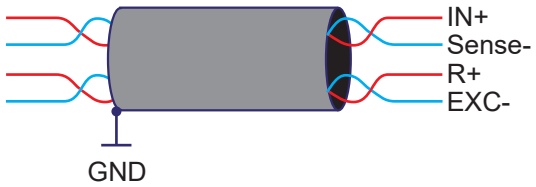
Cables

To keep influence of electromagnetic disturbances as small as possible, shielded twisted pair cables are recommended.

Recommended twisted pairs for **full bridge**, **half bridge** and **voltage** measurement:



For **quarter bridge** mode:



Why to use multi wire technology

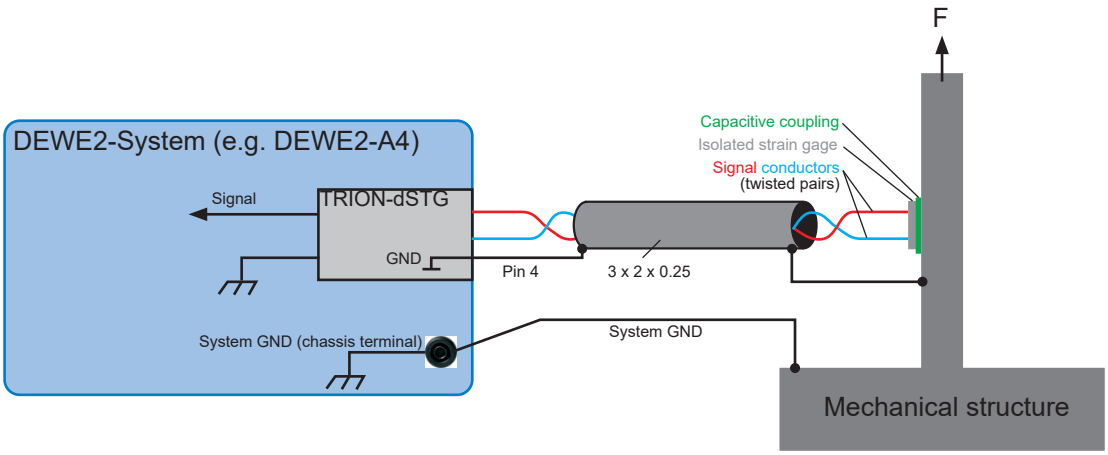
Sensitivity: For sensor wiring typically copper cables are used. For example a 120 Ω full bridge connected with a 4x 0.14 mm² cable will have a sensitivity error of 2.1% just because of the 1.27 Ω wire resistance. By using the 6 wire technology that could be completely compensated.

Temperature drift: Copper has a temperature drift of 0.4 %/°C.This is especially a problem at quarter bridges, because there also the offset changes with the wire resistance. The following table shows the difference between the 3 wiring methods for a 120 Ω strain gage with a 50 m cable 0.25 mm².

	Initial error		Drift because of 10 °C warm-up	
	Offset	Sensitivity	Offset	Sensitivity
2-wire	25183 µm/m	-4,97 %	956 µm/m	-0.18 %
3-wire	0 µm/m	-2.6 %	0µm/m	-0.01 %
4-wire	0 µm/m	0.0 %	0µm/m	0.00 %

Shielding and grounding

Strain gage measurement on a metal structure



It is important that you connect your DEWETRON system ground to the ground potential of your measured object. That is to guarantee that the measurement system is not floating against the measured structure. It could simply be a connection to the metal structure of your proving ground. In case of an automotive application for example it would be a connection to the cars chassis. Only if the DEWETRON system and the measured structure have an earth connection the system grounding line might not be needed.

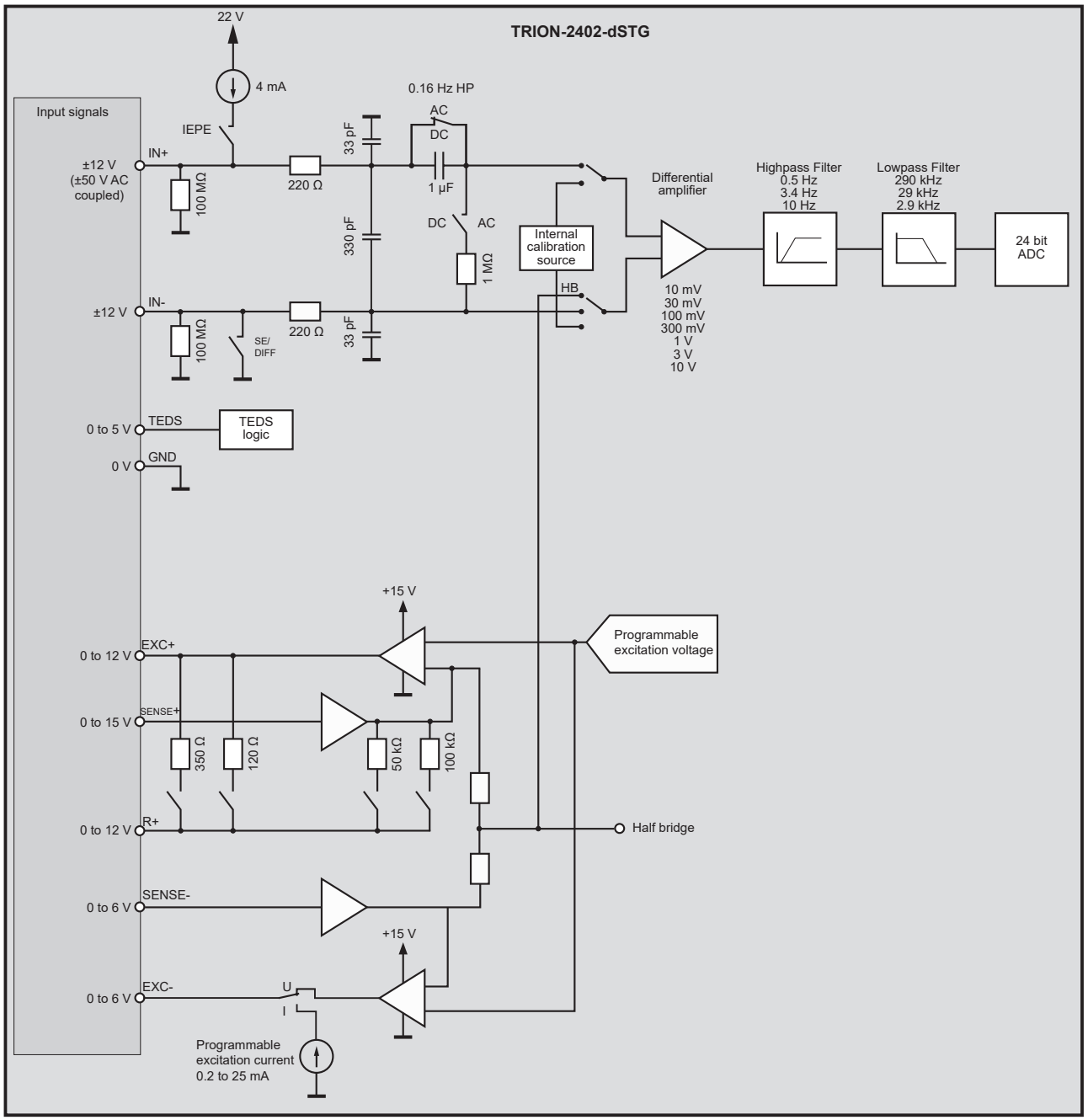
Single Ended / Differential

At ranges below 10 V the input impedance of the TRION-dSTG is very high. The 100 MΩ resistors are usually not enough to balance the differential inputs automatically around the internal GND. As a result you might get wrong or disturbed measurement data if either the measurement instrument or the sensor is floating. To avoid this, you can either switch the input to SE or connect the sensor GND to the DEWE2 GND.

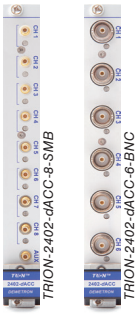
Further information regarding correct single ended / differential measurement please refer to chapter "Single Ended / Differential" of the TRION-dACC module.

Block diagram

Base block diagram of the TRION-2402-dSTG module:



Differential multi-function input module



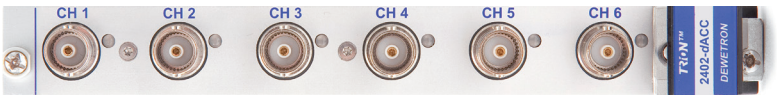
- *Sampling:* *24 bit; 204.8 kS/s per channel*
- *Input types:* *Voltage measurement from ±30 mV to ±70 V_{DC} (46.7 V_{RMS})*
  IEPE®
  Resistance
  Current (using external shunt)
- *Additional feature:* *AUX socket*

TRION-dACC specifications												
Input channels	8 using SMB sockets (TRION-2402-dACC-8-SMB)											
	6 using BNC sockets (TRION-2402-dACC-6-BNC)											
AUX socket (SMB version)	Selectable: Camera trigger, external trigger, CAL-port											
Sampling rate	204.8 kS/s per channel											
Resolution	24 bit											
Rated input voltage (max.)	70 V_{DC} (46.7 V_{RMS})											
Input ranges												
Voltage	±30, ±100, ±300 mV, ±1, ±3 V, ±10 V, ±30 V, ±100 V											
IEPE®	±100, 300 mV, 1 V, 3 V, 10 V											
Resistance	10, 30, 100, 300 Ω, 1, 3, 10, 30, 100, 300, 1000 kΩ											
Current	Depending on external Shunt											
Voltage input accuracy[1]	±0.02 % of reading ± 0.02 % of range ±200 µV											
Gain drift	typical 10 ppm/°C max. 20 ppm/°C											
Offset drift	typical 0.3 µV/°C + 10 ppm of range/°C, max 15 µV/°C + 20 ppm of range/°C											
linearity	typical 0.01 %											
Input impedance												
Range <= 10 V	10 MΩ											
Range > 10 V	2 MΩ											
Input bias current	< 1 nA											
Input configuration	Single ended or differential (programmable)											
Input coupling	DC, AC (0.16 Hz, 0.5 Hz, 3.4 Hz, 10 Hz)											
Sensor fault detection for IEPE®	short circuit and open sensor detection with LED indication											
Excitation current	0.1 to 24 mA_{DC} (programmable, 16 Bit DAC, 2 Ranges)											
Accuracy[1]	0.05% ±2 µA; >20 mA: 10 %											
Drift	15 ppm/°C											
Compliance voltage	23 V											
Output impedance	>10 MΩ											
Supported sensors	IEPE® (up to 24 mA excitation)											
	Resistance											
Counter Channels	2 Counter channels, linked to analog input Channel 1 and Channel 2											
Counter modes	Event counting; Periode; Frequency; Pulsewidth; Dutycycle											
Trigger Level	Trigger and retrigger level freely programmable within analog input range											
Counter input bandwidth	1 MHz											
Typical Signal-to-noise ratio, Spurious free SNR, Effective number of Bits[2]	100 mV range			1 V range			10 V range			100 V range		
	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]
Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]
1 kS/s	97	124	15.8	111	139	18.1	112	140	18.3	112	139	18.3
10 kS/s	90	121	14.7	108	136	17.6	109	138	17.8	107	136	17.5
100 kS/s	87	118	14.2	104	134	17.0	107	134	17.5	104	134	17.0
200 kS/s	80	116[5]/110	13.0	81	131[5]/112	13.2	81	132[5]/110	13.2	81	131[5]/112	13.2
Typical THD	-100 dB											
Typical CMRR ≤10V Range	100 dB @ 50 Hz; 100 dB @ 1 kHz											
>10 to 200 V Range	90 dB @ 50 Hz; 70 dB @ 1 kHz											
Analog anti aliasing filter	2nd order Bessel, automatically set by sample rate											
Sample rate ≤ 1kS/s	2.5 kHz (-3 dB), 1.5 kHz (-1 dB)											
Sample rate ≤ 10kS/s	25 kHz (-3 dB), 15 kHz (-1 dB)											
Sample rate > 10kS/s	250 kHz (-3 dB), 150 kHz (-1 dB)											
Bandwidth (-3 dB digital filter)												
1 kS/s ≤ fs ≤ 51.2 kS/s	0.494 fs											
51.2 kS/s < fs ≤ 102.4 kS/s	0.49 fs											
102.4 kS/s < fs ≤ 204.8 kS/s	0.38 fs											
Crosstalk fin 1 kHz [10 kHz]	120 dB [105 dB]											
Channel to channel phase mismatch	typically <60 nsec between channels using the same range											
Common mode voltage	±200 V_{DC} (input range > 10 V), ±12 V_{DC} (input range ≤ 10 V)											
Over voltage protection	200 V peak input to GND											
Supported TEDS chips	DS2406, DS2430A, DS2431, DS2432, DS2433											
Power consumption[6]												
Voltage mode; no excitation	6 W											
IEPE® mode 4 mA	6.5 W											
IEPE® mode 16 mA	9.5 W											
IEPE® mode 24 mA	11.4 W											
Weight	Appr. 210 g (SMB version), appr.270 g (BNC version)											

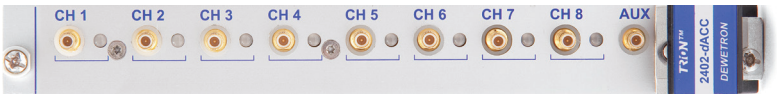
[1] 1 year accuracy 23 °C ±5 °C
[2] LP Filter in auto mode
[3] SFDR excluding harmonics
[4] ENOB calculated from SNR
[5] below 0.22 fs
[6] Consider maximum power supply of your DEWE2 chassis.

TRION-2402-dACC

TRION-2402-dACC model overview

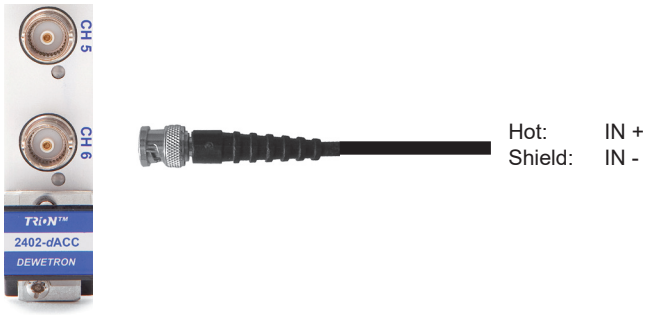


TRION-2402-dACC-6-BNC



TRION-2402-dACC-8-SMB

TRION-2402-dACC-6-BNC module



TRION-2402-dACC-8-SMB module

Optional accessory

TRION-CBL-SMB-OE-05-00

High quality cable from SMB plug to open end, 5 meters

TRION-CBL-SMB-BNC-01-00

High quality adaptor cable from SMB plug to BNC cable-socket, 1 meter

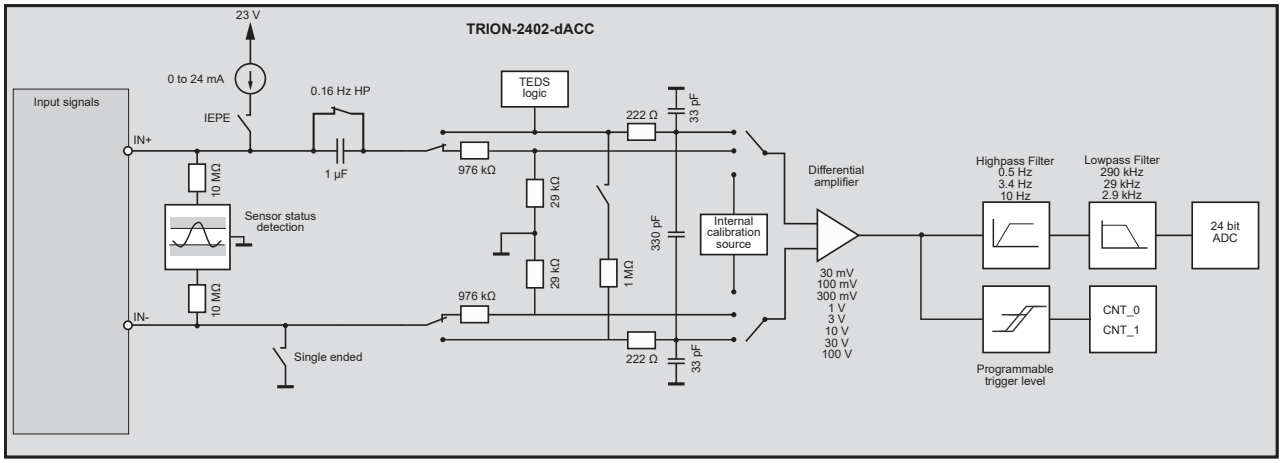
LED function

The LED next to the input connector shows the input state for each channel. In IEPE® mode it becomes red if no sensor is connected or the sensor or the cable has a short circuit. In voltage mode the red colour indicates a channel overload or if the channel is out of the common mode voltage range. A corrected channel state is shown when the LED is green.

Sensor error detection limits					
		IN-		IN+	
		min.	max.	min.	max.
		[V]	[V]	[V]	[V]
Voltage Diff	Range <= 10 V	-12.5	12.5	-12.5	12.5
	Range > 10 V	not available	not available	not available	not available
Voltage SE	Range <= 10 V	-12.5	12.5	-12.5	12.5
	Range > 10 V	not available	not available	not available	not available
IEPE®		-0.8	0.8	2	21
Resistance		not available	not available	not available	not available

Block diagram

Base block diagram of the TRION-2402-dACC module:



AUX terminal (only available at TRION-2402-dACC-8-SMB)

The auxiliary terminal of the TRION-2402-dACC module could be used either for input or output. The following functions are supported:

- **Video trigger output**
  (to synchronize an external camera to the analog data.)
- **Frequency output**
  (Programmable frequency output for synchronizing external hardware.)
- **External Trigger**
  (to start the measurement with an TTL signal.)

Counter functions

The first two channels of the TRION-2402-dACC module supports beside the normal functionality also counter inputs.

The trigger and retrigger level could be programmed within 0 to 100 % of the actual analog input range. This makes the input perfectly suitable for all kind of tacho probes. By activating the IEPE® supply it is even possible using probes without any additional sensor supply, just with a BNC cable.

Supported counter functions are:

- Simple event counting
- Period measurement
- Pulse width
- Frequency
- Duty cycle

For detailed information about this functions please refer to the chapter "Functional description of advanced counter" of the TRION-CNT module.

Caution: It is not possible to change the analog input settings out of the counter dialog. This has to be done in the channel setup of the analog input!

Free variable gain and excitation

Amplifier parameters like gain, excitation voltage, excitation current and sensor offset are free programmable for every channel individually. That allows to perfectly match each input channel to any sensor. Customized programming of the amplifier could be simply done by entering the desired value in the appropriate field.

High pass filter

The TRION-2402-dACC module has four different input high pass filter available for AC coupling:

Frequency	Time constant
0.16 Hz	1 sec
0.5 Hz	320 msec
3.4 Hz	47 msec
10 Hz	16 msec

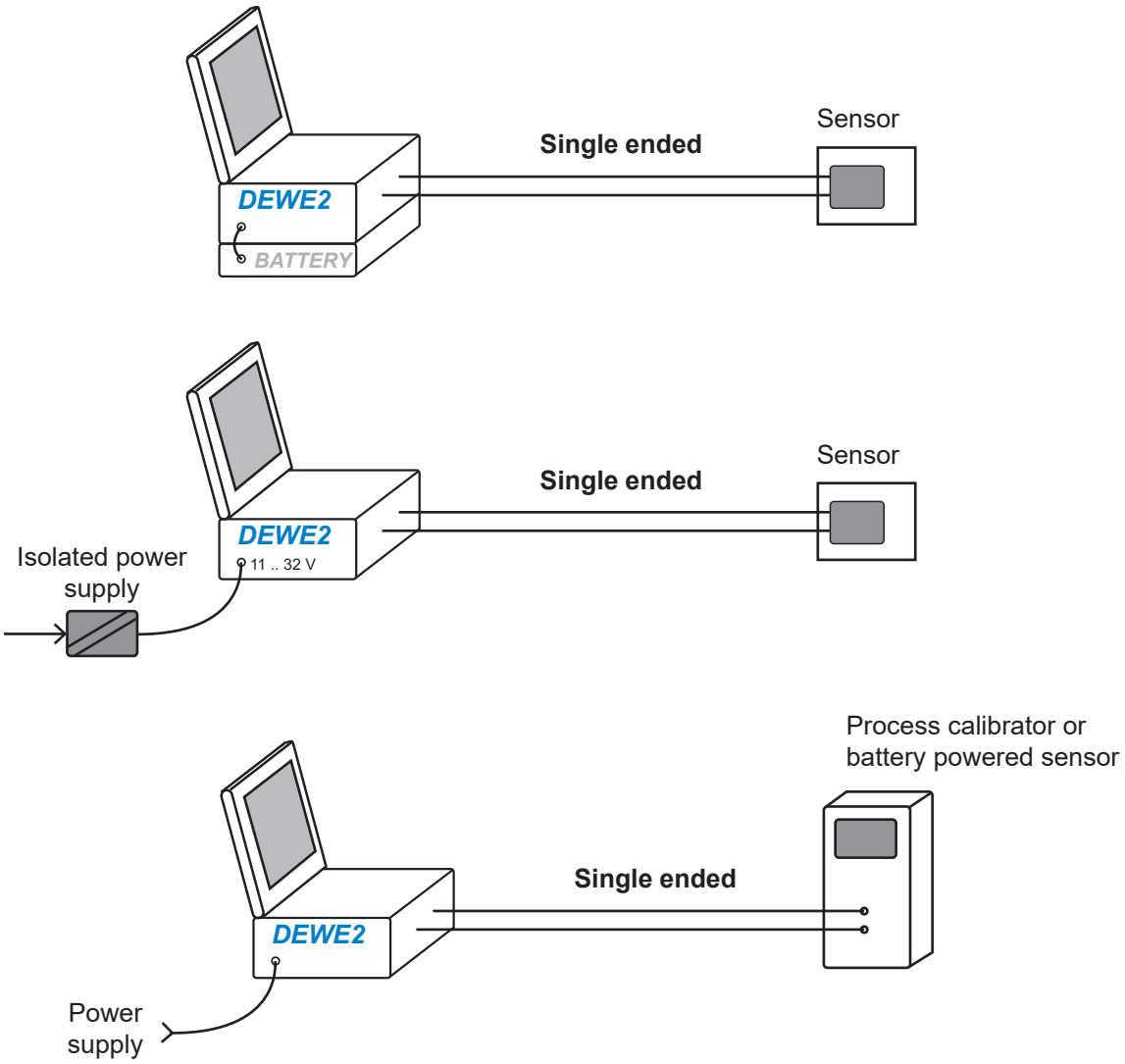
Self test

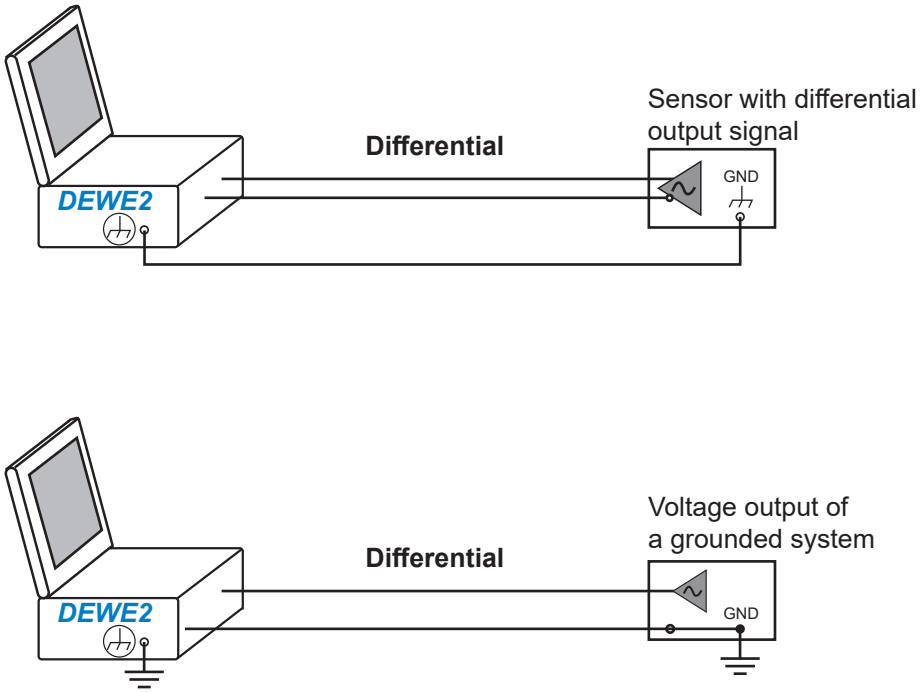
The TRION-2402-dACC module has an integrated special self test circuit. It is a high precision voltage source and a temperature compensated divider. It is used to check the analog input path of the voltage amplifier by applying 0 V and ±98 % of the input range to the input. This test could be performed in the channel setup for the actual range. During the board self test which is available in the DEWE2 Explorer, this test is performed for all ranges and channels automatically.

Single Ended / Differential

The TRION-2402-dACC module could be switched to differential or single ended input. At ranges below 10 V the input impedance of the dACC is very high. The 10 MΩ resistors are usually not enough to balance the differential inputs automatically around the internal GND. As a result you might get wrong or disturbed measurement data if either the measurement instrument or the sensor is floating. To avoid this, you can either switch the input to SE or connect the sensor GND to the DEWE2 GND.

Examples for correct single ended / differential measurement

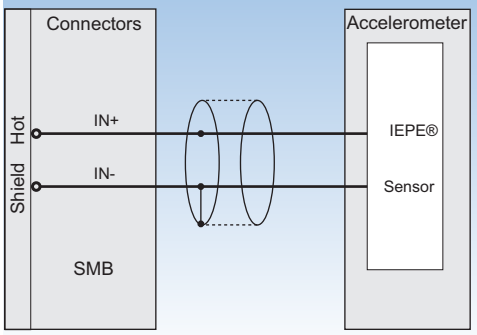




Provides from current loops

Signal connection

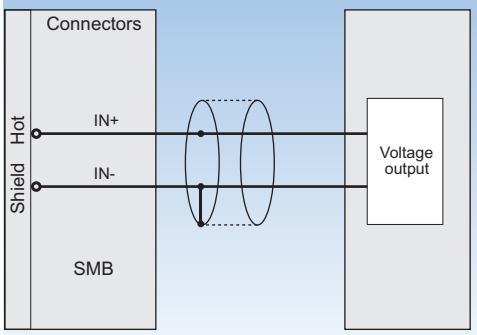
IEPE® signal connection



Software settings for IEPE® sensors

Mode:	IEPE
Range:	10 V
Coupling:	AC 0.5 Hz
Excitation:	4 mA
Type:	Current

Voltage signal connection

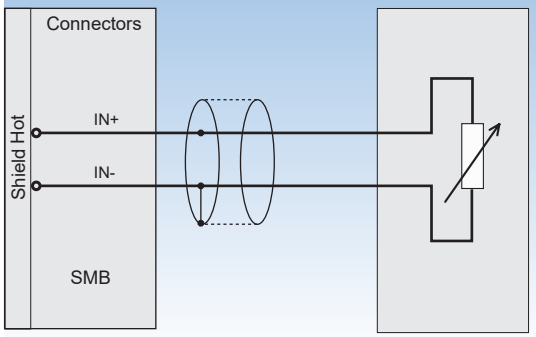


Software settings for voltage measurement

Mode:	Voltage
Range:	10 V
Input type:	Differential
Excitation:	0 V

Caution: Having floating voltage sources like batteries please select “Single ended” as input type!

Resistance signal connection



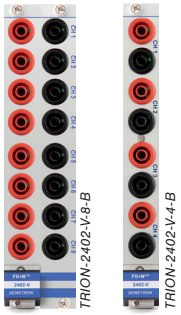
Software settings for resistance measurement

Mode:	Resistance
Range:	10000 Ω

Resistance accuracy table for TRION-2402-dACC

Resistance accuracy table for TRION-2402-dACC				
Range (Ω)	**Excitation current (mA)**	**Voltage range (V)**	**Accuracy**	**Temp drift (ppm /°C)**
1 M	0.01	10	300 Ω ±1 %	100
300 k	0.03	10	75 Ω ±1 %	100
100 k	0.1	10	30 Ω ±1 %	100
30 k	0.2	10	75 Ω ±1 %	100
10 k	0.5	10	3 Ω ±0.5 %	100
3 k	1	3	750 mΩ ±0.5 %	100
1 k	1	1	300 mΩ ±0.5 %	100
300	1	0.3	100 mΩ ±0.5 %	200
100	10	0.1	50 mΩ ±0.5 %	200
30	10	0.3	10 mΩ ±0.5 %	200

Isolated wide range voltage and current module

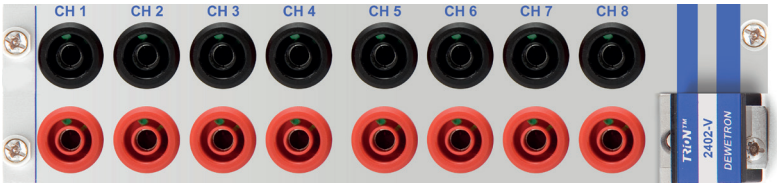


- *Sampling:* *24 bit; 204.8 kS/s per channel*
- *Input types:* *Voltage ranges from ±300 mV to ±1000[2] V_{DC}*
  Current (using external shunt)
- *Isolation:* *4 kV*

TRION-2402-V specifications													
Input channels	4 (TRION-2402-V-4-B, requires 1 slot in DEWE2 instruments)												
	8 (TRION-2402-V-8-B, requires 2 slots in DEWE2 instruments)												
Sampling rate	204.8 kS/s per channel												
Resolution	24 bit												
Input ranges													
Voltage	±0.3 V, ±1 V, ±3 V, ±10 V, ±30 V, ±100 V, ±400 V and ±1000 V[1]												
Current	Depending on external Shunt; e.g. 20 mA, 1 A, 5 A												
Voltage input accuracy	Signal frequency	Accuracy											
Range > 10 V	DC to 1 kHz >1 kHz to 5 kHz >5 kHz to 10 kHz	±0.02 % of reading ±0.02 % of range ±3 mV ±0.3 % of reading ±0.02 % of range ±3 mV ±1 % of reading ±0.02 % of range ±3 mV											
Range ≤ 10 V	DC to 1 kHz >1 kHz to 5 kHz >5 kHz to 10 kHz	±0.02 % of reading ±0.02 % of range ±30 µV ±0.3 % of reading ±0.02 % of range ±30 µV ±1 % of reading ±0.02 % of range ±30 µV											
Offset drift Range > 10 V	typical 10 µV/°C + 10 ppm of range/°C, max 300 µV/°C + 20 ppm of range/°C												
Range ≤ 10 V	typical 0.3 µV/°C + 10 ppm of range/°C, max 3 µV/°C + 20 ppm of range/°C												
Gain drift	typical 10 ppm/°C max. 20 ppm/°C												
Linearity	typical 0.01 %												
Input impedance Range > 10 V	10 MΩ												
Range ≤ 10 V	5 MΩ												
Input bias current	< 100 pA												
Input coupling	DC												
Input configuration	isolated												
Typical Signal-to-noise ratio, spurious-free SNR, Effective number of Bits[2]	1 V range			10 V range			100 V range			1000 V range[1]			
	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	SNR	SFDR[3]	ENOB[4]	
Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	
1 kS/s	113	139	18.5	115	140	18.8	113	139	18.5	115	140	18.8	
10 kS/s	111	137	18.1	113	136	18.5	109	136	17.8	112	138	18.3	
100 kS/s	106	134	17.3	109	134	17.8	106	134	17.3	109	134	17.8	
200 kS/s	98	132	16.0	84	131	13.7	92	132	15.0	95	132	15.5	
200 kS/s (Filter off)	81	132[5]/110	13.2	81	131[5]/112	13.2	81	132[5]/110	13.2	81	131[5]/112	13.2	
Typical THD	-97 dB												
Typical CMRR ≤10 V Range	>150 dB @ DC; 100 dB @ 50 Hz; 100 dB @ 1 kHz												
>10 V Range	>150 dB @ DC; 90 dB @ 50 Hz; 60 dB @ 1 kHz												
Low pass Filter (-3 dB, digital)	10 Hz to 40 % of sample rate freely programmable or OFF												
Characteristic	Bessel or Butterworth												
Filter order	2nd , 4th, 6th, 8th												
Filter setting AUTO	30 % of sample rate with 8th order Bessel												
Analog anti aliasing filter	2nd order Bessel, automatically set by sample rate												
Sample rate ≤ 1kS/s	2.5 kHz (-3 dB), 1.5 kHz (-1 dB)												
Sample rate ≤ 10kS/s	25 kHz (-3 dB), 15 kHz (-1 dB)												
Sample rate > 10kS/s	250 kHz (-3 dB), 150 kHz (-1 dB)												
Bandwidth (-3 dB digital filter)													
1 kS/s ≤ fs ≤ 51.2 kS/s	0.494 fs												
51.2 kS/s < fs ≤ 102.4 kS/s	0.49 fs												
102.4 kS/s < fs ≤ 204.8 kS/s	0.38 fs												
Typ. crosstalk fin 1 kHz [10 kHz]	110 dB [95 dB]												
Channel to channel phase mismatch	typically <60 nsec between channels using the same range												
Measurement category	CAT II 300 V												
Isolation voltage	4 kV channel to channel and channel to ground												
Non-destructive voltage													
Range ≤ 10 V	±1000 V_{DC}												
Range > 10 V	±2000 V_{DC}												
Typical power consumption													
TRION-2402-V-8	8.3 W												
TRION-2402-V-4	4.5 W												
Connector	4 mm safety banana sockets												
Weight	Appr. 270 g (4 ch version), appr. 400 g (8 ch version)												

TRION-2402-V model overview

TRION-2402-V-4-B

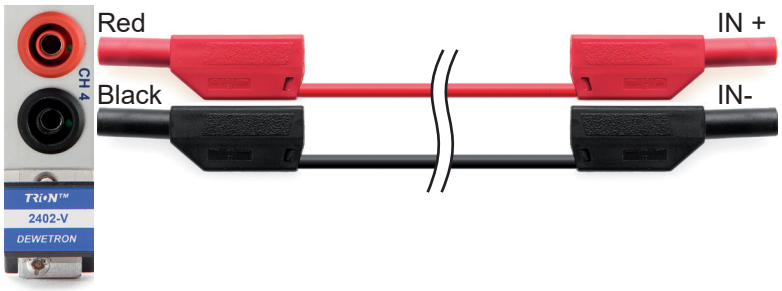


TRION-2402-V-8-B

Signal connection

TRION-2402-V-x-B module

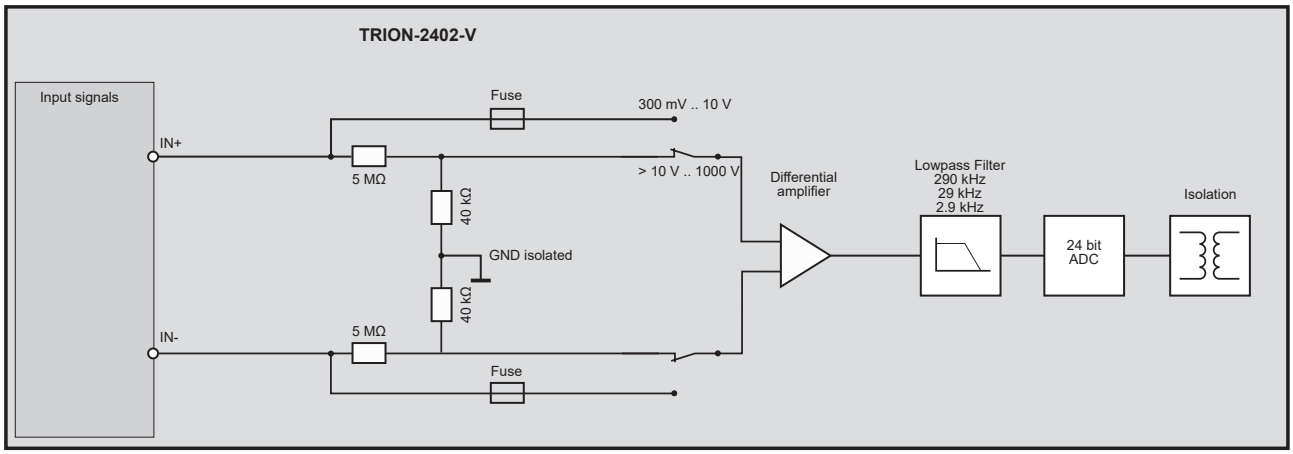
Voltage measurement via banana plug cords



Voltage measurement up to ±1000 V
only with safety banana plug cords!

Block diagram

Base block diagram of the TRION-2402-V module:



General information

The TRION-2402-V module is a highly accurate, isolated, 24-bit voltage digitizer. For ranges above 10 V the input signal is attenuated by a resistance divider. This is a special ceramic divider, which is designed for high linearity even at high voltages and low temperature drift. It is factory trimmed for each channel separately, to maximize the common mode rejection ratio.

For ranges up to 10 V the input amplifier is switched to a direct input path. This allows also measuring low voltages with an excellent signal to noise ratio and accuracy. To protect the input amplifier, this path is equipped with a current limiting circuit that can easily withstand 1000 V_{DC}. So the TRION-2402-V is also very well protected in the small input ranges. The conditioned signal passes a programmable low pass filter before getting to the ADC. Please refer to chapter "*Analog to digital conversion of TRION-2402 series modules*" for more details about bandwidth and filtering.

For AD conversion each channel of the TRION-2402-V uses two 24-bit sigma-delta converters in noise cancelling mode. This means the two ADCs are switched in parallel and the result is the average of the measured values. This results in an additional 3 dB better signal-to-noise ratio without any switching nonlinearities.

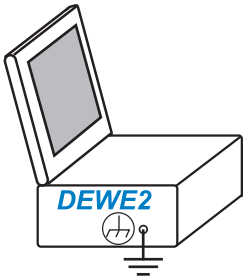
Isolation

The isolation of the module has many advantages:

- It allows very high common mode voltages even in the 300mV range. This is especially required for current measurement with shunt technology.
- High input protection.
- DC offset errors because of ground loops are eliminated.
- Eliminating current loops; Noise reduction

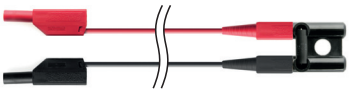
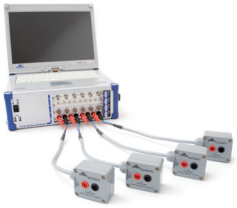
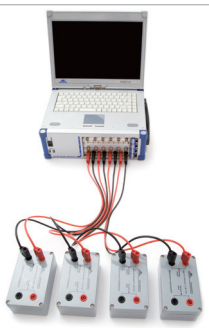
Ground connection

The TRION-2402-V is fully isolated and has high impedance differential inputs, with very high sensitivity. For archieving the best signal to noise ration it is strongly recommended to connect the DEWE2 system to earth or a structural ground potential. This could be for example the chassis of the car or train in case of vehicle measurements. With that simple method you can avoid catching noise signals like the 50/60 Hz interference.



Sometimes this connection is already provided by the power supply cable. If the system runs on battery or with isolated DC power supply the operator should take care of the ground connection.

Optional accessory

Name	Description	Image
SE-CUR-SHUNT-01	20 mA shunt adaptor (50 Ω, ±0.1 %, 1 W); fitting into 4 mm banana jacks with 19 mm distance. Note: Not compatible with TRION-2402-V-4-B module!!	
SE-CUR-SHUNT-01-UNI	20 mA shunt adaptor (50 Ohm, ±0.1 %, 1 W); fitting into 4 mm banana jacks - also for TRION-x-V-4-B	
SE-CUR-SHUNT-04	5 A shunt box (100 mΩ, ±0.1 %, <10 ppm); current input via 2 safety banana jacks, output 2 x 0.3 m cable with banana plugs.	
SE-CUR-SHUNT-05	5 A shunt box (100 mΩ, ±0.1 %, <10 ppm) Current input via 2 safety banana jacks, output via 2 safety banana jacks.	
SE-CUR-SHUNT-06	20 A shunt box (5 mΩ, ±0.1 %, CAT III 500 V), Current input via 2 safety banana jacks, Voltage output via 2 safety banana jacks.	

Isolated TRION™ module for high voltage inputs

- Channels: 4 high voltage channels
- Sampling: up to 1 MS/s
- Resolution: 18-bit
- Input types: Voltage: 1000 V

Module specifications

TRION-1810-HV-4	
Input channels	4 high voltage channels
Resolution	18-bit
Sample Rate	up to 1 MS/sec; 512 MByte onboard data buffer

The following are all accuracy conditions referenced in this section:
Temperature: 23±5°C, Humidity: 40 to 60 % RH, Input waveform: sine wave, Common mode voltage: 0 V, Line filter: Auto, Sample rate: 1 MS/s, Resolution: 24 bit, Power factor: 1, After warm-up. After zero level, 12-month Accuracy ±(Reading error + Range error), Accuracy: Frequency (f) in [kHz]

Voltage input U1, U2, U3, U4	
Input range	1000 V (±2000 V_{PEAK}) CF = 2
Accuracy[1) 2)]	DC: ±0.02 % of reading ±0.02 % of range
	0.5 Hz to 1 kHz: ±0.03 % of reading
	1 kHz to 5 kHz: ±0.15 % of reading
	5 kHz to 10 kHz: ±0.35 % of reading
	10 kHz to 50 kHz: ±0.6 % of reading
	50 kHz to 300 kHz: ±(0.02 % * f) of reading
Gain drift	20 ppm / °C
Offset drift	5 mV / °C
Typical THD	-95 dB
CMRR	>85 dB @ 50 Hz; >60 dB @ 1 kHz; >40 dB @ 100 kHz
Bandwidth	5 MHz
Safety	CAT IV 600 V / CAT III 1000 V
Instantaneous maximum allowable input	4000 V_{PEAK} or 3000 V_{RMS} (1s)
Continuous maximum allowable input	2000 V_{RMS} or 2000 V_{DC}
Input resistance	5 MΩ; 2 pF
Isolation (earth) resistance	100 GΩ; 2.5 pF
Connector	Safety banana sockets

[1)] add 0.02 % of reading with filter settings OFF
[2)] below 1 % of range, add 10 ppm of range

Typical Signal-to-noise ratio, Spurious free SNR, Effective number of Bits[1)]

	SNR	SFDR[2)]	ENOB[3)]
Sample rate	[dB]	[dB]	[Bit]
0.1 kS/s	129	142	21.1
1 kS/s	126	139	20.6
10 kS/s	121	136	19.8
100 kS/s	113	135	18.5
1000 kS/s	103	128	16.9

[1)] LP Filter in auto mode
[2)] SFDR excluding harmonics
[3)] ENOB calculated from SNR

Additional specifications	
Typical channel to channel phase mismatch	<250 ns (0.1° @ 1 kHz, 0.005° @ 50 Hz)
Typical board-to-board phase mismatch	
same board type	<250 ns (0.1° @ 1 kHz, 0.005° @ 50 Hz)
different board type	±1 Sample or 0.2° @ 1 kHz (whichever is higher)
Low pass filter (-3 dB, digital and analog combined)	100 Hz to 1 MHz freely programmable or OFF
Filter order & characteristics	2nd, 4th, 6th, 8th Bessel or Butterworth
Filter delay compensation	up to 15 µs the group delay of the selected filter will be automatically compensated. This works for: 2nd order filter 15 kHz to 1 MHz 4th order filter 30 kHz to 1 MHz 6th order filter 60 kHz to 1 MHz

TRION-1810-HV-4

Signal connection

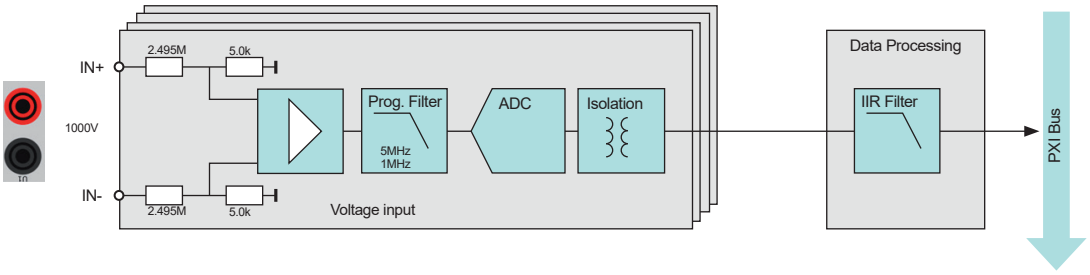
Voltage input

High voltage input for line voltage measurement.

Voltage measurement up to ±2000 V
only with safety banana plug cords!

Block diagram

Base block diagram of the TRION-1810-HV-4 module:



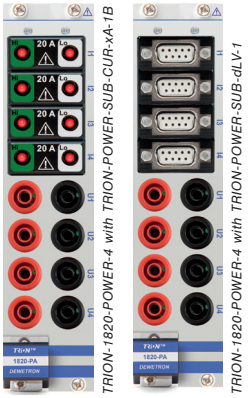
The TRION-1810-HV-4 is a 4 channel high voltage data acquisition module for use in any DEWE2 and DEWE3 chassis.

Voltage input:

Fast sampling, high bandwidth and minimum capacity to earth are just a few outstanding performance qualities of the high voltage inputs. The high input impedance allowes high continous voltage levels with a very low temperature drift. Although small outline, the safety category is on a very high level (CAT III 1000V).

TRION™ module for 4-phase power analysis

- Channels: 4 power channels
- Sampling: up to 2 MS/s
- Resolution: 18-bit
- Input types: Voltage: 1000 V
  Current: 20 A, 1 A, 0.2 A, 5 V



Module specifications

TRION-1820-POWER-4		
Input channels	4 Power channels, each with one voltage and one current input	
Resolution	18-bit	
Sample Rate	up to 2 MS/sec; 512 MByte onboard data buffer	
The following are all accuracy conditions referenced in this section: Temperature: 23±5°C, Humidity: 40 to 60 % RH, Input waveform: sine wave, Common mode voltage: 0 V, Line filter: Auto (8th order Butterworth), Sample rate: 2 MS/s, Resolution: 24 bit, Driver Version >= 3.7, Power factor: 1, After warm-up. After zero level, 12-month Accuracy ±(Reading error + Range error), Accuracy: Frequency (f) in [kHz]		
Voltage input U1, U2, U3, U4		
	Input range	1000 V (±2000 V_{PEAK}) CF = 2
	Accuracy[1) 2)]	DC: ±0.02 % of reading ±0.02 % of range 0.5 Hz to 1 kHz: ±0.03 % of reading 1 kHz to 5 kHz: ±0.15 % of reading 5 kHz to 10 kHz: ±0.35 % of reading 10 kHz to 50 kHz: ±0.6 % of reading 50 kHz to 300 kHz: ±(0.02 % * f) of reading
	Gain drift	20 ppm / °C
	Offset drift	5 mV / °C
	Typical THD	-95 dB
	CMRR	>85 dB @ 50 Hz; >60 dB @ 1 kHz; >40 dB @ 100 kHz
	Bandwidth	5 MHz
	Safety	CAT IV 600 V / CAT III 1000 V
	Instantaneous maximum allowable input	4000 V_{PEAK} or 3000 V_{RMS} (1s)
	Continuous maximum allowable input	2000 V_{RMS} or 2000 V_{DC}
	Input resistance	5 MΩ; 2 pF
	Isolation (earth) resistance	100 GΩ; 2.5 pF
	Connector	Safety banana sockets
[1)] add 0.02 % of reading with filter settings OFF [2)] below 1 % of range, add 10 ppm of range		
Current input I1, I2, I3, I4		
20 A module		TRION-POWER-SUB-CUR-20A-1B
	Range	20 A (±40 A_{PEAK})
	Accuracy[1) 2)]	DC: ±0.02 % of reading ±0.02 % of range[3)] 0.5 Hz to 1 kHz: ±0.03 % of reading 1 kHz to 5 kHz: ±0.15 % of reading 5 kHz to 10 kHz: ±0.35 % of reading 10 kHz to 50 kHz: ±(0.3 % + 0.05 % * f) of reading 50 kHz to 300 kHz: ±(0.10 % * f) of reading
	Safety	CAT II 600 V, unfused
	Bandwidth	300 kHz
	Connector	Safety banana sockets (male)
	Instantaneous maximum allowable input	50 A_{PEAK} or 40 A_{RMS} (1s)
	Continuous maximum allowable input	20 A_{RMS}
	Input resistance	2 mΩ
[1)] For self-generated heat caused by current input, add 0.00008 × I² % of reading + 15 × I² µA to the current accuracy. 'I' is the current reading [A]. The influence from self-generated heat continues until the temperature of the shunt resistor inside the DEWE2-Chassis lowers even if the current input changes to a small value. [2)] below 1 % of range, add 50 ppm of range [3)] add 0.03 % of range with no zero level		

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TRION-1820-POWER-4

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2 A module	TRION-POWER-SUB-CUR-2A-1B
	Range: 2 A (±4 A_{PEAK})
EOL product **This module is replaced by the 1A module below!**	Accuracy[1]: DC: ±0.02 % of reading ±0.02 % of range[2]
	0.5 Hz to 10 kHz: ±0.03 % of reading
	10 kHz to 30 kHz: ±0.1 % of reading
	30 kHz to 200 kHz: ±(0.015 % * f) of reading
	200 kHz to 300 kHz: ±(0.1 % * f) of reading
	Safety: CAT II 600 V, unfused
	Bandwidth: 300 kHz
	Connector: Safety banana sockets (male)
	Instantaneous maximum allowable input: 10 A_{PEAK} or 5 A_{RMS} (1s)
	Continuous maximum allowable input: 3 A_{RMS}
	Input resistance: 50 mΩ

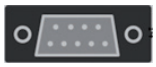
[1] below 1 % of range, add 25 ppm of range
[2] add 0.03 % of range with no zero level

1 A module	TRION-POWER-SUB-CUR-1A-1B
	Range: 1 A (±2 A_{PEAK})
	Accuracy[1]: DC: ±0.02 % of reading ±80 µA[2]
	0.5 Hz to 10 kHz: ±0.03 % of reading
	10 kHz to 30 kHz: ±0.1 % of reading
	30 kHz to 200 kHz: ±(0.015 % * f) of reading
	200 kHz to 300 kHz: ±(0.1 % * f) of reading
	Safety: CAT II 600 V, unfused
	Bandwidth: 300 kHz
	Connector: Safety banana sockets (male)
	Instantaneous maximum allowable input: 4 A_{PEAK} or 2 A_{RMS} (1s)
	Continuous maximum allowable input: 1.1 A_{RMS}
	Input resistance: 500 mΩ

[1] below 1 % of range, add 25 ppm of range
[2] add 0.03 % of range with no zero level

0.2 A module	TRION-POWER-SUB-CUR-02A-1B
	Range: 0.2 A (±0.4 A_{peak})
	Accuracy[1]: DC: ±0.02 % of reading ±0.02 % of range[2]
	0.5 Hz to 10 kHz: ±0.03 % of reading
	10 kHz to 30 kHz: ±0.1 % of reading
	30 kHz to 200 kHz: ±(0.015 % * f) of reading
	200 kHz to 300 kHz: ±(0.1 % * f) of reading
	Safety: CAT II 600 V, unfused
	Bandwidth: 300 kHz
	Connector: Safety banana sockets (male)
	Instantaneous maximum allowable input: 2 A_{PEAK} or 1 A_{RMS} (1s)
	Continuous maximum allowable input: 0.4 A_{RMS}
	Input resistance: 500 mΩ

[1] below 1 % of range, add 25 ppm of range
[2] add 0.03 % of range with no zero level

Clamp input module	TRION-POWER-SUB-dLV-1
	Range: 5 V (±10 V_{peak}) **NOT ISOLATED** ⚠
	Accuracy[1]: DC: ±0.02 % of reading ±0.02 % of range
	0.5 Hz to 5 kHz: ±0.03 % of reading
	5 kHz to 30 kHz: ±(0.01 % * f) of reading
	30 kHz to 50 kHz: ±(0.02 % * f) of reading
	50 kHz to 100 kHz: ±(0.1 % * f) of reading
	Typical THD: -95 dB
	Typical CMRR: 80 dB @ 50 Hz; 80 dB @ 1 kHz; 65 dB @ 10 kHz; 45 dB @ 100 kHz
	Safety: Depending on connected clamp
	Absolute maximum voltage: ±30 V
	Bandwidth: 100 kHz
	Connector: DSUB9
	Input resistance: 1 MΩ
	Sensor supply (+9 V): max. 40 mA

[1] below 1 % of range, add 10 ppm of range

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Power specifications

Accuracy[1] with PF=1	DC:	±0.03 % of reading ±0.03 % of range[2]
	0.5 Hz to 1 kHz:	±0.04 % of reading
	1 kHz to 5 kHz:	±0.2 % of reading
	5 kHz to 10 kHz:	±0.5 % of reading
	10 kHz to 50 kHz:	±(0.5 % + 0.05 % * f) of reading
Influence of power factor	Add 0.01 % × f/50 × √(1/PF²-1)	

[1] voltage and current channel have a minimum input of 1 % of range, otherwise individual uncertainty have to be calculated
[2] add 0.03 % of range with no zero level

Typical Signal-to-noise ratio, Spurious free SNR, Effective number of Bits[1]

	Voltage input 2000 V			20 A module			2 A module			1 A module			0.2 A module			Clamp input module 10 V		
	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]
Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]
0.1 kS/s	129	142	21.1	101	117	16.5	110	125	18.0	131	149	21.5	108	128	17.6	129	146	21.1
1 kS/s	126	139	20.6	100	119	16.3	107	126	17.5	125	149	20.5	107	123	17.5	119	131	19.5
10 kS/s	121	136	19.8	98	113	16.0	105	122	17.1	116	144	19.0	104	121	17.0	109	124	17.8
100 kS/s	113	135	18.5	93	110	15.2	100	120	16.3	106	137	17.3	100	114	16.3	99	119	16.2
1000 kS/s	103	128	16.9	85	110	13.8	91	114	14.8	96	134	15.7	92	114	15.0	94	115	15.3
2000 kS/s	85	106	13.8	84	107	13.7	90	114	14.7	95	130	15.5	90	114	14.7	92	114	15.0

[1] LP Filter in auto mode
[2] SFDR excluding harmonics
[3] ENOB calculated from SNR

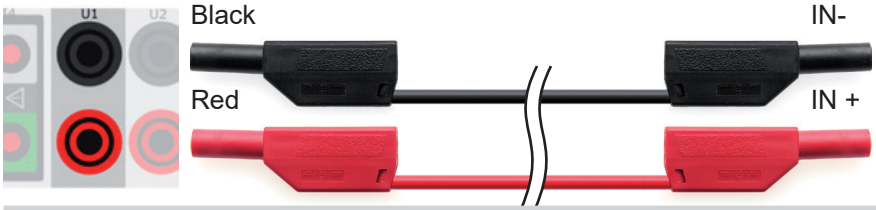
Additional specifications

Typical channel to channel phase mismatch (Voltage-Voltage, Current-Current, Voltage-Current)	<250 ns (0.1° @ 1 kHz, 0.005° @ 50 Hz)
Typical board-to-board phase mismatch	
same board type	<250 ns (0.1° @ 1 kHz, 0.005° @ 50 Hz)
different board type	±1 Sample or 0.2° @ 1 kHz (whichever is higher)
Low pass filter (-3 dB, digital and analog combined)	100 Hz to 1 MHz freely programmable or OFF
Filter order & characteristics	2nd, 4th, 6th, 8th Bessel or Butterworth
Filter delay compensation	up to 15 µs the group delay of the selected filter will be automatically compensated. This works for: 2nd order filter 15 kHz to 1 MHz 4th order filter 30 kHz to 1 MHz 6th order filter 60 kHz to 1 MHz

Signal connection

Voltage input

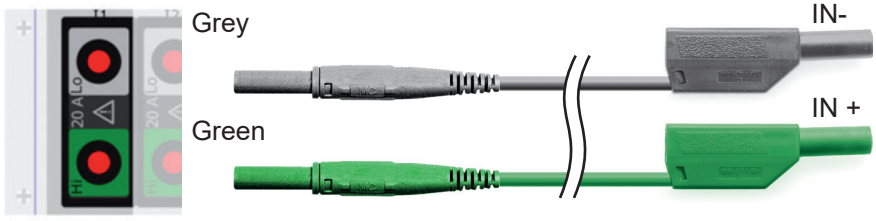
High voltage input for line voltage measurement.



Voltage measurement up to ±2000 V only with safety banana plug cords!

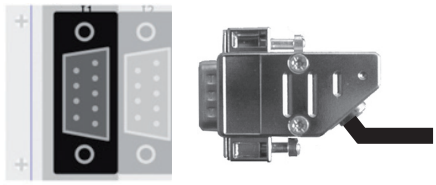
Current input module (TRION-POWER-SUB-CUR-xA-1B)

Direct current input for measuring current directly. This input is isolated and rated with CAT II 600 V. Modules with 20 A, 2 A and 0.2 A nominal current are available.



Clamp input module (TRION-POWER-SUB-dLV-1, for clamps ONLY)

NOT ISOLATED voltage input. This input can be used for isolated current clamps with voltage output.

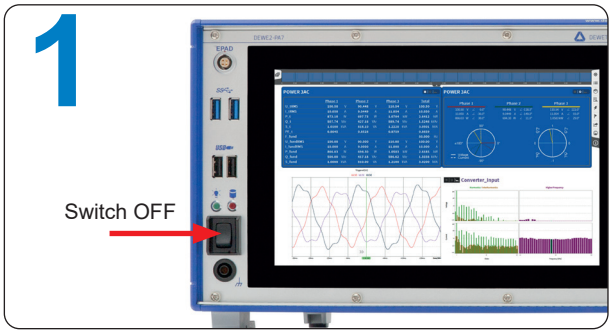


Pin 1: TEDS
Pin 2: IN+
Pin 3: n.c.
Pin 4: GND (not isolated)
Pin 5: +9 V (40 mA max.)
Pin 6: n.c.
Pin 7: IN-
Pin 8: n.c.
Pin 9: -9 V (40 mA max.)

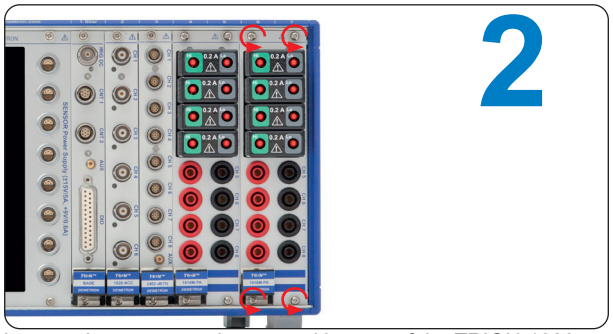
WARNING: *This module is not isolated! Do not connect any other appliances than isolated current transducers with voltage output.*

Exchanging current modules (SUB-modules)

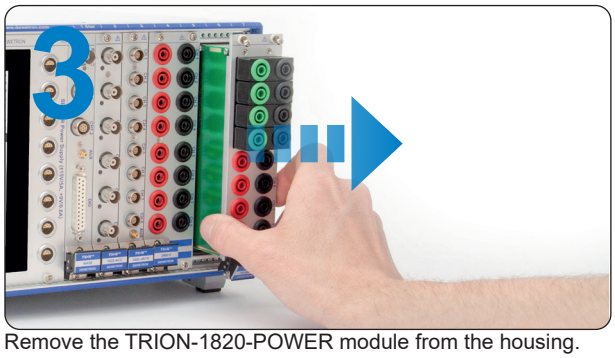
Proper ESD precautions must be taken to avoid any damage to the unit.



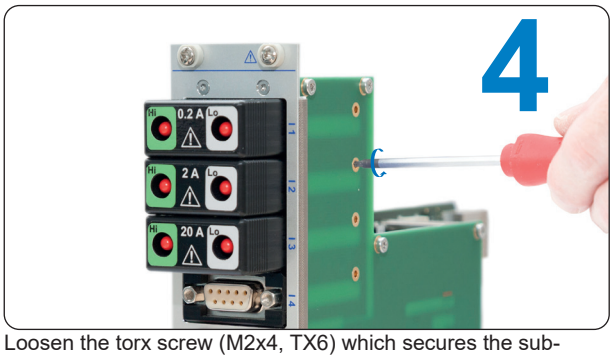
Switch off the instrument and unplug all connected cables including sensors from the TRION™ series modules.



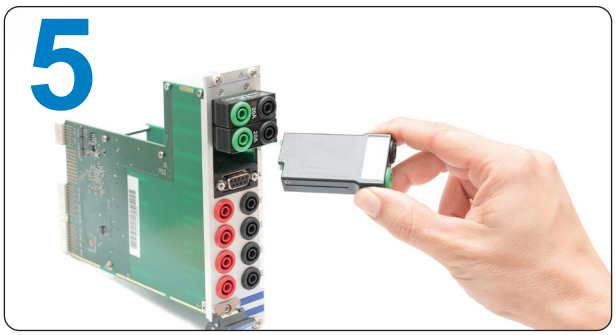
Loosen the screws at the top and bottom of the TRION-1820-POWER-4 module front panel (4x) and pull down the injector/ejector handle to release the module.



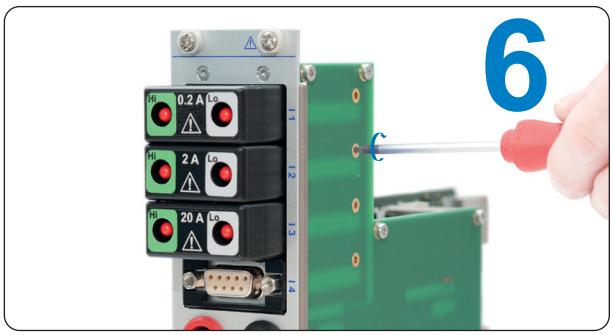
Remove the TRION-1820-POWER module from the housing.



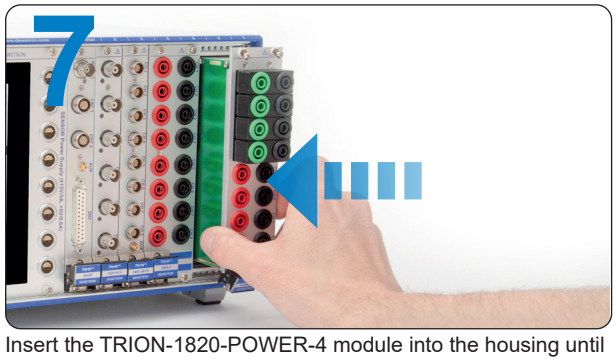
Loosen the torx screw (M2x4, TX6) which secures the sub-module of the channel you want to replace.



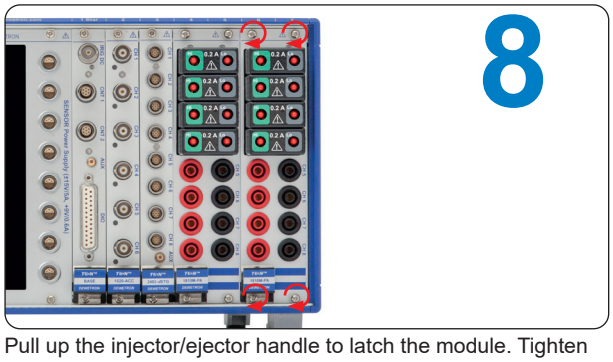
Replace the sub-module.



Secure the replaced sub-module with the torx screw (M2x4, TX6). Max. torque: 0.2 Nm



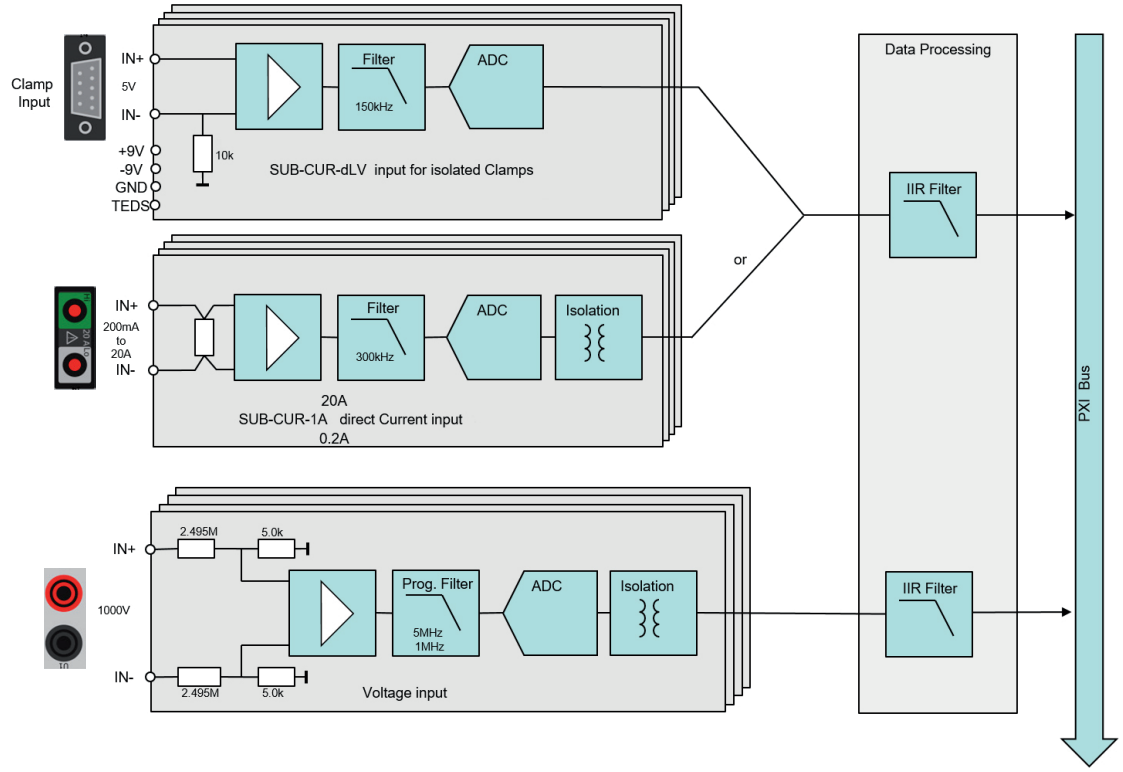
Insert the TRION-1820-POWER-4 module into the housing until a resistance appears.



Pull up the injector/ejector handle to latch the module. Tighten the screws at the top and bottom of the TRION-1820-POWER-4 module front panel (4x) to secure the module.

Block diagram

Base block diagram of the TRION-1820-POWER-4 module:



The TRION-1820-POWER-4 is an ultra-compact, 4-phase power data acquisition module with flexible current inputs for use in any DEWE2-chassis.

Voltage input:

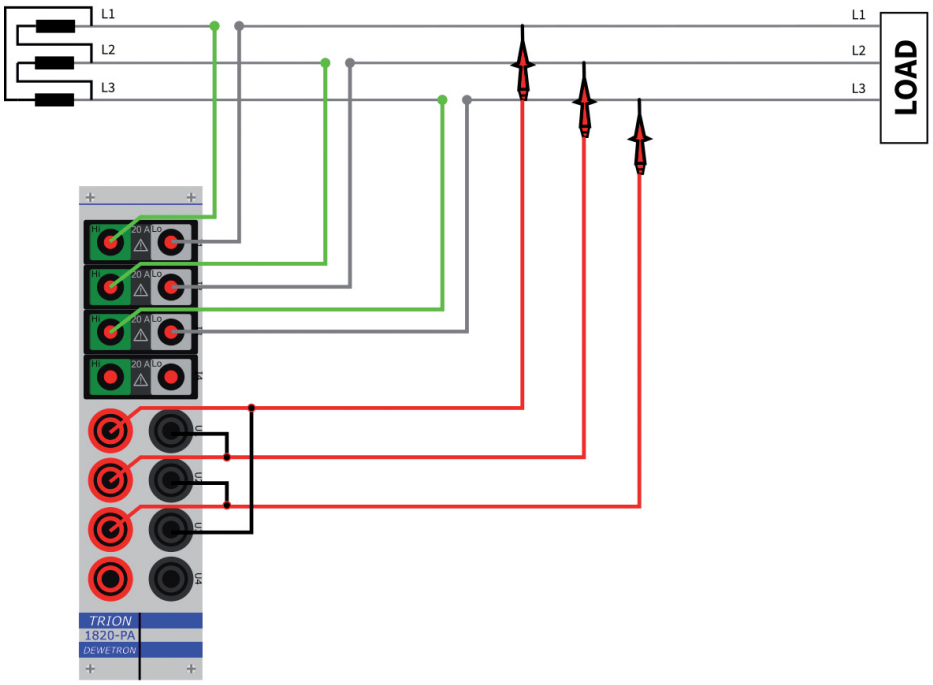
Fast sampling, high bandwidth and minimum capacity to earth are just a few outstanding performance qualities of the high voltage inputs. The high input impedance allowes high continous voltage levels with a very low temperature drift. Although small outline, the safety category is on a very high level (CAT III 1000V).

Current input:

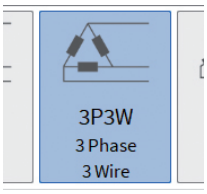
The TRION-1820-POWER-4 has 4 highly flexible current inputs. The 4 slots can be populated with three different direct current measurement modules or with a SUBD-9 module to connect almost any kind of current transducer. Alternatively, this connector can also be used to measure any auxiliary ±10 V signal (e.g. such as windspeed or water flow)

Connection examples

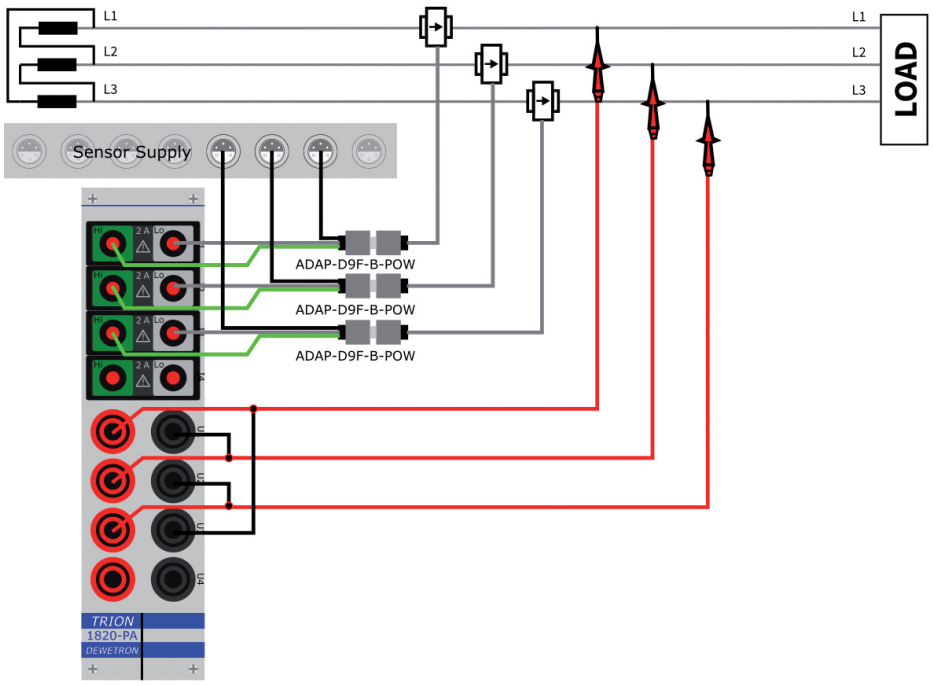
Three phase (3P3W) without neutral line



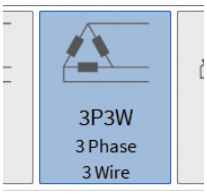
Group selection in OXYGEN Power:



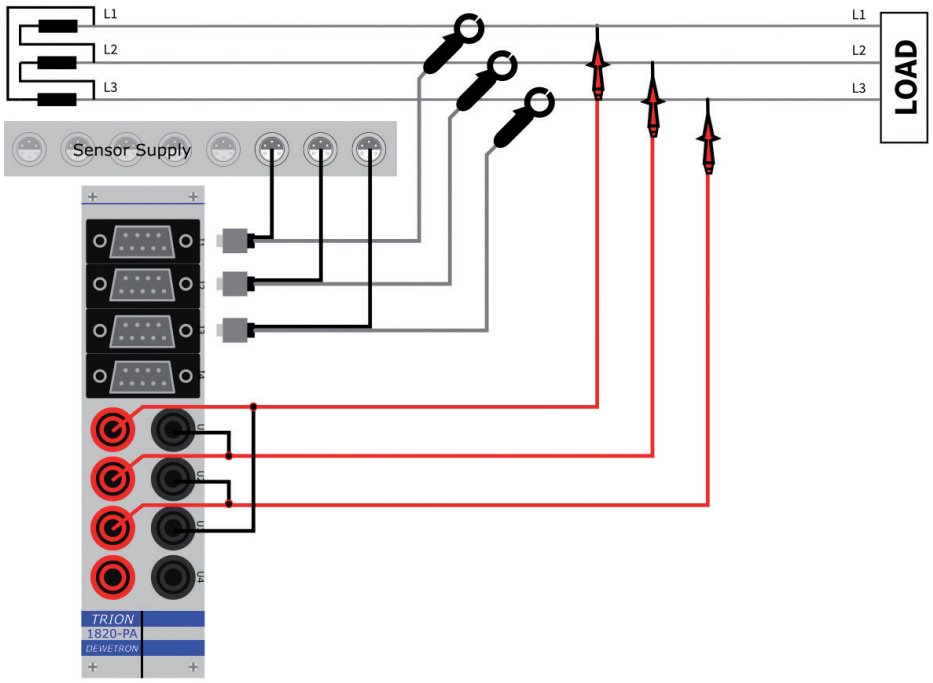
Three phase (3P3W) without neutral line, using current output transducer



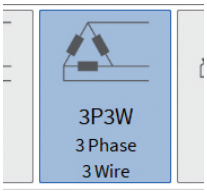
Group selection in OXYGEN Power:



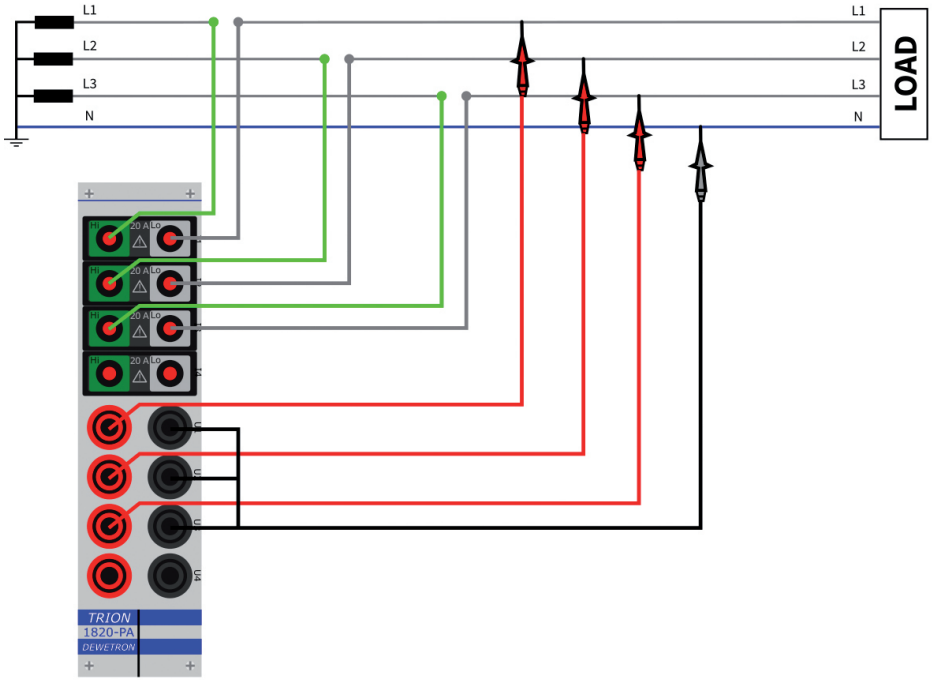
Three phase (3P3W) without neutral line, using voltage output transducers



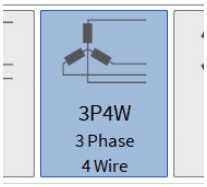
Group selection in OXYGEN Power:



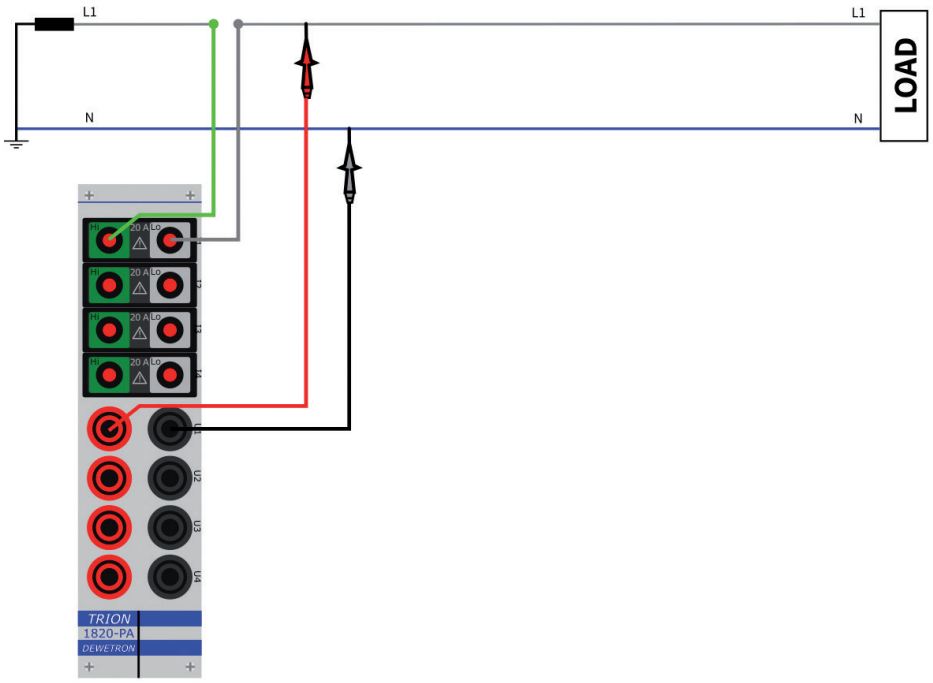
Three phase (3P4W) with neutral line



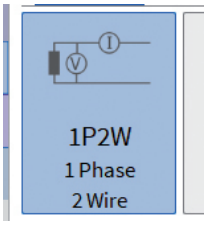
Group selection in OXYGEN Power:



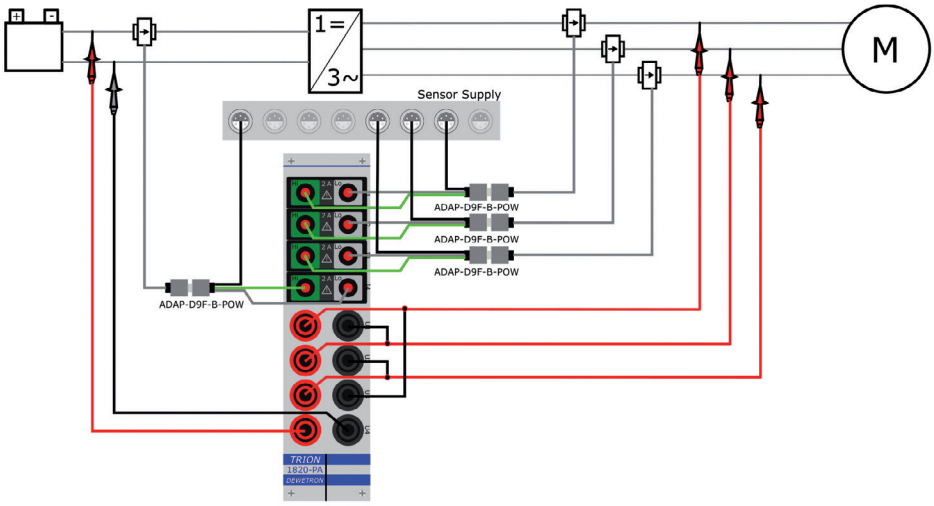
One phase (1P2W)



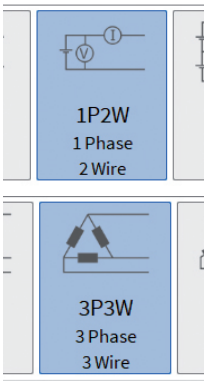
Group selection in OXYGEN Power:



Three phase and one phase (3P3W and 1P2W)



Group selection in OXYGEN Power:



Bessel/Butterworth filter characteristics for Power Analysis

The TRION family is equipped with DSP lowpass filters from 2nd to 8th order in Bessel or Butterworth configuration. The difference between these two filter types can be seen in the following figures.

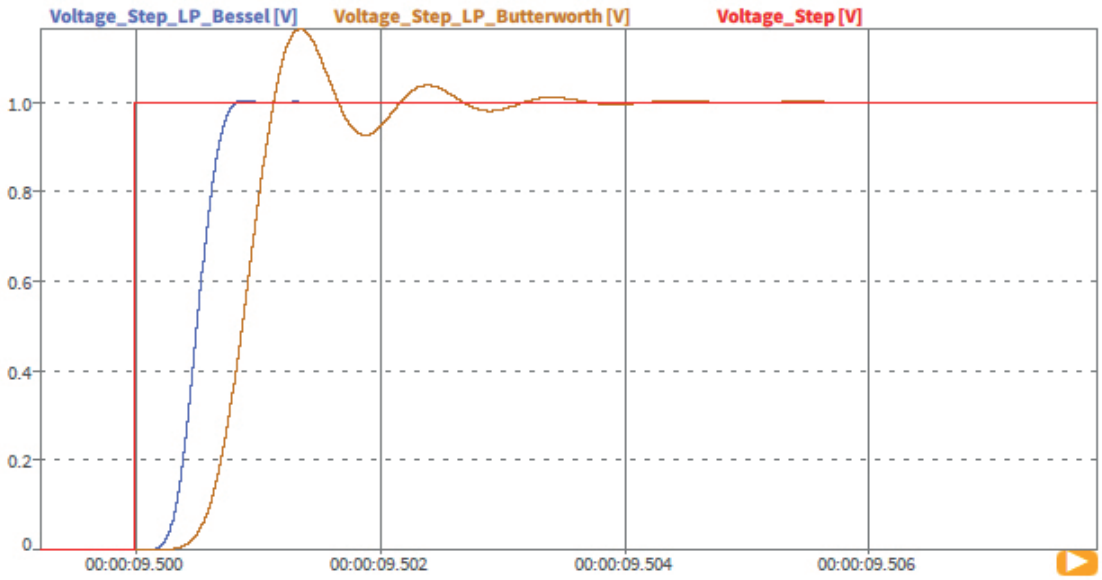


Figure 1: Step response of filter with 1000 Hz cutoff frequency and 8th order.

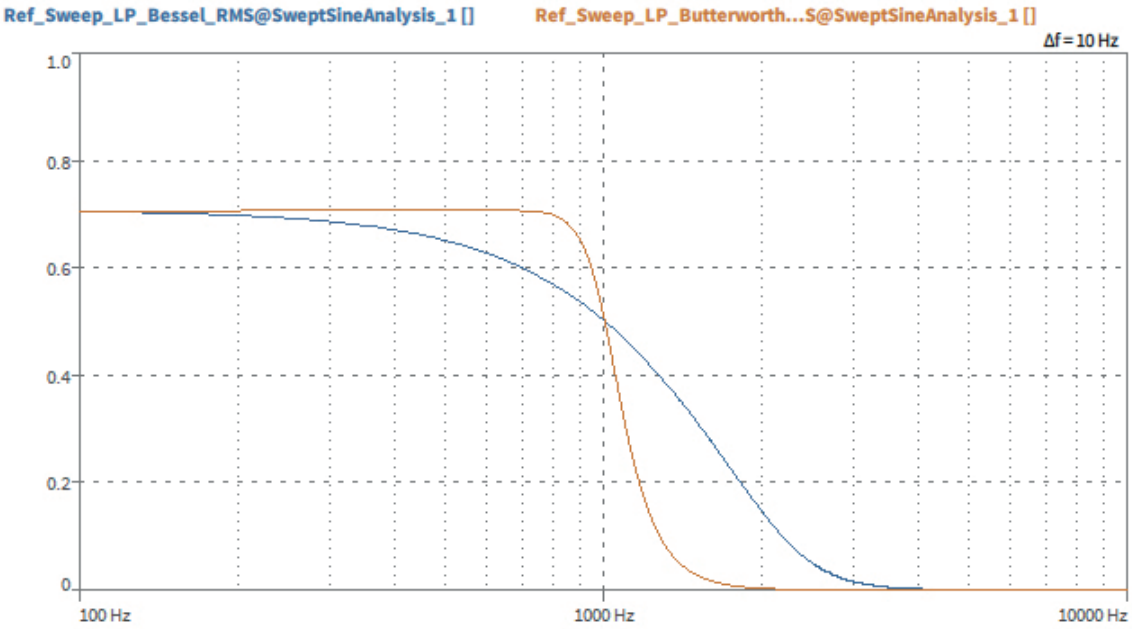
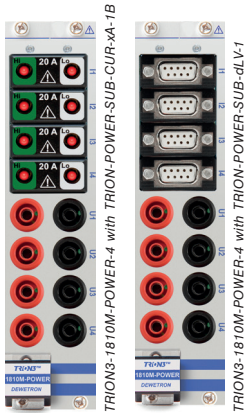


Figure 2: Frequency response of filter with 1000 Hz cutoff frequency and 8th order

	Bessel	Butterworth
Magnitude accuracy e.g. RMS-accuracy	😐	🙂
Step response reproduction e.g. PWM signal observation	🙂	😐

TRION3™ module for 4-phase power analysis

- Channels: 4 power channels
- Sampling: up to 10 MS/s
- Resolution: 18-bit
- Input types: Voltage: 1000 V
  Current: 20 A, 1 A, 0.2 A, 5 V



Module specifications

TRION3-1810M-POWER-4		
Input channels	4 Power channels, each with one voltage and one current input	
Resolution	18-bit	
Sample Rate	up to 10 MS/sec; 512 MByte onboard data buffer	
The following are all accuracy conditions referenced in this section: Temperature: 23±5°C, Humidity: 40 to 60 % RH, Input waveform: sine wave, Common mode voltage: 0 V, Line filter: Auto (8th order Butterworth), Sample rate: 10 MS/s, Resolution: 24 bit, Driver Version >= 3.7, Power factor: 1, After warm-up. After zero level, 12-month Accuracy ±(Reading error + Range error), Accuracy: Frequency (f) in [kHz]		
Voltage input U1, U2, U3, U4		
	Input range	1000 V (±2000 V_{PEAK}) CF = 2
	Accuracy[1) 2)]	DC: ±0.02 % of reading ±0.02 % of range 0.5 Hz to 1 kHz: ±0.03 % of reading 1 kHz to 5 kHz: ±0.15 % of reading 5 kHz to 10 kHz: ±0.35 % of reading 10 kHz to 50 kHz: ±0.6 % of reading 50 kHz to 300 kHz: ±(0.02 % * f) of reading
	Gain drift	20 ppm / °C
	Offset drift	5 mV / °C
	Typical THD	-95 dB
	CMRR	>85 dB @ 50 Hz; >60 dB @ 1 kHz; >40 dB @ 100 kHz
	Bandwidth	5 MHz
	Safety	CAT IV 600 V / CAT III 1000 V
	Instantaneous maximum allowable input	4000 V_{PEAK} or 3000 V_{RMS} (1s)
	Continuous maximum allowable input	2000 V_{RMS} or 2000 V_{DC}
	Input resistance	5 MΩ; 2 pF
	Isolation (earth) resistance	100 GΩ; 2.5 pF
	Connector	Safety banana sockets
[1)] add 0.02 % of reading with filter settings OFF [2)] below 1 % of range, add 10 ppm of range		
Current input I1, I2, I3, I4		
20 A module		TRION-POWER-SUB-CUR-20A-1B
	Range	20 A (±40 A_{PEAK})
	Accuracy[1) 2)]	DC: ±0.02 % of reading ±0.02 % of range[3)] 0.5 Hz to 1 kHz: ±0.03 % of reading 1 kHz to 5 kHz: ±0.15 % of reading 5 kHz to 10 kHz: ±0.35 % of reading 10 kHz to 50 kHz: ±(0.3 % + 0.05 % * f) of reading 50 kHz to 300 kHz: ±(0.10 % * f) of reading
	Safety	CAT II 600 V, unfused
	Bandwidth	300 kHz
	Connector	Safety banana sockets (male)
	Instantaneous maximum allowable input	50 A_{PEAK} or 40 A_{RMS} (1s)
	Continuous maximum allowable input	20 A_{RMS}
	Input resistance	2 mΩ
[1)] For self-generated heat caused by current input, add 0.00008 × I² % of reading + 15 × I² µA to the current accuracy. 'I' is the current reading [A]. The influence from self-generated heat continues until the temperature of the shunt resistor inside the DEWE3-Chassis lowers even if the current input changes to a small value. [2)] below 1 % of range, add 50 ppm of range [3)] add 0.03 % of range with no zero level		

continued on next page ...

TRION3-1810M-POWER-4

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2 A module	TRION-POWER-SUB-CUR-2A-1B	
	Range	2 A (±4 A_{PEAK})
	Accuracy[1]	DC: ±0.02 % of reading ±0.02 % of range[2]
		0.5 Hz to 10 kHz: ±0.03 % of reading
		10 kHz to 30 kHz: ±0.1 % of reading
		30 kHz to 200 kHz: ±(0.015 % * f) of reading
		200 kHz to 300 kHz: ±(0.1 % * f) of reading
	Safety	CAT II 600 V, unfused
	Bandwidth	300 kHz
	Connector	Safety banana sockets (male)
	Instantaneous maximum allowable input	10 A_{PEAK} or 5 A_{RMS} (1s)
	Continuous maximum allowable input	3 A_{RMS}
	Input resistance	50 mΩ

EOL product
This module is replaced by the 1A module below!

[1] below 1 % of range, add 25 ppm of range
[2] add 0.03 % of range with no zero level

1 A module	TRION-POWER-SUB-CUR-1A-1B	
	Range	1 A (±2 A_{PEAK})
	Accuracy[1]	DC: ±0.02 % of reading ±80 µA[2]
		0.5 Hz to 10 kHz: ±0.03 % of reading
		10 kHz to 30 kHz: ±0.1 % of reading
		30 kHz to 200 kHz: ±(0.015 % * f) of reading
		200 kHz to 300 kHz: ±(0.1 % * f) of reading
	Safety	CAT II 600 V, unfused
	Bandwidth	300 kHz
	Connector	Safety banana sockets (male)
	Instantaneous maximum allowable input	4 A_{PEAK} or 2 A_{RMS} (1s)
	Continuous maximum allowable input	1.1 A_{RMS}
	Input resistance	500 mΩ

[1] below 1 % of range, add 25 ppm of range
[2] add 0.03 % of range with no zero level

0.2 A module	TRION-POWER-SUB-CUR-02A-1B	
	Range	0.2 A (±0.4 A_{peak})
	Accuracy[1]	DC: ±0.02 % of reading ±0.02 % of range[2]
		0.5 Hz to 5 kHz: ±0.03 % of reading
		10 kHz to 30 kHz: ±0.1 % of reading
		30 kHz to 200 kHz: ±(0.015 % * f) of reading
		200 kHz to 300 kHz: ±(0.1 % * f) of reading
	Safety	CAT II 600 V, unfused
	Bandwidth	300 kHz
	Connector	Safety banana sockets (male)
	Instantaneous maximum allowable input	2 A_{PEAK} or 1 A_{RMS} (1s)
	Continuous maximum allowable input	0.4 A_{RMS}
	Input resistance	500 mΩ

[1] below 1 % of range, add 25 ppm of range
[2] add 0.03 % of range with no zero level

Clamp input module	TRION-POWER-SUB-dLV-1	
	Range	5 V (±10 V_{peak}) **NOT ISOLATED** ⚠
	Accuracy[1]	DC: ±0.02 % of reading ±0.02 % of range
		0.5 Hz to 5 kHz: ±0.03 % of reading
		5 kHz to 30 kHz: ±(0.01 % * f) of reading
		30 kHz to 50 kHz: ±(0.02 % * f) of reading
		50 kHz to 100 kHz: ±(0.1 % * f) of reading
	Typical THD	-95 dB
	Typical CMRR	80 dB @ 50 Hz; 80 dB @ 1 kHz; 65 dB @ 10 kHz; 45 dB @ 100 kHz
	Safety	Depending on connected clamp
	Absolute maximum voltage	±30 V
	Bandwidth	100 kHz
	Connector	DSUB9
	Input resistance	1 MΩ
	Sensor supply (+9 V)	max. 40 mA

[1] below 1 % of range, add 10 ppm of range

continued on next page ...

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Power specifications

Accuracy[1] with PF=1	DC:	±0.03 % of reading ±0.03 % of range[2]
	0.5 Hz to 1 kHz:	±0.04 % of reading
	1 kHz to 5 kHz:	±0.2 % of reading
	5 kHz to 10 kHz:	±0.5 % of reading
	10 kHz to 50 kHz:	±(0.5 % + 0.05 % * f) of reading
Influence of power factor	Add 0.01 % × f/50 × √(1/PF²-1)	

[1] voltage and current channel have a minimum input of 1 % of range, otherwise individual uncertainty have to be calculated
[2] add 0.03 % of range with no zero level

Typical Signal-to-noise ratio, Spurious free SNR, Effective number of Bits[1]

	Voltage input 2000 V			20 A module			2 A module			1 A module			0.2 A module			Clamp input module 10 V		
	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]	SNR	SFDR[2]	ENOB[3]
Sample rate	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]	[dB]	[dB]	[Bit]
0.1 kS/s	129	142	21.1	105	137	17.2	110	128	18.1	134	149	22.0	106	124	17.3	129	146	21.1
1 kS/s	126	139	20.6	100	130	16.3	94	118	15.4	128	149	21.0	90	115	14.7	119	131	19.5
10 kS/s	121	136	19.8	99	127	16.1	93	117	15.1	119	144	19.5	89	112	14.4	109	124	17.8
100 kS/s	113	135	18.5	94	123	15.4	93	116	15.1	109	137	17.8	88	112	14.3	99	119	16.2
1000 kS/s	103	128	16.9	88	112	14.3	92	116	15.0	99	134	16.2	86	109	13.9	94	115	15.3
2000 kS/s	85	106	13.8	87	110	14.2	92	116	15.0	98	130	16.0	86	109	13.9	92	114	15.0

[1] LP Filter in auto mode
[2] SFDR excluding harmonics
[3] ENOB calculated from SNR

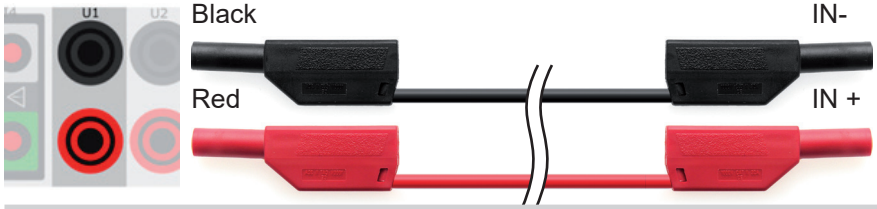
Additional specifications

Typical channel to channel phase mismatch (Voltage-Voltage, Current-Current, Voltage-Current)	<250 ns (0.1° @ 1 kHz, 0.005° @ 50 Hz)
Typical board-to-board phase mismatch	
same board type	<250 ns (0.1° @ 1 kHz, 0.005° @ 50 Hz)
different board type	±1 Sample or 0.2° @ 1 kHz (whichever is higher)
Low pass filter (-3 dB, digital and analog combined)	100 Hz to 1 MHz freely programmable or OFF
Filter order & characteristics	2nd, 4th, 6th, 8th Bessel or Butterworth
Filter delay compensation	up to 15 µs the group delay of the selected filter will be automatically compensated. This works for: 2nd order filter 15 kHz to 1 MHz 4th order filter 30 kHz to 1 MHz 6th order filter 60 kHz to 1 MHz

Signal connection

Voltage input

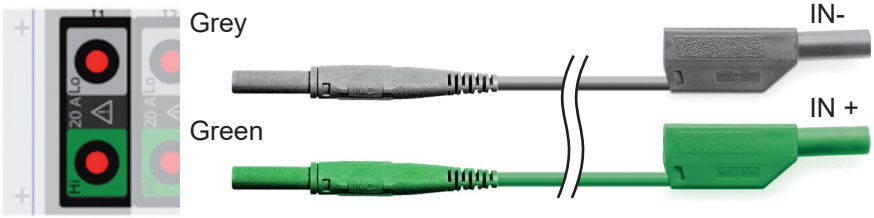
High voltage input for line voltage measurement.



Voltage measurement up to ±2000 V only with safety banana plug cords!

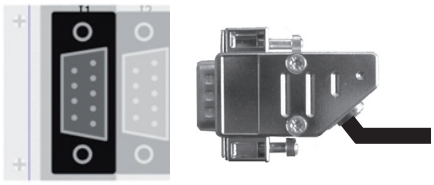
Current input module (TRION-POWER-SUB-CUR-xA-1B)

Direct current input for measuring current directly. This input is isolated and rated with CAT II 600 V. Modules with 20 A, 2 A and 0.2 A nominal current are available.



Clamp input module (TRION-POWER-SUB-dLV-1, for clamps ONLY)

NOT ISOLATED voltage input. This input can be used for isolated current clamps with voltage output.

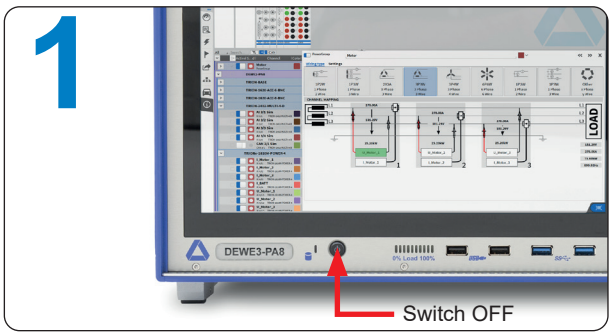


Pin 1: TEDS
Pin 2: IN+
Pin 3: n.c.
Pin 4: GND (not isolated)
Pin 5: +9 V (40 mA max.)
Pin 6: n.c.
Pin 7: IN-
Pin 8: n.c.
Pin 9: -9 V (40 mA max.)

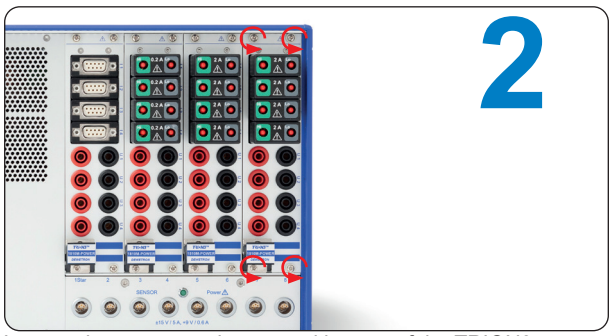
WARNING: This module is not isolated! Do not connect any other appliances than isolated current transducers with voltage output.

Exchanging current modules (SUB-modules)

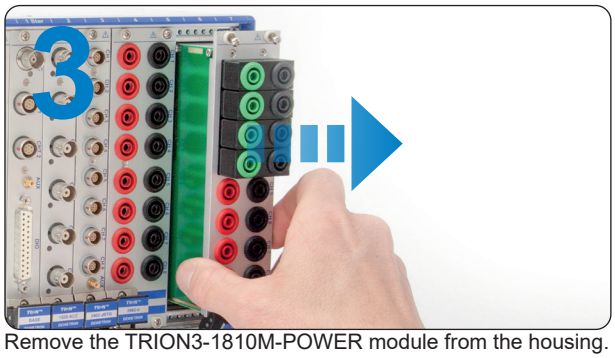
Proper ESD precautions must be taken to avoid any damage to the unit.



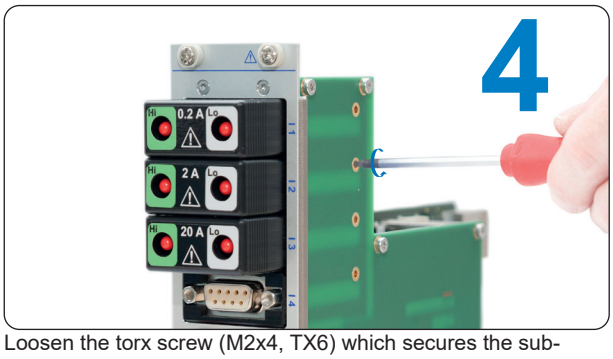
Switch off the instrument and unplug all connected cables including sensors from the TRION™ series modules.



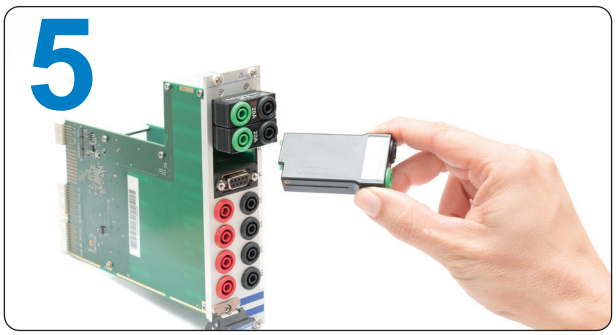
Loosen the screws at the top and bottom of the TRION3-1810M-POWER-4 module front panel (4x) and pull down the injector/ejector handle to release the module.



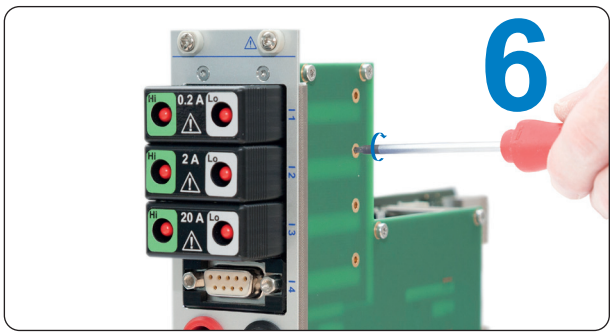
Remove the TRION3-1810M-POWER module from the housing.



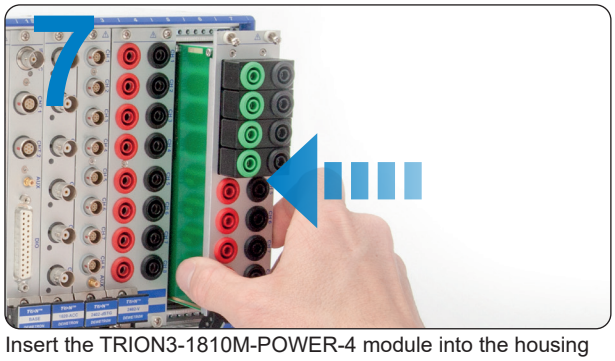
Loosen the torx screw (M2x4, TX6) which secures the sub-module of the channel you want to replace.



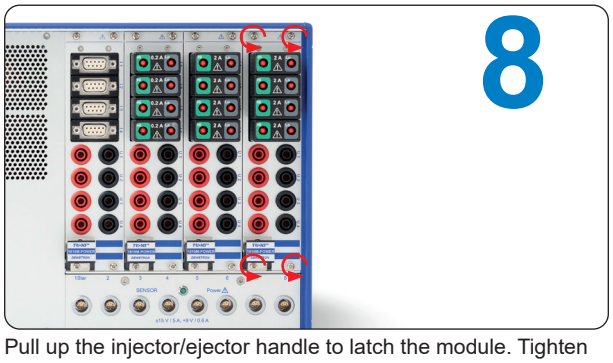
Replace the sub-module.



Secure the replaced sub-module with the torx screw (M2x4, TX6). Max. torque: 0.2 Nm



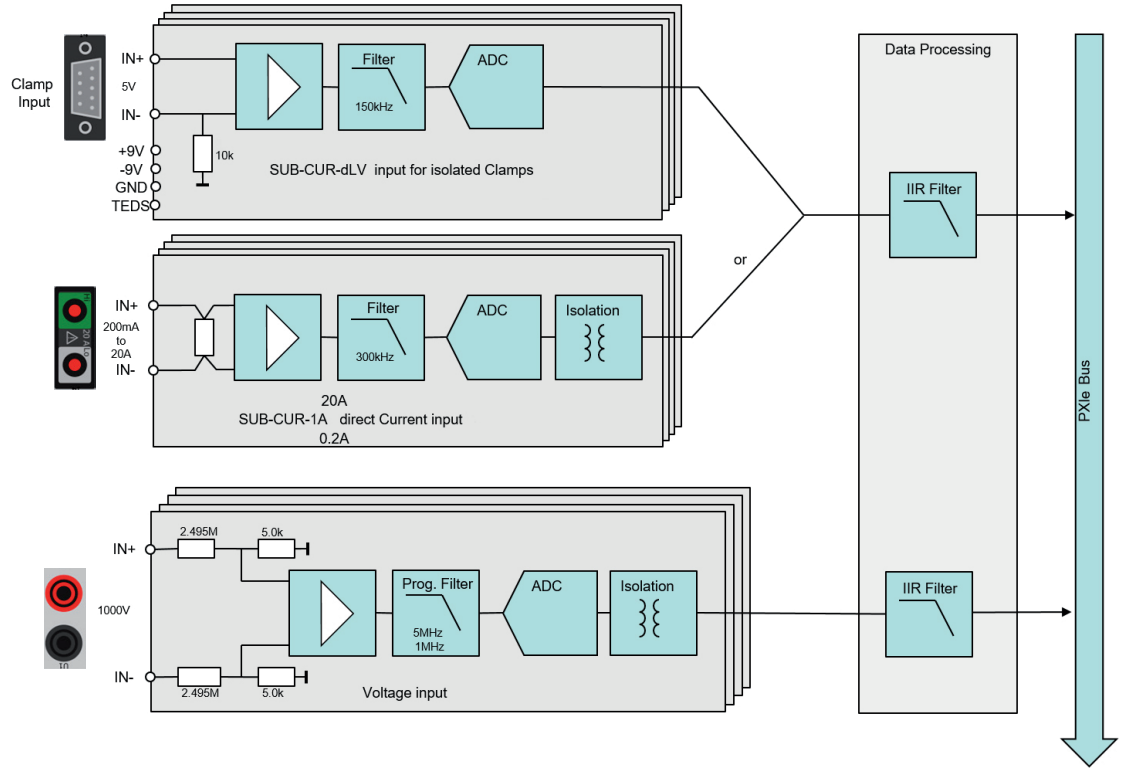
Insert the TRION3-1810M-POWER-4 module into the housing until a resistance appears.



Pull up the injector/ejector handle to latch the module. Tighten the screws at the top and bottom of the TRION3-1810-POWER-4 module front panel (4x) to secure the module.

Block diagram

Base block diagram of the TRION3-1810M-POWER-4 module:



The TRION3-1810M-POWER-4 is an ultra-compact, 4-phase power data acquisition module with flexible current inputs for use in any DEWE3-chassis.

Voltage input:

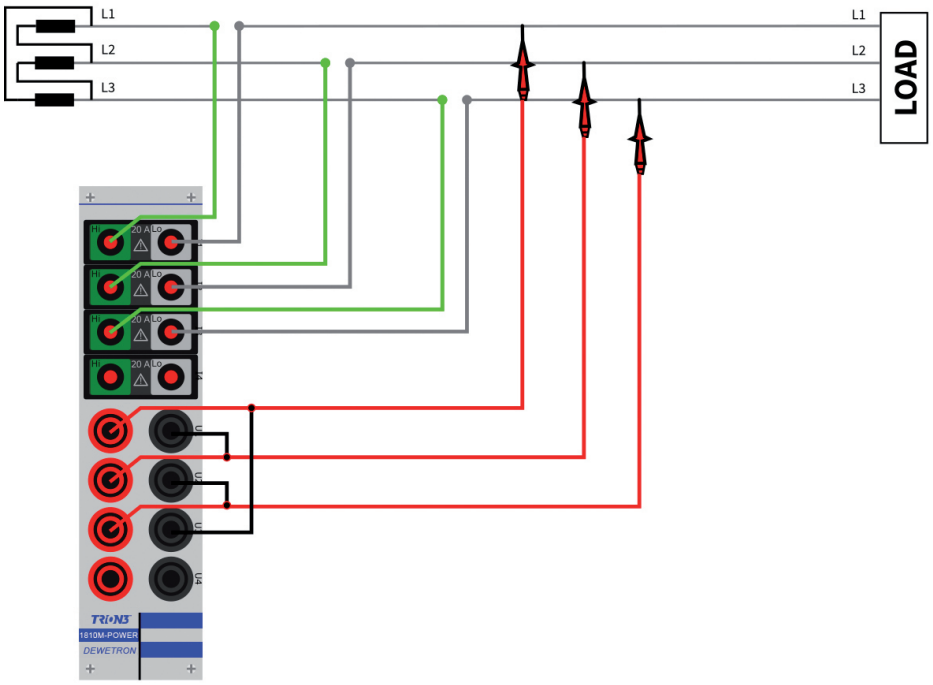
Fast sampling, high bandwidth and minimum capacity to earth are just a few outstanding performance qualities of the high voltage inputs. The high input impedance allowes high continous voltage levels with a very low temperature drift. Although small outline, the safety category is on a very high level (CAT III 1000V).

Current input:

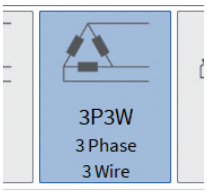
The TRION3-1810M-POWER-4 has 4 highly flexible current inputs. The 4 slots can be populated with three different direct current measurement modules or with a SUBD-9 module to connect almost any kind of current transducer. Alternatively, this connector can also be used to measure any auxiliary ±10 V signal (e.g. such as windspeed or water flow)

Connection examples

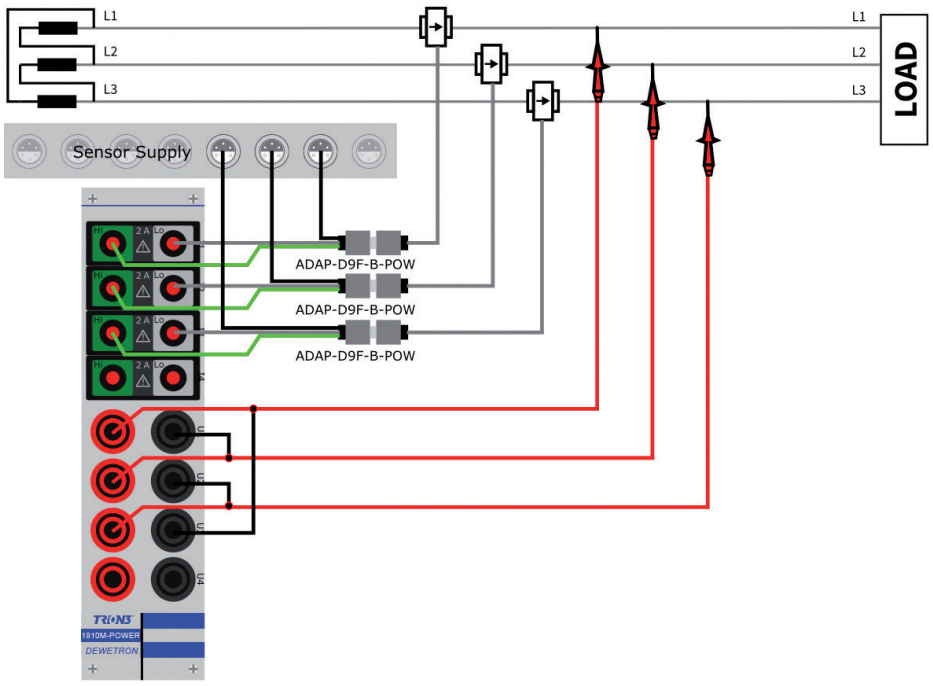
Three phase (3P3W) without neutral line



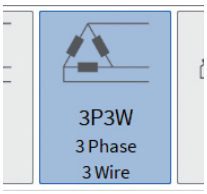
Group selection in OXYGEN Power:



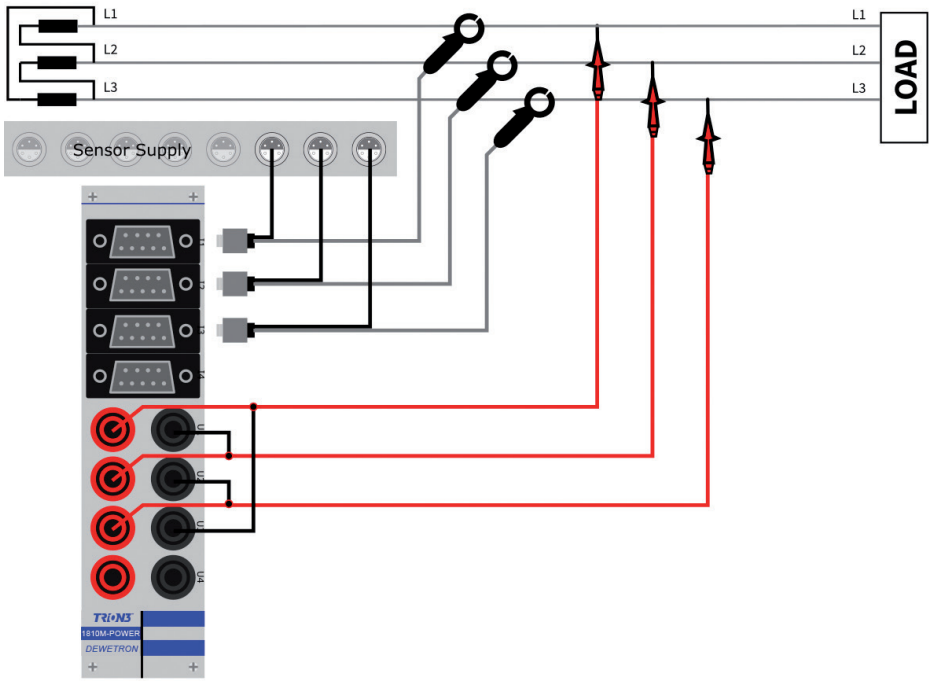
Three phase (3P3W) without neutral line, using current output transducer



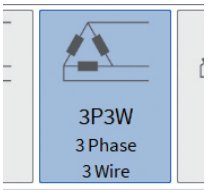
Group selection in OXYGEN Power:



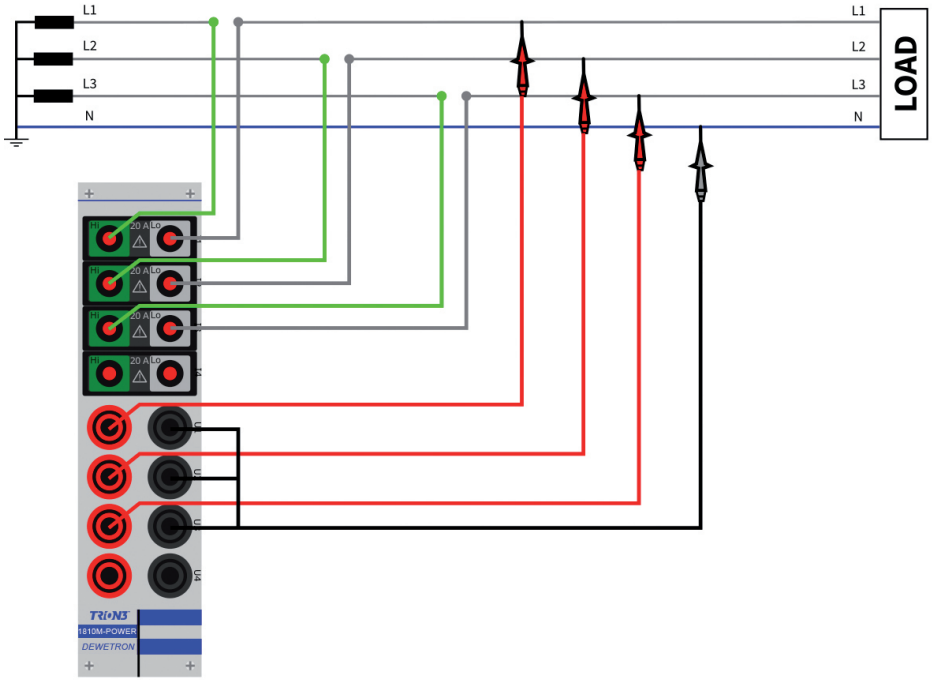
Three phase (3P3W) without neutral line, using voltage output transducers



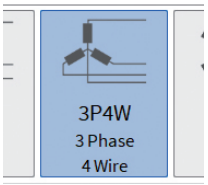
Group selection in OXYGEN Power:



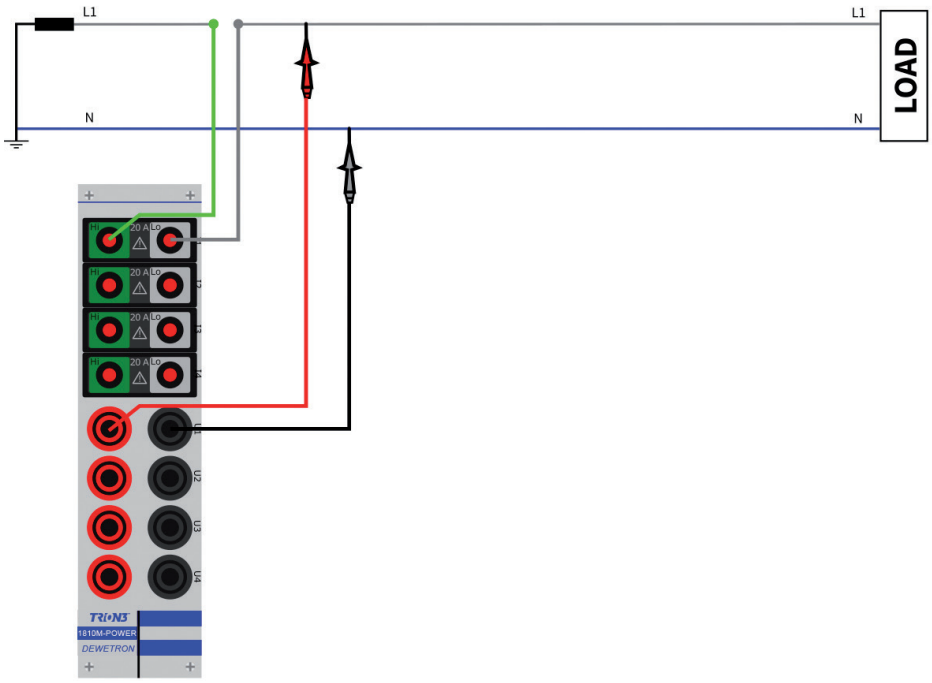
Three phase (3P4W) with neutral line



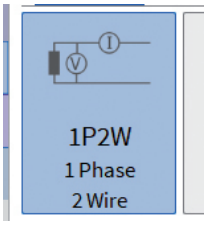
Group selection in OXYGEN Power:



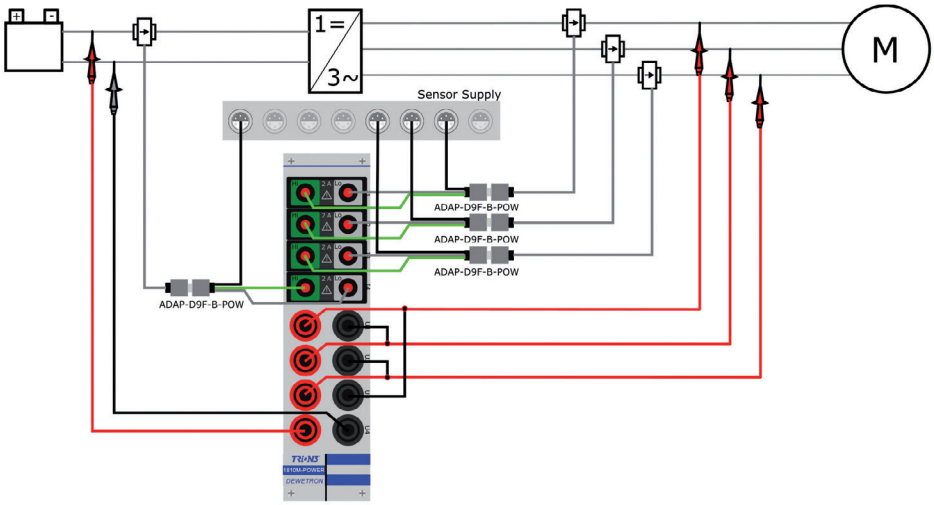
One phase (1P2W)



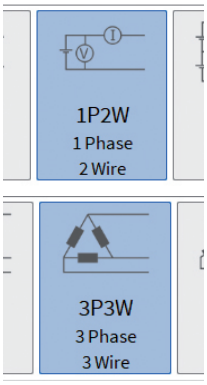
Group selection in OXYGEN Power:



Three phase and one phase (3P3W and 1P2W)



Group selection in OXYGEN Power:



Bessel/Butterworth filter characteristics for Power Analysis

The TRION family is equipped with DSP lowpass filters from 2nd to 8th order in Bessel or Butterworth configuration. The difference between these two filter types can be seen in the following figures.

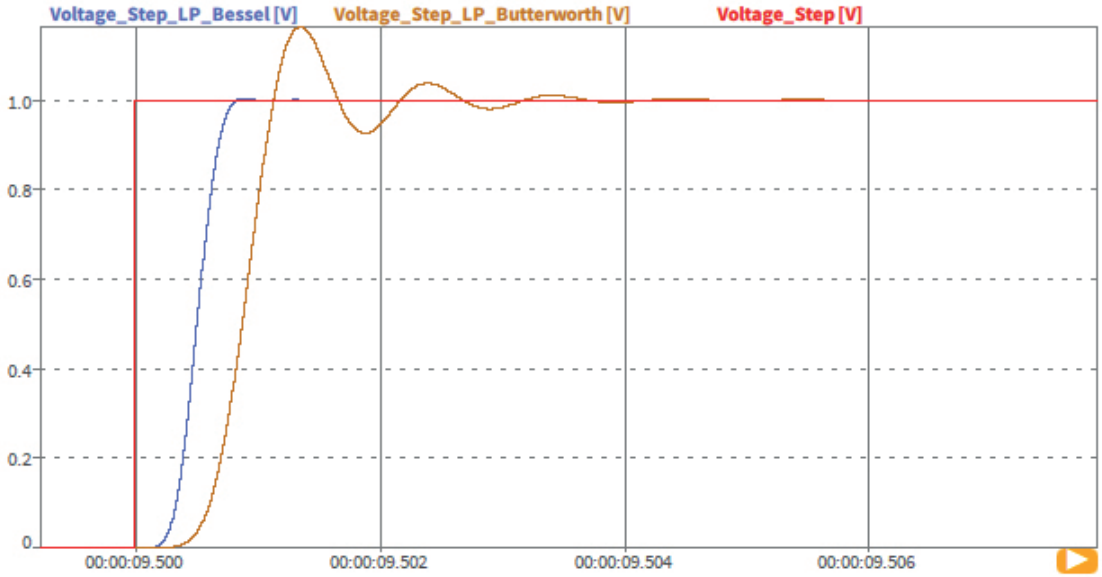


Figure 1: Step response of filter with 1000 Hz cutoff frequency and 8th order.

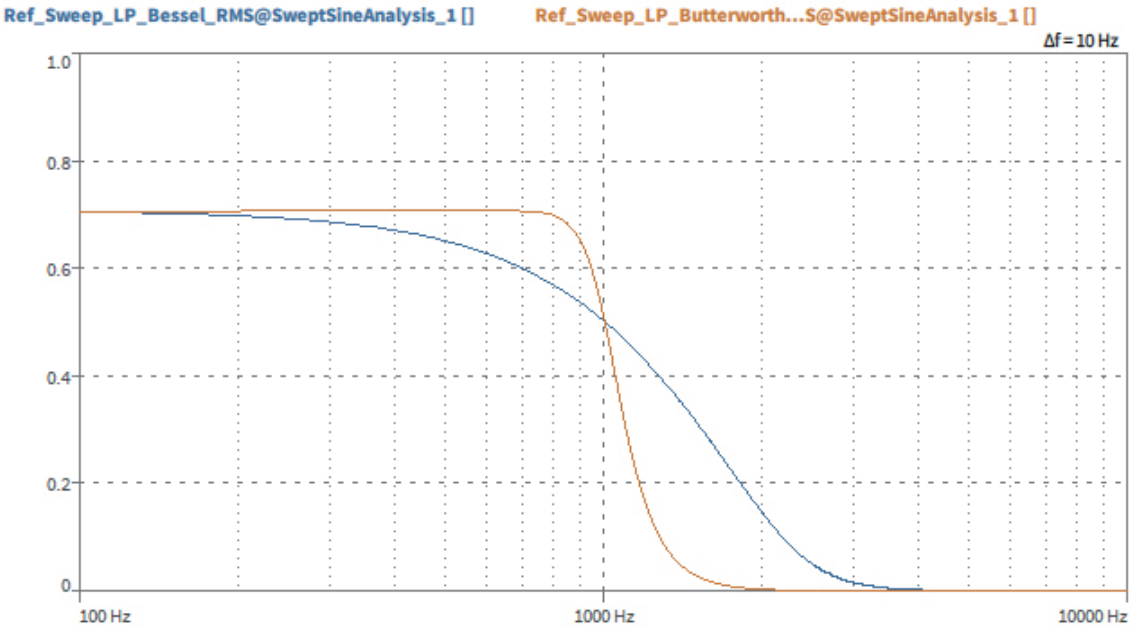


Figure 2: Frequency response of filter with 1000 Hz cutoff frequency and 8th order

	Bessel	Butterworth
Magnitude accuracy e.g. RMS-accuracy	😐	☺
Step response reproduction e.g. PWM signal observation	☺	😐

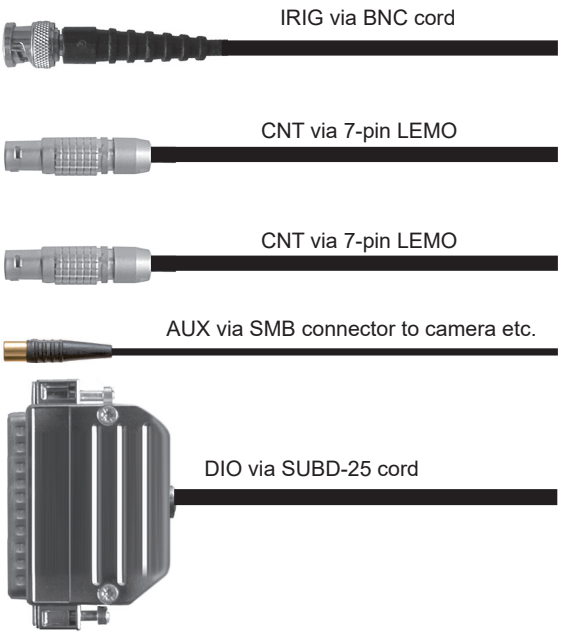
Counter, DIO and synchronization module

- *Counter channels: 2 advanced counter*
- *Digital I/O: 8 DIO and 8 DI*
- *Synchronization: IRIG code B, DC I/O*
- *Additional features: 1 AUX socket (by default set to camera trigger)*

TRION-BASE specifications	
Digital I/O specifications	
Number of channels	8 DIO + 8 DI
Compatibility (Input)	CMOS/TTL; 100 kΩ pull up
Compatibility (Output)	TTL, 20 mA
Overvoltage protection	±30 V permanent, 50 V peak (for 100 msec)
Sensor power supply (module total)	5 V (600 mA) and 12 V (600 mA)
Connector	SUB-D-25 socket
SUBD-25 Sync OUT specifications	
Functionality	Acquistion Clock and Trigger output (can be used to sync two systems/enclosures)
Compatibility (Input)	LVTTL
Compatibility (Output)	LVTTL, 10 mA
Overvoltage protection	±20 V
Counter specifications	
Number of channels	2 advanced counter or 6 digital inputs
Counter modes	
Event counting	Basic event counting, gated counting, up/down counting and encoder mode (X1, X2 and X4)
Waveform timing	Period, frequency, pulse width duty cycle and edge separation
Sensor modes	Encoder (angle and linear)
Compatibility (Input)	CMOS/TTL
Counter resolution	32-bit
Counter time base	80 MHz
Time base accuracy	typ. 10 ppm (defined by the back plane)
Maximum input frequency	10 MHz
Overvoltage protection	±30 V permanent, 50 V peak (100 msec)
Sensor power supply (module total)	5 V (600 mA) and 12 V (600 mA)
Connector	LEMO 1B.307
AUX specifications	
Functionality	Camera trigger, trigger input/output, acquisition clock and programmable clock output
Compatibility (Input)	LVTTL
Compatibility (Output)	LVTTL, 10 mA
Overvoltage protection	±20 V
Connector	SMB socket
Timing specifications	
Input sources	IRIG code B, DC
Output signals	IRIG code B, DC
Input specification	
Compatibility (DC code)	DC Level Shift TTL / CMOS Compatible
Impedance	20 kΩ
Output specification	
Compatibility (DC code)	TTL, 20 mA
Adjustment range	±150 ppm
Clock acc. IRIG locked	without drift
Clock acc. IRIG unlocked	< 1 ppm
Isolation voltage	350 V_{DC}
Connector	BNC socket
General specifications	
Typical power consumption	5 W
Temperature Range	0 - 50 °C
Weight	Appr. 240 g

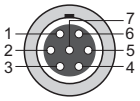
Signal connection

TRION-BASE module



Hot: Signal
Shield: GND Signal (isolated)

1: Input_A
2: Input_B
3: Input_Z
4: GND Power
5: +5 V out
6: +12 V out
7: GND Signal



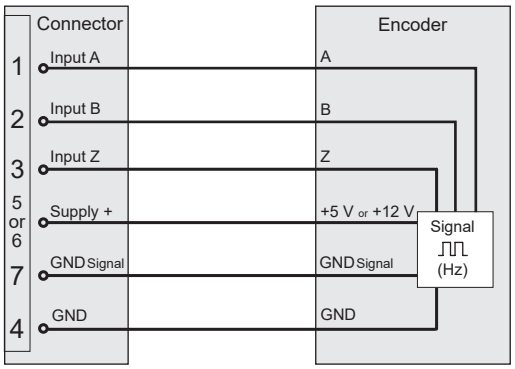
7-pin LEMO connector female
EGG.1B.307.CLL

Hot: OUT +
Shield: OUT -

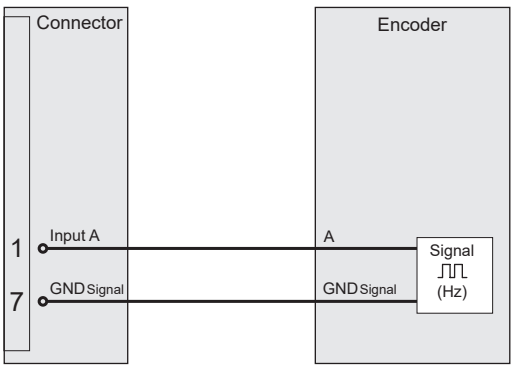
1	DIO 1	14	DI 9
2	DIO 2	15	DI 10
3	DIO 3	16	DI 11
4	DIO 4	17	DI 12
5	DIO 5	18	DI 13
6	DIO 6	19	DI 14
7	DIO 7	20	DI 15
8	DIO 8	21	DI 16
9	DGND	22	DGND
10	DGND	23	DGND
11	Trigger out	24	+ 5V
12	Clock out	25	+12V
13	Sync (reserved)		

Encoder connection

Supplied sensor connection



Non supplied sensor connection



Caution: Since the counter input is isolated from the power supply GND Power and GND Signal must be tied together!

LED function

The IRIG connector has an indication LED flashing either green or red:

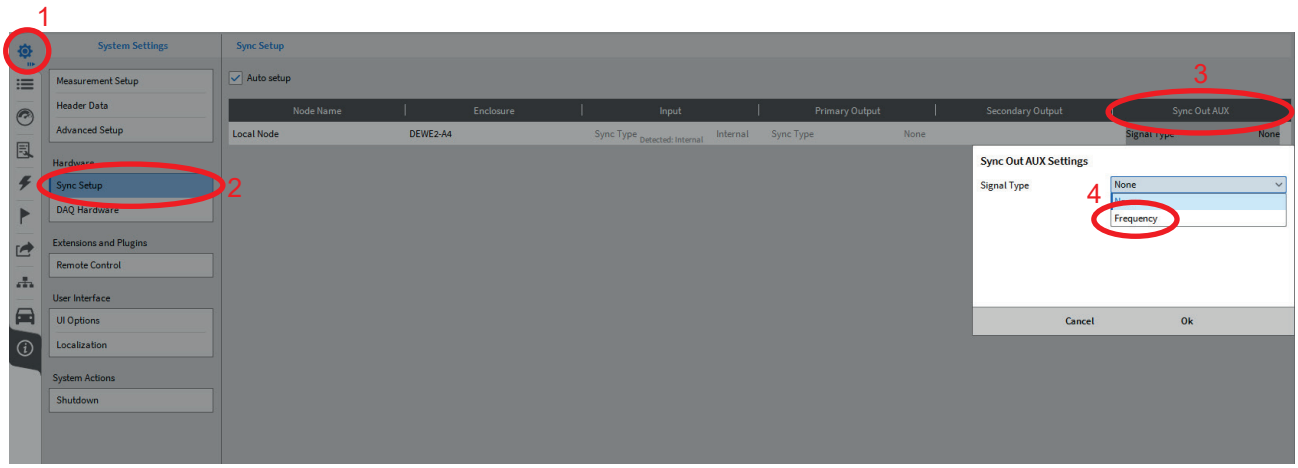
LED indication:

	OFF	ON	Description
GREEN (flashing)	20 %	80 %	SYNC IN not available
RED (flashing)	80 %	20 %	SYNC detected, not locked
GREEN (flashing)	80 %	20 %	SYNC detected and locked

AUX terminal

The auxiliary terminal of the TRION-BASE module could be used as programmable frequency output for synchronizing external hardware.

The output can be set in the Sync Out Aux settings via 'System Settings' -> 'Sync Setup' -> 'Sync Out Aux':



Synchronization options with a TRION-BASE module

To create high channel count systems or for distributed measurements, DEWE2 instruments support multiple synchronization options. The kind of synchronization is depending on how far apart the instruments are from each other.

Synchronization via TRION-SYNC-BUS (CAT IV ethernet cable) is limited to ~100 meters.
Synchronization via IRIG time-code is limited to ~1000 meters.
Synchronization via GPS is not limited.

NOTE: *Further information regarding different synchronization options please refer to chapter "Synchronization of DEWE2" in your DEWE2 Technical reference manual.*

NOTE: *If the system is equipped with a TRION-BASE, TRION-TIMING or TRION-VGPS-20/-100 module, it must be installed in the “star slot”! This is the only slot a module is able to override the system 10MHz clock with its PPS-synced 10MHz, and thus providing the system with a timebase of higher accuracy.*

Advanced counter

The TRION-BASE module supports 2 advanced counter or 6 digital inputs as shown in the module specifications. Further information regarding advanced counters are described in chapter '*Functional description of advanced counters*' of the TRION-CNT module.

Notes

System timing and synchronization module

- PTP / IEEE 1588
- GPS, GLONASS, GALILEO, BeiDou
- IRIG
- PPS
- 8x DIO, 1x counter, 1x AUX

Module specifications

TRION-TIMING-V3 specifications		
Synchronization input modes		PTP / IEEE 1588, GPS, IRIG, PPS (pulse per second)
Features		1 Programable frequency output (10 to 1 000 000Hz) 1 Advanced counter input 8 Digital I/O
PTP / IEEE 1588		
IP Mode		Multicast
Protocoll		UDP / IPv4; ETH
Delay Mechanism		End to End; Peer to Peer
IP-Address Method		DHCP
RJ-45 Ethernet		10 / 100 Mbit Ethernet connection; only for synchronization, no data transfer possible.
Programmable Correction Limit		10 ns to 500 ms
GPS specifications		
Supported GNSS signals		GPS / SBAS L1, GLONASS, GALILEO, BeiDou
Number of channels		35
PPS accuracy		100 ns
Refreshrate		1 Hz, 5 Hz, 10 Hz
Position accuracy (Horizontal CEP)		
Autonomous		1.5 m
Differential		1.0 m
Input Connector GPS		SMA for GPS antenna
IRIG input specifications		
Supported codes:		IRIG code A or B; AM or DC
Compatibility (AM code):		0.5 Vp-p to 10 Vp-p
Ratio (AM):		3:1 ±10 %
Compatibility (DC code):		DC Level Shift TTL / CMOS compatible
Impedance:		20 kΩ
Digital I/O specifications		
Number of channels		8
Compatibility (input)		CMOS/TTL
Compatibility (output)		TTL, 20 mA
Overvoltage protection		±30 V
Connector		SUB-D-15 socket
Counter specifications		
Number of channels		1 advanced counter or 3 digital inputs
Counter modes	Event counting	Basic event counting, gated counting, up/down counting and encoder mode (X1, X2 and X4)
	Waveform timing	Period, frequency, pulse width, duty cycle and edge separation
	Sensor modes	Encoder (angle and linear), gear tooth with/without zero, gear tooth with missing/double teeth
Input signal compatibility		CMOS/TTL
Counter resolution		32-bit
Counter time base		80 MHz
Maximum input frequency		10 MHz
Overvoltage protection		±30 V permanent, 50 V peak (for 100 msec)
Sensor power supply		5 V (600 mA) and 12 V (600 mA)
Connector		On same SUB-D-15 socket as Digital I/O

continued on next page ...

TRION-TIMING-V3

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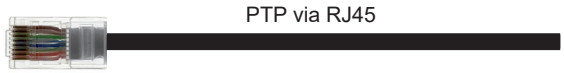
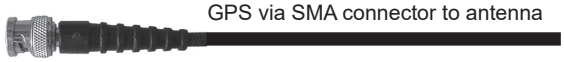
AUX specifications	
Functionality	Camera trigger, trigger input/output, acquisition clock and programmable clock output
Compatibility (input)	LVTTL
Compatibility (output)	LVTTL, 10 mA
Overvoltage protection	±20 V
Connector	SMB socket
General specifications	
Typical power consumption	5 W
Temperature Range	0 - 50 °C
Weight	Appr. 240 g

Signal connection

TRION-TIMING-V3 module



Hot: IN +
Shield: IN -



Hot: IN +
Shield: IN -

Hot: OUT +
Shield: OUT -

1	DIO 1	9	CNT_Source
2	DIO 2	10	CNT_AUX
3	DIO 3	11	CNT_Gate
4	DIO 4	12	DGND
5	DIO 5	13	DGND
6	DIO 6	14	+5 V
7	DIO 7	15	+12 V
8	DIO 8		

LED function

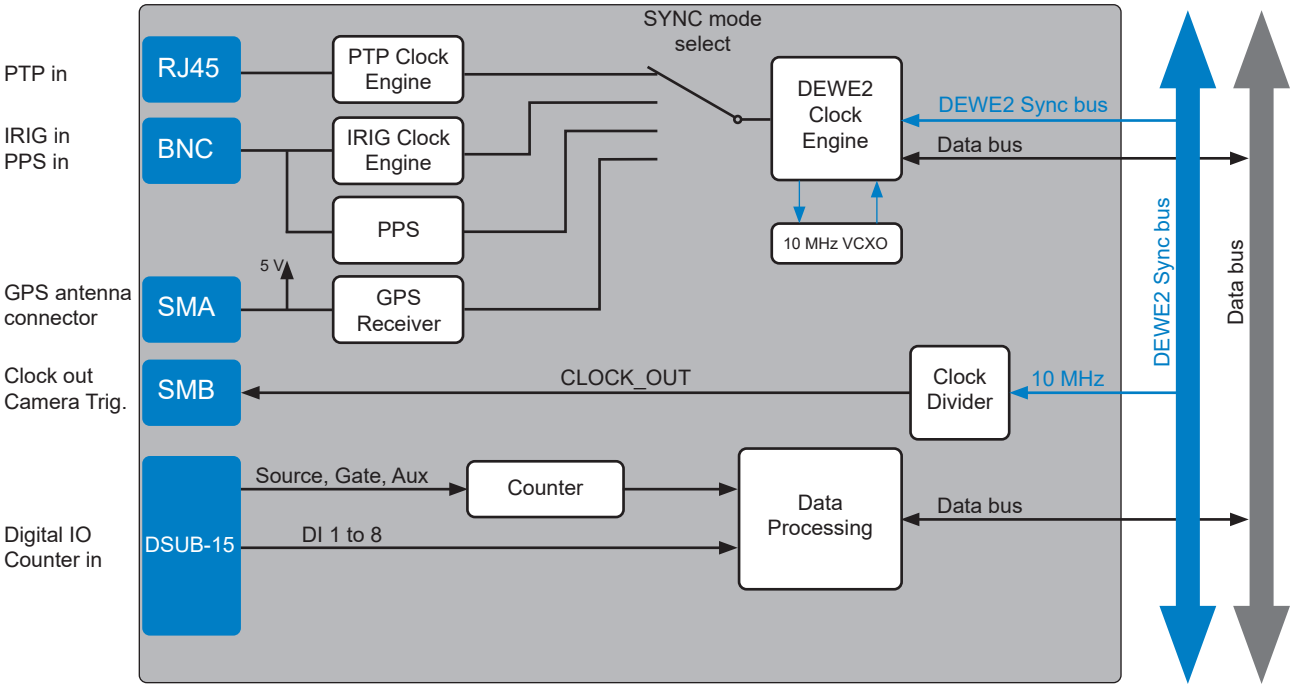
The 4 LEDs indicate the active synchronization source and current sync status.

Steady green:	source is selected and working stable
PPS LED green flashing:	The PPS LED flashes with the internal generated PPS, immediately when a working source is connected.
Red:	source is selected but is not yet synced.

Depending on the selected input that can have several reasons:

> Source might not be connected, or is deactivated. (E.g. IRIG generator is turned off)
> IRIG: wrong IRIG code is selected.
> GPS: not enough satellites found because of antenna position.
> GPS: GPS fix takes up to 1 minute and is not yet established.

Simplified schematic

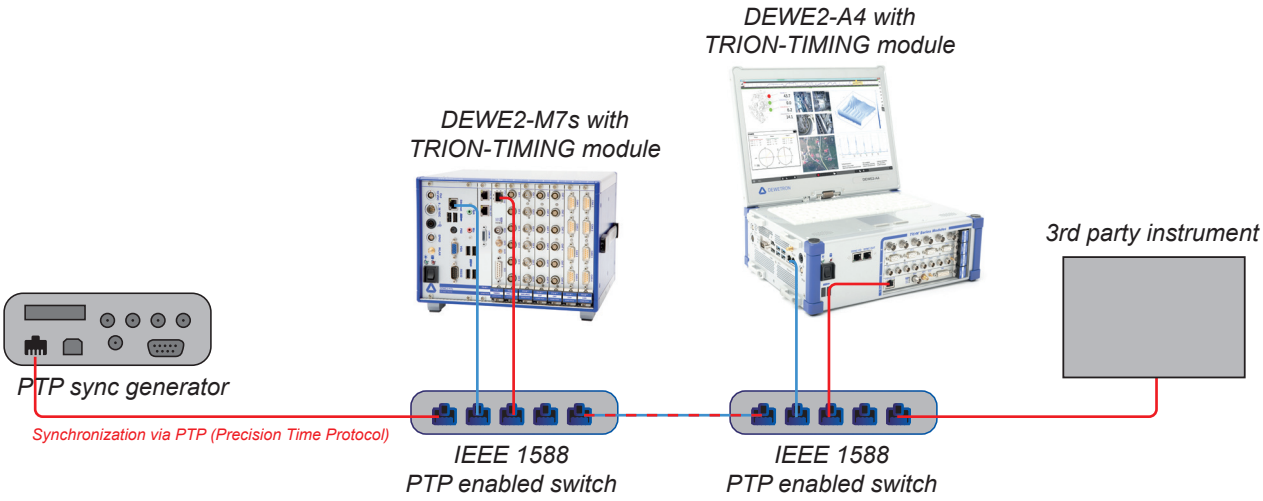


TRION-TIMING-V3

PTP / IEEE 1588

Typical topology

Example: Two or more instruments from DEWETRON or 3rd party instruments synchronized via PTP.
Data transmission via Ethernet and local data storage possible.



Settings

Protocol	UDP/IPv4	(default)
	TCP/IPv4	coming soon
	ETH-IEEE 802.3	coming soon
Delay mechanism	End-to-end	
	Peer-to-peer	coming soon
IP-Address method	DHCP	
	Auto-IP	coming soon
	Fix-IP address	coming soon
RJ-45 Ethernet	10/100 Mbit Ethernet connection; *CAUTION: only for synchronization, no data transfer possible.*	
Programmable correction limit	10 ns to 500 ms	

DEWE2 Clock Engine Operation

The TRION-TIMING-V3 is designed for continuously measuring data, even if the external time base source is temporary not available. Especially in GPS mode that could easily happen. Reason for that is the weather sensitive GPS reception. One cloud might be enough to interrupt the synchronization for a while. In that case the TRION-TIMING-V3 generates a notifying event and continues measuring on its internal time base. This internal time base has been adjusted to the external reference while the sync was stable.

That minimizes the drift in free-run mode. Typically it is far below 1 ppm. Only when the environmental conditions change dramatically during a longer non-synced period of time, it might go up to a maximum of 10 ppm.

When the synchronization has established again the TRION-TIMING checks if the internal time base error is still below the pre-programmed restart limit. If yes, it starts resyncing by slightly changing the time-base until the time stamps matches again exactly. That prevents from gaps in the data file due to resync. That might take a while because the maximum readjusting speed is 100 ppm. If for some reason a hard resync is needed the restart limit could be set to a low value. In that case the datafile will be interrupted.

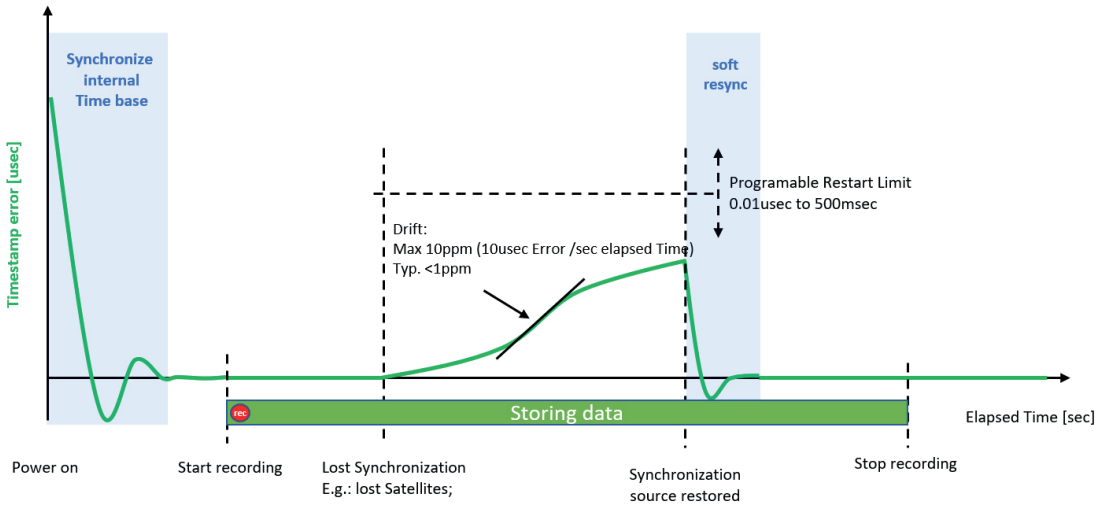


Figure 1: Gapless recording

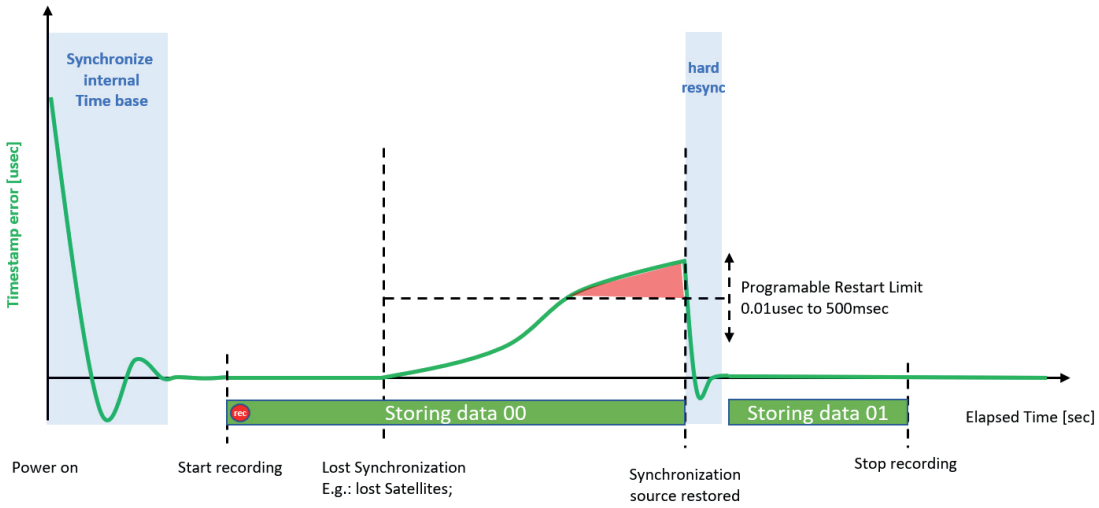


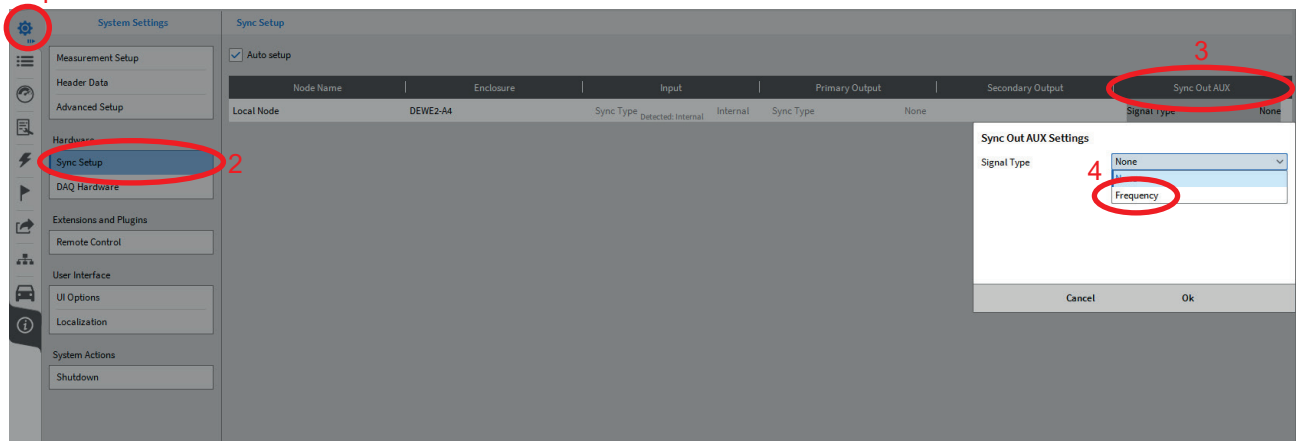
Figure 2: Forced restart after restored synchronization

NOTE: *If the system is equipped with a TRION-BASE, TRION-TIMING or TRION-VGPS-20/-100 module, it must be installed in the “star slot”! This is the only slot a module is able to override the system 10MHz clock with its PPS-synced 10MHz, and thus providing the system with a timebase of higher accuracy.*

AUX terminal

The auxiliary terminal of the TRION-TIMING-V3 module could be used as programmable frequency output for synchronizing external hardware.

The output can be set in the Sync Out Aux settings via 'System Settings' -> 'Sync Setup' -> 'Sync Out Aux':



Advanced counter

The TRION-TIMING module supports a advanced counter or 3 digital inputs via the 15-pin SUB-D connector shown in the module specifications. Further information regarding advanced counters are described in chapter '*Functional description of advanced counter*' of the TRION-CNT module.

Optional accessory

TRION-CBL-CAMTRG-03-00

Camera trigger cable to synchronize a DEWE-CAM-GIGE via an AUX socket of TRION™ modules, 3 meters.

GPS-ANT-1

GPS antenna for TRION-TIMING, for fixed installation.
Only supports GPS L1.

TIMING-ANT-1

IP67 compliant, magnetic GPS antenna for TRION-TIMING for mobile applications. Support of GPS L1, GLONASS G1, BeiDou B1, Galileo E1, SBAS (WAAS, EGNOS & MSAS).

5m cable, SMA plug.

Position, speed and displacement module

- 20/100 Hz GPS receiver
- Supports differential GPS (SBAS) and GLONASS as a standard
- Optional Galileo & BeiDou capability
- GPS or IRIG timing, 8x DIO, 1x counter, 1x AUX
- PTP / IEEE 1588
- Isolation: 350 V_{DC}

GPS antenna

Module specifications

TRION-VGPS-20/100-V3 specifications	
GPS specifications[1]	
Supported GNSS signals	
GPS	L1 C/A, L1C, L2C, L2P, L5
SBAS	L1, L5
GLONASS[2]	L1 C/A, L2 C/A, L2P, L3, L5
Galileo[3] (optional)	E1, E5 AltBOC, E5a, E5b, E6
BeiDou[4] (optional)	B1I, B1C, B2I, B2C, B3I
Number of channels	555
Horizontal position accuracy	
Single Point L1	1.5 m
Single Point L1/L2	1.2 m
SBAS	60 cm
Refreshrate	
TRION-VGPS-20-V3	20 Hz
TRION-VGPS-100-V3	100 Hz
Time to first fix	
Cold start[5] [6]	<40 s (typical)
Hot start[6] [7]	<19 s (typical)
Signal lost recovery	
L1	<0.5 s (typical)
L2	<1.0 s (typical)
Time accuracy[8]	20 ns RMS
Velocity accuracy	<0.03 m/s RMS
Velocity limit[9]	515 m/s
GPS antenna	included (5m cable); supports GPS L1, GLONASS G1, BeiDou B1, Galileo E1, SBAS (WAAS, EGNOS & MSAS)
Input Connector GPS	SMA for GPS antenna
PTP / IEEE 1588	
IP Mode	Multicast
Protocoll	UDP / IPv4; ETH
Delay Mechanism	End to End; Peer to Peer
IP-Address Method	DHCP
RJ-45 Ethernet	10 / 100 Mbit Ethernet connection; only for synchronization, no data transfer possible.
Programmable Correction Limit	10 ns to 500 ms
IRIG input specifications	
Supported codes:	IRIG code A or B; AM or DC
Compatibility (AM code):	0.5 Vp-p to 10 Vp-p
Ratio (AM):	3:1 ±10 %
Compatibility (DC code):	DC Level Shift TTL / CMOS compatible
Impedance:	20 kΩ
Digital I/O specifications	
Number of channels	8
Compatibility (input)	CMOS/TTL
Compatibility (output)	TTL, 20 mA
Overvoltage protection	±30 V
Connector	SUB-D-15 socket

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TRION-VGPS-20/100-V3

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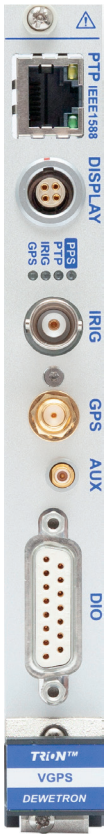
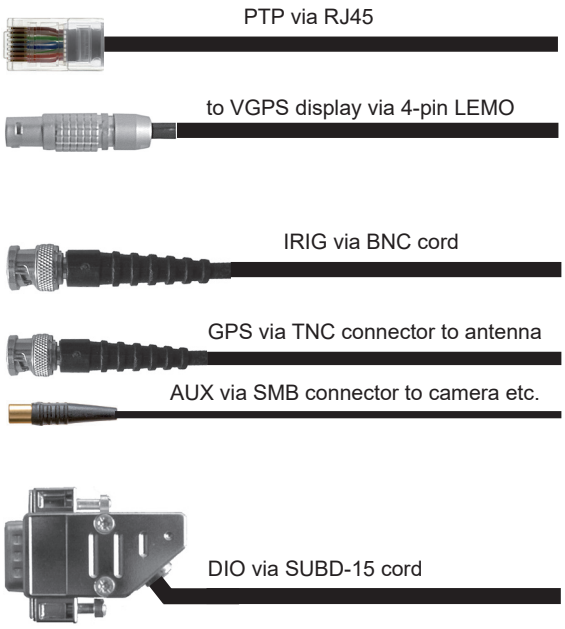
Counter specifications		
Number of channels		1 advanced counter or 3 digital inputs
Counter modes	Event counting	Basic event counting, gated counting, up/down counting and encoder mode (X1, X2 and X4)
	Waveform timing	Period, frequency, pulse width, duty cycle and edge separation
	Sensor modes	Encoder (angle and linear), gear tooth with/without zero, gear tooth with missing/double teeth
Input signal compatibility		CMOS/TTL
Counter resolution		32-bit
Counter time base		80 MHz
Time base accuracy		typ. 10 ppm (defined by the backplane)
Maximum input frequency		10 MHz
Overvoltage protection		±30 V permanent, 50 V peak (for 100 msec)
Sensor power supply		5 V (600 mA) and 12 V (600 mA)
Connector		On same SUB-D-15 socket as Digital I/O
AUX specifications		
Functionality		Camera trigger, trigger input/output, acquisition clock and programmable clock output
Compatibility (input)		LVTTL
Compatibility (output)		LVTTL, 10 mA
Overvoltage protection		±20 V
Connector		SMB socket
General specifications		
Typical power consumption		5 W
Temperature Range		0 - 50 °C
Weight		Appr. 240 g

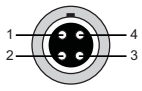
[1] Typical values. Performance specifications subject to GNSS system characteristics, Signal-In-Space (SIS) operational degradation, ionospheric and tropospheric conditions, satellite geometry, baseline length, multipath effects and the presence of intentional or unintentional interference sources.
[2] Hardware ready for L3 and L5.
[3] E1bc support only. Hardware ready for E6bc.
[4] Designed for BeiDou Phase 2 and 3, B1 and B2 compatibility. B3 conditionally supported and subject to change.
[5] Typical value. No almanac or ephemerides and no approximate position or time.
[6] Available in Q2 2019.
[7] Typical value. Almanac and recent ephemerides saved and approximate position and time entered.
[8] Time accuracy does not include biases due to RF or antenna delay.
[9] Export licensing restricts operation to a maximum of 515 metres per second, message output impacted above 500 m/s.

Signal connection

TRION-VGPS-20/100-V3 module







4-pin Lemo connector (female)
EXG.1B.304.HLN

Pin assignment
1: +5 V_{DC}
2: GND
3: TXD
4: RXD

Hot: IN +
Shield: IN -

Hot: IN +
Shield: IN -

Hot: OUT +
Shield: OUT -

1	DIO 1	9	CNT_Source
2	DIO 2	10	CNT_AUX
3	DIO 3	11	CNT_Gate
4	DIO 4	12	DGND
5	DIO 5	13	DGND
6	DIO 6	14	+5 V
7	DIO 7	15	+12 V
8	DIO 8		

LED function

The 4 LEDs indicate the active synchronization source and current sync status.

Steady green: source is selected and working stable

PPS LED green flashing: The PPS LED flashes with the internal generated PPS, immediately when a working source is connected.

Red: source is selected but is not yet synced.

Depending on the selected input that can have several reasons:

> Source might not be connected, or is deactivated. (E.g. IRIG generator is turned off)
> IRIG: wrong IRIG code is selected.
> GPS: not enough satellites found because of antenna position.
> GPS: GPS fix takes up to 1 minute and is not yet established.

Software settings for TRION-VGPS-20/100-V3

GPS receiver settings

Following parameters are selectable:

Receiver Dynamic

is used to adjust the receiver dynamics to that of an application. It is used to optimally tune receiver parameters.

Software setting	Description
Normal (default)	Receiver is being carried by a person with velocity less than 11 km/h (3 m/s)
High	Receiver is in a stable land vehicle with velocity less than 110 km/h (30 m/s)
Highest	Receiver is in an aircraft or a land vehicle, for example a high-speed train, with velocity greater than 110 km/h (30 m/s). This is also the most suitable dynamic for a jittery vehicle at any speed
Automatic	Receiver monitors dynamics and adapts behavior accordingly

Position smoothing

is used to enable, disable or reset the Pseudorange/Delta-Phase (PDP) filter. The main advantages of the PDP implementation are:

> Smooths a jumpy position
> Bridges outages in satellite coverage (the solution is degraded from normal but there is at least a reasonable solution without gaps)

Software setting	Description
Disabled	Disable the PDP filter
Normal (default)	Enable the PDP filter. Auto detect dynamics mode
Glide	Receiver is in an aircraft or a land vehicle, for example a high-speed train, with velocity greater than 110 km/h (30 m/s). This is also the most suitable dynamic for a jittery vehicle at any speed. GLIDE is a mode of the PDP filter that optimizes the position for consistency over time rather than absolute accuracy.

Satellite Based Augmentation System

is used to dictate how the receiver tracks and uses correction data from Satellite Based Augmentation Systems (SBAS). SBAS improves the accuracy and reliability of GNSS information by correcting signal measurement errors and by providing information about the accuracy, integrity, continuity and availability of its signals.

Software setting	Description
Disabled	GPS and GLONASS satellites used, but no SBAS satellites *attention: GLONASS satellites are disabled for any other SBAS setting!*
Auto (default)	Automatically determines satellite system to use and prevents the receiver from using satellites outside of the service area
Any	Uses any and all SBAS satellites found
WAAS	Uses only "**W**ide **A**rea **A**ugmentation **S**ystem" satellites available in the United States
EGNOS	Uses only "**E**uropean **G**eostationary **N**avigation **O**verlay **S**ervice" satellites available in Europe
MSAS	Uses only "**M**ulti-functional **S**atellite **A**ugmentation **S**ystem" satellites available in Japan
GAGAN	Uses only "**G**PS **A**ided **G**eo **A**ugmented **N**avigation" satellites available in India
QZSS	Uses only Quasi-Zenit-Satellite-System signals available in Japan

Velocity Type

configures the source of the velocity that is used.

Software setting	Description
Position	Use the velocity from the same positioning filter that is being used
Doppler (default)	using Doppler-derived velocities. It is the highest-availability, lowest-latency velocity available from the receiver. Due to its low latency, it is also the noisiest velocity.

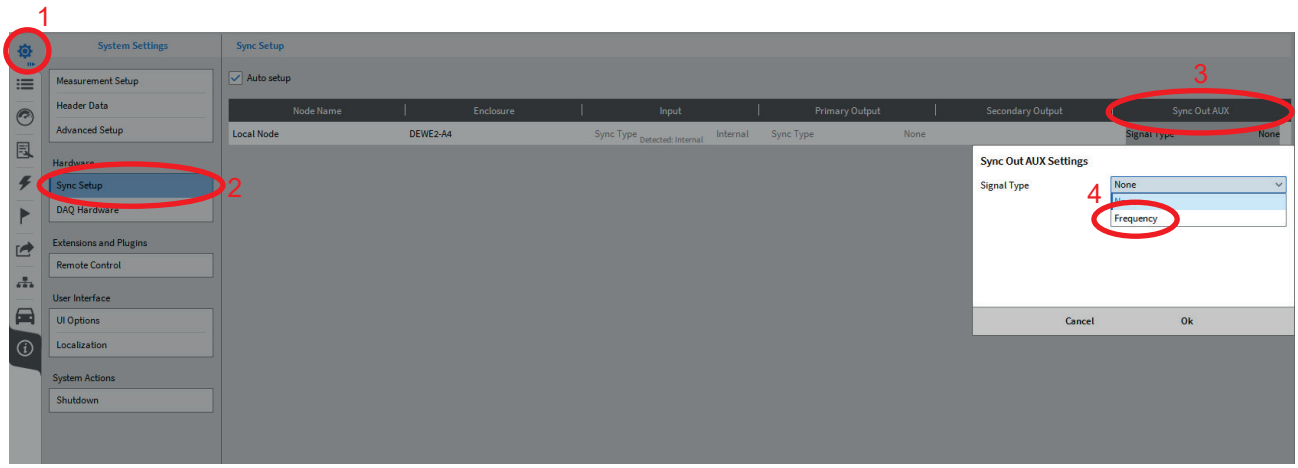
PTP / IEEE 1588

The TRION-VGPS-20/100-V3 module supports synchronization via PTP (Precision Time Protocol). Further information regarding synchronization via PTP are described in chapter '*PTP / IEEE 1588*' of the TRION-TIMING-V3 module.

AUX terminal

The auxiliary terminal of the TRION-VGPS-20/100-V3 module could be used as programmable frequency output for synchronizing external hardware.

The output can be set in the Sync Out Aux settings via 'System Settings' -> 'Sync Setup' -> 'Sync Out Aux':



Optional accessory

TRION-CBL-CAMTRG-03-00

Camera trigger cable to synchronize a DEWE-CAM-GIGE via an AUX socket of TRION™ modules, 3 meters.

Synchronization options with a TRION-VGPS-20/-100 module

To create high channel count systems or for distributed measurements, DEWE2 instruments support multiple synchronization options. The kind of synchronization is depending on how far apart the instruments are from each other.

Synchronization via IRIG time-code is limited to ~1000 meters.
Synchronization via GPS is not limited.

NOTE: *If the system is equipped with a TRION-BASE, TRION-TIMING or TRION-VGPS-20/-100 module, it must be installed in the “star slot”! This is the only slot a module is able to override the system 10MHz clock with its PPS-synced 10MHz, and thus providing the system with a timebase of higher accuracy.*

Advanced counter

The TRION-VGPS module supports a advanced counter or 3 digital inputs via the 15-pin SUB-D connector shown in the module specifications. Further information regarding advanced counters are described in chapter '*Functional description of advanced counters*' of the TRION-CNT module.

Isolated advanced counter module

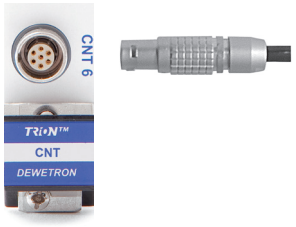
- 80 MHz time base
- 2 MS/s per channel
- Event, waveform timing and sensor mode
- Programmable threshold and AC/DC coupling

Module specifications

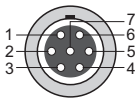
TRION-CNT specifications	
Input channels	6 advanced counter or 18 digital inputs (TRION-CNT-6-L1B)
Counter modes	
Event counting	Basic event counting, gated counting, up/down counting and encoder mode (X1, X2 and X4)
Waveform timing	Period, frequency, pulse width duty cycle and edge separation
Sensor modes	Encoder (angle and linear)
Input signal characteristic	
Rated input voltage (max.) Compatibility	70 V_{DC} (46.7 V_{PEAK}) Adjustable trigger levels
Configuration	Isolation (±500 VDC) for each input channel
Input coupling	DC and AC (1Hz) AC for input A only
Input impedance (ground referenced)	1 MΩ / 5 pF
Sampling rate	2 MS/s per channel
Bandwidth (-3dB)	5 MHz
Trigger adjustment range	0 to 50 V
Trigger resolution	12 mV
Trigger level accuracy	±20 mV ±1% of threshold/retrigger level
Overvoltage protection	±100 V continuous
Max. DC level @AC coupling	±50 V continuous
Counter resolution	32-bit
Counter time base	80 MHz
Time base accuracy	typ. 10 ppm (defined by the backplane)
Sensor power supply (per module)	5 V (600 mA) and 12 V (600 mA), not isolated
Typcial power consumption without sensor supply	5 W
Weight	Appr. 240 g

Signal connection

Measurement via LEMO cord



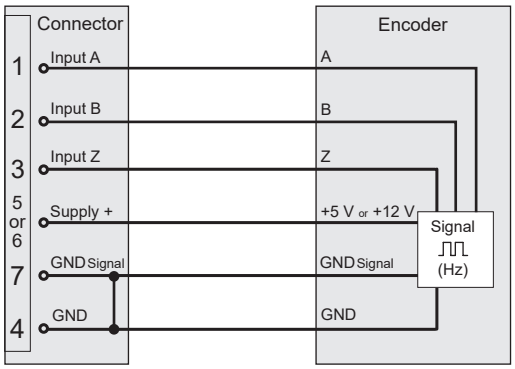
1: Input_A (isolated, white)
2: Input_B (isolated, brown)
3: Input_Z (isolated, green)
4: GND Power (blue)
5: +5 V out (red)
6: +12 V out (yellow)
7: GND Signal (isolated, grey)



7-pin LEMO connector female
EGG.1B.307.CLL

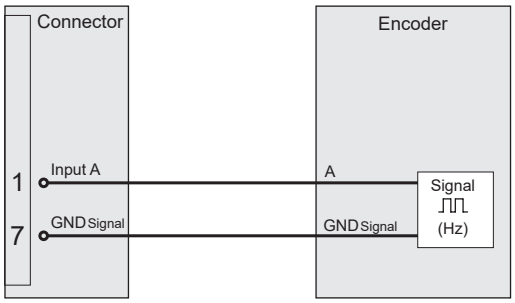
Encoder connection

Supplied sensor connection



CAUTION: *Since the counter input is isolated from the power supply GND Power and GND Signal must be tied together!*

Non supplied sensor connection



Optional accessory

TRION-CBL-L7OE-05-00

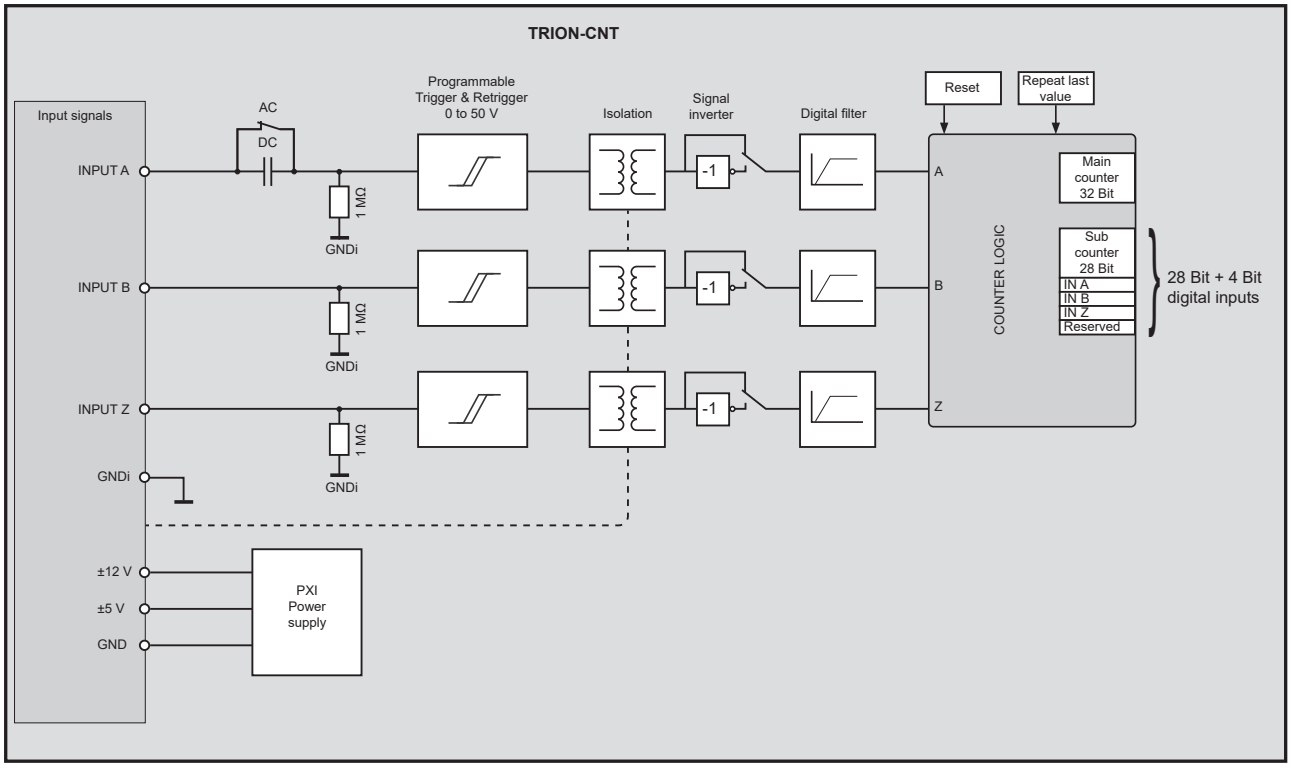
High quality cable from LEMO 1B.307 plug to open end, 5 meters for TRION-CNT-6-LEMO modules.

TRION-CBL-CAMTRG-03-00

Camera trigger cable to synchronize a DEWE-CAM-GIGE via an AUX socket of TRION™ modules, 3 meters.

Functional description of advanced counter

Block diagram



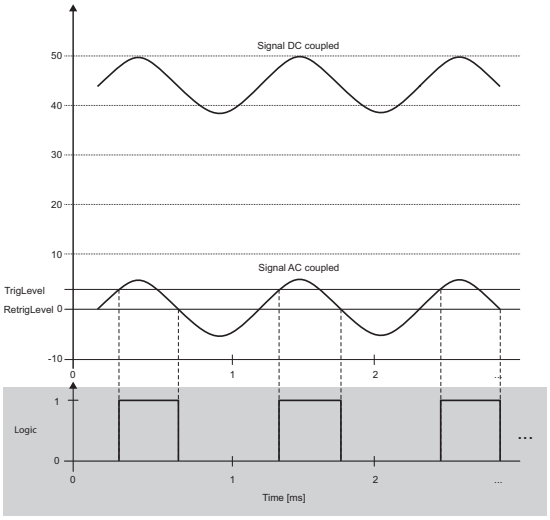
AC/DC coupling

The Input A can be switched either to DC or AC coupling. The AC coupling removes the DC offset and is typically used for inductive pic-up sensors.

Programmable trigger level

The TRION-CNT has free programmable trigger and retrigger levels for every input channel. The trigger voltage could be programmed between 0 and 50 V.

The diagram below illustrates the functionality of the settable trigger levels.



Isolation & Power supply

The digital inputs of the TRION-CNT and the GNDi are isolated up to 500 V_{DC} .The 500 V are for channel to channel and channel to chassis isolation. The 5 V and 12 V power supply are not isolated and directly connected to the system power supply. The current consumption is limited to 600 mA for the 12 V and the 5 V supply for the complete TRION-CNT and not for a single channel.

Caution: Do not short circuit the power supply pins. This might cause a system power down!

Signal Inversion

Each input signal could be digitally inverted before applying it to the counter logic.

Digital filter

Each counter and digital input has a digital filter, which can be set to various gate times. If the gate time is set to “Off”, no filter is on the input signal.

The filter circuit samples the input signal on each rising edge of the internal time base. If the input signal maintains his state for at least the gate time, the new state is propagated. As an effect the signal transition is shifted by the gate time.

Figure 1 demonstrates the function of the filter.

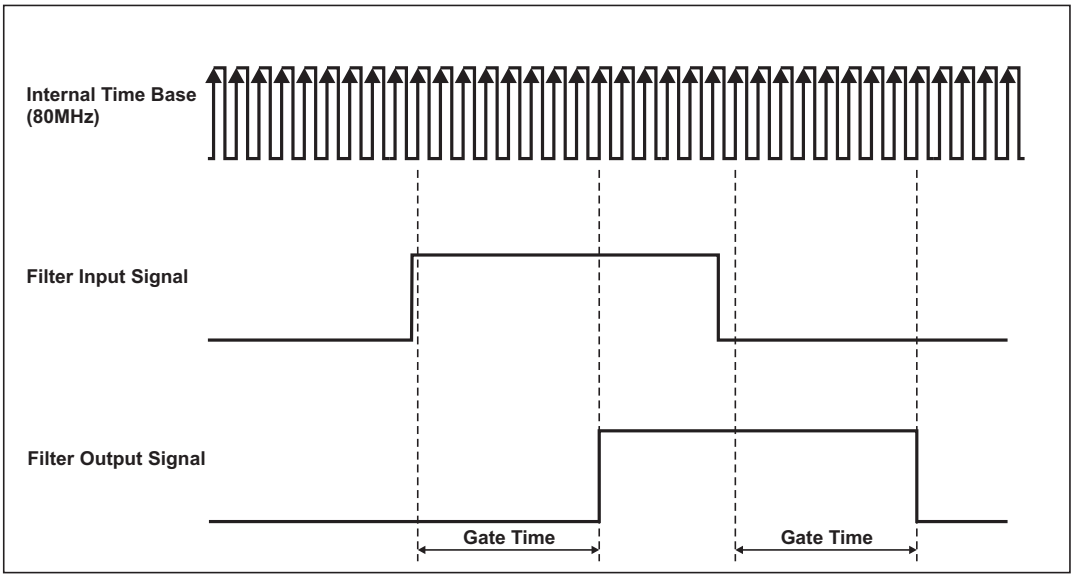


Figure 1: Filters

The intent of the filter is to eliminate unstable states, e.g. glitches, jitter, …, which may appear on the input signal, as shown in Figure 2.

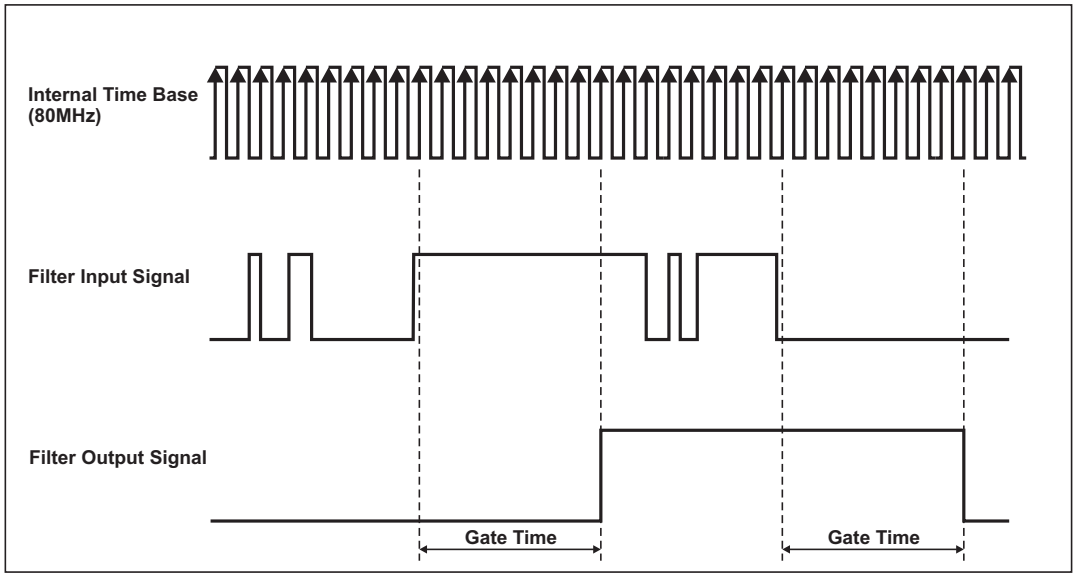


Figure 2: Input signal with jitter

It can be chosen between eight filter settings: Off, 100 ns, 200 ns, 500 ns, 1 µs, 2 µs, 4 µs and 5 µs. Two examples of filter settings are described.

The 100 ns filter will pass all pulse widths (high and low) that are 100 ns or longer. It will block all pulse widths that are 75 ns or shorter. The 5 µs filter will pass all pulse widths (high and low) that are 5 µs or longer and will block all pulse widths that are 4.975 µs or shorter.

The internal sampling clock (time base) is 80 MHz, so the period time amounts 12.5 ns. Pulse widths between gate time minus two internal time base period times may or may not pass, depending on the phase of the input signal with respect to the internal time base.

Properties of all filter settings:

Filter settings	Pulse width to pass	Pulse width to be blocked
100 ns	100 ns	75 ns
200 ns	200 ns	175 ns
500 ns	500 ns	475 ns
1 µs	1 µs	975 ns
2 µs	2µs	1.975 µs
4 µs	4 µs	3.975 µs
5 µs	5 µs	4.975 µs
Off	-	-

Filter Gate Times

Reset

Usually all counters are reset at the start of data acquisition, i.e. the counter value is set to zero at the start of data acquisition. In some applications this is not required. An angle encoder for example is adjusted to the physical zero position at the beginning of a test procedure. By resetting the counter at every start of the measurement this adjustment get lost. Without this reset the counter is also active if the acquisition is interrupted between the test cycles. As a result the counter types out the absolute angle position at the measurement output all the time.

Repeat last value

Especially in every kind of input period time measurement mode (also pulse width or two pulse edge separation measurement) there may be new information between two samples. Also measuring the line frequency of about 50 Hz with a sample rate of 10 kSamples/sec means, that only after every 200th measurement new input frequency information is available. Another example is the measurement on rotating machines if the sensor output frequency is lower than the sample rate. Depending on the application the TRION-CNT module can choose between two different output data settings:

- Repeat last value: last measured cycle time is taken until a new measured cycle time is available.
- Make zero value: as soon as no input information is available the output is set to Zero.

Counter logic

As shown in the blockdiagram above each counter block is equipped with three inputs. With this three inputs the following applications can be done:

- Event Counting
- Gated Event Counting
- Period Time Measurement
- Pulse Width Measurement
- Two Pulse Edge Separation
- Quadrature Encoder (X1, X2, X4, A-Up/B-Down)
- Up/Down Counter

Event counting

In Event Counting the counter counts the number of pulses that occur on input A/B. At every acquisition clock the counter value is read without disturbing the counting process.

Figure 3 shows an example of event counting where the counter counts eight events on *Input A or B*. *Synchronized Value* is the value read by the TRION-CNT module at *Acquisition Clock* (encircled numbers in the figure, e.g. ①, ②).

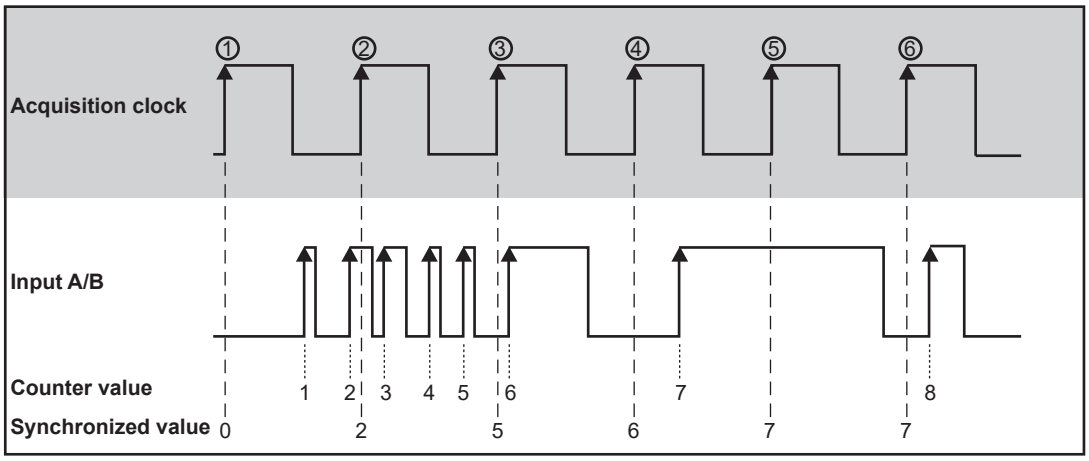


Figure 3: Event Counting

If counting at falling edges is necessary, the input signal has to be inverted. This can be done directly in the software by selecting inverted input.

Gated event counting

Gated Event Counting is similar to Event Counting except that the counting process is gated. When *Input B* is active, the counter counts pulses which occur on counter source. When *Input B* is inactive the counter retains the current count value. At every *Acquisition Clock* the value is read.

Figure 5 shows an example of Gated Event Counting where the counter counts three events on *Input A*. At ① and ② the counter value is zero, because the input B signal is inactive. At acquisition clock ③, ④ and ⑤ the actual counter value is read out. At ⑥ the same value as at ⑤ is typed out.

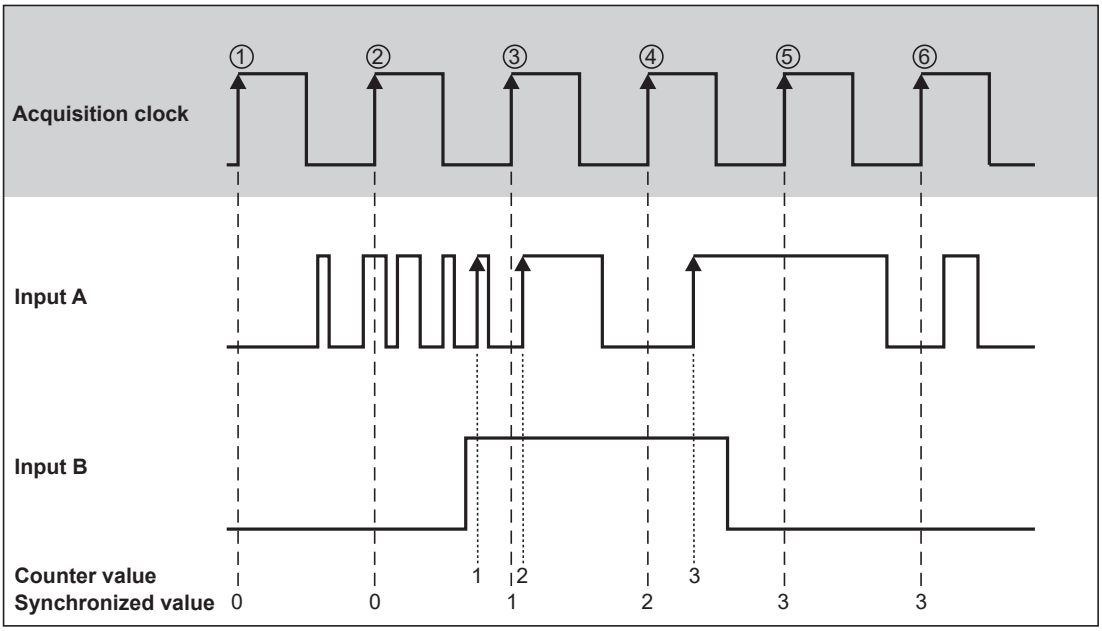


Figure 5: Gated Event Counting

It is also possible, as in Event Counting, to invert the input signals.

Period time measurement

In Period Time Measurement the counter uses the internal time base to measure the period time of the signal present on *Input A*. The counter counts the rising edges of the internal time base which occurs between two rising edges on *Input A*. At the completion of the period interval the counter value is stored in a register and the counter starts counting from zero. At every *Acquisition Clock* (①, ②, …, ⑥) the register value is read out.

Figure 7 shows a Period Time Measurement.

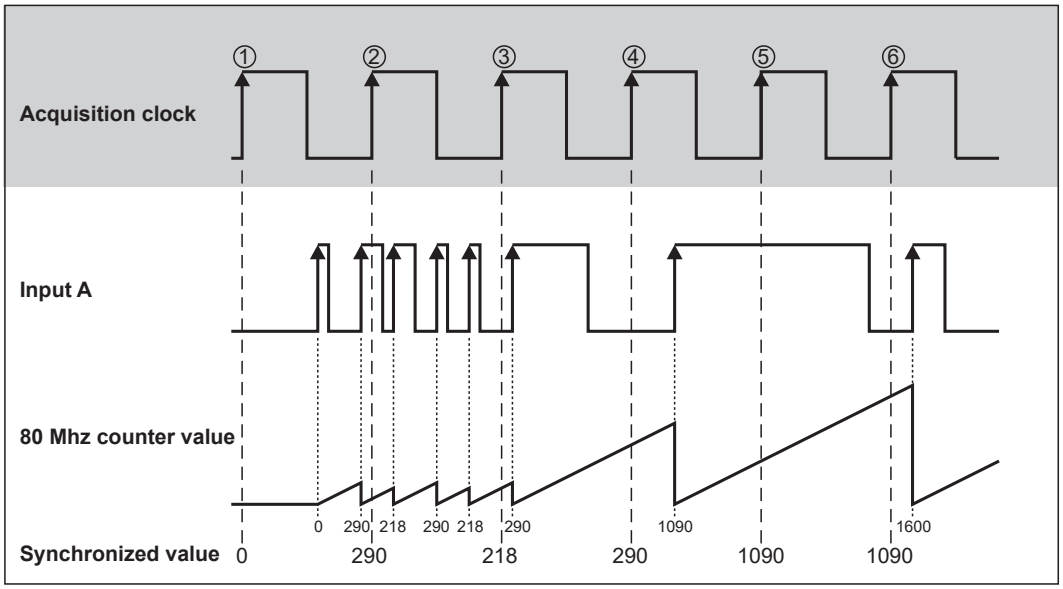


Figure 7: Period Time Measurement

Pulse width measurement

In Pulse Width Measurement the counter uses the internal time base to measure the pulse width of the signal present on *Input A*. The counter counts the rising edges of the internal time base after a rising edge occurs on Input A. At the falling edge on *Input A* the counter value is stored in a register and the counter is set to zero. With the next rising edge on *Input A* the counter starts counting again. At every *Acquisition Clock* (①, ②, …, ⑥) the register value is read out.

Figure 9 shows a pulse width measurement.

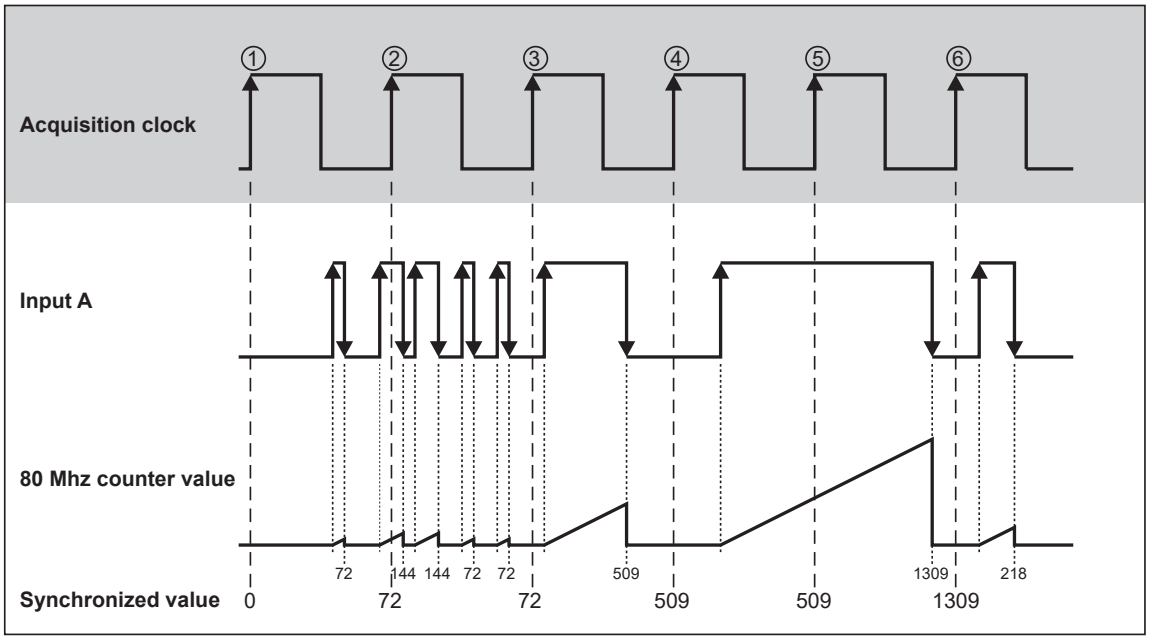


Figure 9: Pulse Width Measurement

For measuring the low time of the signal, the input signal has to be inverted on the TRION-CNT module.

Two pulse edge separation measurement

The two pulse edge separation measurement is similar to the pulse width measurement, except that there are two input signals: *Input A* and *Input B*. After a rising edge has occurred on *Input A* the counter counts rising edges of the internal time base. Additional edges on input A are ignored. After a rising edge has occurred on *Input B* the counter stops counting and the value is stored in a register. At the next rising edge on *Input A* the counter starts counting from zero again. At every *Acquisition Clock* (①, ②, …, ⑥) the register value is read out.

Figure 11 shows an example of Two Pulse Edge Separation Measurement.

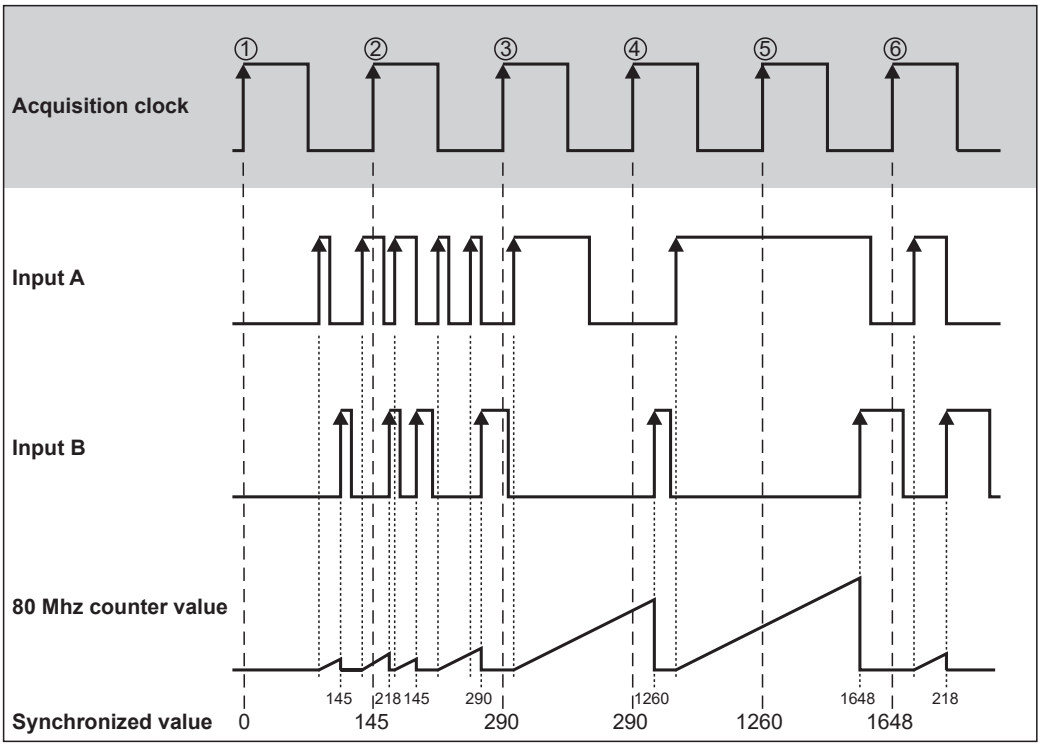


Figure 11: Two Pulse Edge Separation Measurement

If the input signals are inverted the counter takes the falling edges for counting.

Encoder

Motion encoders have usually three channels: channel A, B and Z. Channel A and channel B are providing the square signals for the counter, and have a phase shift of 90°. With this phase shift the decoder is able to recognize the rotation direction of the motion encoder. The third channel types out one pulse at a certain position at each revolution. This pulse is used to set the counter to zero. The amount of counts per cycle at a given motion encoder depends on the type of decoding: X1, X2, X4. All three types are provided by the TRION-CNT module. Some motion encoders have two outputs, which are working in a different way. Either channel A or channel B providing the square signal, depending on the direction of the rotation. Also this type is supplied by the TRION-CNT module.

In the first case X1 decoding is explained. When *Input A* leads *Input B* in a quadrature cycle, the counter increments on rising edges of *Input A*. When *Input B* leads *Input A* in a quadrature cycle, the counter decrements on the falling edges of *Input A*. At every *Acquisition Clock* (①, ②, …, ⑨) the counter value is read out.

Figure 13 shows the resulting increments and decrements for X1 encoding.

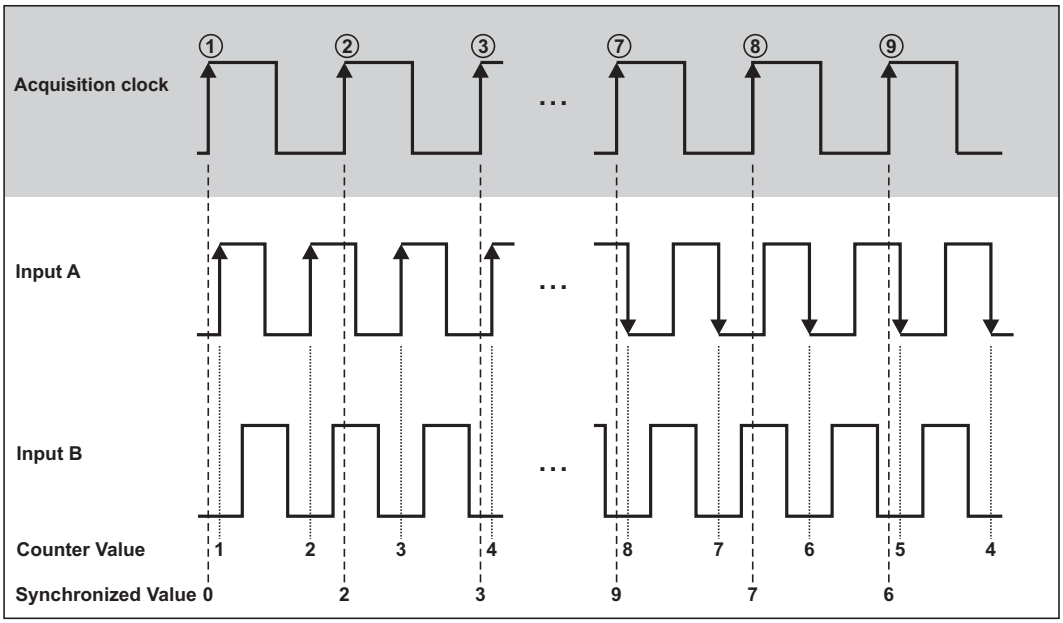


Figure 13: Quadrature Encoder X1 Mode

For X2 encoding the rising edges and the falling edges of *Input A* are used to increment or decrement. The counter increments if *Input A* leads *Input B* and decrements if *Input B* leads *Input A*. This is shown in Figure 14.

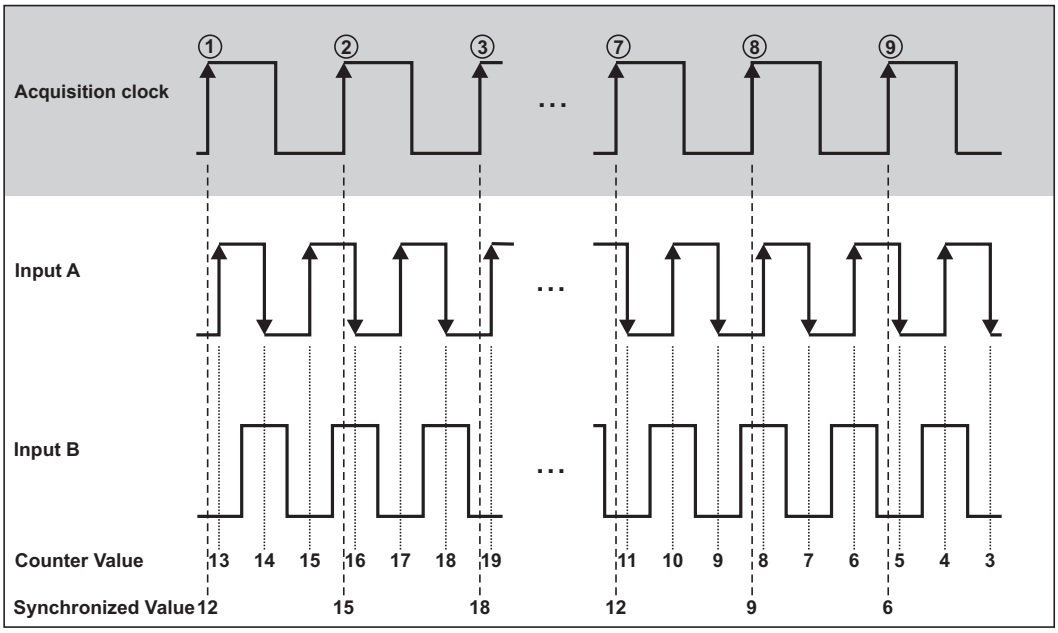


Figure 14: Quadrature Encoder X2 Mode

Similarly, the counter increments or decrements on each edge of *Input A* and *Input B* for X4 decoding. The condition for increment and decrement is the same as for X1 and X2.

Figure 20 shows the results for X4 encoding.

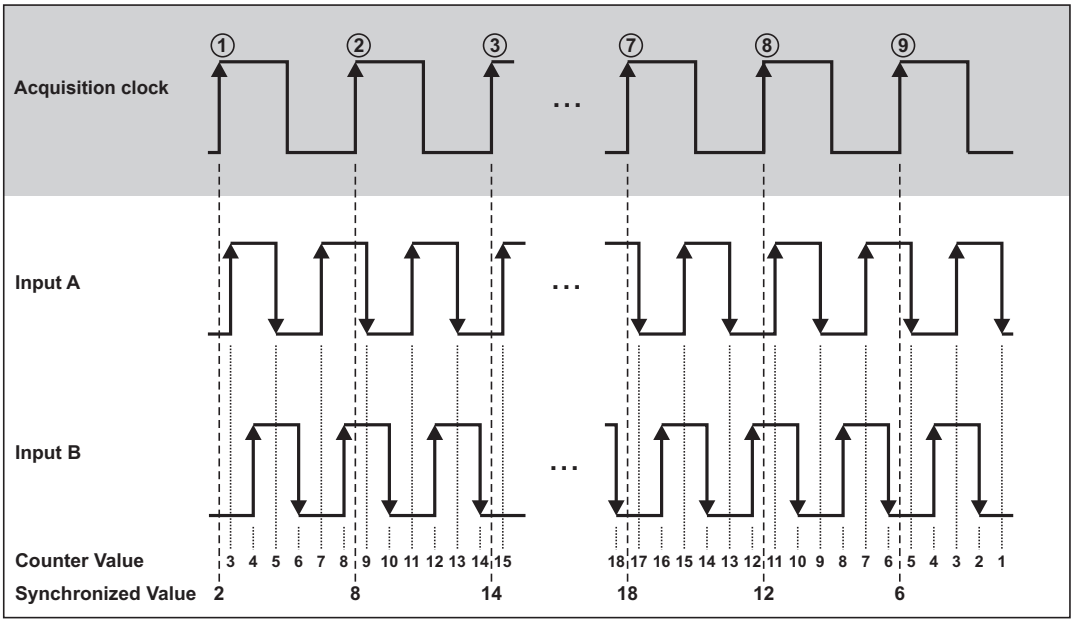


Figure 15: Quadrature Encoder X4 Mode

The third channel *Input Z*, which is also referred as the index channel, causes the counter to be reloaded with zero in a specific phase of the quadrature cycle.

Figure 16 shows the results for X1 encoding with input Z.

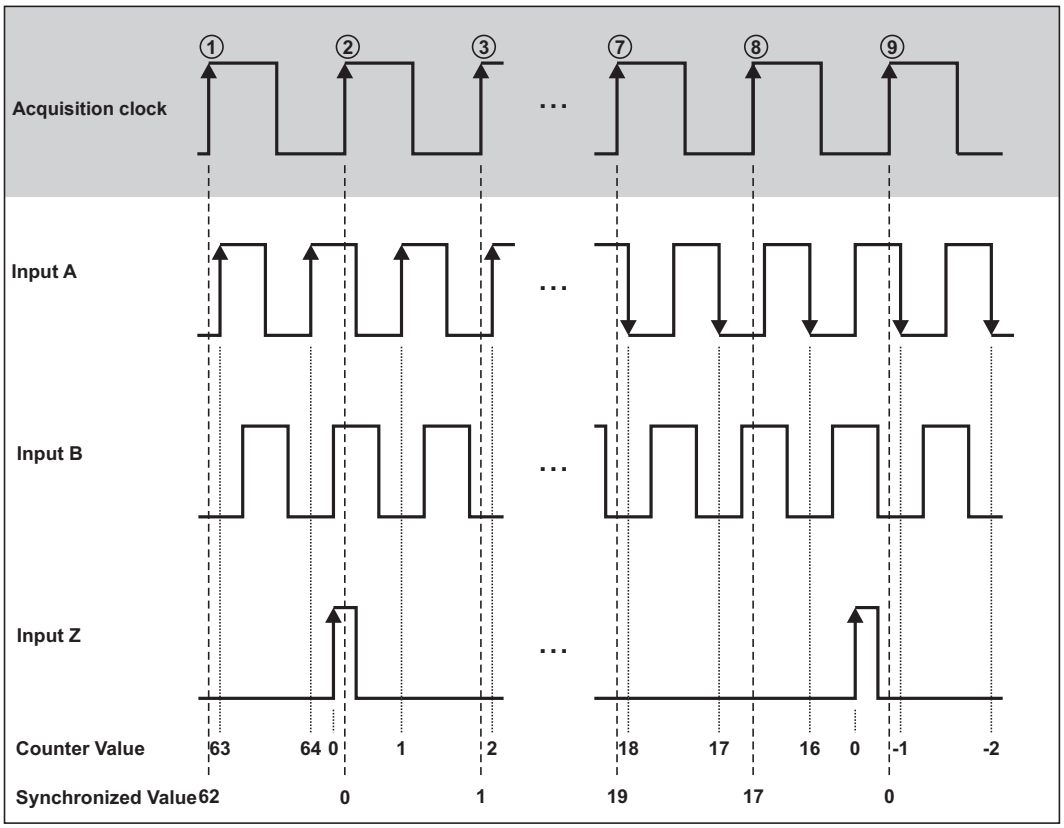


Figure 16: Quadrature Encoder with channel Z

Up/Down

The A-Up/B-Down Encoder supports two inputs, A and B. A pulse on *Input A* increments the counter on its rising edges. A pulse on *Input B* decrements the counter on its rising edges. At every *Acquisition Clock* (①, ② , …, ⑨) the counter value is read out.

This situation is shown in Figure 22.

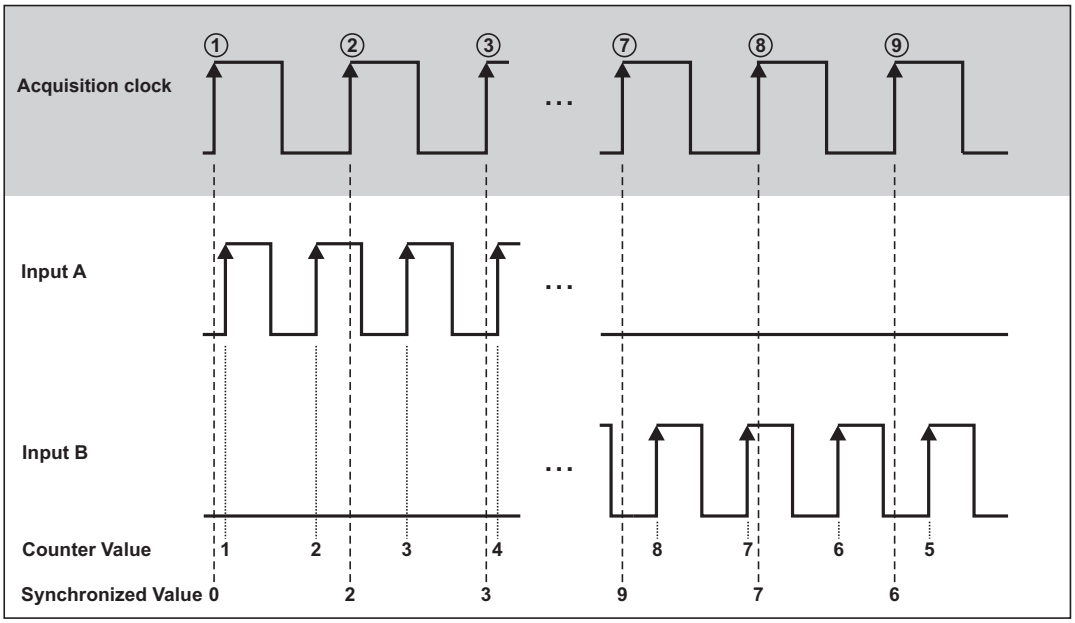


Figure 18: A-Up/B-Down Encoder

Frequency measurement

In general it is possible to take the inverse of a period measurement to get the frequency of the input signal. If the period time measurement is done an inaccuracy of counted internal time base cycles of ±1 cycle appears, because the counted cycles of the internal time base depends on the phase of the input signal with respect to the internal time base. For long period times, and therewith low frequencies, the measurement error is negligible. At high frequencies, and therewith short period times, few cycles are counted. In this case the error of ±1 cycle becomes significant.

Input Frequency	Number of internal time base cycles	Measurement error of -1 cycle	Measurement error of +1 cycle	Calculated frequency with error of -1 cycle	Calculated frequency with error of +1 cycle
40 kHz	2000	1999	2001	39,98 kHz	40,02 kHz
10 MHz	8	7	9	8,75 MHz	11,25 MHz

Accuracy at period time measurement

For higher precision result a combination of main and sub counter is used internally for getting higher precision at the frequency measurement. The main counter is running on event counting (or encoder mode). The sub counter measures the time between. The sub counter measures exactly the time of the input event with a resolution of 12.5 nsec relative to the acquisition clock. At every rising edge on *Input A* the counter value of the sub counter is stored in a register. At every *Acquisition Clock* (①, ②, …, ⑥) the values of both counters are read out.

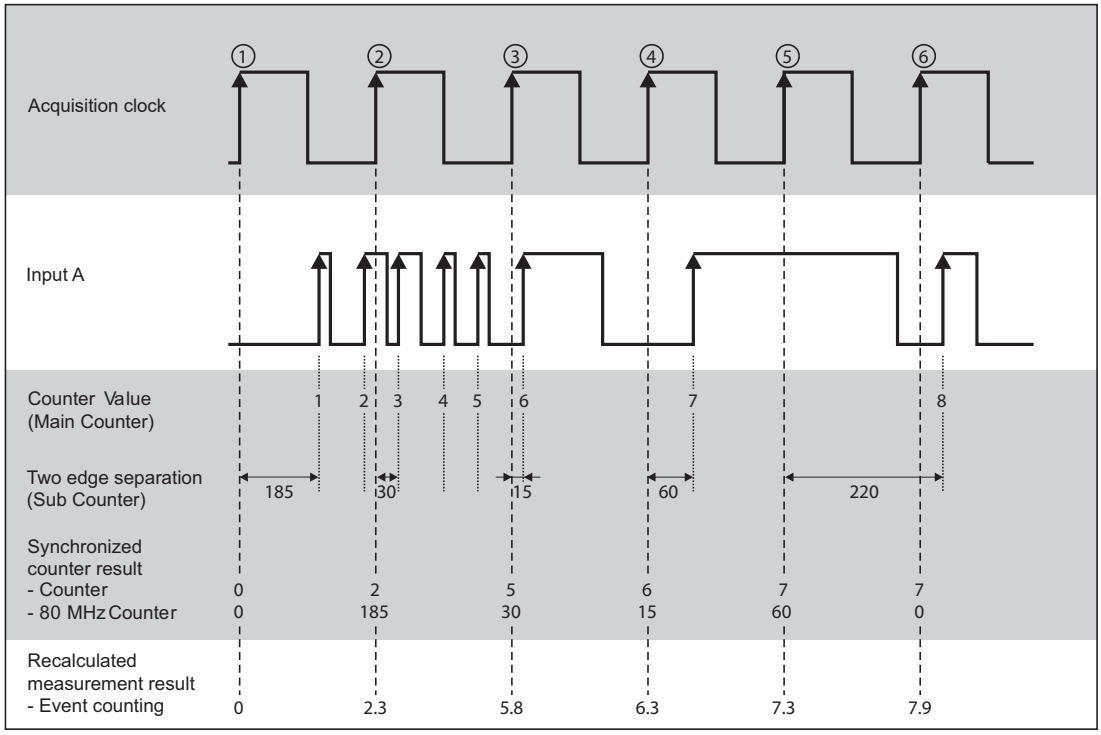
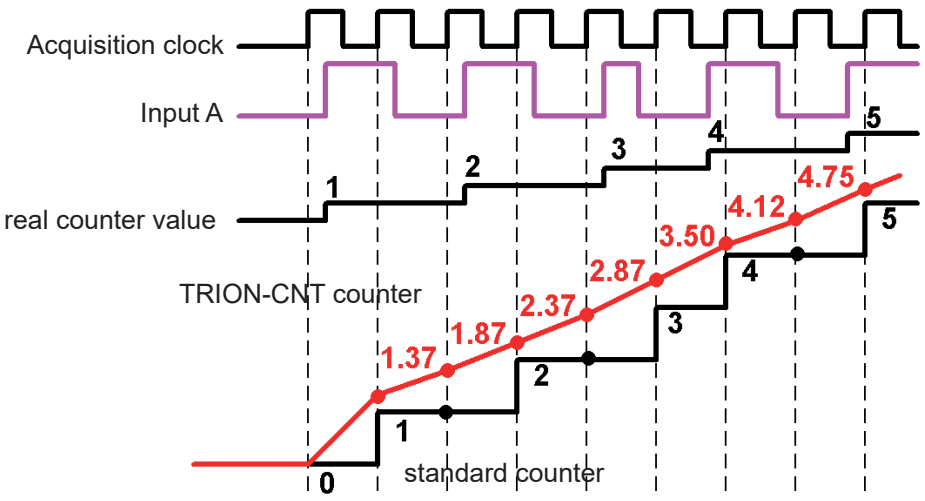


Figure 19: Frequency Measurement

With these both measurement results not only the frequency can be calculated in a precise way. Also the event counter result could be shown in fractions because the exact time when the event occurs at the input is known. The event counting result is recalculated with interpolation to the exact sample point like shown in the diagram above.

In the next figure, the difference of the measurement result is shown. While a standard counter input shows the value up to one sample delayed, the counter input of the TRION-CNT calculates the counter result at the exact sample point.



For low frequency input signals the frequency also can be obtained by measure the period time and take its inverse without more inaccuracy.

Isolated digital input module

- 48 isolated digital inputs (TRION-DI-48)
- 204.8 kS/s per channel

Module specifications

TRION-DI-48 specifications	
Input channels	48 isolated digital inputs (TRION-DI-48)
Input modes	digital input (discrete)
Sampling rate	2 MS/s
Input signal characteristic	
Compatibility	CMOS
Configuration	Isolated input
Input low level	U_{IN} < 1.8 V
Input high level	U_{IN} > 3.2 V
Input high current @ 5 V UIN	< 3.5 mA
Input high current @ 30 V UIN	< 7 mA
Propagation delay	< 160 nsec
Bandwidth	3 MHz
Overvoltage protection	35 V continuous (65 V peak)
Isolation voltage (channel to channel)	100 V
Isolation voltage (input to output)	250 V
Input connector	2 x 50 pin mini centronics
Sensor power supply (per module)	5 V (600 mA), not isolated
Typical power consumption	5 W
Weight	Appr.190 g

TRION-DI-48

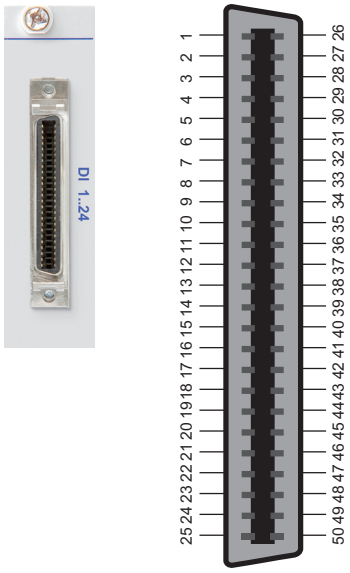
TRION-DI-48 model overview

TRION-DI-48

Signal connection

TRION-DI-48 module

Measurement via mini-centronics cord



50-pin mini-centronics socket

Digital inputs (DI 1..24)

1:	DI1+	26:	DI1-
2:	DI2+	27:	DI2-
3:	DI3+	28:	DI3-
4:	DI4+	29:	DI4-
5:	DI5+	30:	DI5-
6:	DI6+	31:	DI6-
7:	DI7+	32:	DI7-
8:	DI8+	33:	DI8-
9:	DI9+	34:	DI9-
10:	DI10+	35:	DI10-
11:	DI11+	36:	DI11-
12:	DI12+	37:	DI12-
13:	DI13+	38:	DI13-
14:	DI14+	39:	DI14-
15:	DI15+	40:	DI15-
16:	DI16+	41:	DI16-
17:	DI17+	42:	DI17-
18:	DI18+	43:	DI18-
19:	DI19+	44:	DI19-
20:	DI20+	45:	DI20-
21:	DI21+	46:	DI21-
22:	DI22+	47:	DI22-
23:	DI23+	48:	DI23-
24:	DI24+	49:	DI24-
25:	+5 V	50:	GND

Digital inputs (DI 25..48)

1:	DI25+	26:	DI25-
2:	DI26+	27:	DI26-
3:	DI27+	28:	DI27-
4:	DI28+	29:	DI28-
5:	DI29+	30:	DI29-
6:	DI30+	31:	DI30-
7:	DI31+	32:	DI31-
8:	DI32+	33:	DI32-
9:	DI33+	34:	DI33-
10:	DI34+	35:	DI34-
11:	DI35+	36:	DI35-
12:	DI36+	37:	DI36-
13:	DI37+	38:	DI37-
14:	DI38+	39:	DI38-
15:	DI39+	40:	DI39-
16:	DI40+	41:	DI40-
17:	DI41+	42:	DI41-
18:	DI42+	43:	DI42-
19:	DI43+	44:	DI43-
20:	DI44+	45:	DI44-
21:	DI45+	46:	DI45-
22:	DI46+	47:	DI46-
23:	DI47+	48:	DI47-
24:	DI48+	49:	DI48-
25:	+5 V	50:	GND

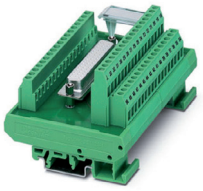
Optional accessory

TRION-CB24-B

24 channel break-out box with 4 mm banana jacks. 1 m cable, terminated with 50-pin mini-centronics plug.
Note: Two boxes are required for all 48 channels!

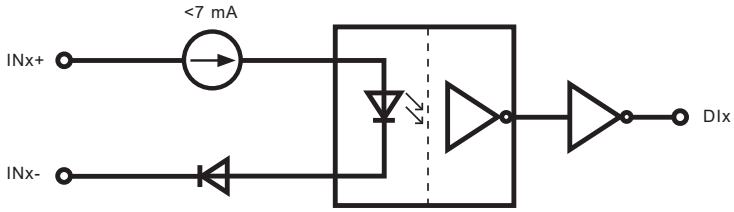
TRION-CB24-SC

24 channel screw-terminal block, unshielded 1 m cable, terminated with 50-pin mini-centronics plug.
Note: Two blocks are required for all 48 channels!



Block diagram

Base block diagram of the input circuit of the TRION-DI-48 module:



Notes

CAN interface module

- Isolated high speed CAN 2.0B
- up to 1 Mbit/s with programmable termination
- supports OBDII, J1939, CAN output

Module specifications

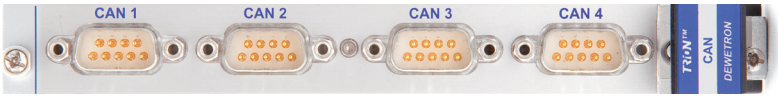
TRION-CAN specifications	
Input channels	2 with DSUB 9 connector (TRION-CAN-2-D)
	4 with DSUB 9 connector (TRION-CAN-4-D)
Specification	CAN 2.0B
Physical Layer	High Speed, (low speed and single wire with optional converter)
Listen only mode	Supported
Termination	Programmable: High Impedance or 120 Ω
Galvanic isolation	500 V_{DC}
Bus pin fault protection	±36 V
ESD Protection	12 kV (HBM)
CAN Transceiver	SNHVD235
Sensor power supply (per module)	5 V (600 mA) and 12 V (600 mA)
Typical power consumption without Sensor/CPAD supply	5 W

The TRION-CAN module is either suited with two or four high speed CAN interfaces. All ports are compatible with the CAN 2.0B specification. The CAN transceiver has a bus-pin fault protection of up to ±36 V.

The main application for these CAN-ports is acquiring CAN data alongside with analog data. Although the CAN data is asynchronous to the analog data, the TRION-CAN module guarantees perfect synchronization. Each incoming CAN message is directly time stamped to the analog sample count number before the data is transferred to the application software.

When only CAN data should be acquired the “Listen Only-mode” can be used. In this mode the TRION-CAN module generates no data on CAN even if the CAN-baud-rate is not correctly selected.
This mode is not working when a direct connection to a sensor is used. In a point-to-point connection the “Listen Only” mode has to be deactivated at the CAN-Interface (see chapter “Listen Only-mode” & “point to point connection” for further information).

TRION-CAN model overview



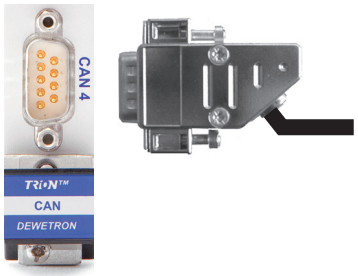
TRION-CAN-4-D

TRION-CAN-2-D

Signal connection

TRION-CAN-x-D module

Measurement via SUB-D cord



Pin 1: +5 V out
Pin 2: CANx Low (isolated)
Pin 3: GNDx CAN (isolated)
Pin 4: NC
Pin 5: NC
Pin 6: GND Power
Pin 7: CANx High (isolated)
Pin 8: NC
Pin 9: +12 V out

High-speed CAN

The high-speed CAN is a differential bus where complementary signals are sent over two wires. The voltage difference between the two wires defines the logical state of the bus. The differential CAN receiver monitors this voltage difference and outputs the bus state with a single-ended output signal.

The high-speed CAN bus topology as well as the possible cable lengths and the recommended termination resistors are specified in the standards ISO-11898 and CiA 102.
The high-speed CAN bus supports bitrates of up to 1 Mbit/s (or >125 kbit/s).
The schematic below will give you an overview of the high-speed CAN bus topology and the termination resistor placement.

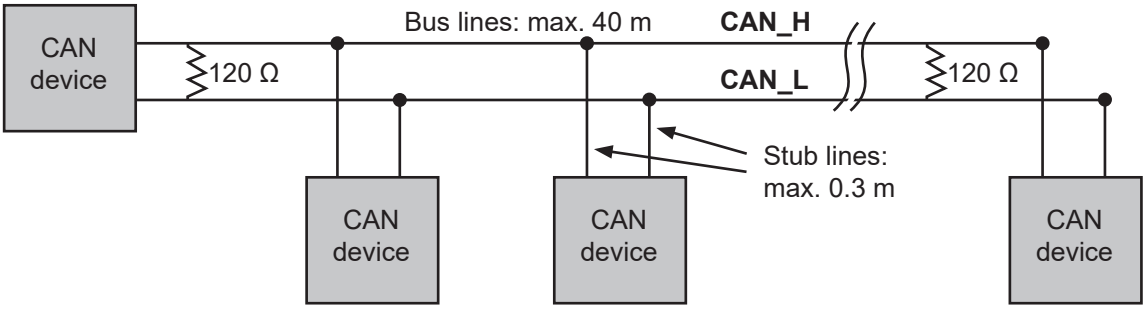


Figure 1: High-speed CAN

Cable lengths for high-speed CAN bus according to ISO-11898 & CiA 102

The cabling characteristics and the desired bit transmission rate affect the allowable cable length. ISO-11898 standard specifies a maximum bus length of 40 m and a maximum stub length of 0.3 m with a maximum of 30 nodes for a bitrate of 1 Mbit/s. However, with careful design, users can have longer cables, longer stub lengths, and many more nodes to a bus. A large number of nodes requires a transceiver with high input impedance and each node should be analyzed for signal integrity problems.

Characteristics of two-wire differential bus:

- Impedance: 108 Ω min., 120 Ω nominal, 132 Ω max.
- Length-related resistance: 70 mΩ/m nominal
- Nominal specific propagation delay: 5 ns/m nominal

Further information please see ISO-11898 and CiA 102 specifications.

Termination

CAN_H and CAN_L are transmission lines. If the transmission line is not terminated, each signal line causes reflections which can cause communication failures therefore both ends of the cable have to be terminated. If multiple devices are connected only the devices at the ends of the cable need to be terminated. Recommended termination resistors in a high-speed CAN bus topology (according to ISO-11898): 120 Ω. The TRION-CAN module offers a programmable termination resistance, either high impedance or 120 Ω.

Low-speed CAN / fault-tolerant (with opt. adaptor "ADAP-CAN-LS-HS")

With an optional additional adapter (ADAP-CAN-LS-HS) it is also possible to run the TRION-CAN module in a low-speed CAN bus. The low-speed CAN is also a differential bus, but the transmission lines (CAN_H & CAN_L) are not connecteyd via a termination resistor. Low-speed CAN is not interference-prone and keeps working even if a transmission line is corrupt (switches to single-wire mode). The schematic below will give you an overview of a low-speed / fault-tolerant CAN bus topology and termination resistor placement.

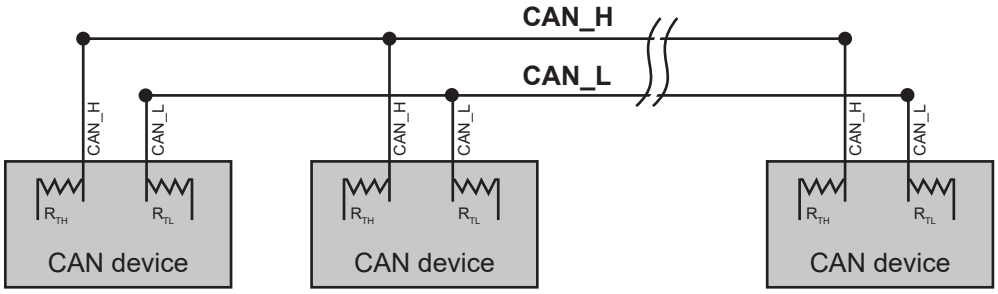


Figure 2: Low-speed / fault-tolerant CAN

The low-speed / fault-tolerant CAN is specified in ISO 11898-3 (fault-tolerant CAN). The influence of signal reflections in this topology is not as big, but when long bus lines are used, the influence increases. This makes the use of an open bus line possible, which means low-speed CAN transceiver can be used for networks with very low power consumption and the bus topology is no longer limited to a linear structure.

Low-speed CAN bus supports bitrates of up to 125 kbit/s.

Cable requirements for low-speed / fault-tolerant CAN bus according to ISO-11898-3

Cable characteristics for low-speed / fault-tolerant CAN devices:

- Length-related resistance: 90 mΩ/m nominal
- Length-related capacitance (CAN_L and ground, CAN_H and ground, CAN_L and CAN_H): 30 pF/m nominal

The maximum number of devices depends on the electrical characteristics of the devices on the network. According to ISO-11898-3 it is possible to connect up to 32 devices to the bus. It is possible to connect a higher number of devices if the electrical characteristics of the devices do not degrade signal quality below low-speed / fault-tolerant signal level specifications.

Termination

Each device on the low-speed CAN network needs a termination for each transmission line:
RTH for CAN_H and RTL for CAN_L. This configuration allows the SNHVD235 transceiver to recover from bus faults. Before connecting the TRION-CAN module to a low-speed / fault-tolerant network it is important to determine the overall termination.

In general, if just small CAN networks are required or individual components are tested you should select the 560 Ω option at the low-speed adaptor to gain an overall impedance of ~120 Ω (parallel connection of the resistors). If the existing low-speed CAN network already has the desired overall impedance of ~120 Ω select the 5.66 kΩ option on the adaptor. This won't influence the overall impendance that much.

LED indication

The ADAP-CAN-LS-HS has two leds indicating the following states:

green: Power (power supply +5 V)
red: Error (error in low-speed CAN network)

The error LED indicates one of the following low-speed CAN errors:

- CAN_H transmission line interrupted
- CAN_L transmission line interrupted
- Short-circuit between CAN_H & GND
- Short-circuit between CAN_H & VCC
- Short-circuit between CAN_L & GND
- Short-circuit between CAN_L & VCC
- Short-circuit between CAN_H & CAN_L

Single-wire CAN (with opt. adaptor "ADAP-CAN-SW-HS")

In single-wire CAN mode, the communication takes place via just one bus line with a nominal data rate of up to 33.3 kbit/s (83.3 kbit/s in high-speed mode for diagnostics) according to the standard SAE J2411. This standard defines up to 32 devices per network. An unshielded single wire is defined as the bus medium. A linear bus topology is not necessary.

In single-wire mode each CAN port has a high impedance local bus load between the CAN_H and RTH pins of the transceiver to provide protection against loss of ground (see Figure 2: Low-speed / fault-tolerated CAN).

Cable lengths / requirements for single-wire CAN bus according to SAE J2411

The cable length between two ECU nodes should not exceed 60 m.
Due to the fact that the total system cable length, bus loading of each node and clock tolerance are all interrelated it is the users responsibility to factor in all the above mentioned parameters when designing a single-wire CAN network. The SAE J2411 specification includes some recommendations that will help building up a single-wire CAN network.

Termination

The ADAP-CAN-SW-HS adaptor is internally terminated with a high impedance load resistor of 5.1 kΩ as specified by SAE J2411.

LED indication

The ADAP-CAN-SW-HS has two leds indicating the following states:

red: power supply (+5 V)
orange: Reception / transmission of wake-up signals on the single-wire CAN bus

Single-wire CAN operation modes

With the optional adaptor "ADAP-CAN-SW-HS" it is possible to operate in three different single-wire CAN modes. A slide switch on the side of the adaptor determines the different modes of operation:

Switch position	Mode	Description
	Normal	Up to 33.3 kbit/s, with waveshaping
	High-speed	Up to 83.3 kbit/s, without waveshaping
	Wake-up	Like normal mode, but with increased signal levels

Normal mode

This mode is used for normal operation. Bit rates up to 33.3 kbit/s are supported. The output of signals on the Single-wire CAN bus is done with waveshaping. The voltage slew rate and the shape of the rising edge as well as the beginning of the falling edge are controlled. This behavior contributes to the minimization of EM emissions.

High-speed single wire CAN mode

The adaptor provides a high-speed mode for the transfer of software or diagnostic data, for example. Bit rates up to 83.3 kbit/s can be used. In contrast to the normal mode, the waveshaping function is deactivated, i.e. the bus driver is switched on and off as fast as possible, in order to be able to reach higher bit rates. However, the electromagnetic compatibility (EMC) consequently is reduced in comparison to the normal mode. The high-speed mode is only used in special cases and shouldn't be used for regular operation of a single-wire CAN bus.

Wake-up mode

In this mode transmission is done with an increased voltage level in comparison to the normal mode. An activation of all "sleeping" bus nodes in the network results from it. Sleeping bus nodes ignore normal 4 V levels and only react to levels with higher voltage (12 V). Because the adaptor itself does not have a sleep mode, incoming signals are all interpreted in the same manner independently of their level (normal or wake-up). The red/orange LED indicates a received or transmitted signal with wake-up level. After detection the LED switches off with some delay. Thus signals of short duration are also signalled.

Point-to-Point connection

An additional approach to using CAN low-speed networks with fault-tolerant functionality is specified in the ISO 11992 standard. It defines a point-to-point connection for use in e.g. towing vehicles and their trailers. For one vehicle with one trailer, a point-to-point connection is defined (transceiver - receiver).

The nominal data rate is 125 kbit/s with a maximum bus line length of 40 m. The standard defines the bus error management and the supply voltage of 12 V or 24 V. An unshielded twisted pair of wires is defined as the bus medium.

The TRION-CAN module supports 12 V sensor supply voltage.

Cabling example: CAN with sensor supply (point-to-point connection)

Caution: Return path of the power supply (GND) is Pin 6 (GND Power). Do not use Pin 3 (GNDx CAN)!

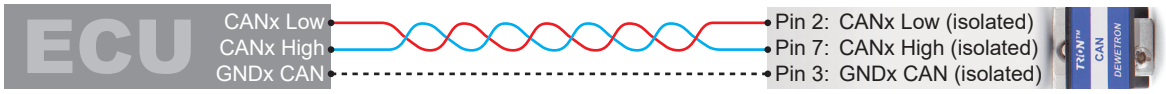
Listen-only mode

The SNHVD235 (transceiver of the TRION-CAN module) supports a listen-only-loopback feature which allows the local node controller to synchronize its baud rate with the CAN bus baudrate. In autobaud / listen-only mode, the transceiver output is placed in a high-impedance state while the receivers bus input remains active.

With the listen-only mode activated, normal bus activity can be monitored by the device. However, if an error frame is generated by the local CAN controller, it is not transmitted to the bus.
Since in listen-only mode the module has no transmit function this feature must not be used in a point-to-point connection.

This mode is designed to evaluate the correct baud rates of existing networks. Once this mode has been selected, assume a desired baud rate, then wait for a message to be transmitted by another node on the bus. If the wrong baud rate has been selected, an error message is generated by the host CAN controller. However, since the transmit function of the device has been disabled, no other devices receive the error message. If an error is generated, reset the CAN device with another baud rate and wait to receive another message. When an error free message has been received, the correct baud rate has been detected.

Cabling example: CAN monitoring (listen-only / autobaud mode)



Note: GNDx CAN is an optional connection when measuring under harsh environment.

Optional accessory

TRION-CBL-D9-OE-05-00

High quality cable from DSUB-9 socket to open end, 5 meters for TRION-CAN-x-D modules.

ADAP-CAN-LS-HS

DSUB-9 adaptor from low-speed CAN to high-speed CAN.

ADAP-CAN-SW-HS

DSUB-9 adaptor from single-wire CAN to high-speed CAN.

ADAP-LIN-CAN

DSUB-9 isolated LIN to CAN adaptor. Requests data from LIN as a LIN-master and outputs the data to CAN.

EtherCAT-Slave interface module

- Measurement data output
- Up to 500 Samples/s
- Up to 100 channels simultaneous

Module specifications

TRION-EtherCAT-1-SLAVE	
I/O Connector	1 EtherCAT Slave IN, RJ45 1 EtherCAT Slave OUT, RJ45
Communication Standard	Ethernet, 10BASE-T/100BASE-TX
Galvanic Insulation	1000 V_{DC}
Compatible Chassis	DEWE2-A4/A7/A13 DEWE2-PA7 DEWE2-M4/M7/M13(s) DEWE2e-PA8
Required Software	DEWETRON OXYGEN 3.4 or higher Hilscher Driver
Required Operating System	6-pin LEMO EPL.1S.306
Measurement Data Update Rate	500 S/s (typ.)
I/O Delay	500 ms (typ.)
Manufacturer Part Number	Hilscher CFIX 80-RE

Note: This module is designed to be used in DEWE2-A, -PA and M-series instruments and is not compatible with F-series and TRIONet.

The TRION-EtherCAT-1-SLAVE is a communication interface module for the DEWE2-series. It can be used in combination with DEWETRON OXYGEN software to provide measurement data in an EtherCAT network via PDO mechanism.

Up to 100 user selectable measurement channels can be provided, combined with a high accurate timestamp. The measurement data is provided via IEEE 754 Floating-Point Number. The configuration is available via a separate ESI-File for use on the EtherCAT master.

Additionally, the input SDOs allow a selection of a measurement setup, the measurement (recording) can be triggered by a PDO control word.

Notes

Analog output module

- 2 channel TRION™ analog output module, based on NI PXI-6251
- Voltage range max. ±10 V
- 16-bit resolution
- Update rate max. 2.8 MS/s

Module specifications

TRION-1628-AO-2	
Analog output channels	2
DAC resolution	16 bits
Output range	±10 V, ±5 V
DNL	±1 LSB
Monotonicity	16 bit guaranteed
Accuracy	
Residual gain error	75 ppm of reading @ ±10 V 85 ppm of reading @ ±5 V
Gain temperature coefficient	17 ppm/°C @ ±10 V 8 ppm/°C @ ±5V
Reference temperature coefficient	1
Residual offset error	40 ppm of range
Offset temperature coefficient	2 ppm of range/°C
INL error	64 ppm of range
Absolute accuracy @ full scale 1	2.08 mV @ ±10 V 1.045 mV @ ±5 V
Maximum update rate	
1 channel	2.86 MS/s
2 channels	1.00 MS/s
Timing accuracy	50 ppm of sample rate
Timing resolution	50 ns
Output coupling	DC
Output impedance	0.2 Ω
Output current drive	±5 mA
Overdrive protection	±25 V
Overdrive current	20 mA
Power-on state	±5 mV2
Power-on glitch	1.5 V peak for 1.5 s
Output FIFO size	8,191 samples shared among channels used
Settling time, full scale step 15 ppm (1 LSB)	2 µs
Slew rate	20 V/µs

TRION-1628-AO-2

TRION-1628-AO-2 analog output model overview

TRION-1628-AO-2

Signal connection

TRION-1628-AO-2 module

68-pin VHDCI connector

Note: *Standard connector of NI-PXI 6251 series is NOT supported!!*

Power supply module

- Compact power supply for 4 active current clamps
- Ultra-low noise
- Power indicator to avoid overload
- Short circuit protected

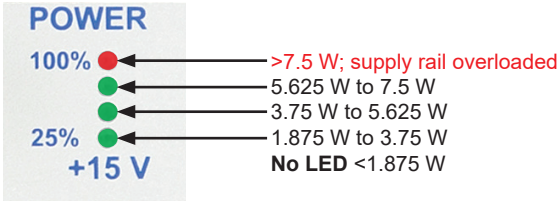
Module specifications

TRION-PSU-15W	
Configuration	Single slot module; 4 supply connectors with common voltage rails
Supported clamps	4 × CLAMP-200-DC 4 × CLAMP-500-DC 4 × CLAMP-1000-DC Other sensors might be connected, as long as they are not exceeding the max. power supply range
+15 V	±1 %; maximum 0.5 A cumulated at all 4 channels (7.5 W); short circuit protected
-15 V	±1 %; maximum 0.5 A cumulated at all 4 channels (7.5 W); short circuit protected
+9 V	±1 %; maximum 0.2 A cumulated at all 4 channels (1.8 W); short circuit protected
Connector	6-pin LEMO EPL.1S.306
CAUTION ⚠ Do not apply external voltage to the module!	

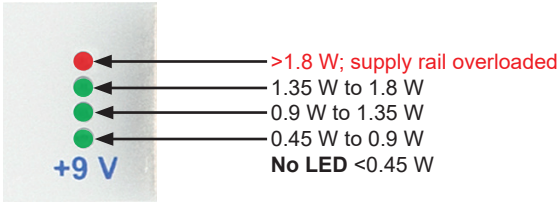
Power limits

The TRION-PSU-15W comes with 4 LEDs for power indication to avoid overload.

+15 V/ -15 V



+9 V



Supply rail overload

Depending on the total power consumption, there are two recommended solutions:

> use an additional TRION-PSU-15W and connect half of the sensors to the second module.

> use an external DW2-CLAMP-DC-POWER-8 for sensors with high power consumption (up to 150 W sensor supply)

Optional accessory: DW2-CLAMP-DC-POWER-8

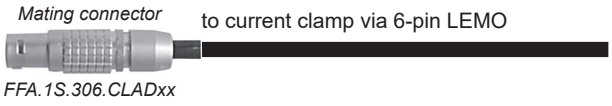
CAUTION: *The power consumption of many current clamps depends on the measured signal. The system might be within supply range when measured current is low, but exceeds if the nominal current is measured. Carefully check the sensor datasheet! Check power indication during measurement, not only in standby!*

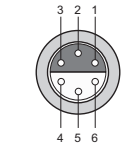
The operation of supported clamps listed in the table, is guaranteed within the complete input range!

NOTE: *This simple power supply TRION module is neither detected by the measurement system, nor displayed or listed in the DEWE2Explorer.*

Signal connection

TRION-PSU-15W module





Lemo EPL.1S.306.HLN

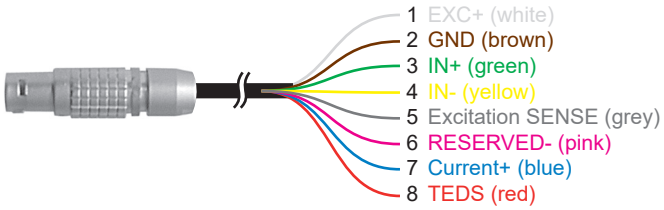
Pin assignment
1: +15 V
2: -15 V
3: +9 V
4: DGND
5: n.c.
6: DGND

General accessories and options for all TRION™ modules

TRION-x-LV-6-L1B module & TRION-1620-ACC-6-L1B module

TRION-CBL-L1B8-OE-05-00

High quality cable from Lemo 1B.308 plug to open end, 5 meters



TRION-CBL-L1B8-BNC-0.5-00

High quality cable from Lemo 1B.308 plug to BNC connector, 0.5 meters

LEMO-FGG.1B.308.CLAD52Z

LEMO FGG.1B.308 mating connector, for cable diameter 4.2 to 5.2 mm

LEMO-FGG.1B.308.CLAD62Z

LEMO FGG.1B.308 mating connector, for cable diameter 5.2 to 6.2 mm

LEMO-FGG.1B.308.CLAD72Z

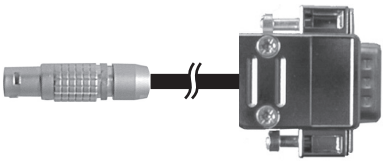
LEMO FGG.1B.308 mating connector, for cable diameter 6.2 to 7.2 mm

TRION-ADAP-CLAMP-L1B8-D9F

High quality adaptor cable from Lemo 1B.308 plug to SUBD-9 socket, 0.5 meters.

For connecting a PNA-CLAMP-1800-DC or SE-CUR-HALL-6 sensor to TRION-16xx-LV-6-L1B modules.

Other sensors on request.



TRION-2402-dACC-x module

TRION-CBL-SMB-OE-05-00

High quality cable from SMB plug to open end, 5 meters

TRION-CBL-SMB-BNC-01-00

High quality adaptor cable from SMB plug to BNC cable-socket, 1 meter

ACCESSORY & OPTIONS

TRION-CAN module

TRION-CBL-D9-OE-05-00

High quality cable from DSUB-9 socket to open end, 5 meters for TRION-CAN-x-D modules.

ADAP-CAN-LS-HS

DSUB-9 adaptor from low-speed CAN to high-speed CAN.

ADAP-CAN-SW-HS

DSUB-9 adaptor from single-wire CAN to high-speed CAN.

ADAP-LIN-CAN

DSUB-9 isolated LIN to CAN adaptor. Requests data from LIN as a LIN-master and outputs the data to CAN.

TRION-CNT / TRION-BASE module

TRION-CBL-L1B7-OE-05-00

High quality cable from LEMO 1B.307 plug to open end, 5 meters for TRION-CNT-6-LEMO and TRION-BASE modules.

LEMO-FGG.1B.307.CLAD42Z

LEMO FGG.1B.307 mating connector,
for cable diameter 3.1 to 4.2 mm

LEMO-FGG.1B.307.CLAD52Z

LEMO FGG.1B.307 mating connector,
for cable diameter 4.2 to 5.2 mm

LEMO-FGG.1B.307.CLAD62Z

LEMO FGG.1B.307 mating connector,
for cable diameter 5.2 to 6.2 mm

LEMO-FGG.1B.307.CLAD72Z

LEMO FGG.1B.307 mating connector,
for cable diameter 6.2 to 7.2 mm

TRION-CBL-CAMTRG-03-00

Camera trigger cable to synchronize a DEWE-CAM-GIGE via an AUX socket of TRION™ modules,
3 meters.

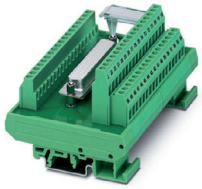
TRION-DI-48 module

TRION-CB24-B

24 channel break-out box with 4 mm banana jacks. 1 m cable, terminated with 50-pin mini-centronics plug.
Note: Two boxes are required for all 48 channels!

TRION-CB24-SC

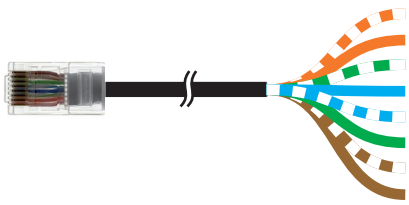
24 channel screw-terminal block, unshielded 1 m cable, terminated with 50-pin mini-centronics plug.
Note: Two blocks are required for all 48 channels!



TRION-2402-dSTG-8-RJ module

TRION-CBL-RJ-OE-05-00

High quality cable from RJ45 plug to open end, 5 meters

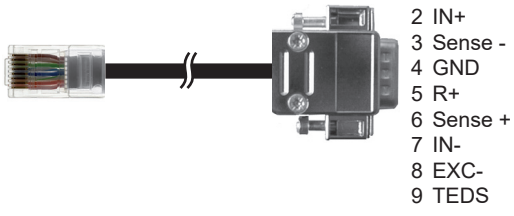


TRION-CBL-RJ-BNC-01-00

High quality cable from RJ45 plug to BNC socket, 1 meter

TRION-CBL-RJ-D9-01-00

High quality adaptor cable from RJ45 plug to DSUB-9 socket, 1 meter

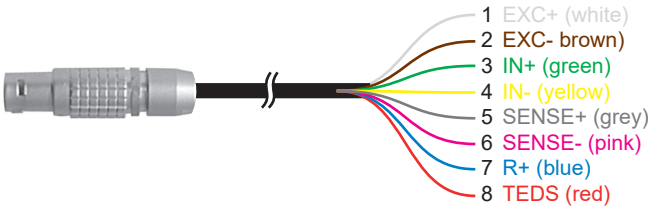


ACCESSORY & OPTIONS

TRION-2402-dSTG-6-L1B module

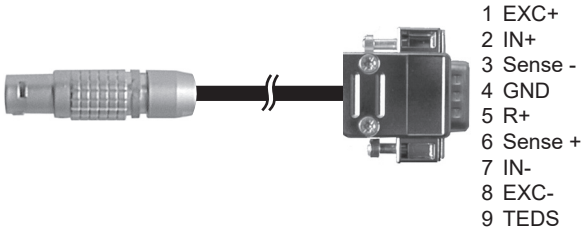
TRION-CBL-L1B8-OE-05-00

High quality cable from Lemo 1B.308 plug to open end, 5 meters



TRION-CBL-L1B8-D9-0.5-00

High quality adaptor cable from LEMO 1B.308 plug to DSUB-9 socket, 0.5 meters



LEMO-FGG.1B.308.CLAD52Z

LEMO FGG.1B.308 mating connector, for cable diameter 4.2 to 5.2 mm

LEMO-FGG.1B.308.CLAD62Z

LEMO FGG.1B.308 mating connector, for cable diameter 5.2 to 6.2 mm

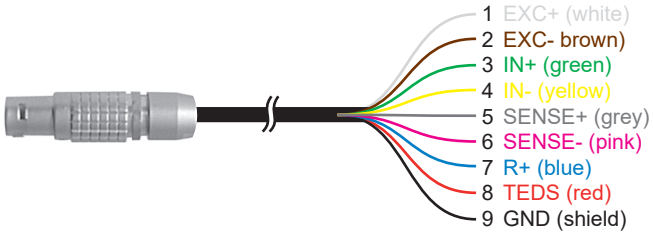
LEMO-FGG.1B.308.CLAD72Z

LEMO FGG.1B.308 mating connector, for cable diameter 6.2 to 7.2 mm

TRION-2402-dSTG-8-L0B module

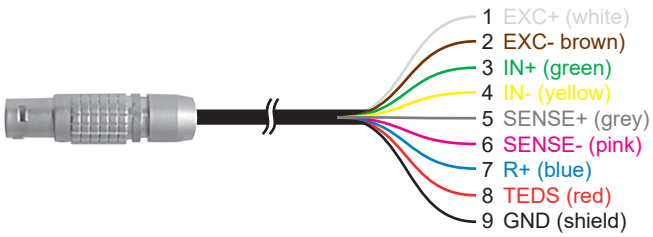
TRION-CBL-L0B9-OE-05-00

High quality cable from Lemo 0B.309 plug to open end, 5 meters.



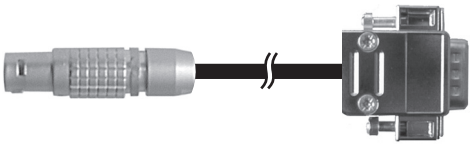
TRION-CBL-L0B9-OE-01-00

High quality cable from Lemo 0B.309 plug to open end, 1 meters.



TRION-CBL-L0B9-D9-0.5-00

High quality adaptor cable from Lemo 0B.309 plug to SUBD-9 socket, 0.5 meters.



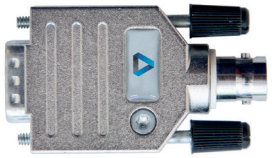
1 EXC+
2 IN+
3 SENSE-
4 GND
5 R+
6 SENSE+
7 IN-
8 EXC-
9 TEDS

TRION-2402-dSTG-8-D module

ADAP-D9M-BNCF

Adapter to directly connect voltage or IEPE® signals to TRION-2402-dSTG-8-D.

1 EXC+
2 IN+
3 Sense -
4 GND
5 R+
6 Sense +
7 IN-
8 EXC-
9 TEDS



Hot: IN+
Shield: IN-

TRION-TIMING

TRION-CBL-CAMTRG-03-00

Camera trigger cable to synchronize a DEWE-CAM-GIGE via an AUX socket of TRION™ modules, 3 meters.

GPS-ANT-1

GPS antenna for TRION-TIMING, for fixed installation.
Only supports GPS L1.

TIMING-ANT-1

IP67 compliant, magnetic GPS antenna for TRION-TIMING for mobile applications. Support of GPS L1, GLONASS G1, BeiDou B1, Galileo E1, SBAS (WAAS, EGNOS & MSAS).

5m cable, SMA plug.

TRION-VGPS-20/-100 module

TRION-CBL-CAMTRG-03-00

Camera trigger cable to synchronize a DEWE-CAM-GIGE via an AUX socket of TRION™ modules, 3 meters.

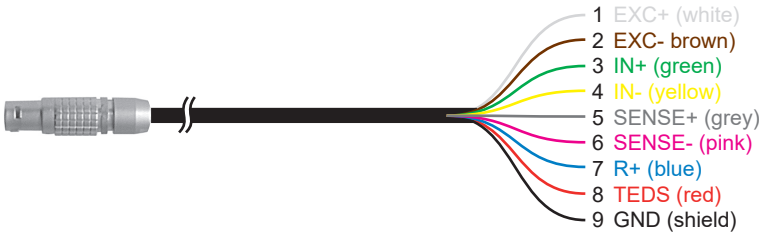
ACCESSORY & OPTIONS

TRION3-18xx-MULTI-8-L0B module

TRION-CBL-L0B9-OE-xx-00

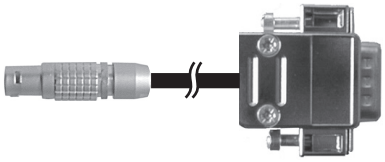
01 = 1 meter
05 = 5 meters

High quality cable from Lemo 0B.309 plug to open end, 1 or 5 meters.



TRION-CBL-L0B9-D9-0.5-02

High quality adaptor cable from Lemo 0B.309 plug to SUBD-9 socket, 0.5 meters.



1 EXC+
2 IN+
3 SENSE+
4 GNDi
5 R+
6 SENSE-
7 IN-
8 EXC-
9 TEDS
H Shield

For connecting any MSI-BR series adaptors

TRION-CBL-L0B9-BNC-0.5-02

High quality adaptor cable from LEMO 0B.309 plug to BNC cable jack, 0.5 meters

For connecting voltage signals and IEPE® sensors to TRION3-18xx-MULTI-8-L0B modules.

Hot: IN +
Shield: IN -

TRION-CBL-L0B9-CAN-0.5

Adaptor cable from LEMO 0B.309 plug to D-SUB-9 plug for CAN, 0.5 meters

For TRION3-18xx-MULTI-8-L0B modules channel 1 only!

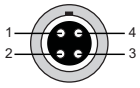
Pin 1: NC
Pin 2: CAN Low (isolated)
Pin 3: GNDx CAN (isolated)
Pin 4: NC
Pin 5: NC
Pin 6: GND Power
Pin 7: CAN High (isolated)
Pin 8: NC
Pin 9: +12 V out

TRION-CBL-L0B9-CPAD-01-00

Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION3-18xx-MULTI-8-L0B (CH 1).

For connecting 1x CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

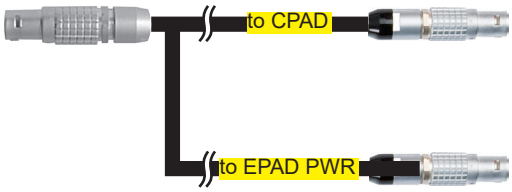
Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-L0B9-CPAD-01-01

Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

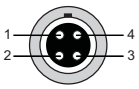
For connecting the first CPAD-series module of a module-chain to a TRION3-18xx-MULTI-8-L0B.



Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND

Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

For connecting more than one CPAD series module



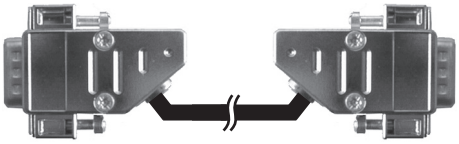
4-pin LEMO connector male FGG.1B.304.CLL

TRION-2402-MULTI-4-D module

TRION-CBL-D9-CAN-0.5

Adaptor cable from D-SUB-9 plug to D-SUB-9 plug for CAN, 0.5 meters.

For TRION-2402-MULTI-4-D modules.

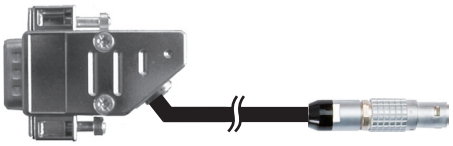


Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

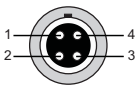
TRION-CBL-D9-CPAD-01-00

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION(3)-18xx-MULTI-4-D (CH 1).



For connecting 1x CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

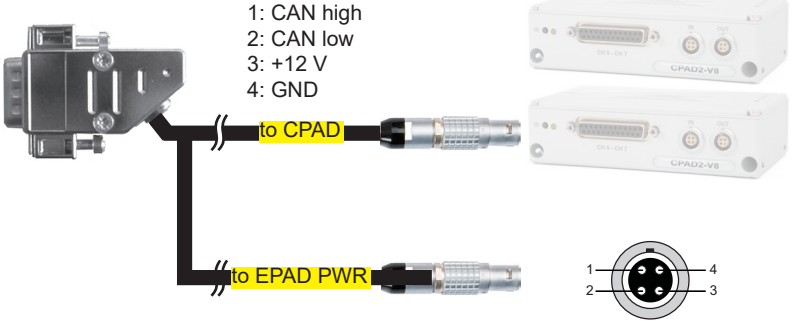
Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-D9-CPAD-01-01

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

For connecting the first CPAD-series module of a module-chain to a TRION(3)-18xx-MULTI-4-D.



Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

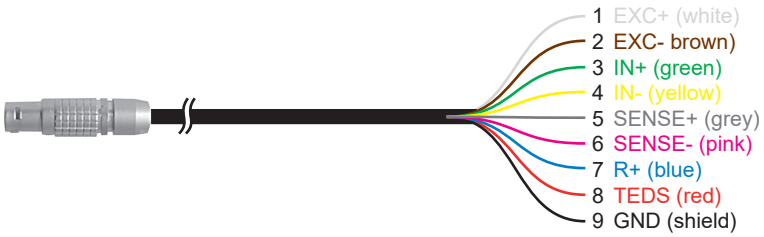
4-pin LEMO connector male FGG.1B.304.CLL

TRION-2402-MULTI-8-L0B module

TRION-CBL-L0B9-OE-xx-00

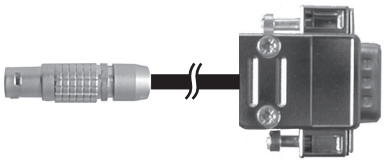
01 = 1 meter
05 = 5 meters

High quality cable from Lemo 0B.309 plug to open end, 1 or 5 meters.



TRION-CBL-L0B9-D9-0.5-01

High quality adaptor cable from Lemo 0B.309 plug to SUBD-9 socket, 0.5 meters.



1 EXC+
2 IN+
3 SENSE+
4 GND
5 R+
6 SENSE-
7 IN-
8 EXC-
9 TEDS

For connecting any MSI-BR series adaptors

TRION-CBL-L0B9-BNC-0.5-01

High quality adaptor cable from LEMO 0B.309 plug to BNC cable jack, 0.5 meters

For connecting IEPE® sensors to TRION-2402-MULTI-8-L0B modules.

NOTE: Do not use this cable for connecting voltage signals!

TRION-CBL-L0B9-BNC-0.5-02

High quality adaptor cable from LEMO 0B.309 plug to BNC cable jack, 0.5 meters

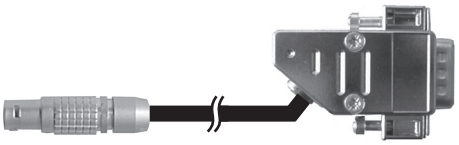
For connecting voltage signals to TRION-2402-MULTI-8-L0B modules.

NOTE: Do not use this cable for connecting IEPE® sensors!

TRION-CBL-L0B9-CAN-0.5

Adaptor cable from LEMO 0B.309 plug to D-SUB-9 plug for CAN, 0.5 meters

For TRION-2402-MULTI-8-L0B modules channel 1 only!



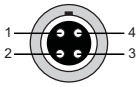
Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

TRION-CBL-L0B9-CPAD-01-00

Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION-2402-MULTI-8-L0B (CH 1).

For connecting 1x CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-L0B9-CPAD-01-01

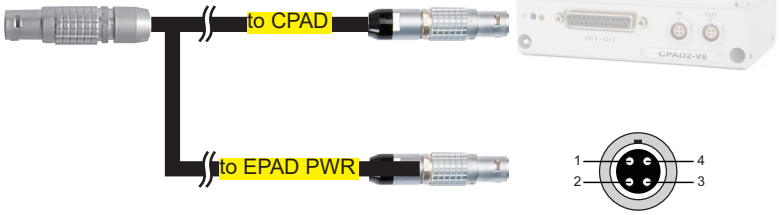
Adaptor cable from LEMO 0B.309 plug to CPAD-series modules, 1 meter.

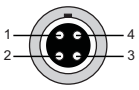
Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

For connecting the first CPAD-series module of a module-chain to a TRION-2402-MULTI-8-L0B.

Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND

For connecting more than one CPAD series module





4-pin LEMO connector male FGG.1B.304.CLL

Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

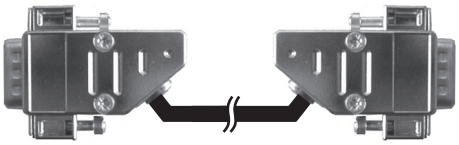
ACCESSORY & OPTIONS

TRION-2402-MULTI-4-D module

TRION-CBL-D9-CAN-0.5

Adaptor cable from D-SUB-9 plug to D-SUB-9 plug for CAN, 0.5 meters.

For TRION-2402-MULTI-4-D modules.

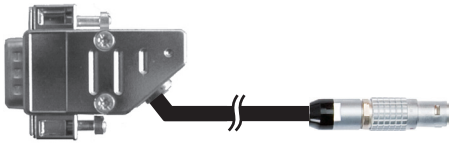


Pin 1:	NC
Pin 2:	CAN Low (isolated)
Pin 3:	GNDx CAN (isolated)
Pin 4:	NC
Pin 5:	NC
Pin 6:	GND Power
Pin 7:	CAN High (isolated)
Pin 8:	NC
Pin 9:	+12 V out

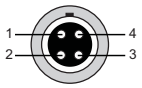
TRION-CBL-D9-CPAD-01-00

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

For connecting exactly one (!) CPAD-series module to a TRION-2402-MULTI-4-D (CH 1).



For connecting 1x CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

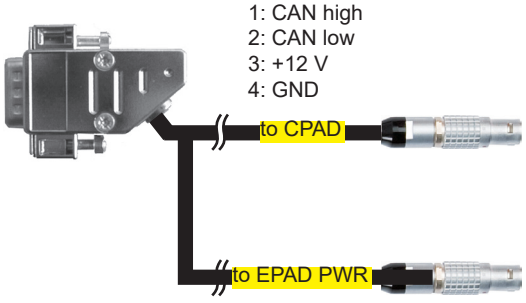
Pin assignment CPAD
1: CAN high
2: CAN low
3: Power supply (+)
4: GND

TRION-CBL-D9-CPAD-01-01

Adaptor cable from D-SUB-9 plug to CPAD-series modules, 1 meter.

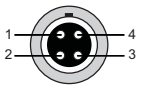
Additional LEMO FGG.1B.304 plug (EPAD) for CPAD-module power supply.

For connecting the first CPAD-series module of a module-chain to a TRION-2402-MULTI-4-D.



Pin assignment to CPAD series module
1: CAN high
2: CAN low
3: +12 V
4: GND

For connecting more than one CPAD series module



4-pin LEMO connector male FGG.1B.304.CLL

Additional power supply (e.g. to EPAD connector of instrument)
1: n.c.
2: n.c.
3: +12 V
4: GND

TRION-1600/1802-dLV module

TRION-x-dLV-CB16-D9

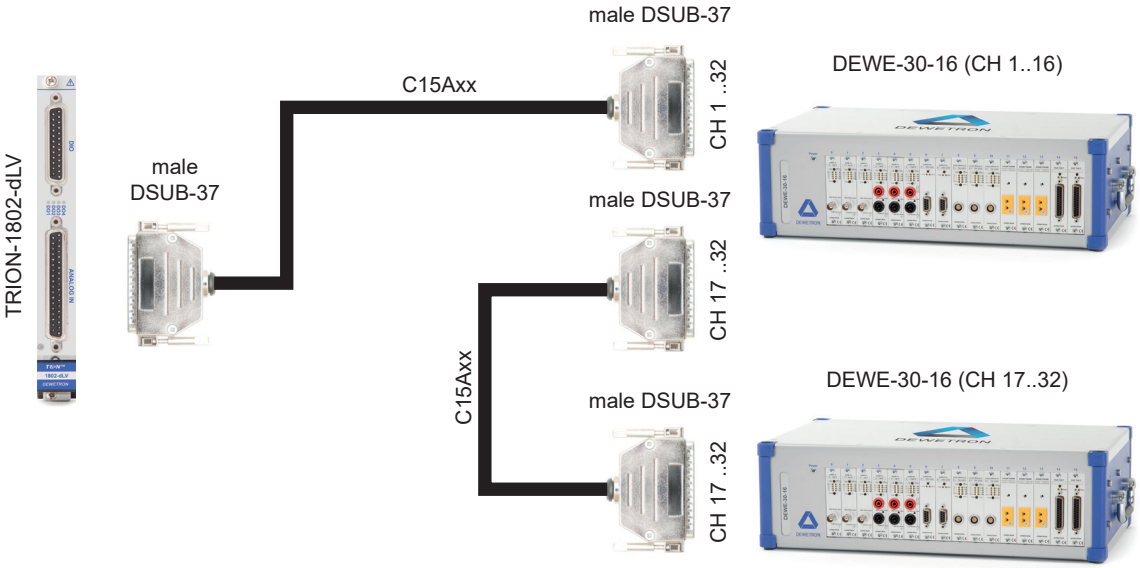
- 16 channel sensor connection box
- Precision ±5 V excitation voltage with remote sense per channel
- MSI support (Modular Smart Interface)
- Auxiliary sensor supply

MODULAR SMART INTERFACES	INPUT	SENSOR EXCITATION	BANDWIDTH (max.) consider limit of used TRION module	ACCURACY (typ.)	SENSOR CONNECTION
MSI2-STG	Bridge-type sensors Full-bridge, half-bridge, quarter bridge 120 Ω and 350 Ω	5 V and 10 V	100 kHz	±0.1 %	Miniature spring terminals
MSI2-LVDT	LVDT and RVDT sensors, 5 wire connection	AC supply 1 to 3.5 V at 2.5, 10 or 20 kHz	1 kHz	±0.1 %	Soldering pads
MSI-BR-ACC	IEPE® sensors, typ. accelerometer, microphone	4 mA	1.4 Hz to 100 kHz	±0.2 %	BNC
MSI2-CH-5 MSI2-CH-100	Charge type sensors up to 100 000 pC	n/a	0.08 Hz to 45 kHz	±0.5 %	BNC
MSI2-TH-K MSI2-TH-J MSI2-TH-T MSI2-TH-x	Thermocouple sensors Standard models for type K, J, T, others on request	n/a	DC to 100 kHz	±1 °C	Mini TC socket
MSI-BR-V-200	Voltage up to 200 V	n/a	DC to 100 kHz	±0.1 %	BNC
MSI-BR-RTD	RTD sensors PT100, Pt200, Pt500, PT1000, PT2000; 2, 3 and 4 wire connection	1.25 mA	DC to 10 kHz	±0.1 %	Binder 712 series 5-pin socket
MSI2-250R-20mA	4 to 20 mA sensors	5 to 48 V AUX PWR	DC to 100 kHz	±0.1 %	Miniature spring terminals

For further information please refer to the '*TRION-x-dLV-CB16-D9 / MSI2 Series*' technical reference manual.
https://ccc.dewetron.com/dl/TRION-x-dLV-CB16-D9

C15Axx

Cable for connecting 16 or 32 channels to a TRION-1802-dLV module in **single ended** configuration.
Two C15Axx cables are required for 32 channels.



ACCESSORY & OPTIONS

TRION-2402-V module

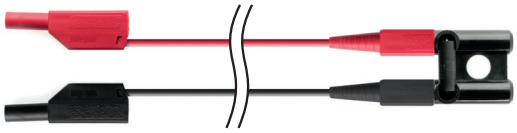
SE-CUR-SHUNT-01

20 mA shunt adaptor (50 Ω, ±0.1 %, 1 W)
Fitting into 4 mm banana jacks with 19 mm distance.
Note: Not compatible with TRION-2402-V-4-B module!!

SE-CUR-SHUNT-01
(not compatible with TRION-x-V-4 modules)

SE-CUR-SHUNT-01-UNI

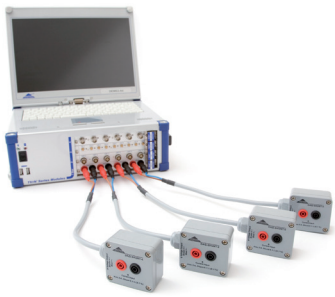
20 mA shunt adaptor (50 Ohm, ±0.1 %, 1 W);
Fitting into 4 mm banana jacks -
also for TRION-x-V-4-B



SE-CUR-SHUNT-01-UNI
(50 cm cable length, also for TRION-x-V-4 modules)

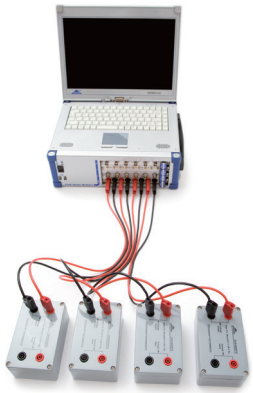
SE-CUR-SHUNT-04

5 A shunt box (100 mΩ, ±0.1 %, <10 ppm)
Current input via 2 safety banana jacks, output 2x 0.3 m cable with banana plugs.



SE-CUR-SHUNT-05

5 A shunt box (100 mΩ, ±0.1 %, <10 ppm)
Current input via 2 safety banana jacks, output via 2 safety banana jacks.



SE-CUR-SHUNT-06

20 A shunt box (5 mΩ, ±0.1 %, CAT III 500 V), Current input via 2 safety banana jacks, Voltage output via 2 safety banana jacks.

General

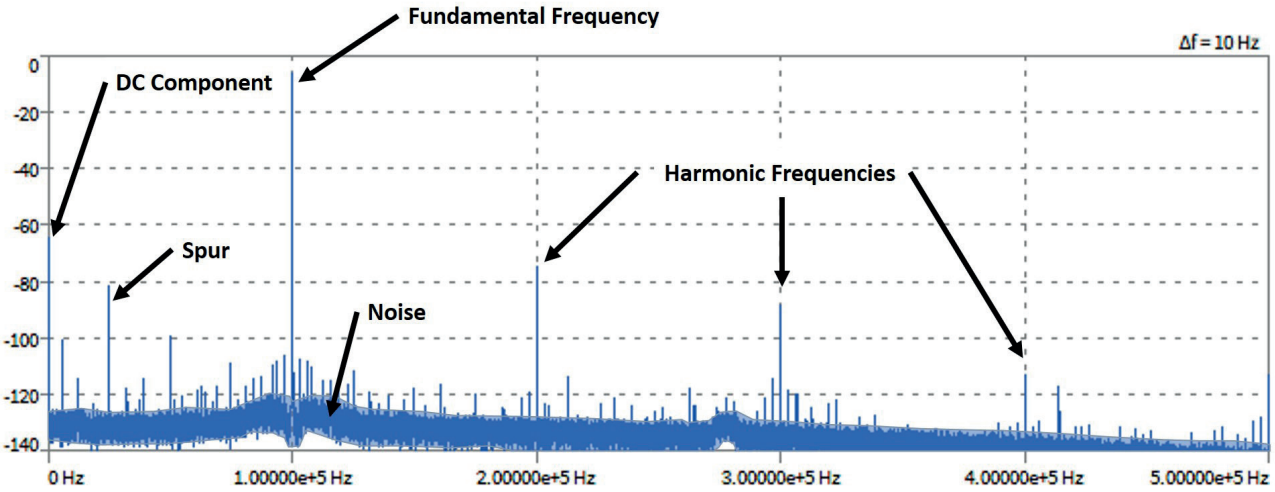


Figure 1: *FFT chart of TRION-1620-ACC*

Fundamental Frequency

The fundamental frequency is the component with the lowest frequency of a periodic test-signal. In the case of an ideal sine wave, the only frequency which would appear is the fundamental frequency.

Harmonic Frequencies

Harmonic frequencies or Harmonics, as they are sometime called, are frequencies that are multiples of the fundamental frequency. These disturbances are most likely caused by amplifier components and the function generator itself, which is used for testing. This is the reason why only special ultra-low noise function generators should be used for measurement.

Spurs

These are frequency components that appear in signals due to the electrical components of the instrument, but which are not harmonics.

Noise

Noise includes all voltage and frequency components in the signal which are present during measurement or generation but not present in the ideal or present signal, except for spurs and harmonics.

DC Component

The DC Component is a spur with a frequency of 0 Hz.

APPENDIX

Testing

General Information about testing

For the measurement of the SNR, SFDR, THD and CMRR the eighth-order Bessel low-pass filter is set in Auto-mode, if a filter is available and if it is not stated otherwise. While measuring the SNR and SFDR a short circuit is placed at the channel of testing. At the measurement of THD a sine wave is used as the input signal for testing. At a sample rate higher than 100 kS/s, a frequency of 1 kHz and an amplitude of 70% of the maximum input range is used. To measure the CMRR, both inputs are provided with the same sinusoidal signal which amplitude should be adjusted to a value that utilizes the range used for measurement. The CMRR is then directly measured from the FFT chart. As a function generator only ultra-low distortion function generators should be used. In our case we use the Model D360 Ultra Low Distortion Function Generator from Stanford Research Systems.

SNR

The SNR, or Signal-to-noise ratio, is the ratio of the input power value to the root-mean-square value of the noise power value. The RMS of the noise power excludes the fundamental frequency, all harmonics and spurs and the DC component.

To calculate the SNR, the AC_{RMS} is measured with a short circuit on the input channel. Afterwards the SNR is calculated depending on the range, used during measurement, with the equation below. Every AC_{RMS} measurement, which is taken over the period of 1 second, is done five times and the mean of these measurements is used in the equation.

$$SNR\ [dB] = \left| 20 \times log_{10} \left(\frac{AC_{RMS}\ [V] \times \sqrt{2}}{range\ [V]} \right) \right|$$

Equation 1: *calculating the SNR from AC_{RMS} depending on the range*

Example for TRION-1802/1600-dLV-32 at 10 V range and a sample rate of 50 kS/s:

$$SNR\ [dB] = \left| 20 \times log_{10} \left(\frac{(5.6068\ V \times 10^{-5}) \times \sqrt{2}}{10\ V} \right) \right| = 102\ dB$$

Example 1: *calculation of SNR*

SFDR

The SFDR, or spurious free SNR or spurious free dynamic range can be defined as the free range between the signal amplitude of the fundamental frequency and the spur with the heights power value, excluding all harmonics and the DC component as shown in Figure 2.

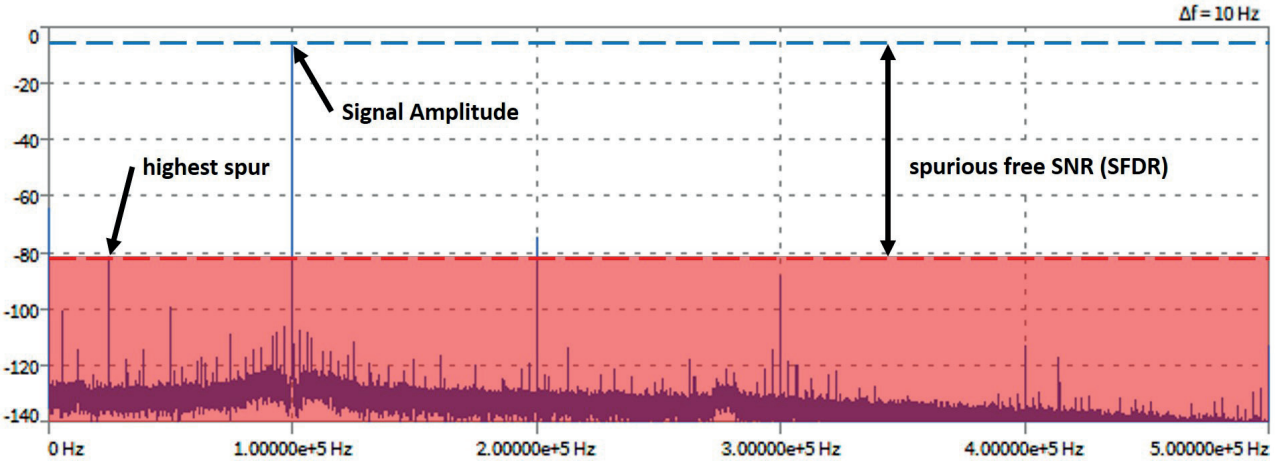


Figure 2: *SFDR in a FFT chart with input signal (alternative method)*

The SFDR can also be measured as the highest spur with a short circuit on the input channel. This method, as used by DEWETRON, automatically excludes all harmonics and defines the SFDR as the highest spur seen in the FFT chart, excluding the DC component, as shown in Figure 3.

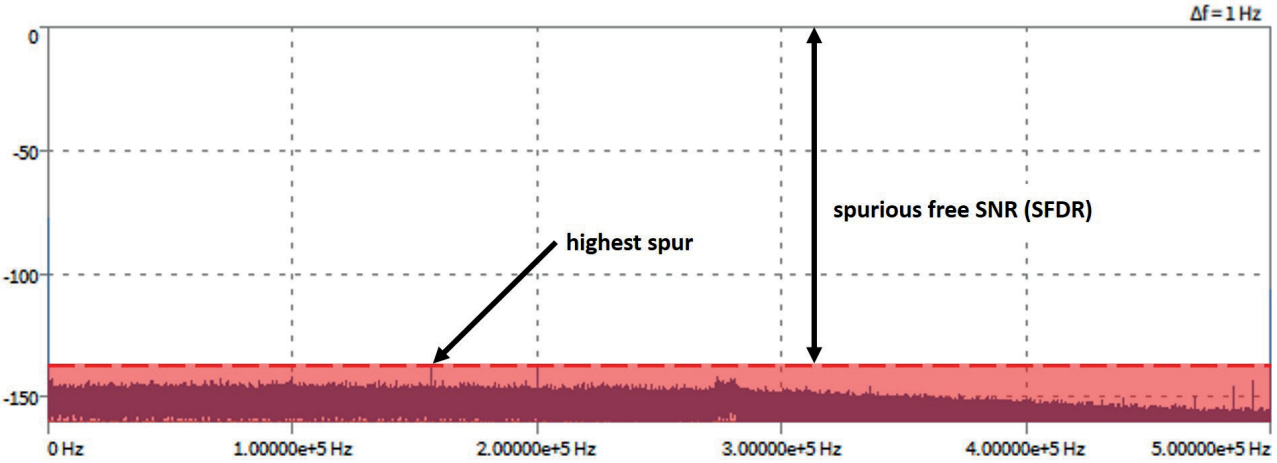


Figure 3: *SFDR in a FFT chart without input signal (DEWETRON)*

APPENDIX

ENOB

The ENOB, or Effective Number of Bits, is a characteristic value that relates the SNR with bits of resolution, a common specification of data converters. The ENOB is directly calculated from the SNR.

$$ENOB\ [Bit] = \frac{SNR\ [dB] - 1.76\ dB}{6.02\ dB}$$

Equation 2: *calculating the ENOB from SNR*

Example for TRION-1802/1600-dLV-32 at 10 V range and a sample rate of 50 kS/s:

$$ENOB\ [Bit] = \frac{102\ dB - 1.76\ dB}{6.02\ dB} = 16.65\ Bit \approx 16.7\ Bit$$

Example 2: *calculation of ENOB*

THD

The THD, or total harmonic distortion, is defined as the root-mean-square value of the first five harmonics of the fundamental frequency compared to the fundamental frequency. It is possible to calculate the THD as it is shown in equation 3 (with harmonics stated in [dB]) and equation 4 (with harmonics stated in [V]), if the amplitudes of the harmonics are expressed with respect to the input frequency. Figure 5 shows how the harmonics are measured from the FFT, when the maximum peak in the FFT is equal to 0 dB.

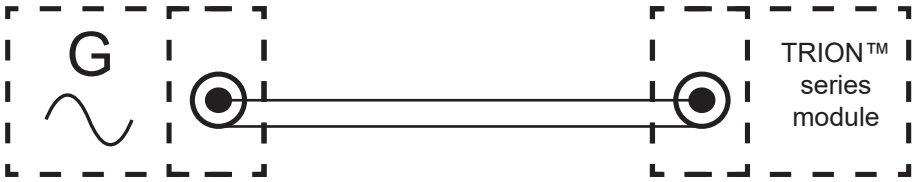


Figure 4: *schematic circuit diagram of THD measurement*

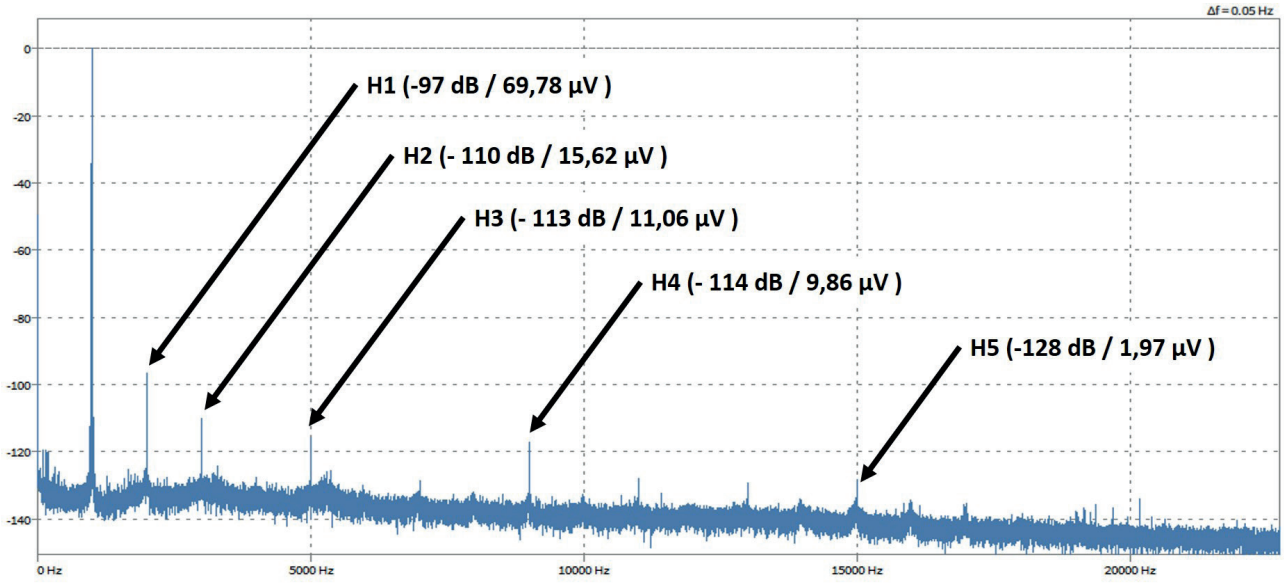


Figure 5: THD measurement of TRION-1802/1600-dLV-32 (@ 50 kS/s) where U_r = 4,94 V equals 0 dB

$$THD\ [dB] = 10 \times log_{10}\left(10^{\frac{H_1\ [dB]}{10\ dB}} + 10^{\frac{H_2\ [dB]}{10\ dB}} + 10^{\frac{H_3\ [dB]}{10\ dB}} + 10^{\frac{H_4\ [dB]}{10\ dB}} + 10^{\frac{H_5\ [dB]}{10\ dB}}\right)$$

Equation 3: *Calculating the THD from harmonics [dB] -*
H: Harmonics [dB]

$$THD\ [dB] = 20 \times log_{10}\left(\frac{\sqrt{U_{H1}{}^2\ [V] + U_{H2}{}^2\ [V] + U_{H3}{}^2\ [V] + U_{H4}{}^2\ [V] + U_{H5}{}^2\ [V]}}{U_f\ [V]}\right)$$

Equation 4: *calculating the THD from harmonics [V] -*
U_H: Harmonics [V], U_f: Fundamental (amplitude of test signal) [V]

Example for TRION-1802/1600-dLV-32 at 10 V range and a sample rate of 50 kS/s with U_f = 4.94 V; (data from figure 5):

$$THD = 10 \times log_{10}\left(10^{\frac{-97\ dB}{10\ dB}} + 10^{\frac{-110\ dB}{10\ dB}} + 10^{\frac{-113\ dB}{10\ dB}} + 10^{\frac{-114\ dB}{10\ dB}} + 10^{\frac{-128\ dB}{10\ dB}}\right) = -96.6\ dB$$

Example 3: *calculation of THD from harmonics [dB]*

$$THD = 20 \times log_{10}\left(\frac{\sqrt{(69.78\ V \times 10^{-6})^2 + (15.62\ V \times 10^{-6})^2 + (11.06\ V \times 10^{-6})^2 + (9.86\ V \times 10^{-6})^2 + (1.97\ V \times 10^{-6})^2}}{4.94\ V}\right) = -96.6\ dB$$

Example 4: *calculation of THD from harmonics [V]*

CMRR

The CMRR or common-mode rejection ratio of an ADC in differential mode (ADC input voltage is the difference between the two inputs) is the capability to filter out the input signal which is common to both inputs. It is often the case that noise is common to both terminals while the relevant information is contained in the voltage difference between the two inputs. A high CMRR results in a good noise rejection common to both terminals while the relevant signal information is preserved. To measure the CMRR the same input signal is applied to both inputs, as seen in figure 6 and afterward directly measured from the FFT chart as seen in figure 7.

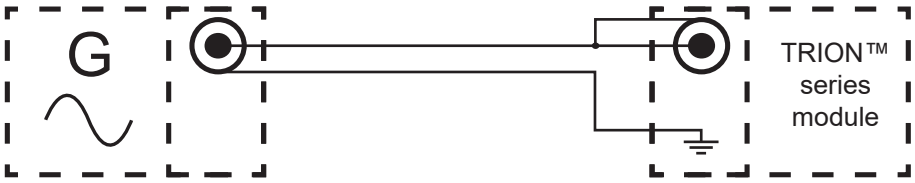


Figure 6: *schematic circuit diagram of CMRR measurement*

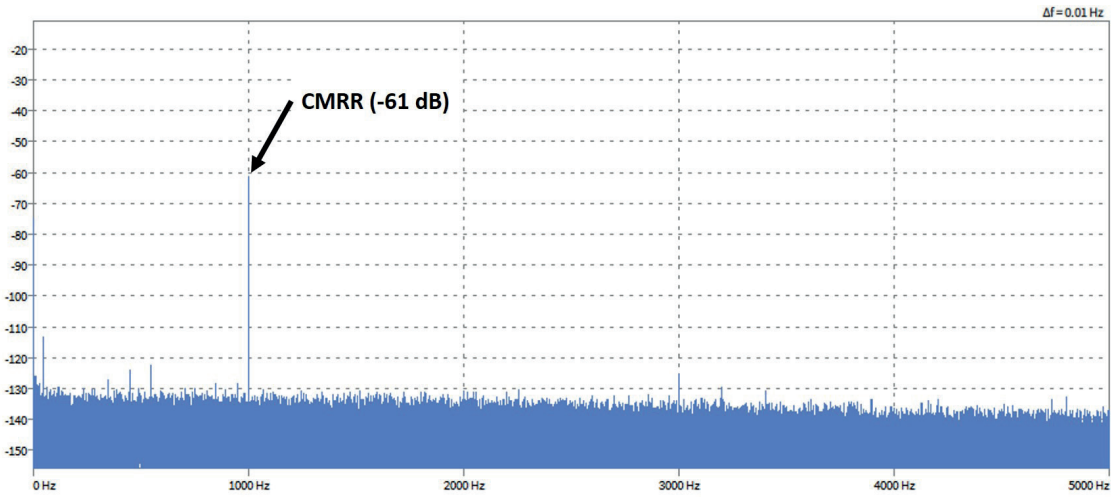


Figure 7: *measurement of CMRR in FFT chart of TRION-1620-ACC (>2 V range @ 1 kHz)*

Figure 8 shows the CMRR response, depending on the input frequency, of all four voltage channels of the TRION-1820-POWER-4.

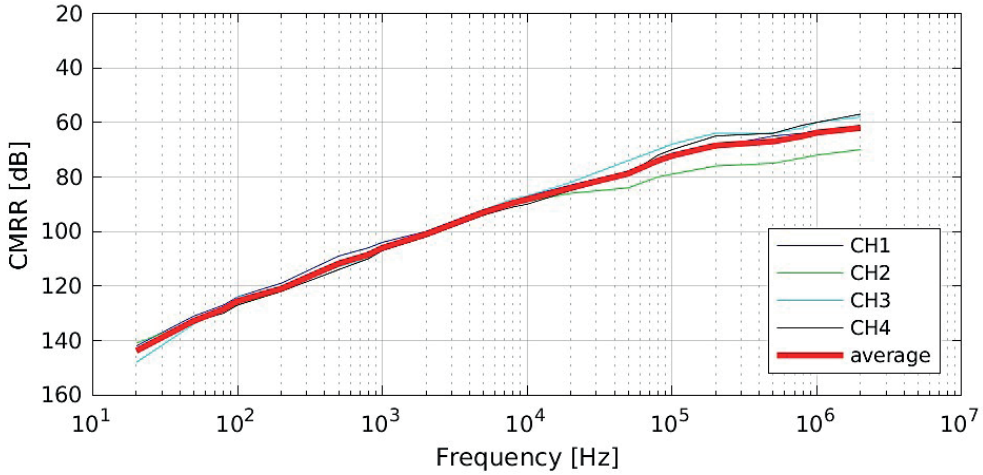


Figure 8: *CMRR over frequency response of a TRION-1820-POWER-4 (voltage inputs)*
(not a directly measured FFT chart)

Phase mismatch

We state all our phase mismatch values Δt in the unit nanoseconds [ns]. To convert the phase mismatch in the unit degree [deg] equation 5 is needed, as shown below. As the test-frequency f, we typically use 1 kHz.

$$\varphi\,[deg] = \frac{\Delta t\,[ns] \times 360\ deg \times f\,[Hz]}{10^9} \quad \Leftrightarrow \quad \Delta t\,[ns] = \frac{\varphi\,[deg]}{360\ deg \times f\,[Hz]} \times 10^9$$

Equation 5: *calculating phase mismatch from [ns] in [deg] and conversely*

Example for TRION-1802/1600-dLV-32 at 10 V range at 1 kHz test signal measured between CH1 and CH2:

$$\varphi\,[deg] = \frac{18.33\ ns \times 360\ deg \times 1000\ Hz}{10^9} \approx 0.0066\ deg \Leftrightarrow \Delta t\,[ns] = \frac{0.0066\ deg}{360\ deg \times 1000\ Hz} \times 10^9$$

Example 5: *calculating phase mismatch from [ns] in [deg] and conversely*

CE-Certificate of Conformity

Manufacturer: **DEWETRON GmbH**

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e-mail: sales@dewetron.com
http://www.dewetron.com

Name of product: **TRION™/TRION3™ Modules**

Kind of product: *Sensor connection box for modular smart interfaces*

The product meets the regulations of the following EC-directives:

2014/35/EU
"Directive of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of electrical equipment designed for use within certain voltage limits"

2014/30/EU
"Directive of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to electromagnetic compatibility (recast)"

The accordance is proved by the observance of the following standards:

LV	**Safety**	IEC 61010-1:2011	
EMC	**Emissions**	EN 61000-6-4	EN 55011 Class B
EMC	**Immunity**	EN 61000-6-2	Group standard

Graz, December 05, 2016
Place / Date of the CE-marking

Ing. Thomas Propst / Manager Total Quality

NOTES